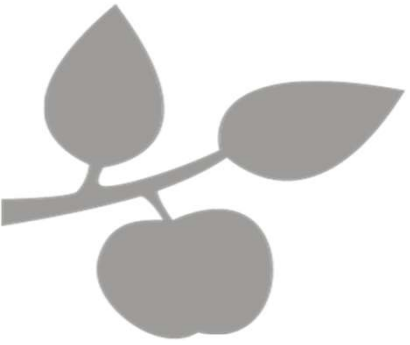




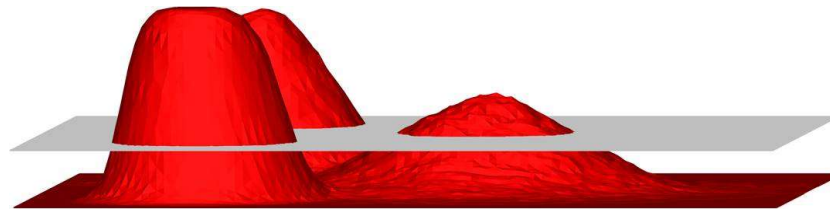
DM583

Data Mining

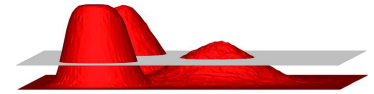
Density-Based Clustering

Density-Based Modelling

- Classic approach to data analysis both in statistics and CS
- Goal is to model an underlying multivariate data distribution $f(\mathbf{x})$ (p.d.f.)
 - Pattern extraction
 - Inference
 - Prediction

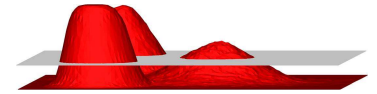


Density-Based Modelling



- Classic approach to data analysis both in statistics and CS
- Goal is to model an underlying multivariate data distribution $f(\mathbf{x})$ (p.d.f.)
 - Pattern extraction, inference, prediction
- Basic Building Block: **Density Estimate**
 - Parametric Approaches (e.g., EM-GMMs)
 - **Non-Parametric Approaches**
 - More prevailing in the “Big Data” context and CS community
 - Do not assume any particular distribution or constrained spatial patterns
 - E.g., Kernel-Based density estimate, K-Nearest Neighbours (KNN) density estimate

Density-Based Modelling



- Classic approach to data analysis both in statistics and CS
- Goal is to model an underlying multivariate data distribution $f(\mathbf{x})$ (p.d.f.)
 - Pattern extraction, inference, prediction
- Applications:
 - **Clustering**
 - Outlier Detection
 - Semi-Supervised Classification
 - ...

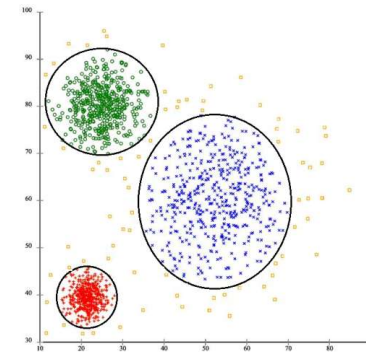
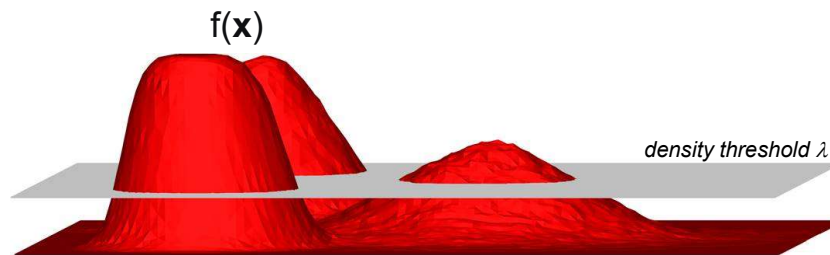
Density-Based Clustering

- **Density-Based Clustering Paradigm**
 - Clusters are contiguous, high-density regions of the data space
 - These regions are separated by low-density regions
 - This follows the classic statistical principle of **density-contour clusters**
 - It is an alternative to the prototype-based clustering paradigm
 - E.g., K-means and variants

Density-Contour Clusters

- **Classic statistical principle** [J.A. Hartigan, “*Clustering Algorithms*”, J. Wiley & Sons, 1975]

- *maximal connected subsets* of the level set $\{\mathbf{x} \mid f(\mathbf{x}) \geq \lambda\}$

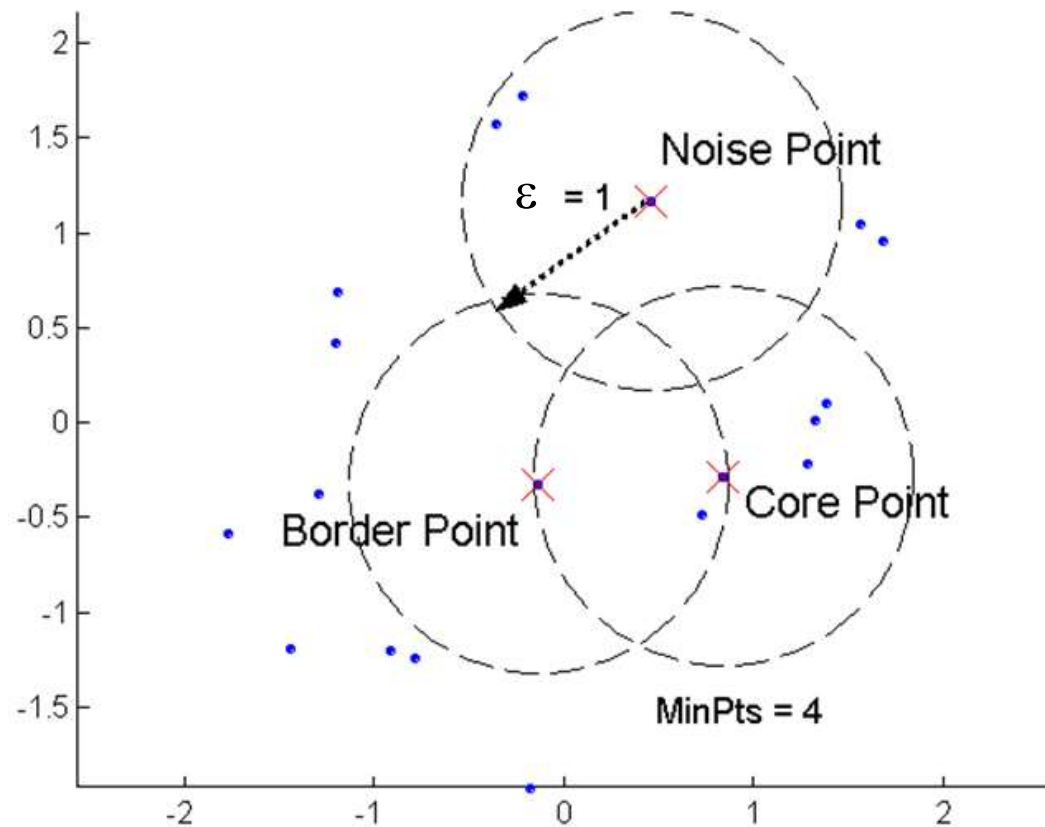


- “dense regions separated by low-density regions”
- basic principle behind many density-based algorithms
 - One of the most famous and widely used is DBSCAN

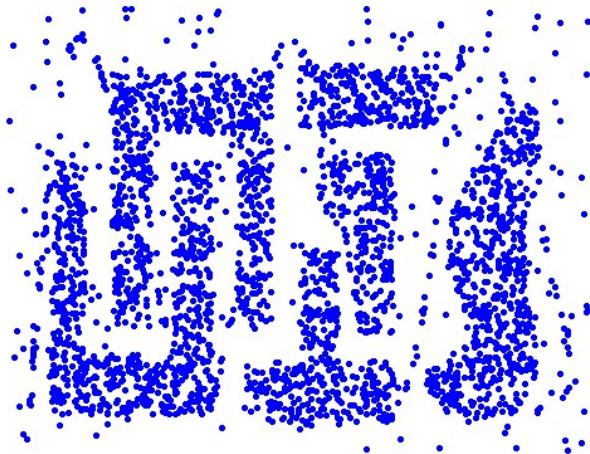
DBSCAN

- DBSCAN is a *non-parametric* density-based clustering algorithm
 - It implicitly uses a (coarse) non-parametric density estimate
 - number of points within a specified radius ε from the query point
- Definitions:
 - A point is a **core point** if it has at least a specified number of points (**MinPts**) within the radius ε (including the point itself)
 - these are points that are in the interior of a cluster
 - A **border point** has fewer than MinPts within ε , but is in the neighborhood (within the radius) of at least one core point
 - A **noise point** is neither a core point nor a border point

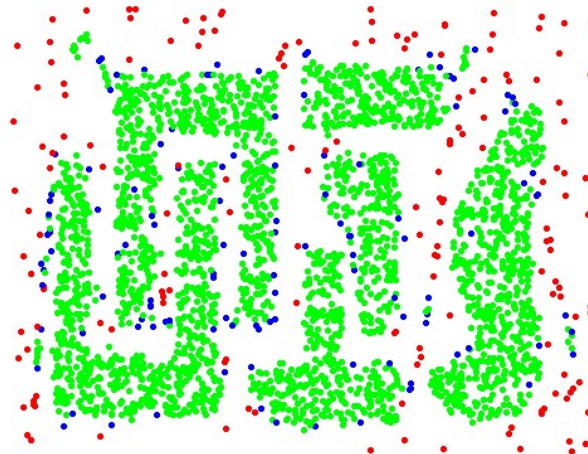
DBSCAN: Core, Border, and Noise Points



DBSCAN: Core, Border and Noise Points



Original Points



Point types: **core**,
border and **noise**

$\epsilon = 10$, MinPts = 4

The DBSCAN Approach

Neighborhood and Core Points

Define a ball of radius ϵ around a point $\mathbf{x}$, called the ϵ -neighborhood of $\mathbf{x}$:

$$N_\epsilon(\mathbf{x}) = B_d(\mathbf{x}, \epsilon) = \{\mathbf{y} \mid \delta(\mathbf{x}, \mathbf{y}) \leq \epsilon\}$$

Here $\delta(\mathbf{x}, \mathbf{y})$ represents the distance between points $\mathbf{x}$ and $\mathbf{y}$. which is usually assumed to be the Euclidean (**note**: the algorithm works with any distance function though)

We say that $\mathbf{x}$ is a *core point* if there are at least *minpts* points in its ϵ -neighborhood, i.e., if $|N_\epsilon(\mathbf{x})| \geq \text{minpts}$.

A *border point* does not meet the *minpts* threshold, i.e., $|N_\epsilon(\mathbf{x})| < \text{minpts}$, but it belongs to the ϵ -neighborhood of some core point $\mathbf{z}$, that is, $\mathbf{x} \in N_\epsilon(\mathbf{z})$.

If a point is neither a core nor a border point, then it is called a *noise point* or an outlier.

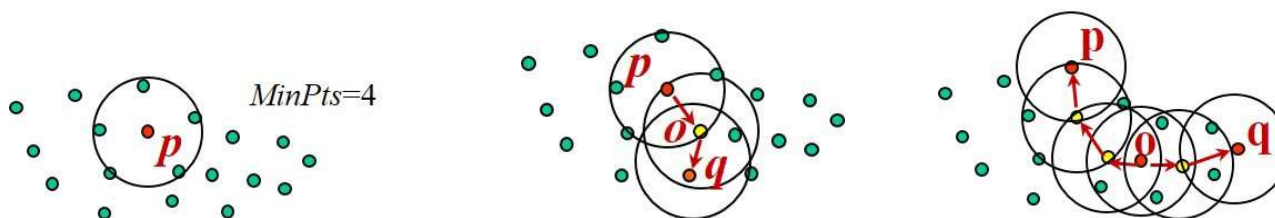
The DBSCAN Approach

Reachability and Density-based Cluster

A point x is *directly density reachable* from another point y if $x \in N_\epsilon(y)$ and y is a core point.

A point x is *density reachable* from y if there exists a chain of points, $x_0, x_1, \dots, x_l$, such that $y = x_0$ and $x = x_l$, and x_i is directly density reachable from x_{i-1} for all $i = 1, \dots, l$. In other words, there is set of core points leading from y to x .

Two points x and y are *density connected* if there exists a core point z , such that both x and y are density reachable from z .



The DBSCAN Approach

Reachability and Density-based Cluster

A point $\mathbf{x}$ is *directly density reachable* from another point $\mathbf{y}$ if $\mathbf{x} \in N_\epsilon(\mathbf{y})$ and $\mathbf{y}$ is a core point.

A point $\mathbf{x}$ is *density reachable* from $\mathbf{y}$ if there exists a chain of points, $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_l$, such that $\mathbf{y} = \mathbf{x}_0$ and $\mathbf{x} = \mathbf{x}_l$, and $\mathbf{x}_i$ is directly density reachable from $\mathbf{x}_{i-1}$ for all $i = 1, \dots, l$. In other words, there is set of core points leading from $\mathbf{y}$ to $\mathbf{x}$.

Two points $\mathbf{x}$ and $\mathbf{y}$ are *density connected* if there exists a core point $\mathbf{z}$, such that both $\mathbf{x}$ and $\mathbf{y}$ are density reachable from $\mathbf{z}$.

A *density-based cluster* is defined as a maximal set of density connected points.

DBSCAN Density-based Clustering Algorithm

DBSCAN computes the ϵ -neighborhood $N_\epsilon(\mathbf{x}_i)$ for each point $\mathbf{x}_i$ in the dataset $\mathbf{D}$, and checks if it is a core point. It also sets the cluster id $id(\mathbf{x}_i) = \emptyset$ for all points, indicating that they are not assigned to any cluster.

Starting from each unassigned core point, the method recursively finds all its density connected points, which are assigned to the same cluster.

Some border point may be reachable from core points in more than one cluster; they may either be arbitrarily assigned to one of the clusters or to all of them (if overlapping clusters are allowed).

Those points that do not belong to any cluster are treated as outliers or noise.

DBSCAN Algorithm

```
dbscan ( $\mathbf{D}$ ,  $\epsilon$ ,  $minpts$ ):  
1  $Core \leftarrow \emptyset$   
2 foreach  $\mathbf{x}_i \in \mathbf{D}$  do // Find the core points  
3   Compute  $N_\epsilon(\mathbf{x}_i)$   
4    $id(\mathbf{x}_i) \leftarrow \emptyset$  // cluster id for  $\mathbf{x}_i$   
5   if  $N_\epsilon(\mathbf{x}_i) \geq minpts$  then  $Core \leftarrow Core \cup \{\mathbf{x}_i\}$   
6  $k \leftarrow 0$  // cluster id  
7 foreach  $\mathbf{x}_i \in Core$ , such that  $id(\mathbf{x}_i) = \emptyset$  do  
8    $k \leftarrow k + 1$   
9    $id(\mathbf{x}_i) \leftarrow k$  // assign  $\mathbf{x}_i$  to cluster id  $k$   
10  DensityConnected ( $\mathbf{x}_i, k$ )  
11  $\mathcal{C} \leftarrow \{C_i\}_{i=1}^k$ , where  $C_i \leftarrow \{\mathbf{x} \in \mathbf{D} \mid id(\mathbf{x}) = i\}$   
12  $Noise \leftarrow \{\mathbf{x} \in \mathbf{D} \mid id(\mathbf{x}) = \emptyset\}$   
13  $Border \leftarrow \mathbf{D} \setminus \{Core \cup Noise\}$   
14 return  $\mathcal{C}, Core, Border, Noise$ 
```

```
DensityConnected ( $\mathbf{x}, k$ ):  
15 foreach  $\mathbf{y} \in N_\epsilon(\mathbf{x})$  do  
16    $id(\mathbf{y}) \leftarrow k$  // assign  $\mathbf{y}$  to cluster id  $k$   
17   if  $\mathbf{y} \in Core$  then DensityConnected ( $\mathbf{y}, k$ )
```

DBSCAN Algorithm

```
dbscan ( $\mathbf{D}$ ,  $\epsilon$ ,  $minpts$ ):  
1  $Core \leftarrow \emptyset$   
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13  $Border \leftarrow \mathbf{D} \setminus \{Core \cup Noise\}$   
14 return  $\mathcal{C}, Core, Border, Noise$ 
```

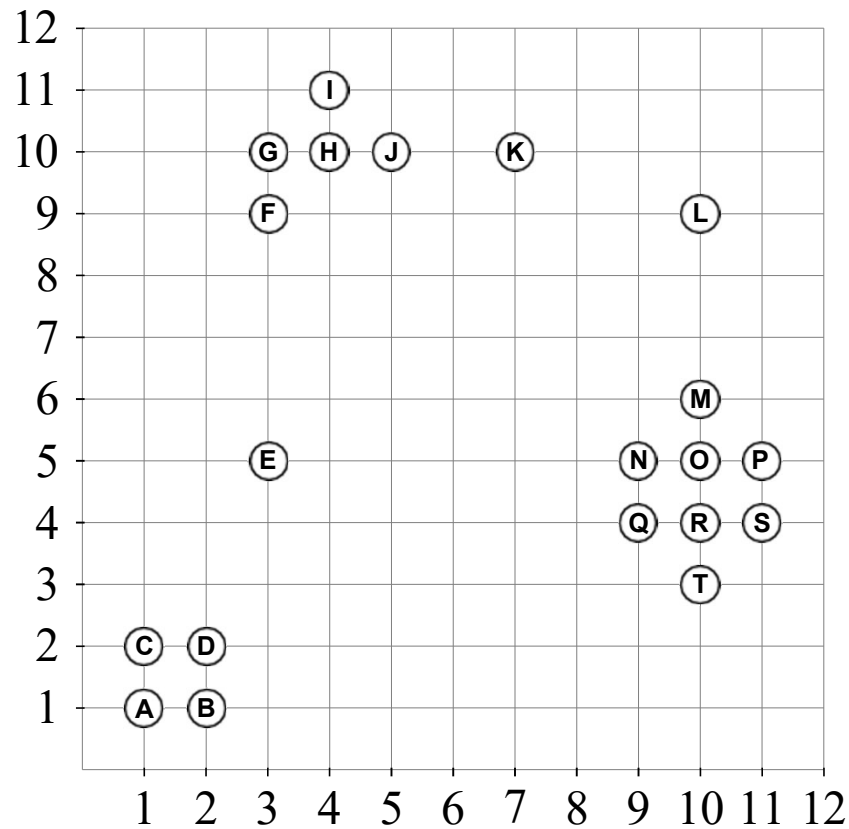
```
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15 foreach  $\mathbf{y} \in N_\epsilon(\mathbf{x})$  do  
16    $id(\mathbf{y}) \leftarrow k$  // assign  $\mathbf{y}$  to cluster id  $k$   
17   if  $\mathbf{y} \in Core$  then DensityConnected ( $\mathbf{y}, k$ )
```

NOTE: This is a **recursive** implementation. Different **iterative** implementations are also possible

Example 1 (Notice the Use of **Manhattan Distance**)

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

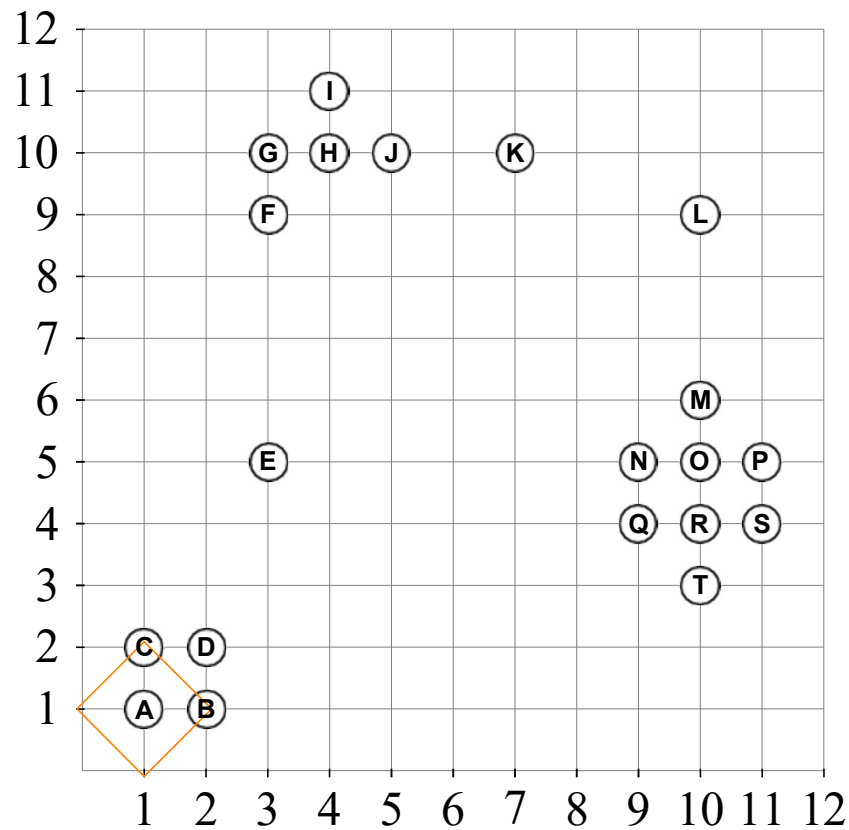
$$\text{minPts} = 2$$

Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

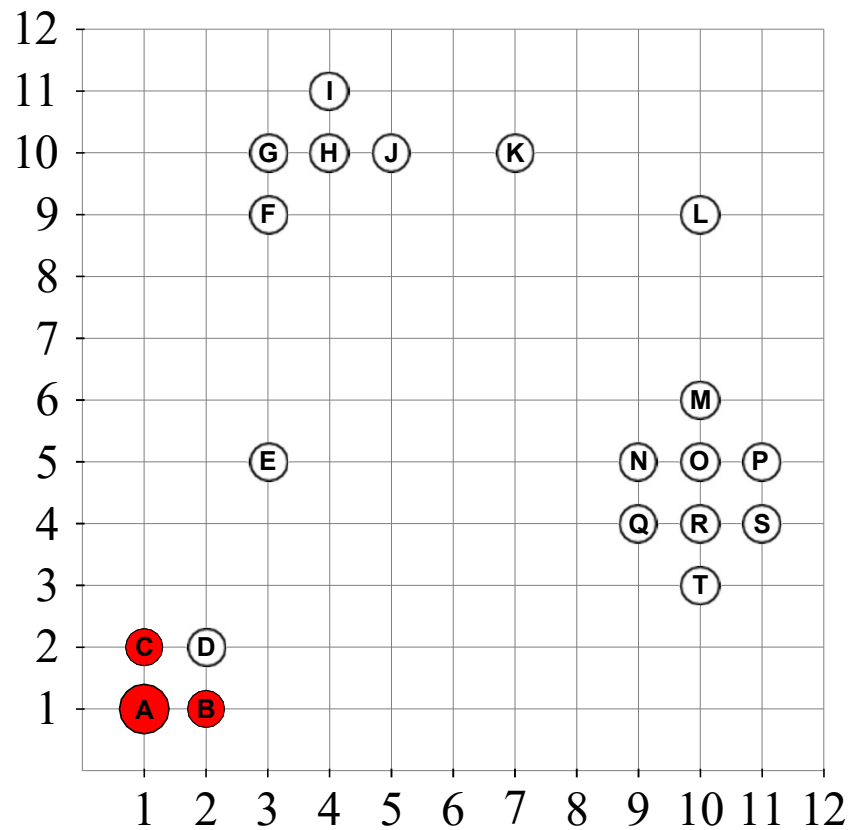
$$\text{minPts} = 2$$

Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

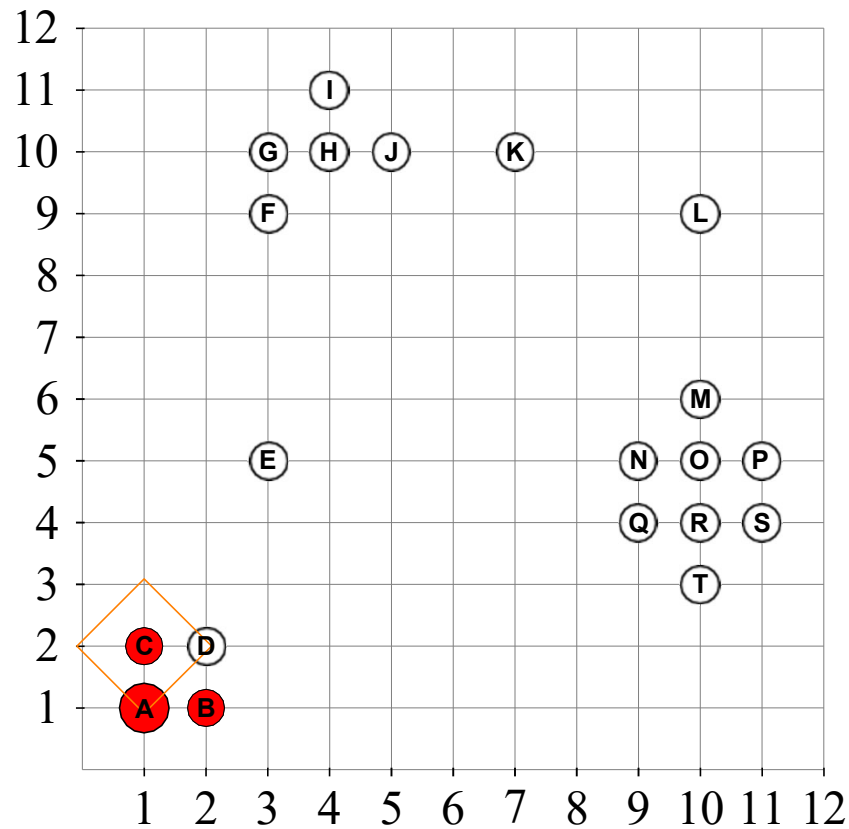
$$\text{minPts} = 2$$

Seed list:
BC

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

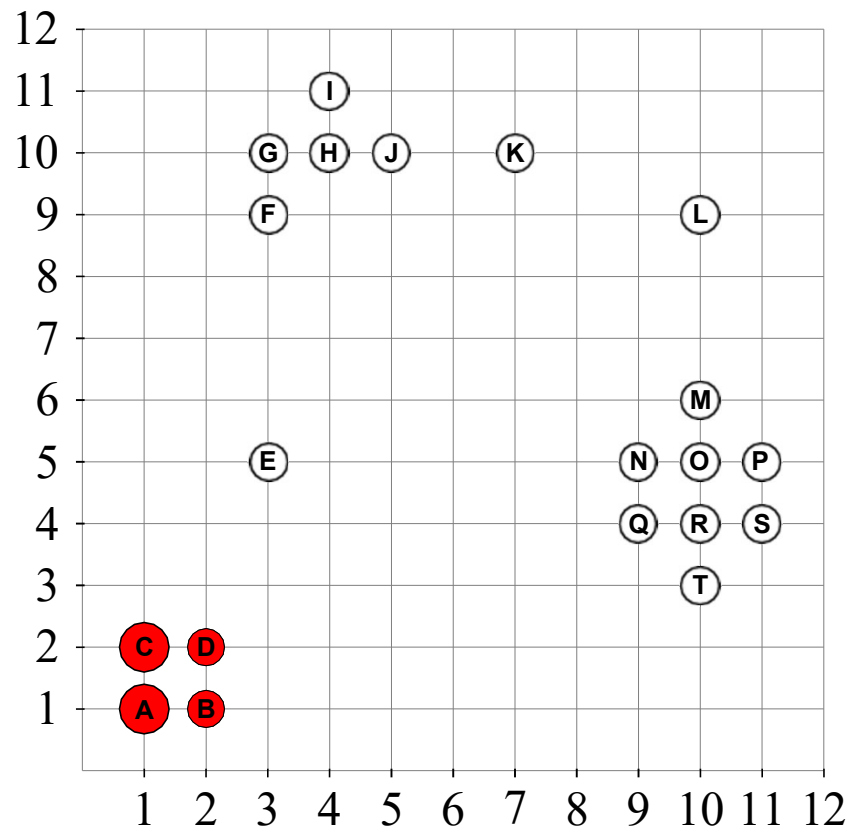
Seed list:

B

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

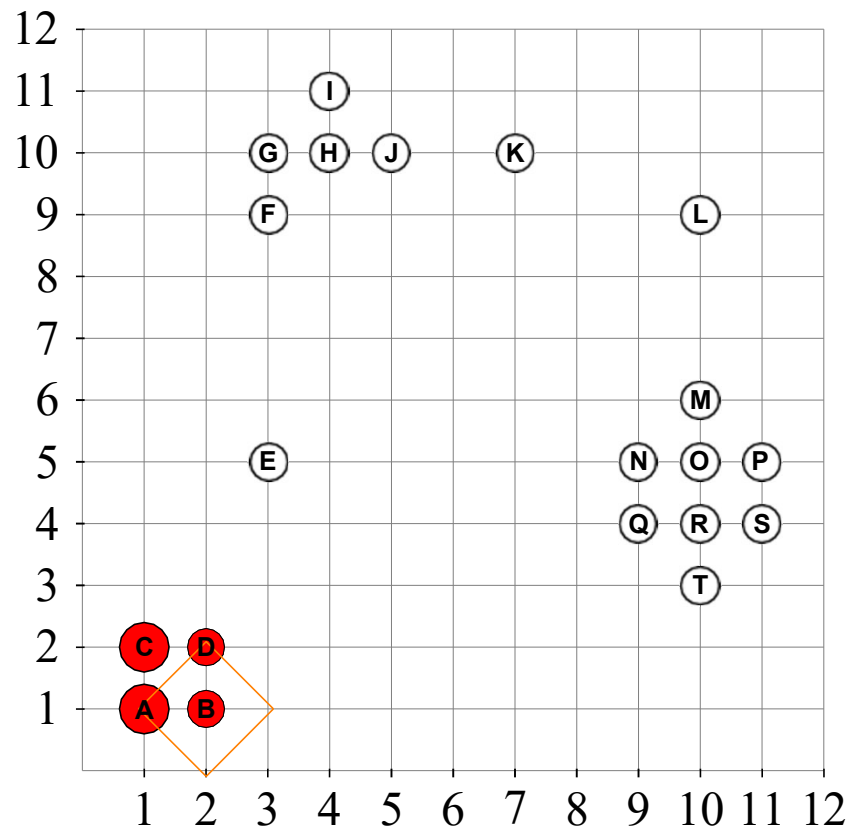
Seed list:

BD

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

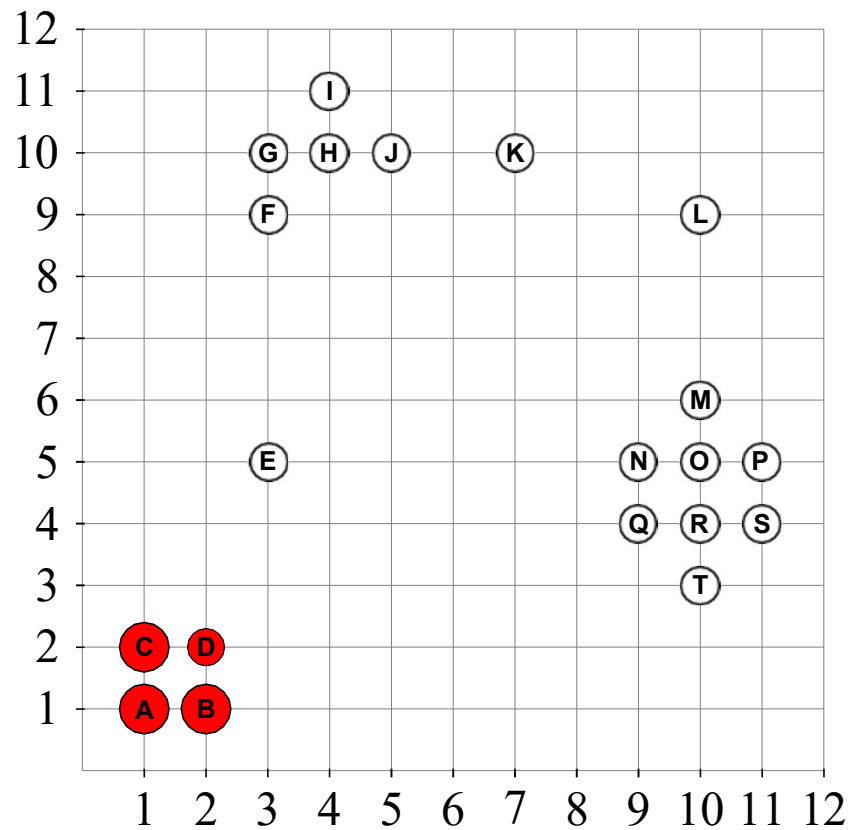
Seed list:

D

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

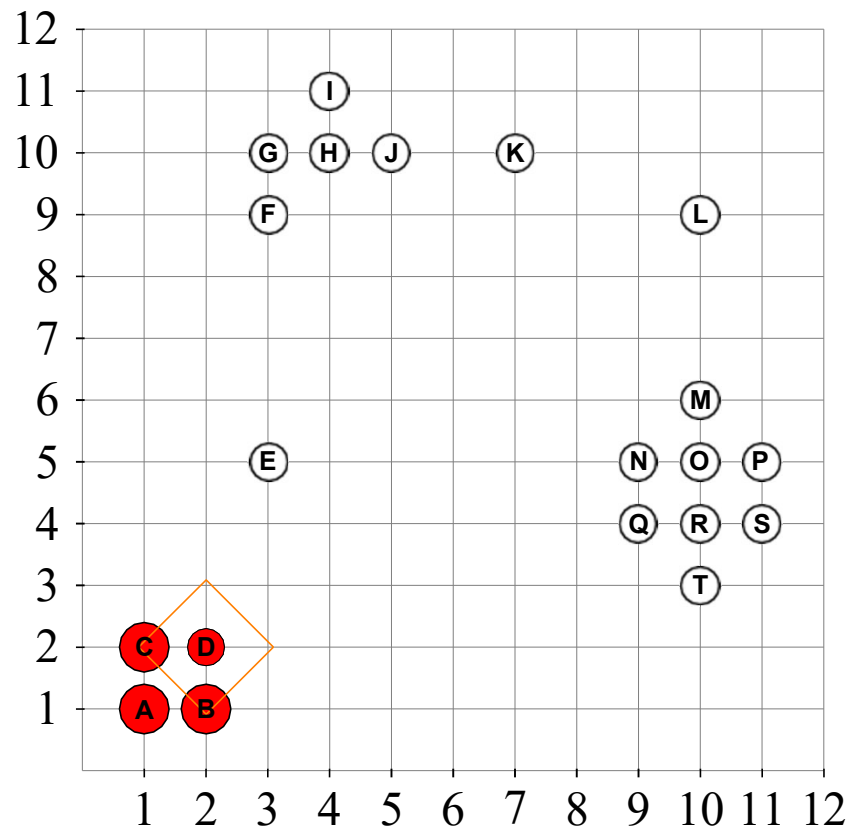
Seed list:

D

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

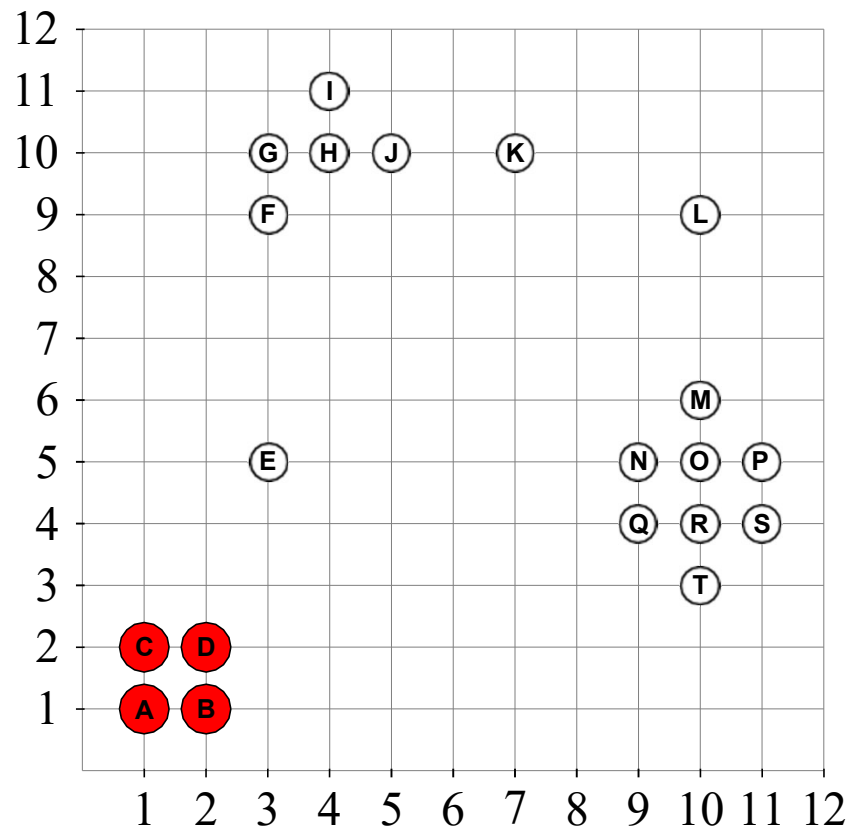
$$\text{minPts} = 2$$

Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

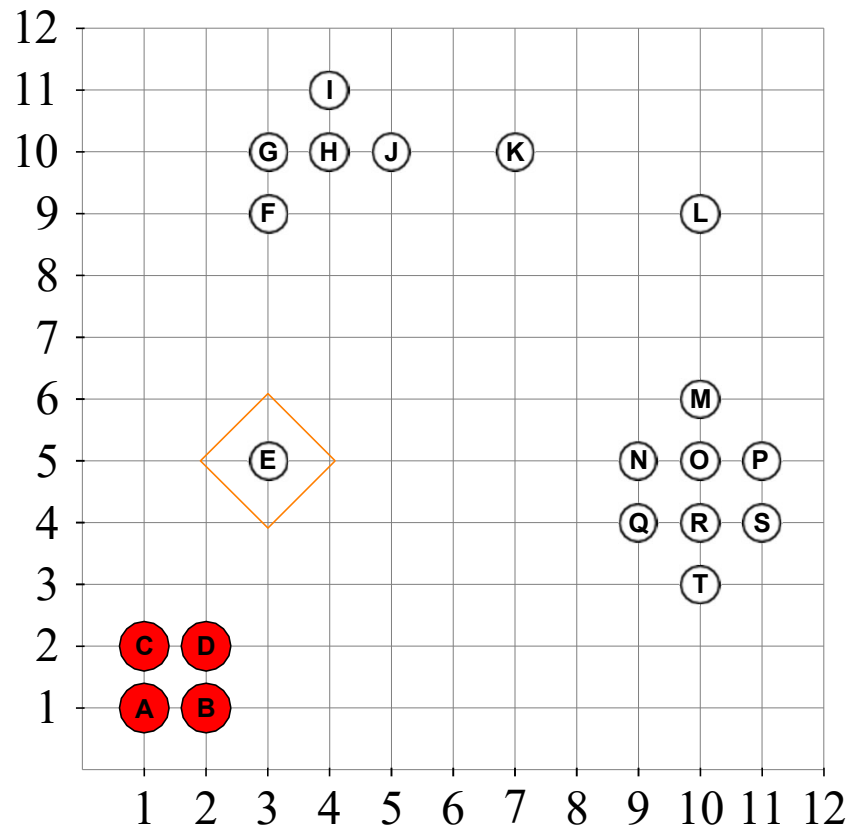
$$\text{minPts} = 2$$

Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

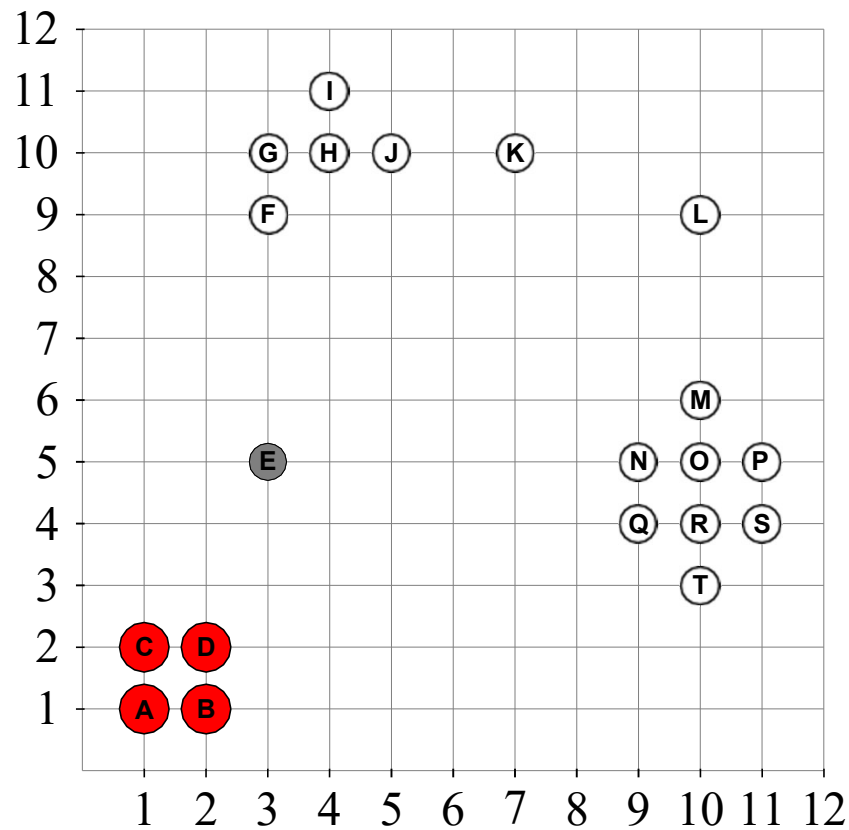
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Seed list:

Example 1

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DS804

Arthur Zimek



$$\varepsilon = 1.1$$

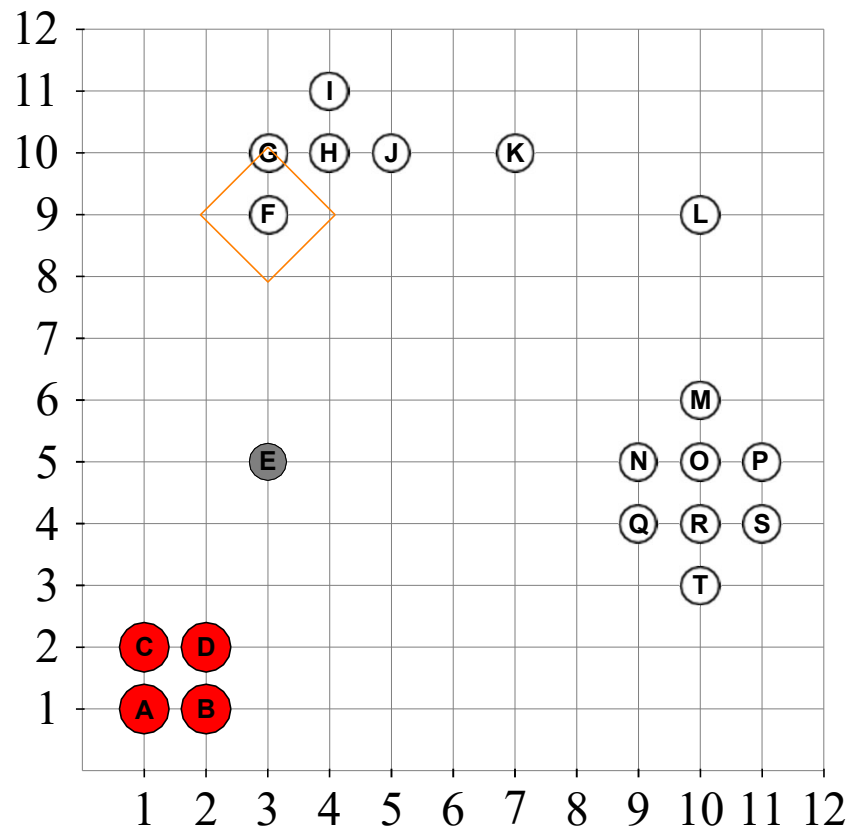
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Seed list:

Example 1

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DS804

Arthur Zimek



$$\varepsilon = 1.1$$

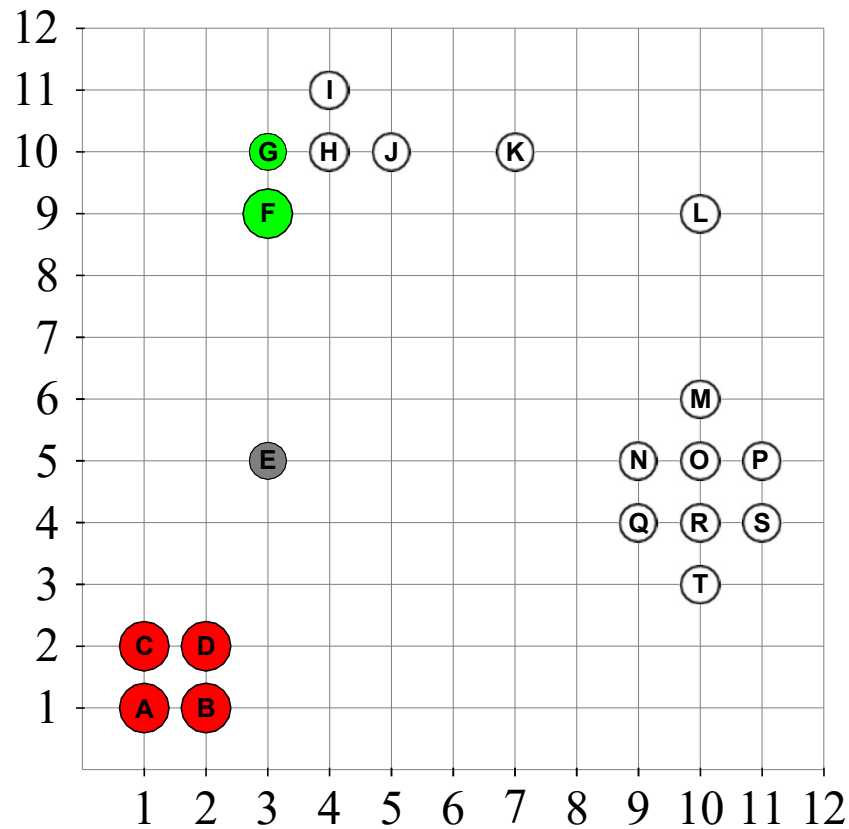
$$\text{minPts} = 2$$

Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

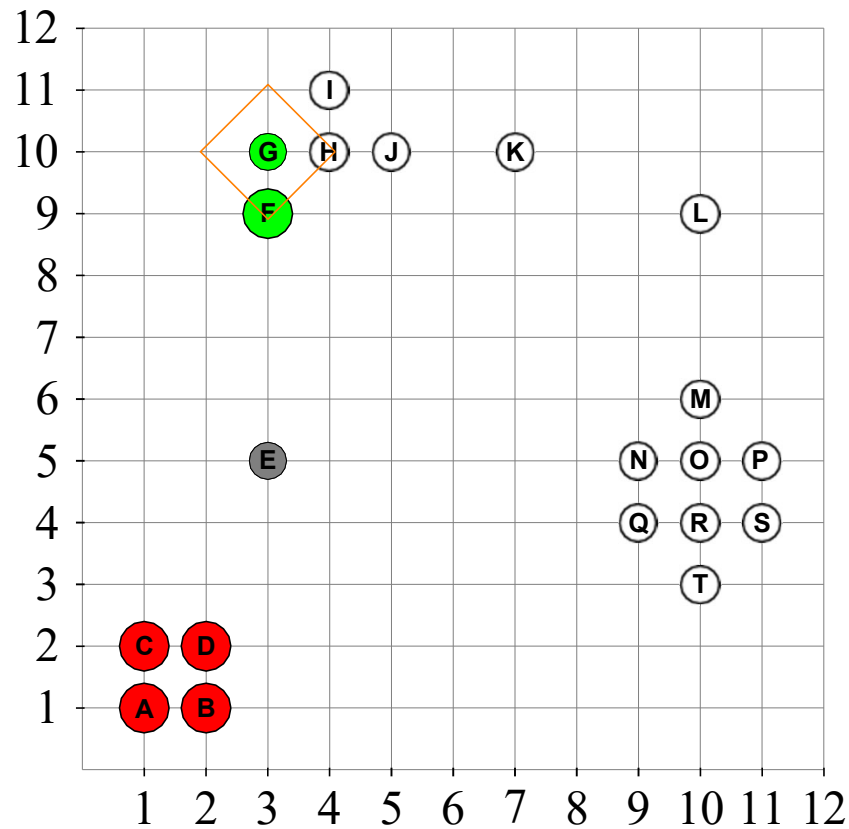
Seed list:

G

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

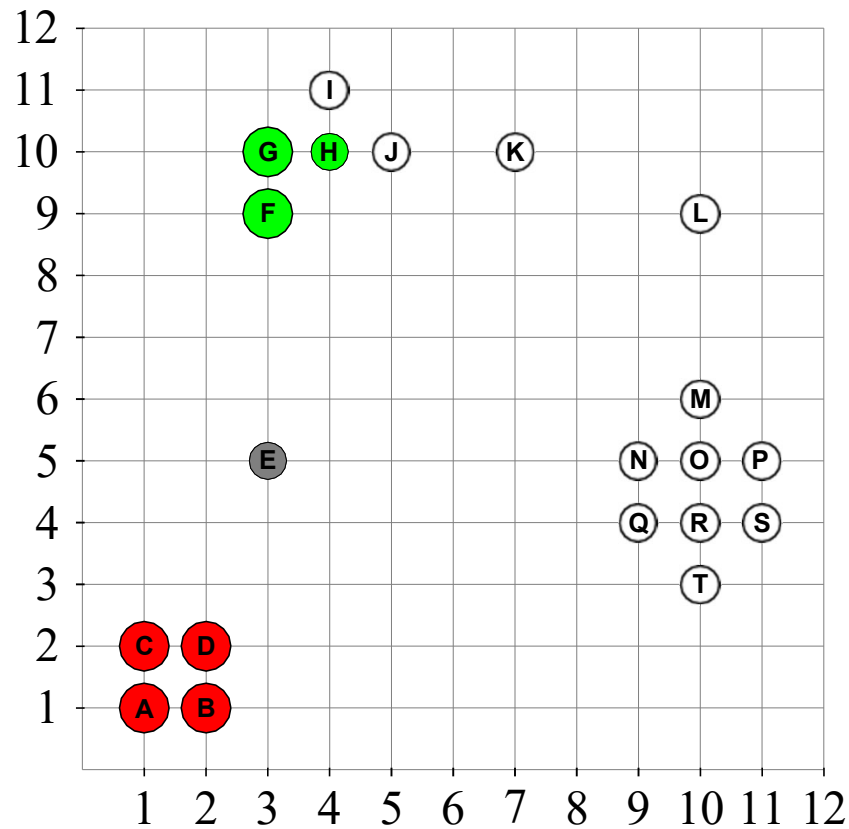
$$\text{minPts} = 2$$

Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

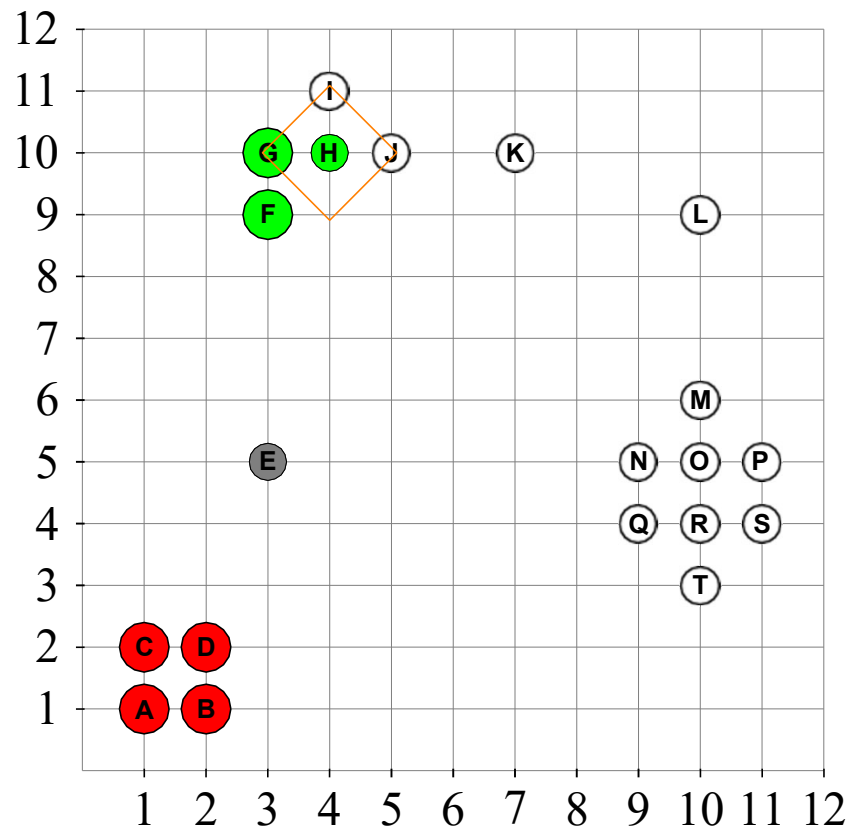
Seed list:

H

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

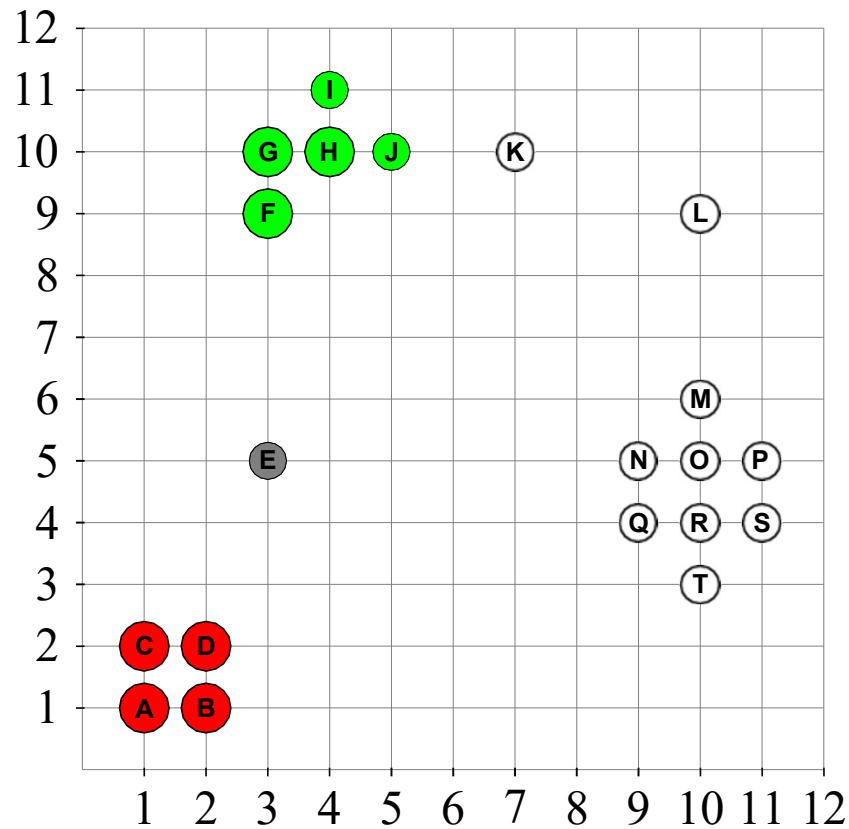
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Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

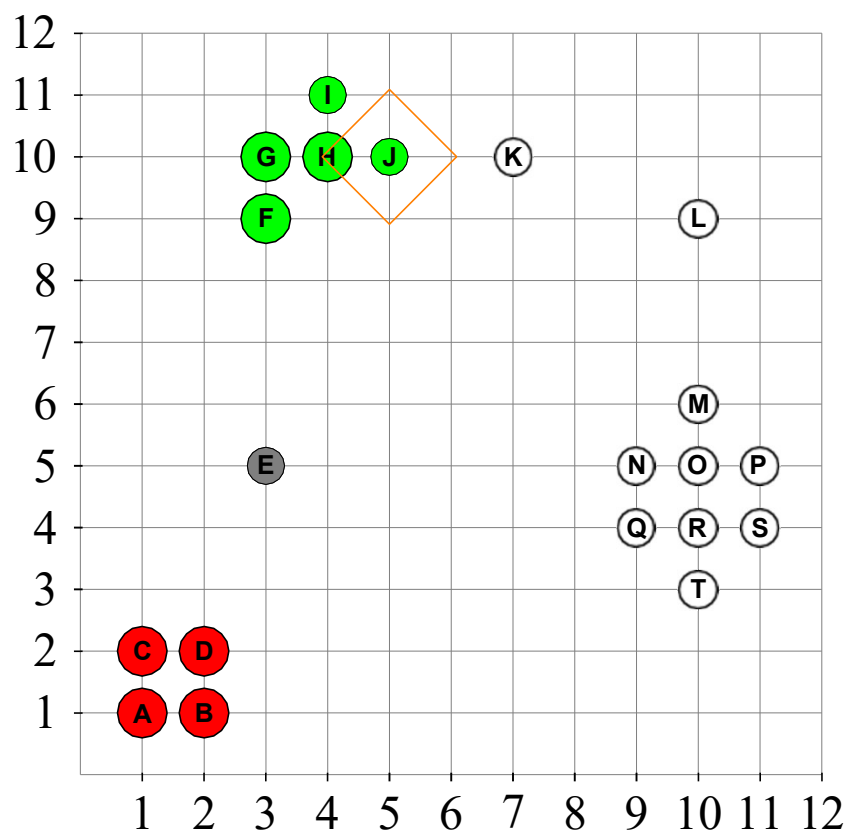
Seed list:

IJ

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

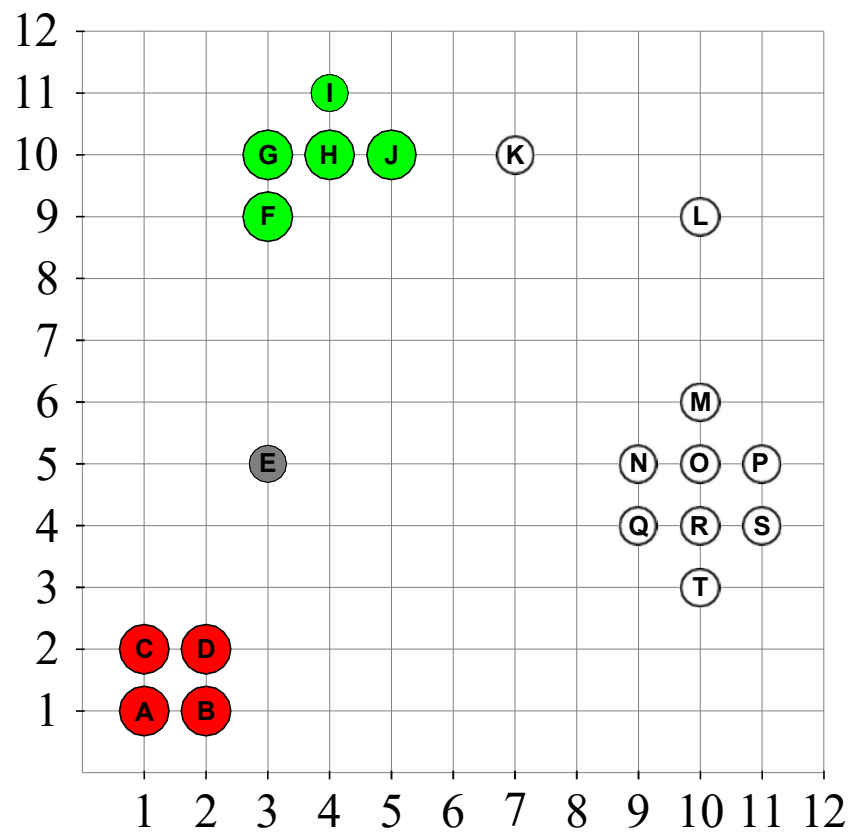
Seed list:

I

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

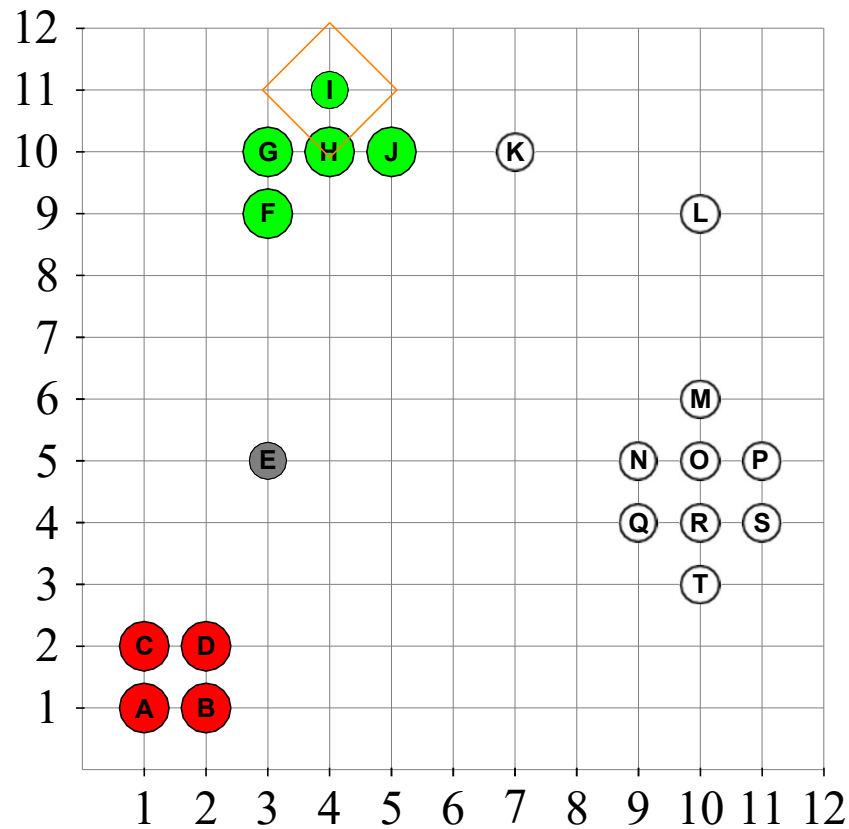
Seed list:

I

Example 1

DM868 DM870
DS804

Arthur Zimek



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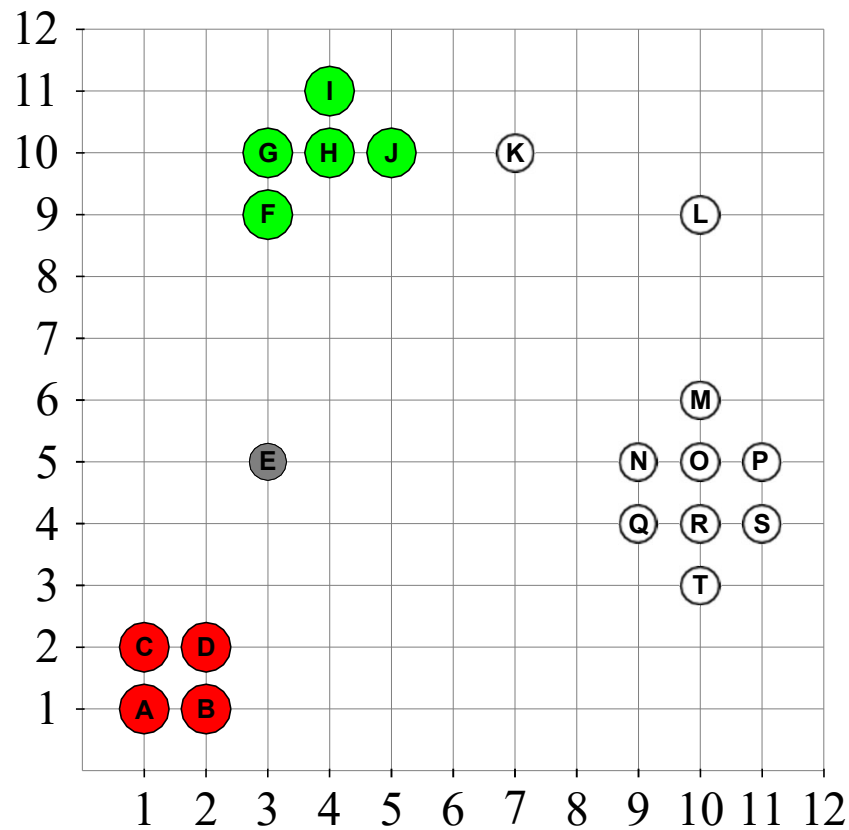
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Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

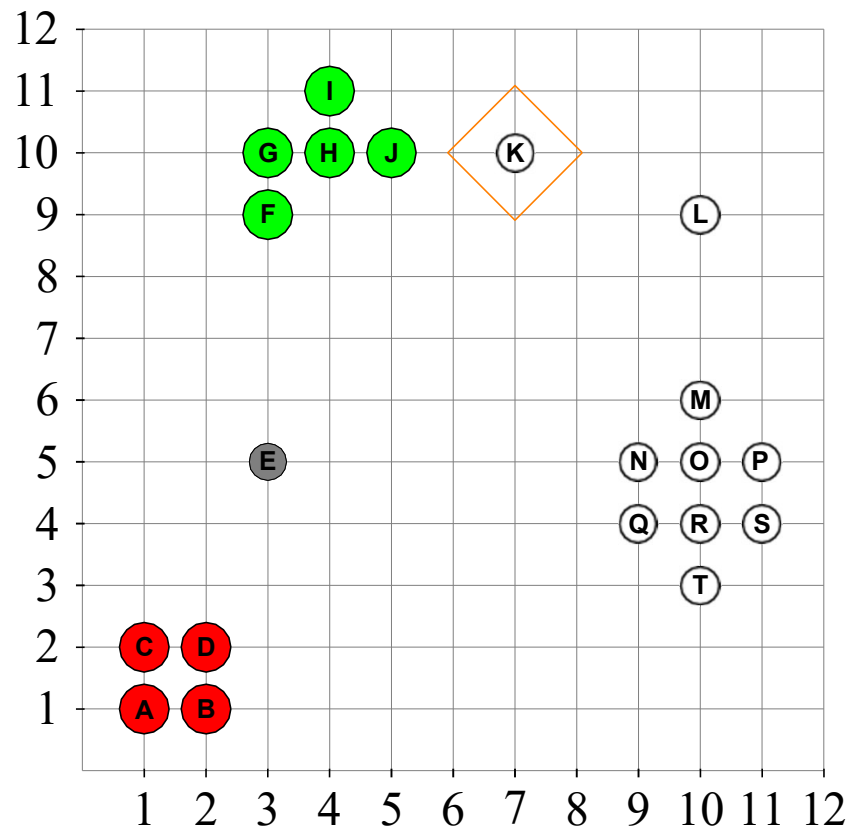
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Seed list:

Example 1

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DS804

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$$\varepsilon = 1.1$$

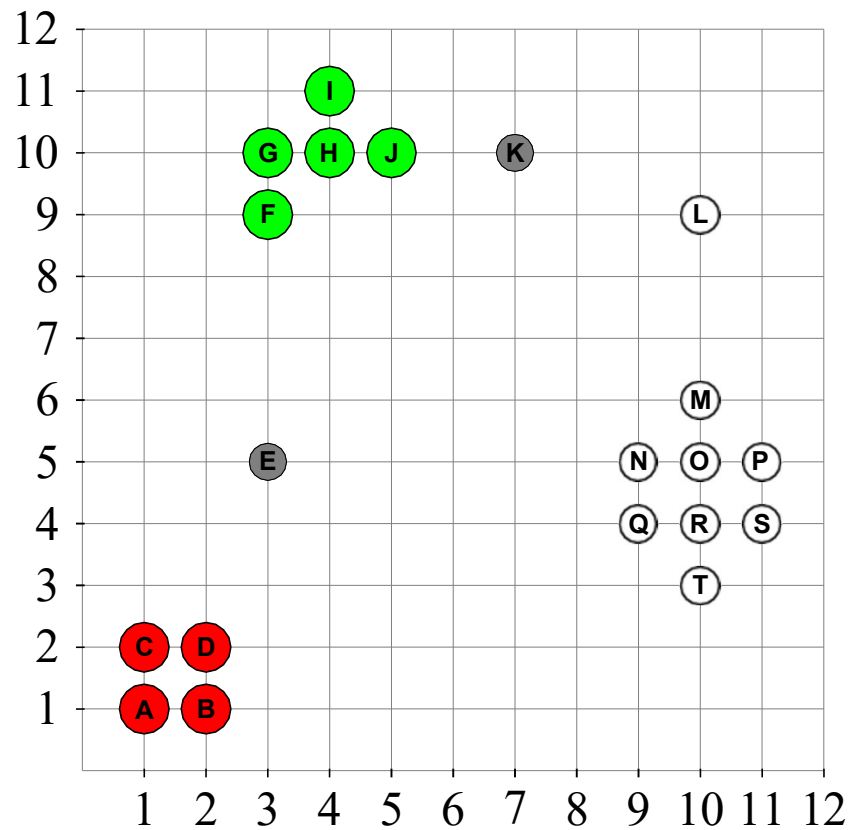
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Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

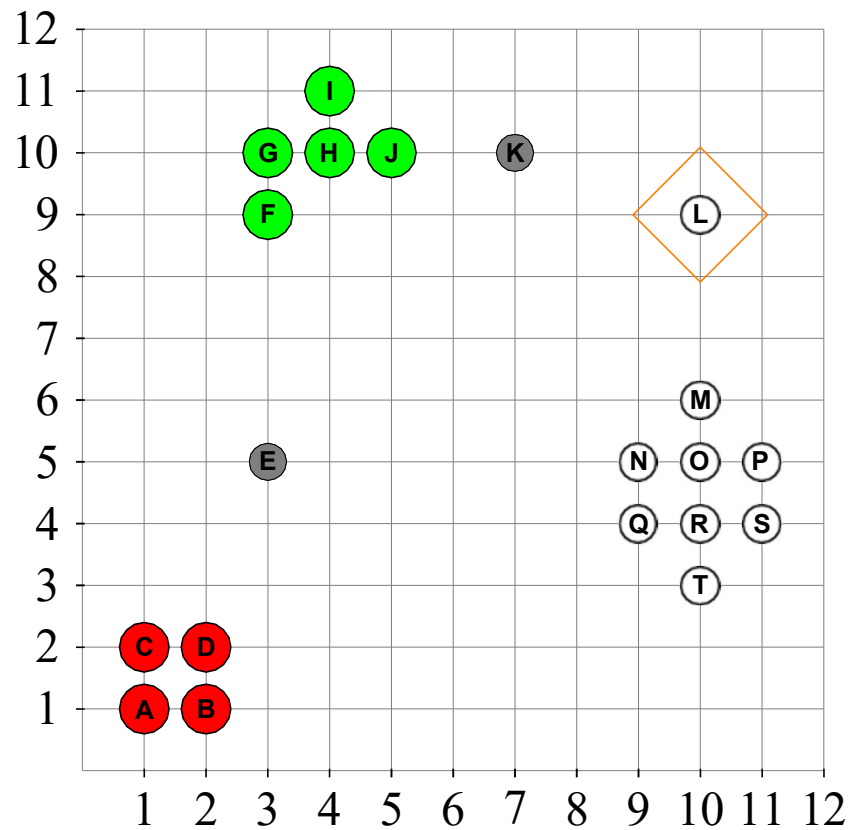
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Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

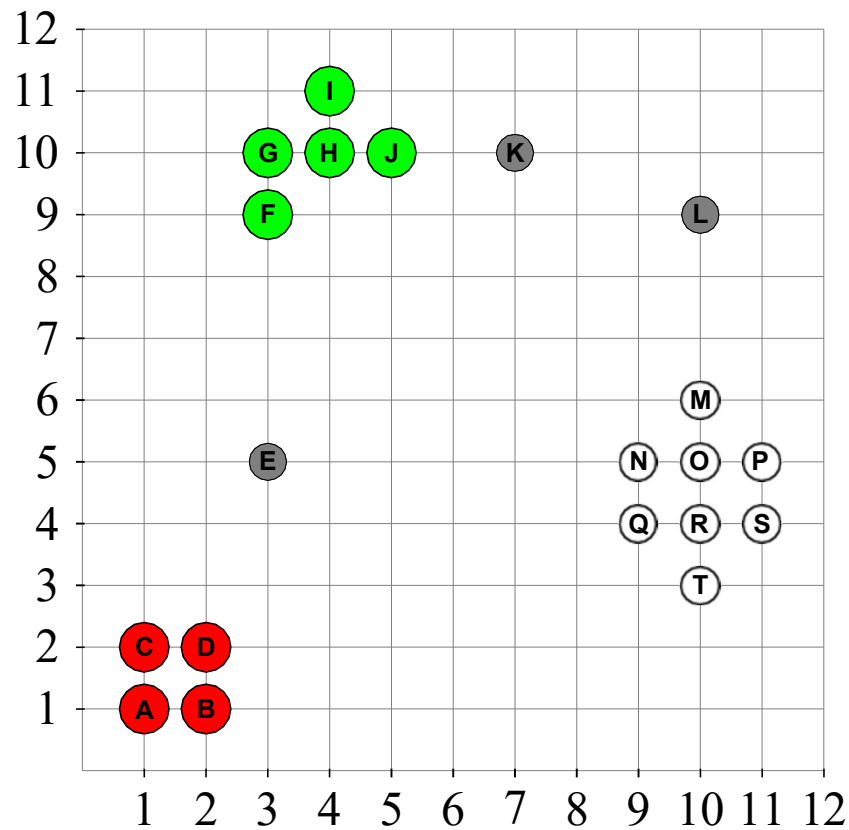
$$\text{minPts} = 2$$

Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

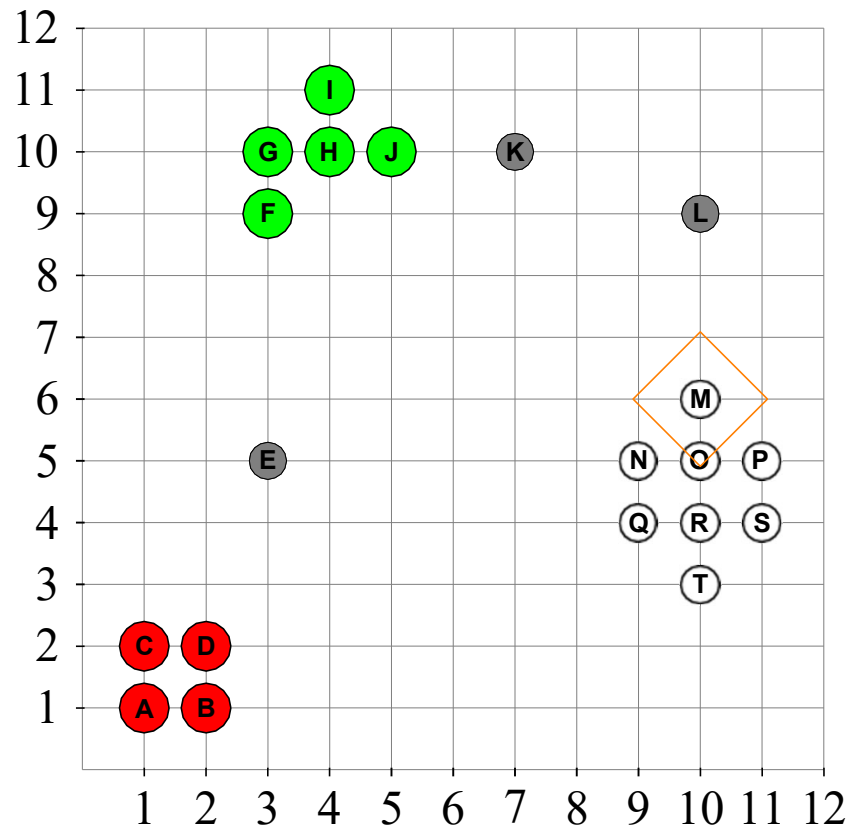
$$\text{minPts} = 2$$

Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

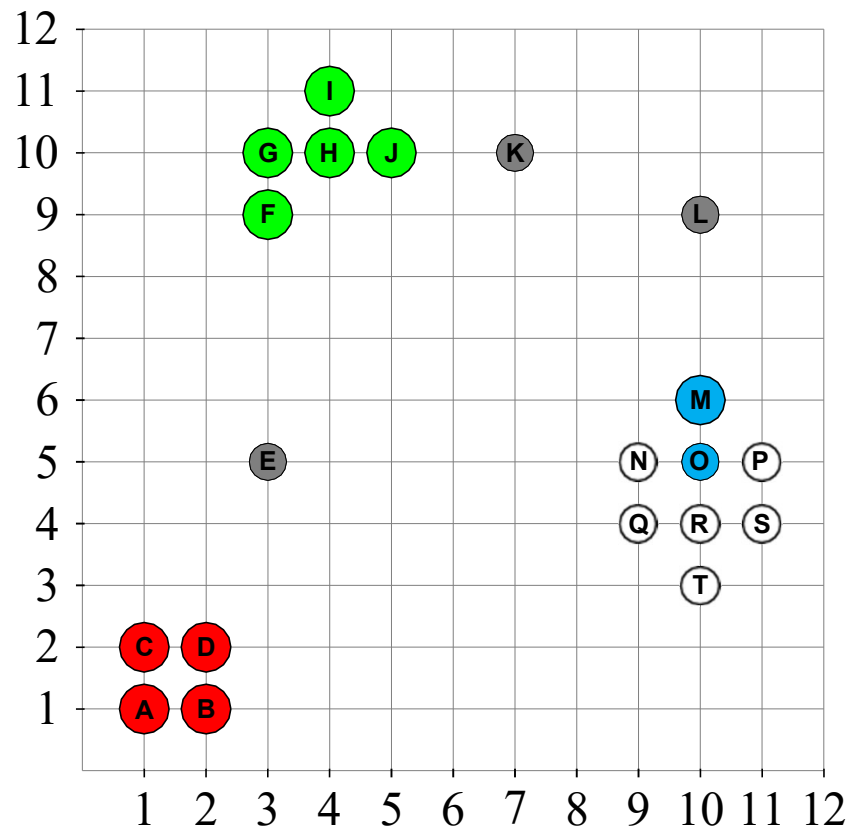
$$\text{minPts} = 2$$

Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

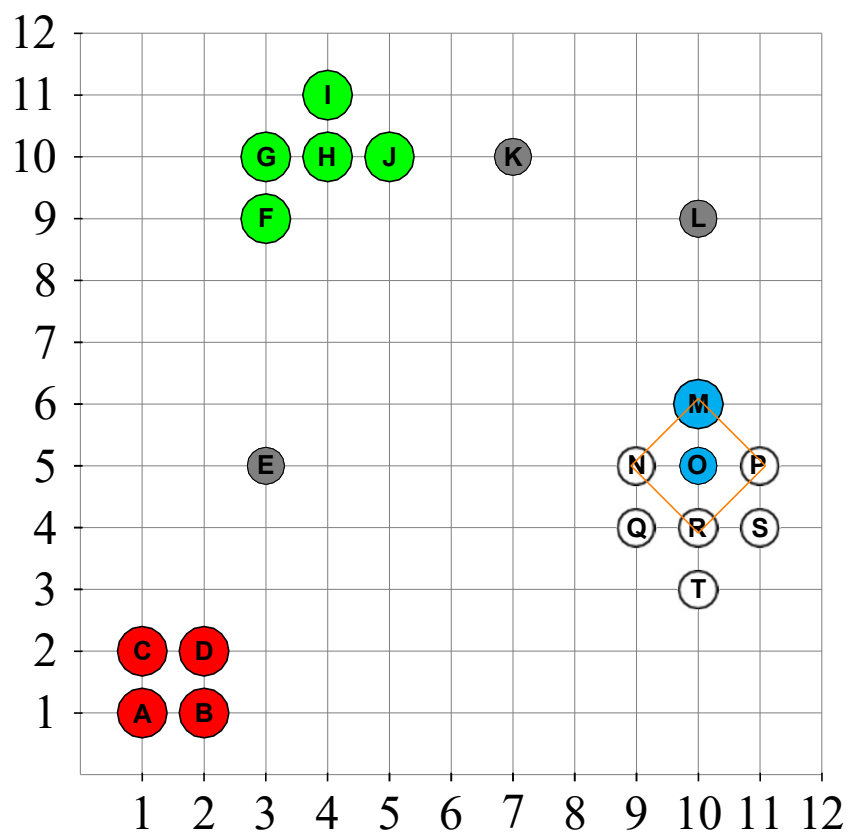
Seed list:

O

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

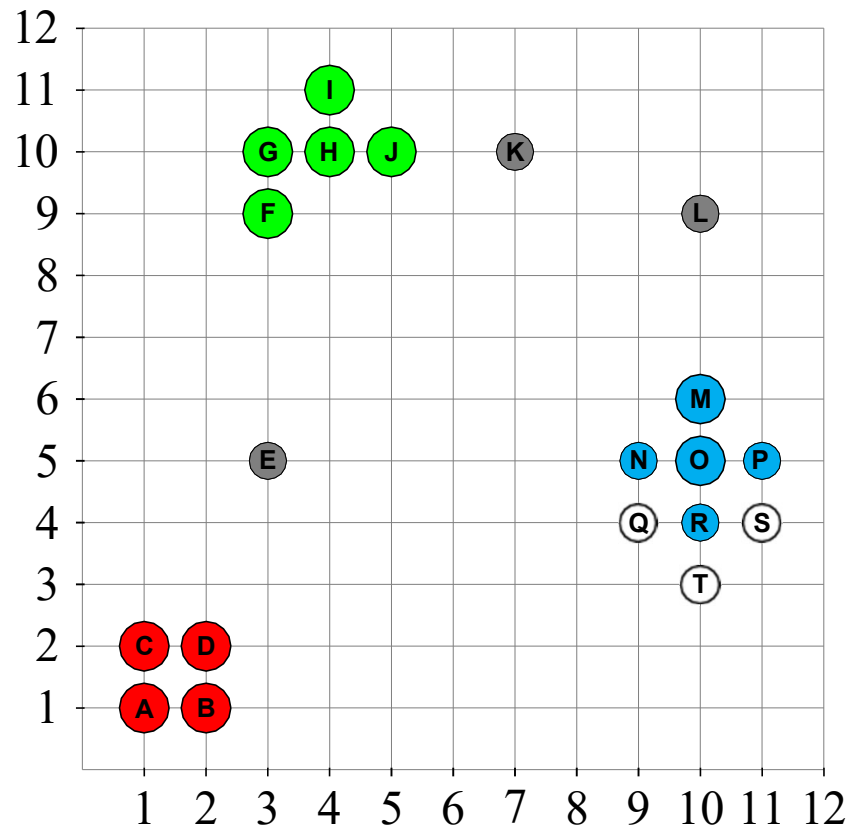
$$\text{minPts} = 2$$

Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

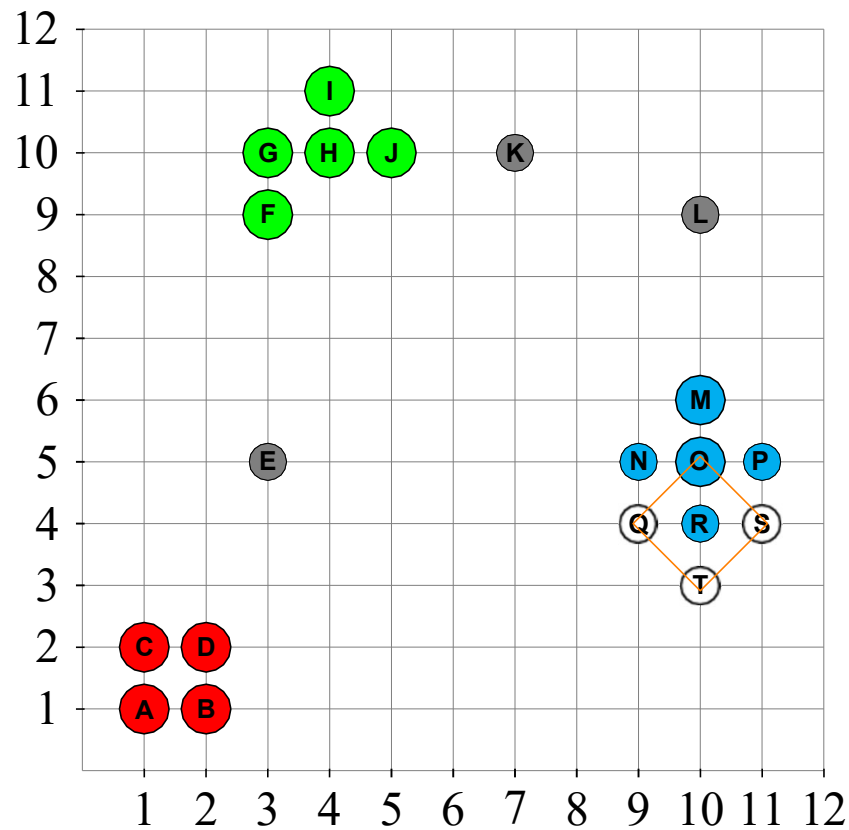
$$\text{minPts} = 2$$

Seed list:
NPR

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

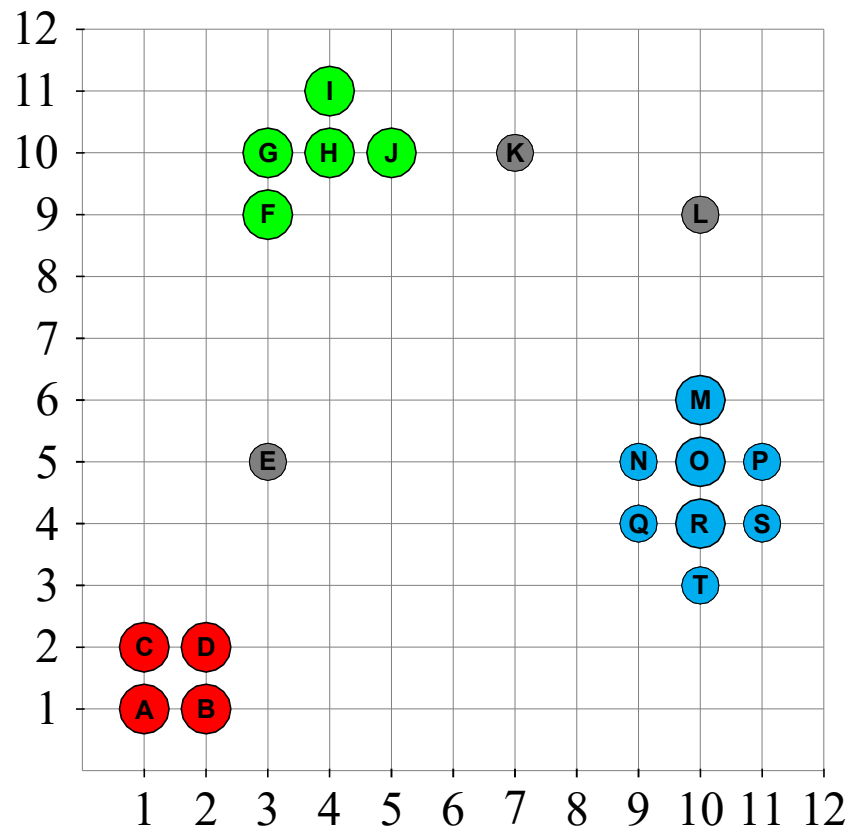
$$\text{minPts} = 2$$

Seed list:
NP

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

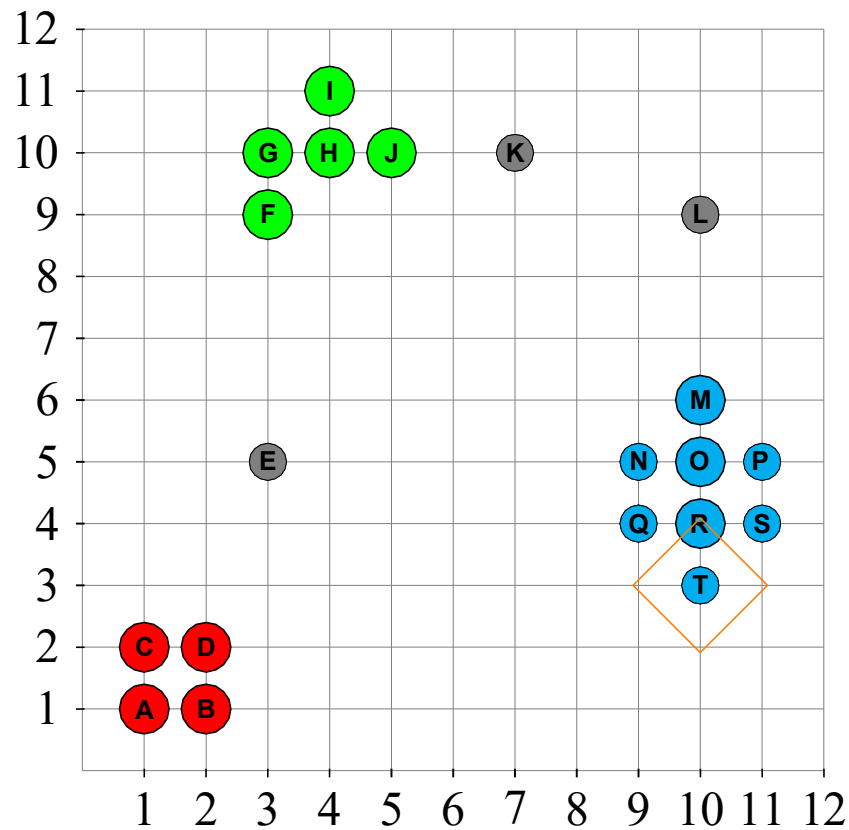
$$\text{minPts} = 2$$

Seed list:
NPQST

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

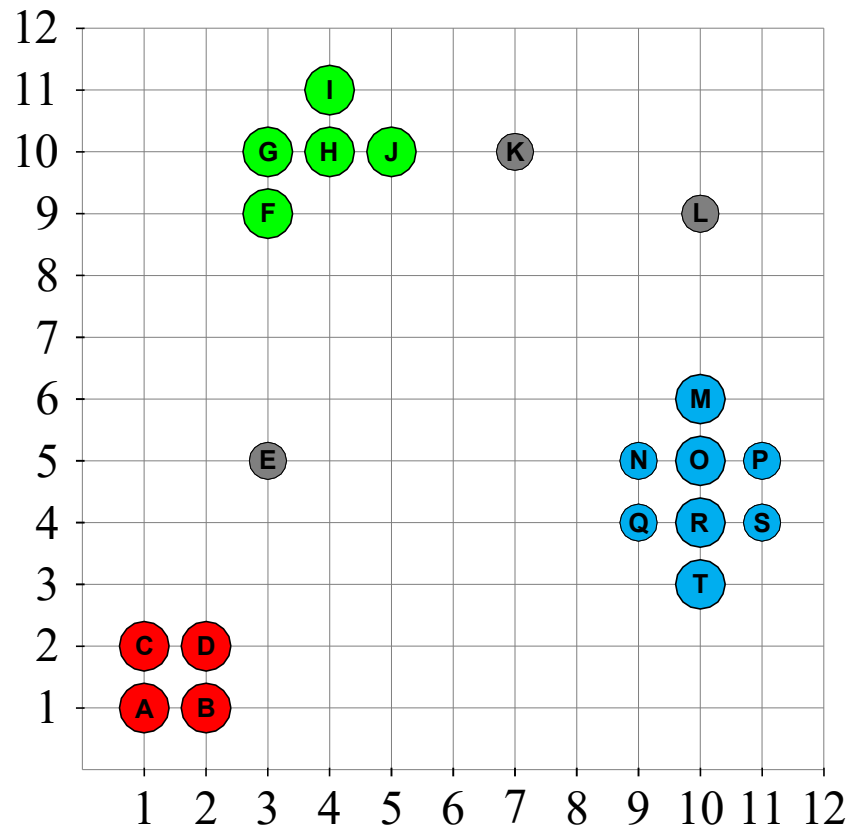
$$\text{minPts} = 2$$

Seed list:
NPQS

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

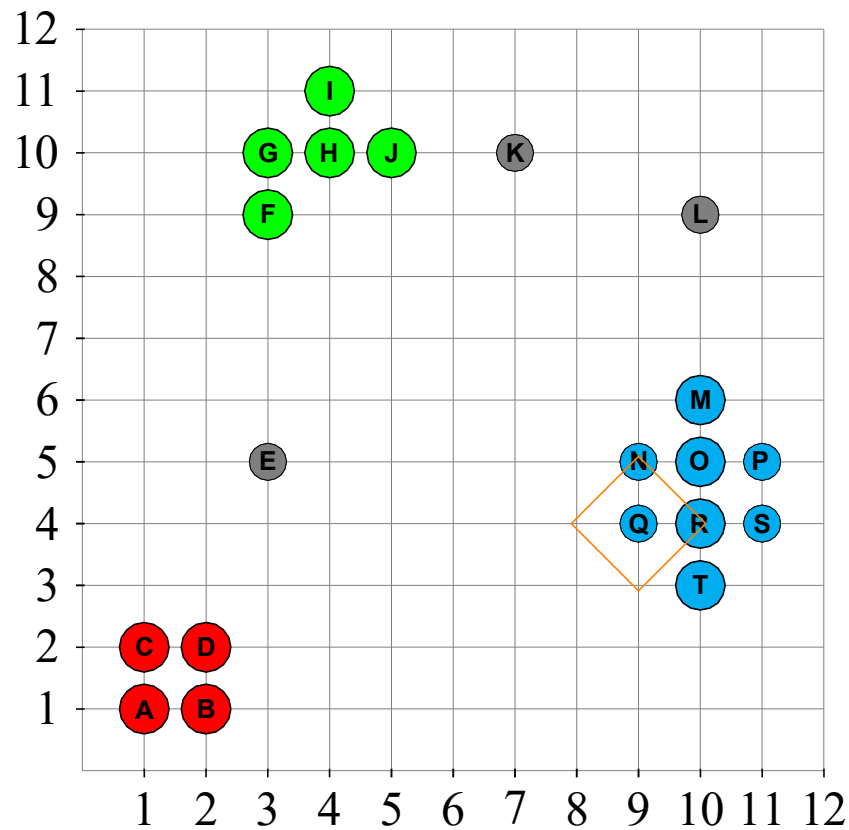
$$\text{minPts} = 2$$

Seed list:
NPQS

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

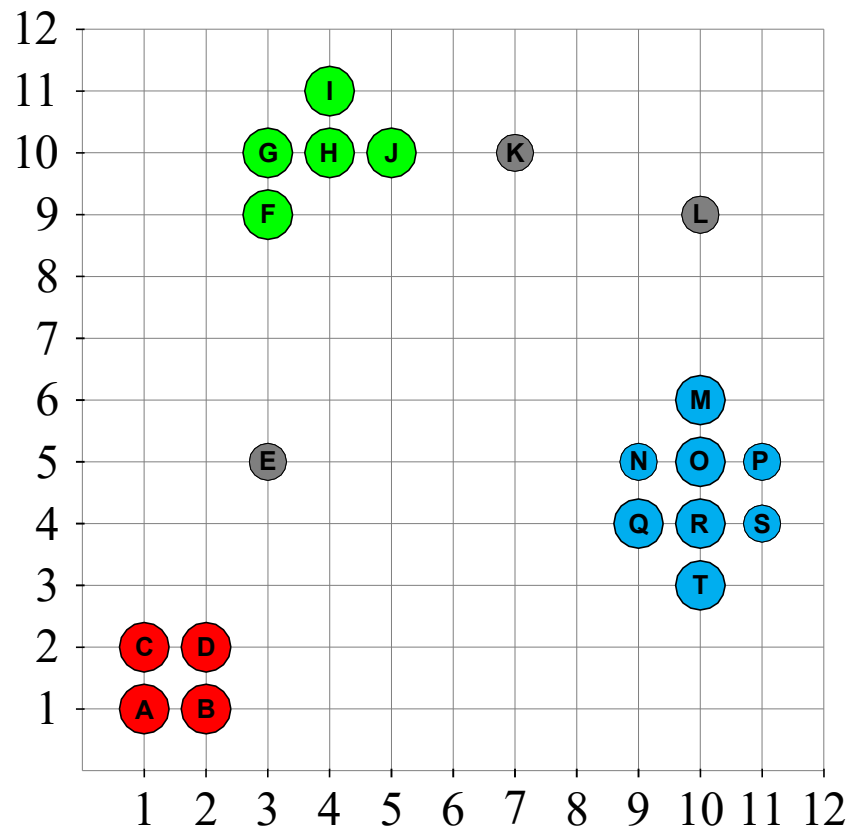
$$\text{minPts} = 2$$

Seed list:
NPS

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

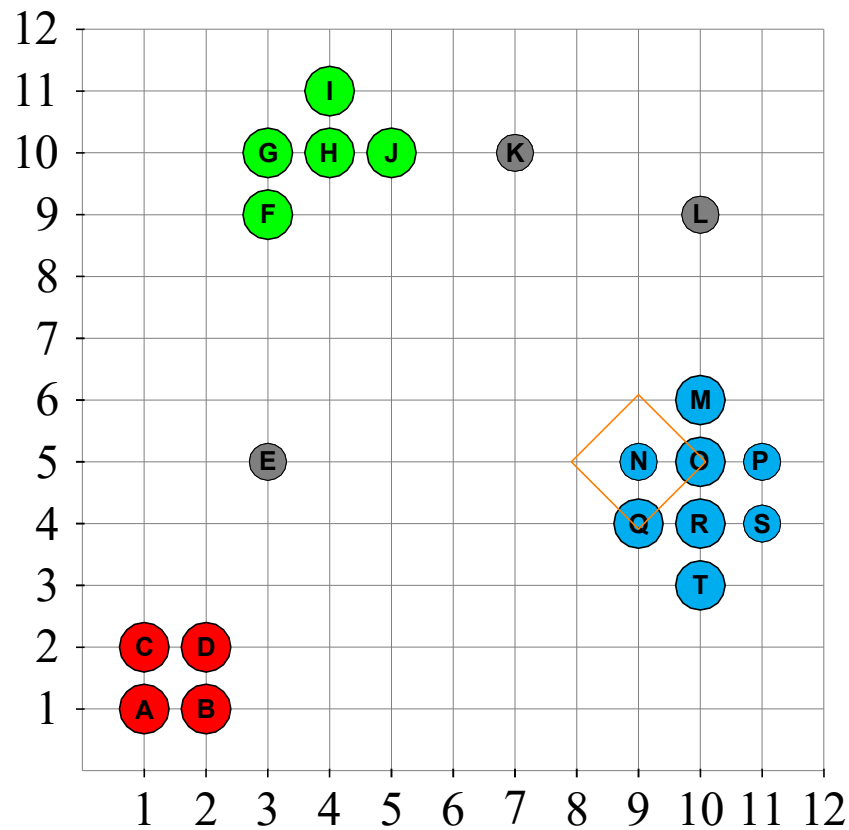
$$\text{minPts} = 2$$

Seed list:
NPS

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

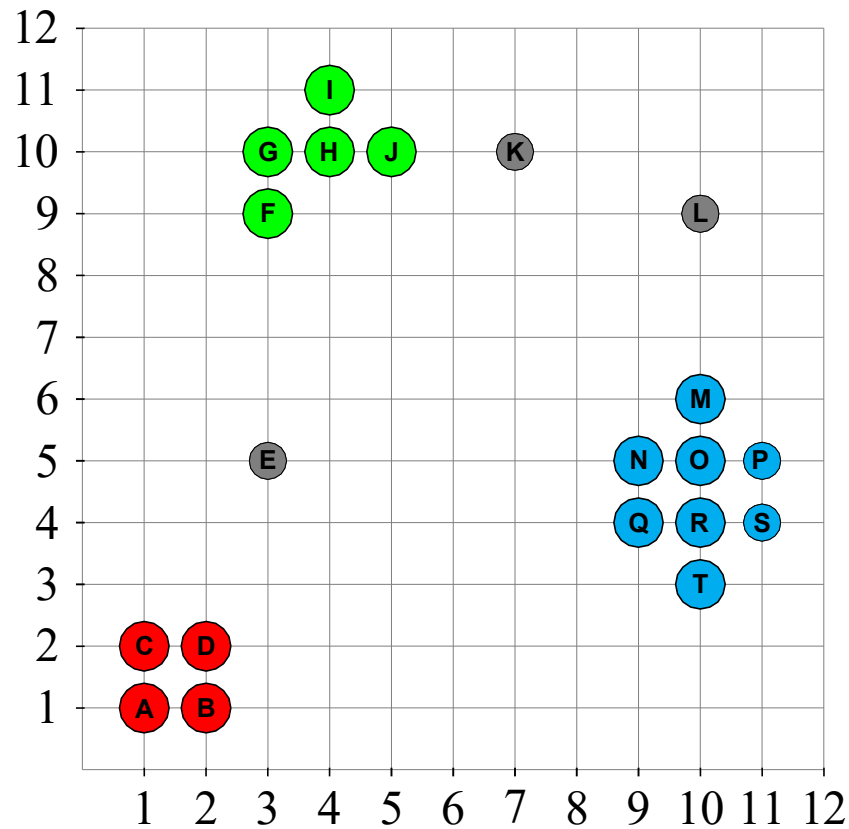
$$\text{minPts} = 2$$

Seed list:
PS

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

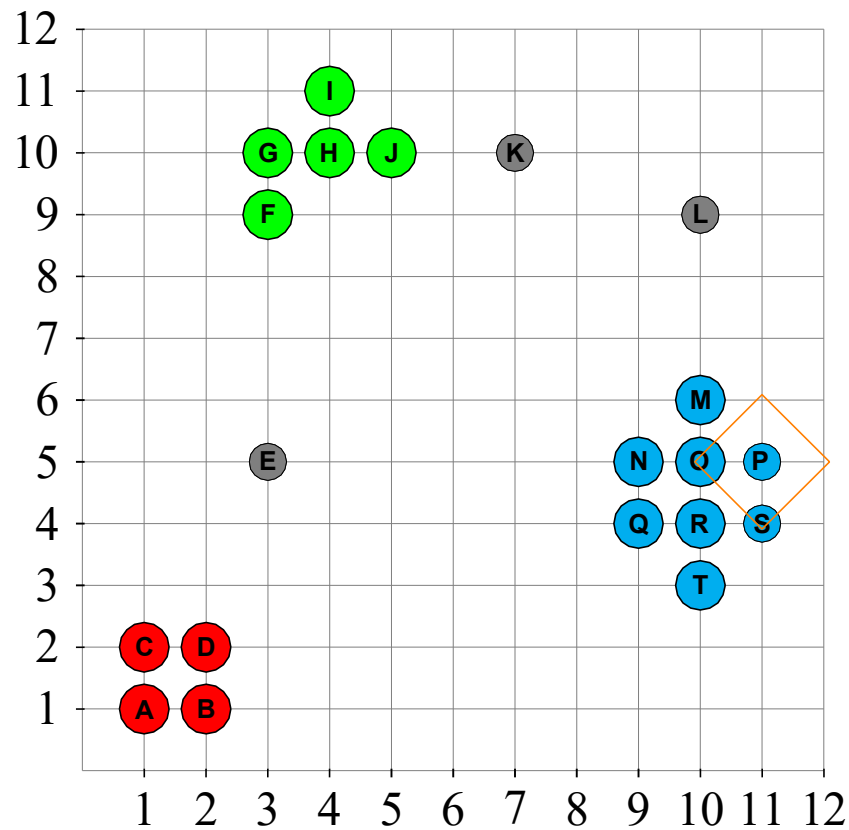
$$\text{minPts} = 2$$

Seed list:
PS

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

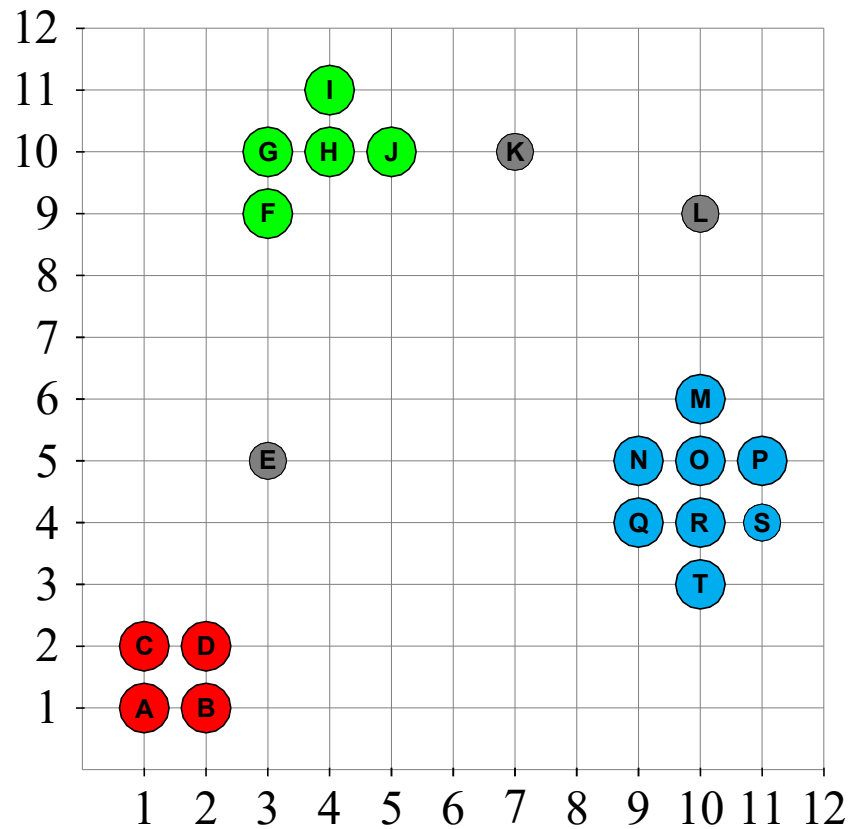
Seed list:

S

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

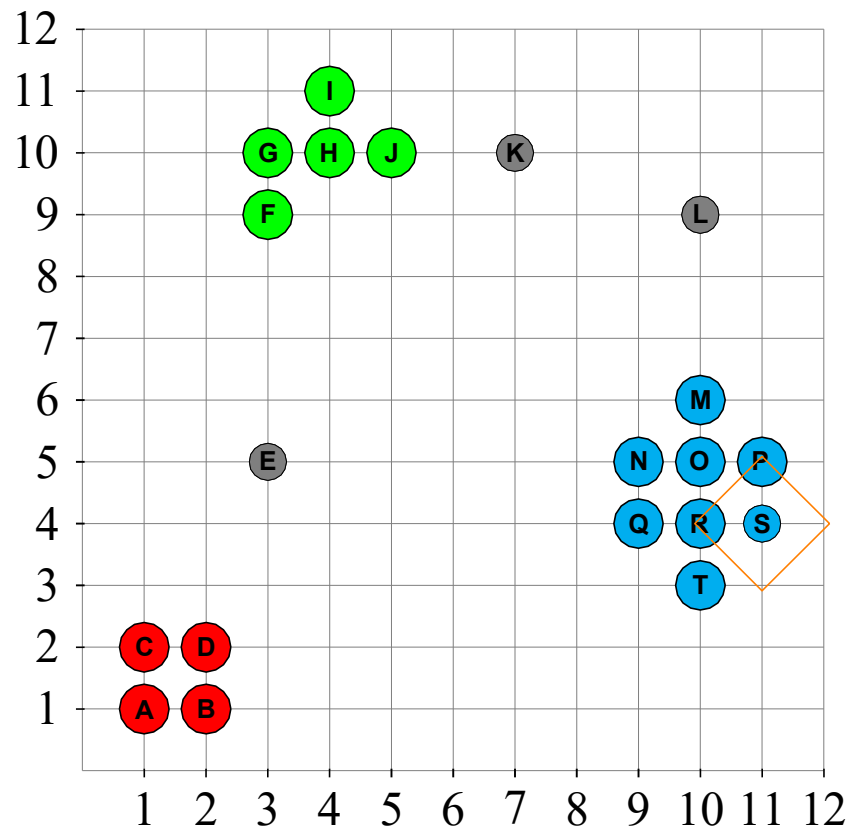
Seed list:

S

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

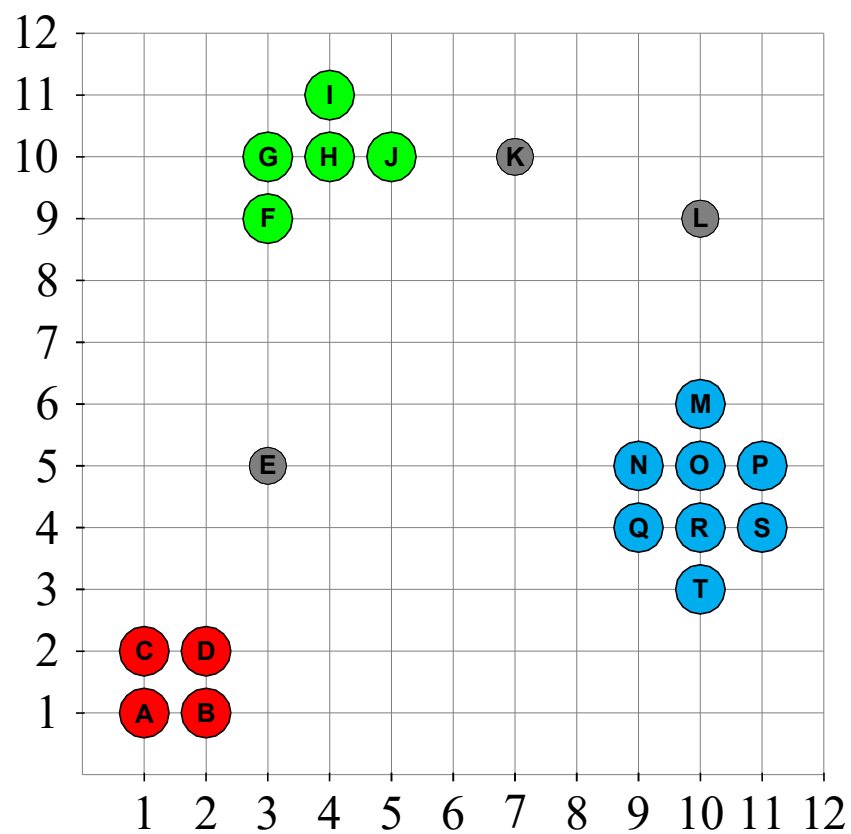
$$\text{minPts} = 2$$

Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

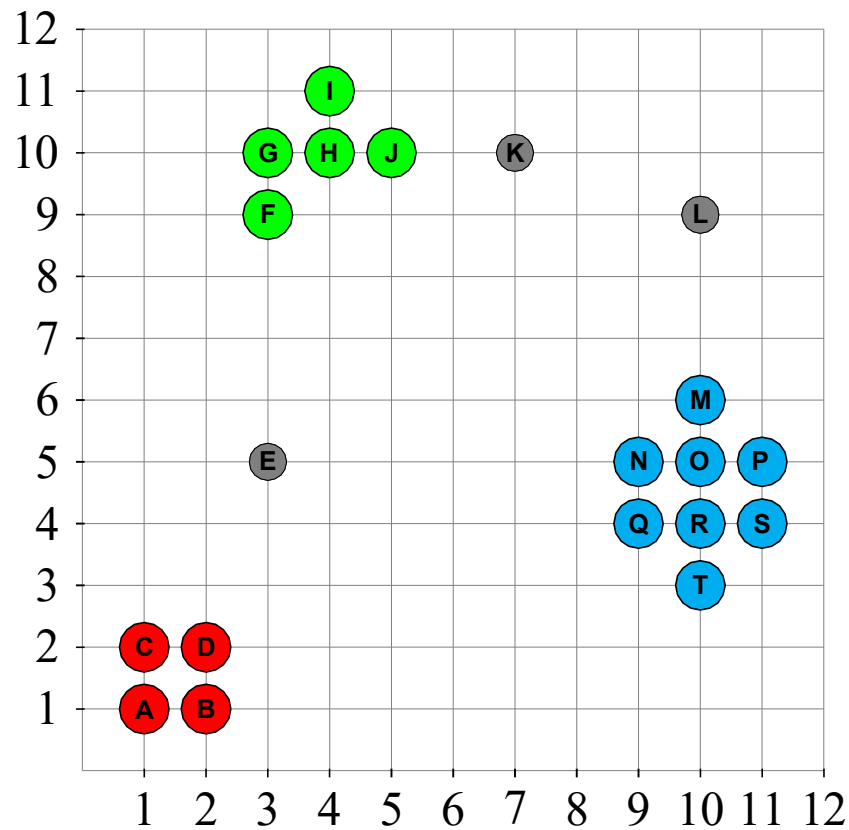
$$\text{minPts} = 2$$

Seed list:

Example 1

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 2$$

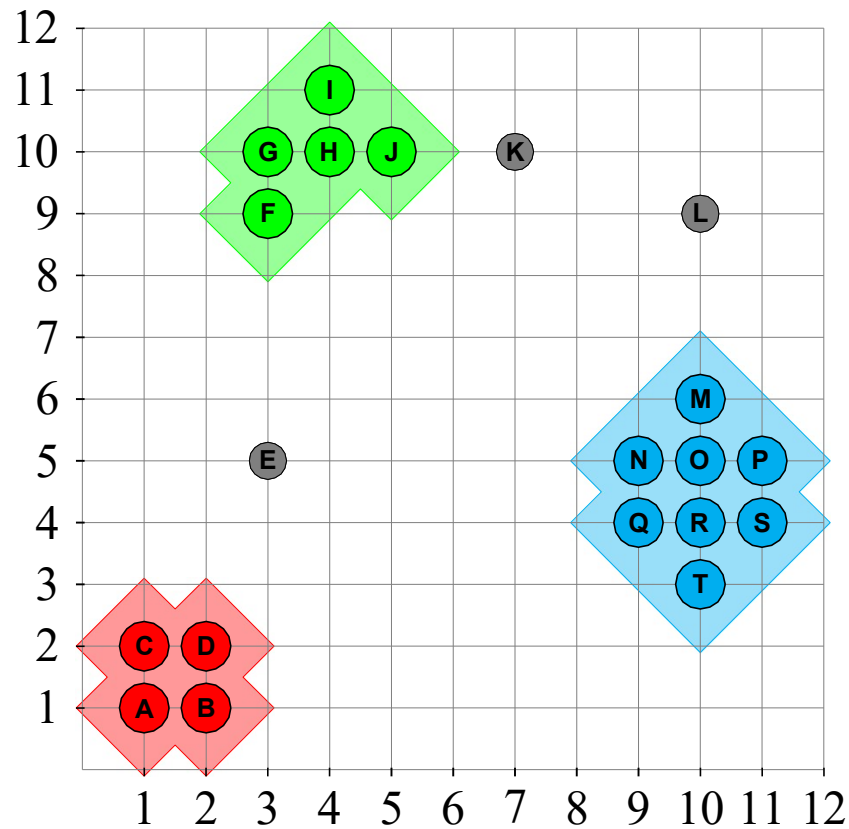
Seed list:

No border
points!

Example 1

DM868 DM870
DS804

Arthur Zimek



$\varepsilon = 1.1$

$minPts = 2$

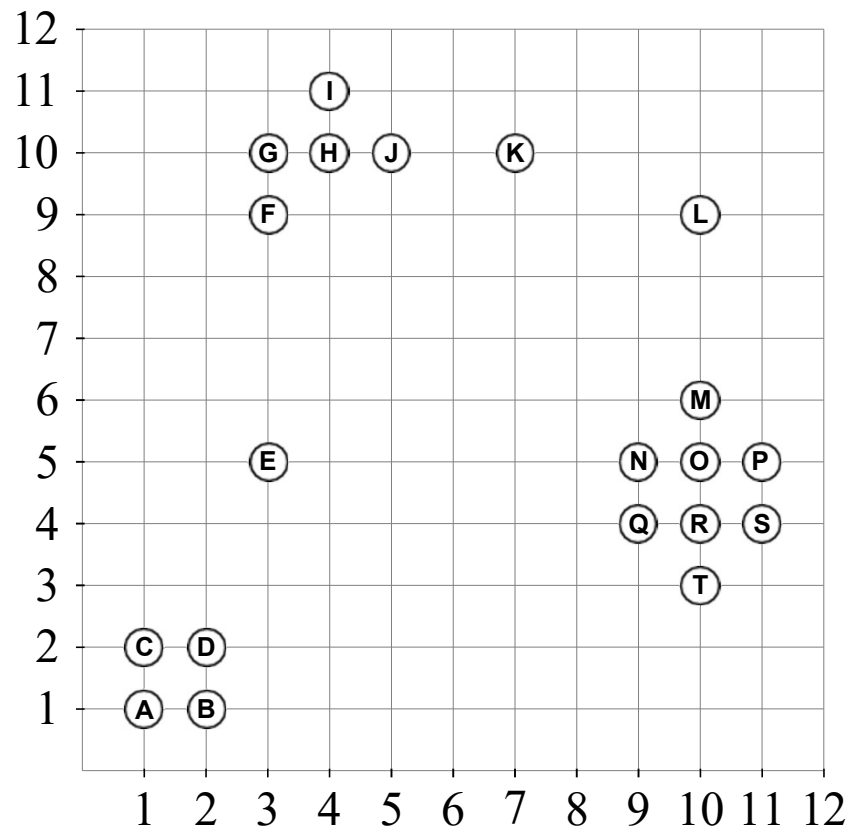
Seed list:

No border
points!

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

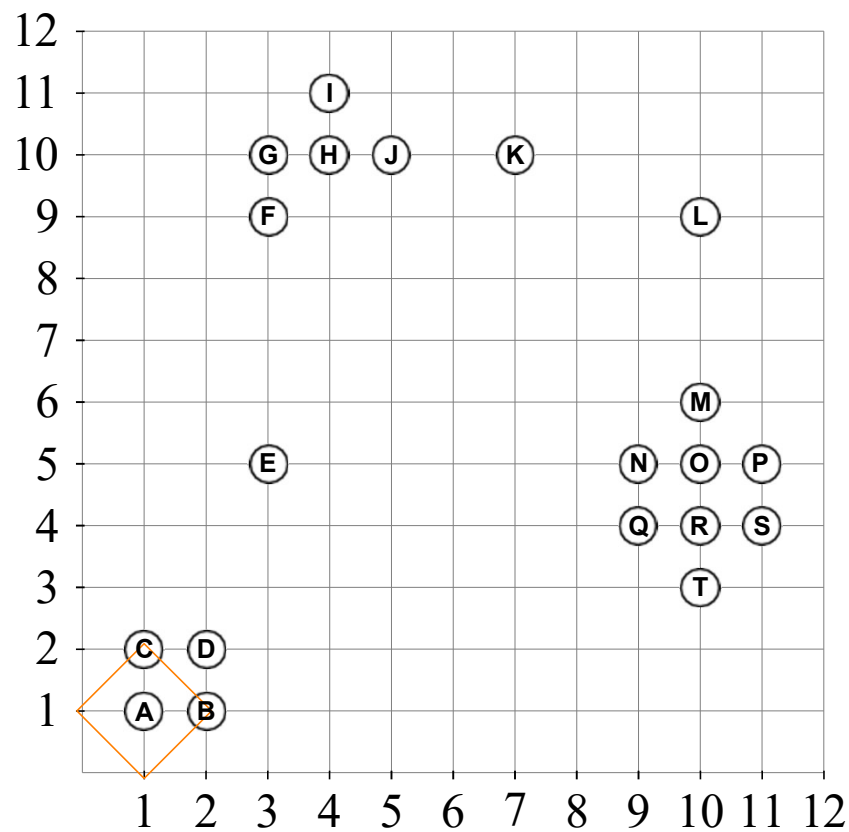
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

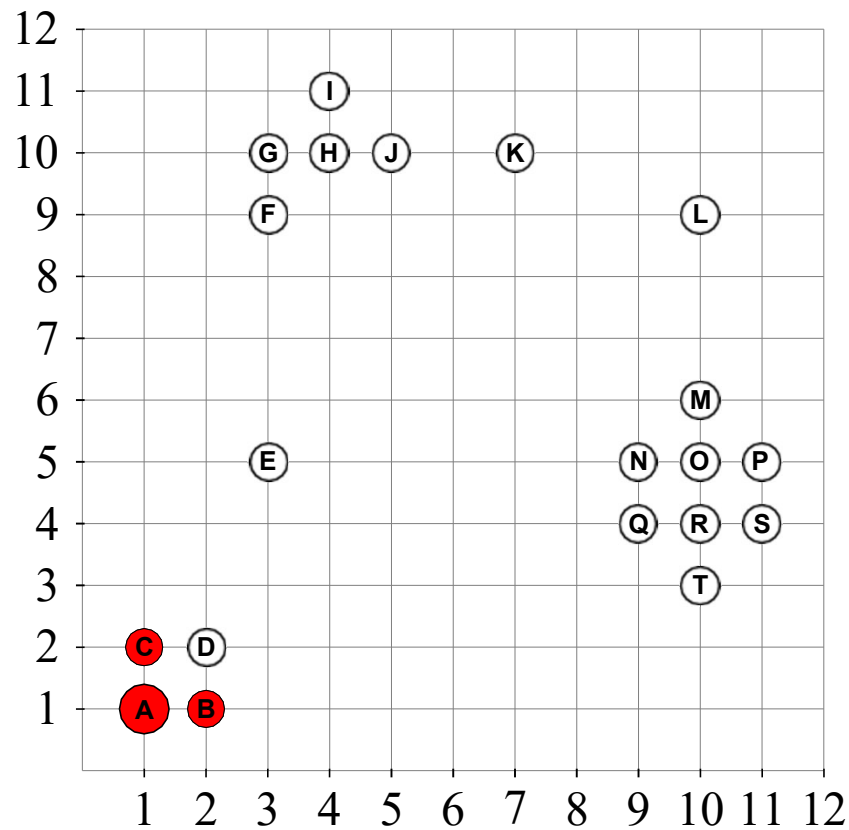
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

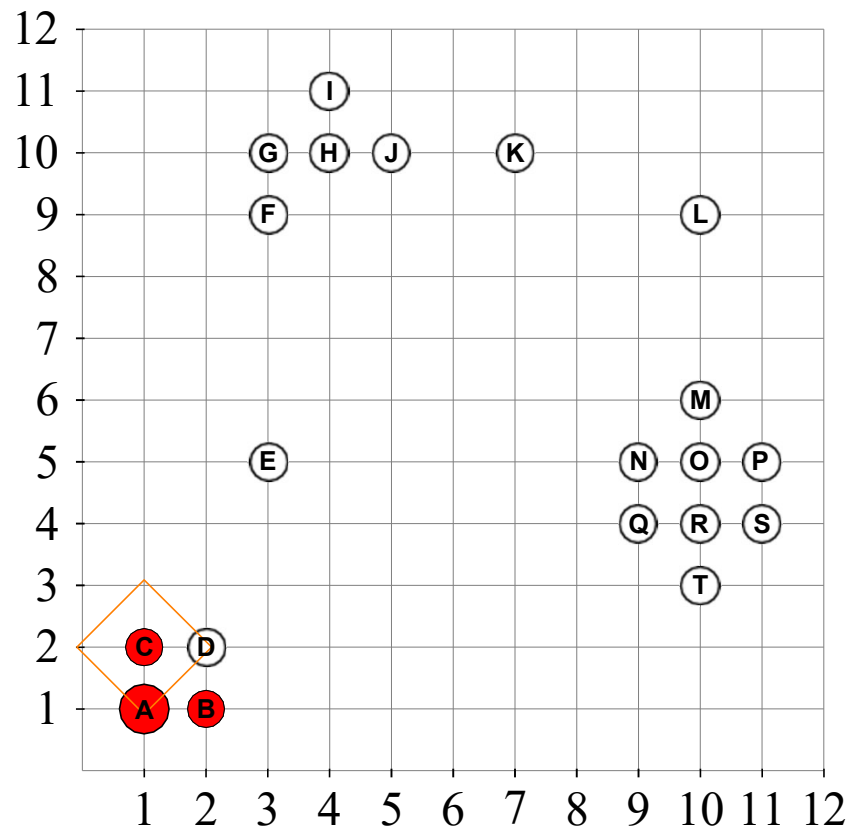
$$\text{minPts} = 3$$

Seed list:
BC

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 3$$

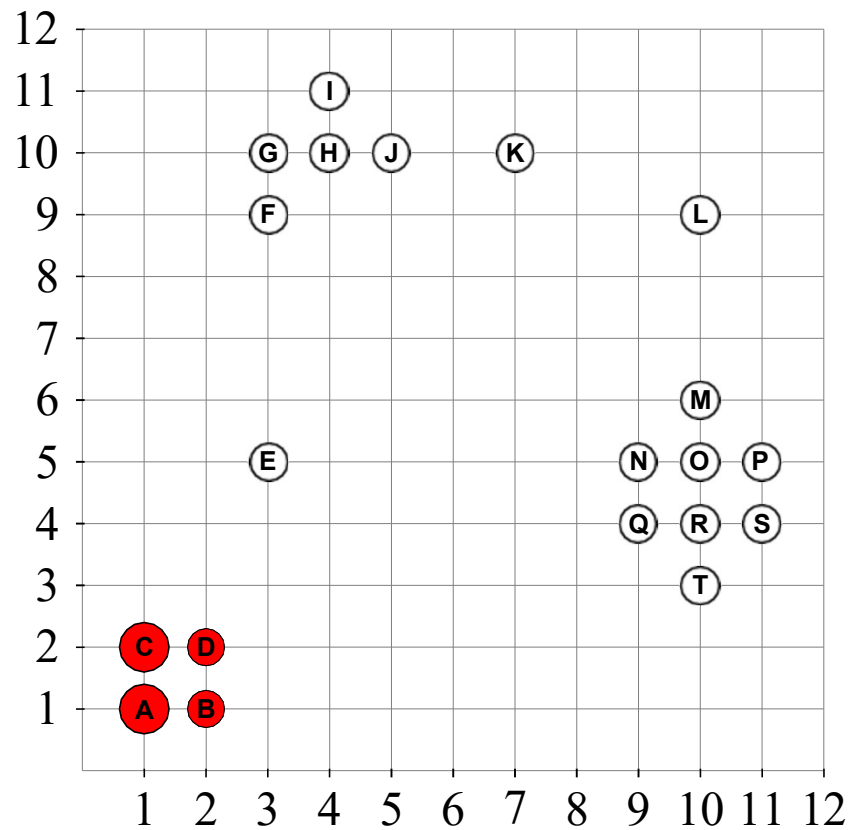
Seed list:

B

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 3$$

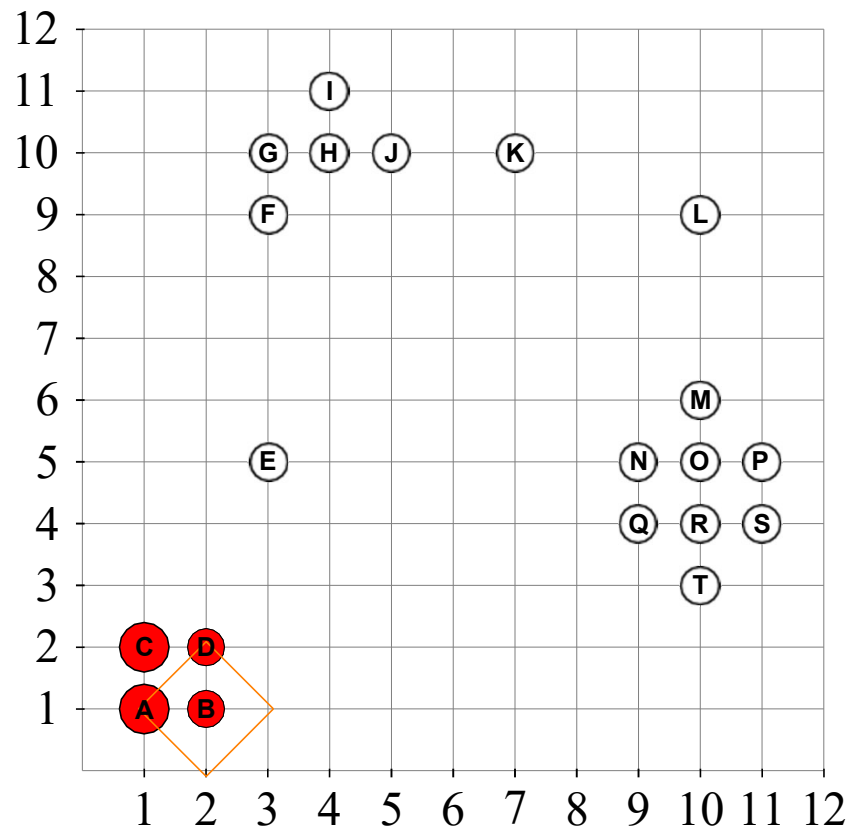
Seed list:

BD

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 3$$

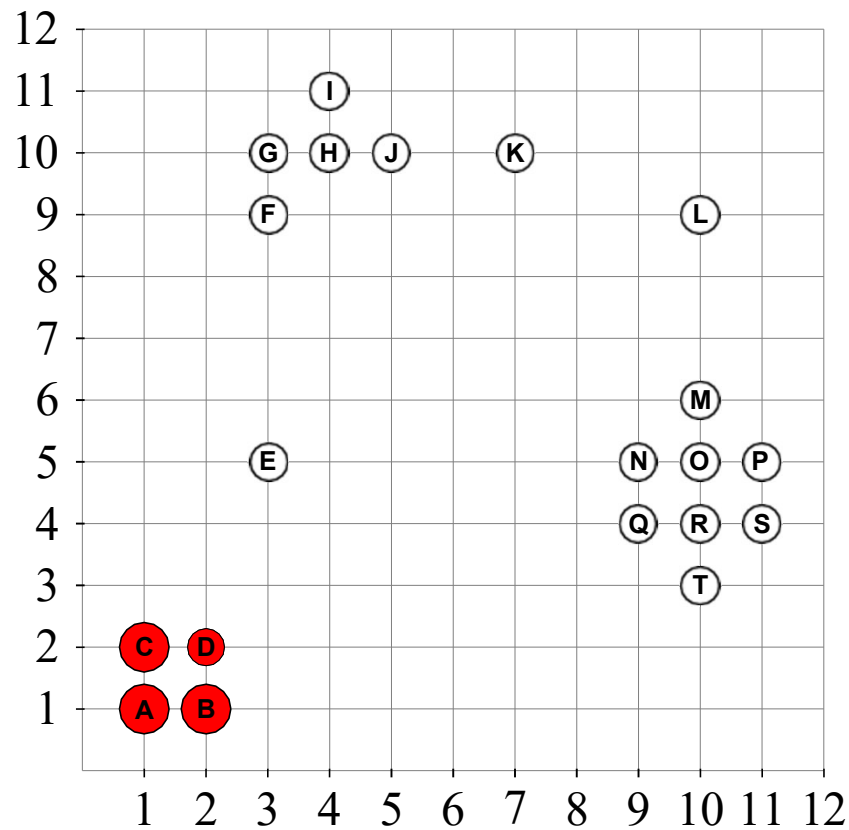
Seed list:

D

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 3$$

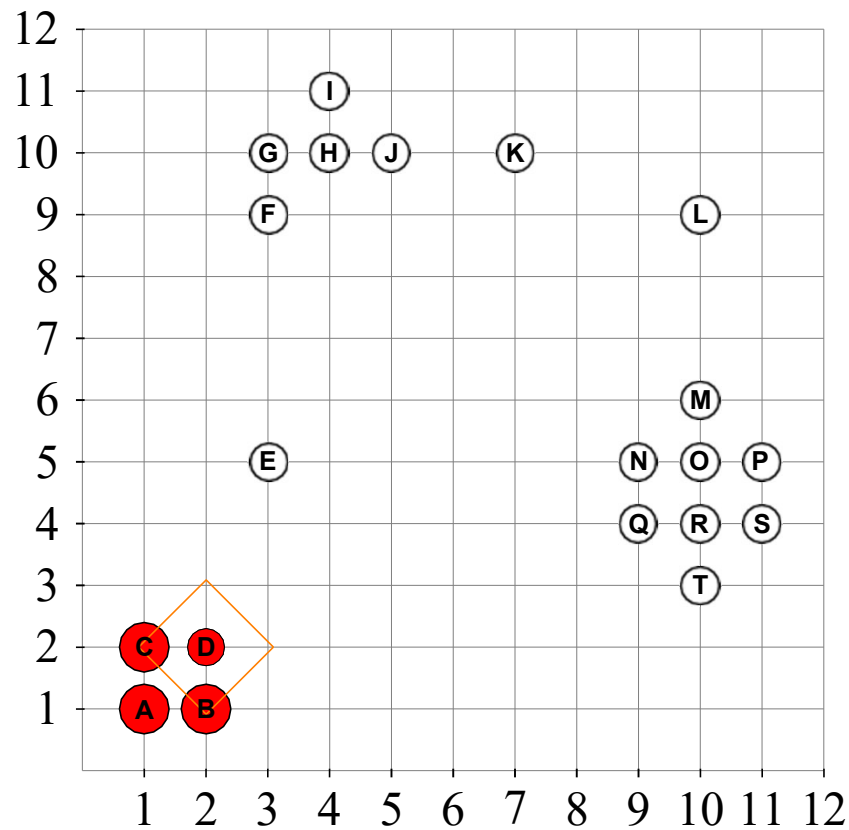
Seed list:

D

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

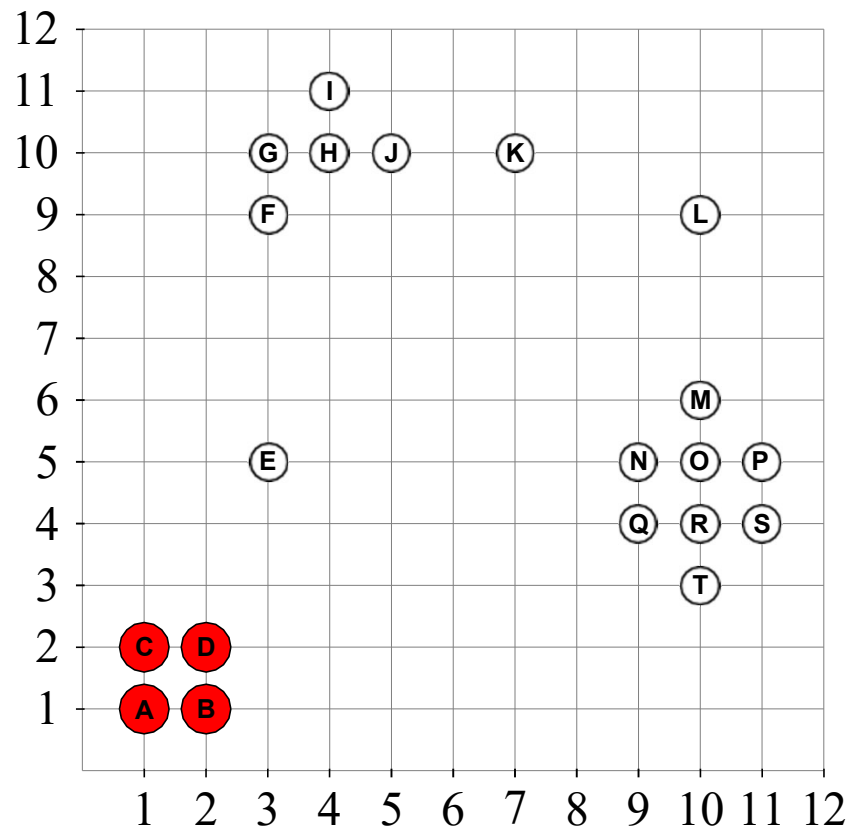
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

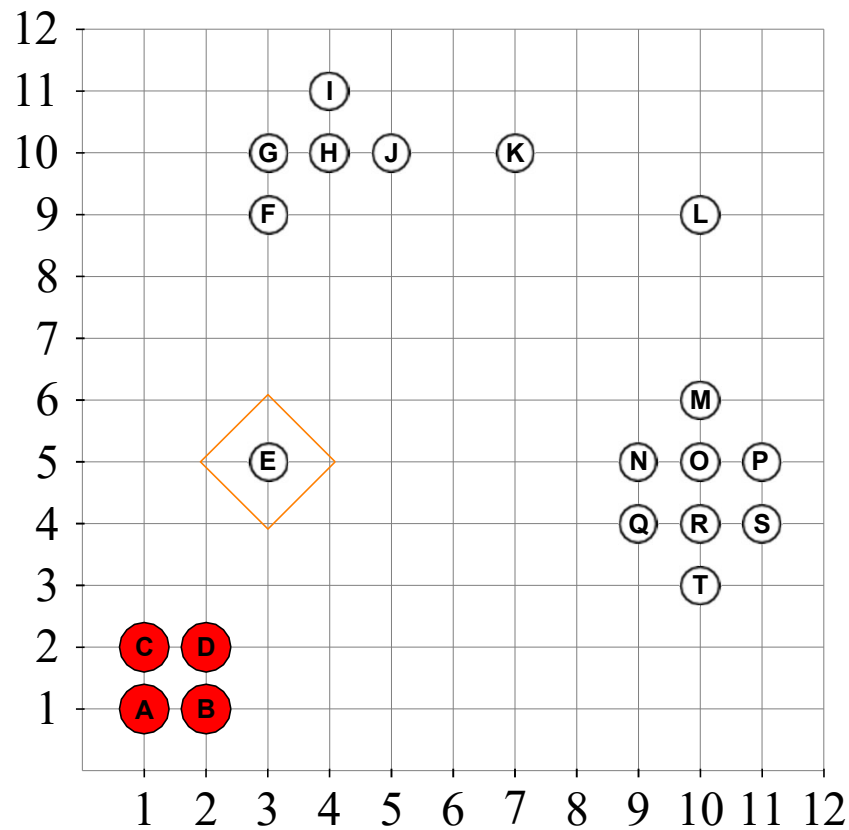
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

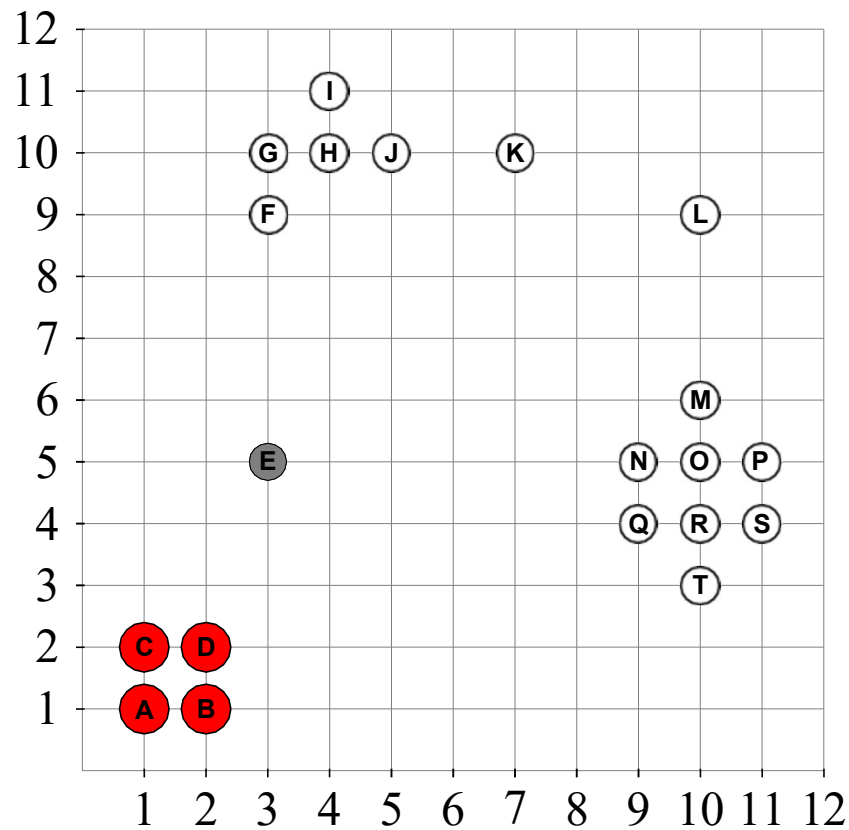
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

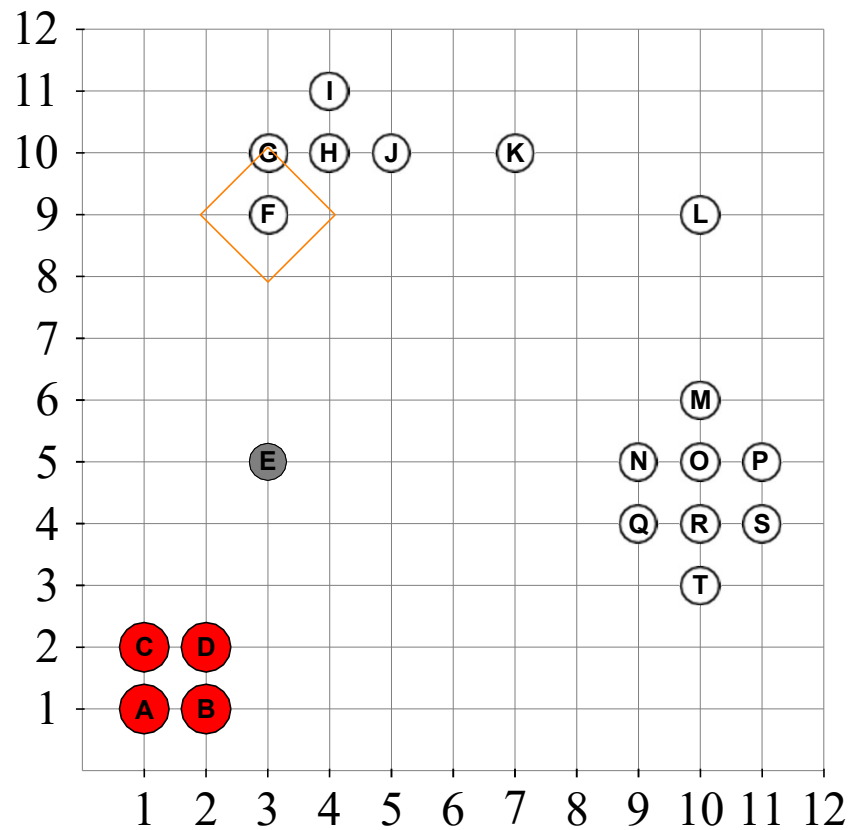
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

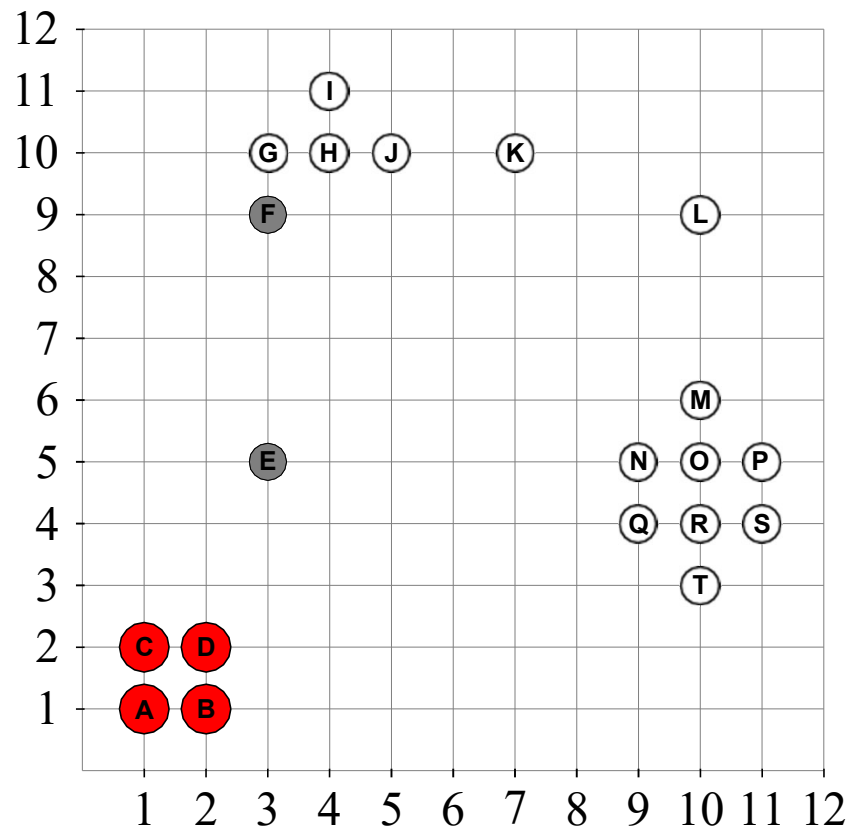
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

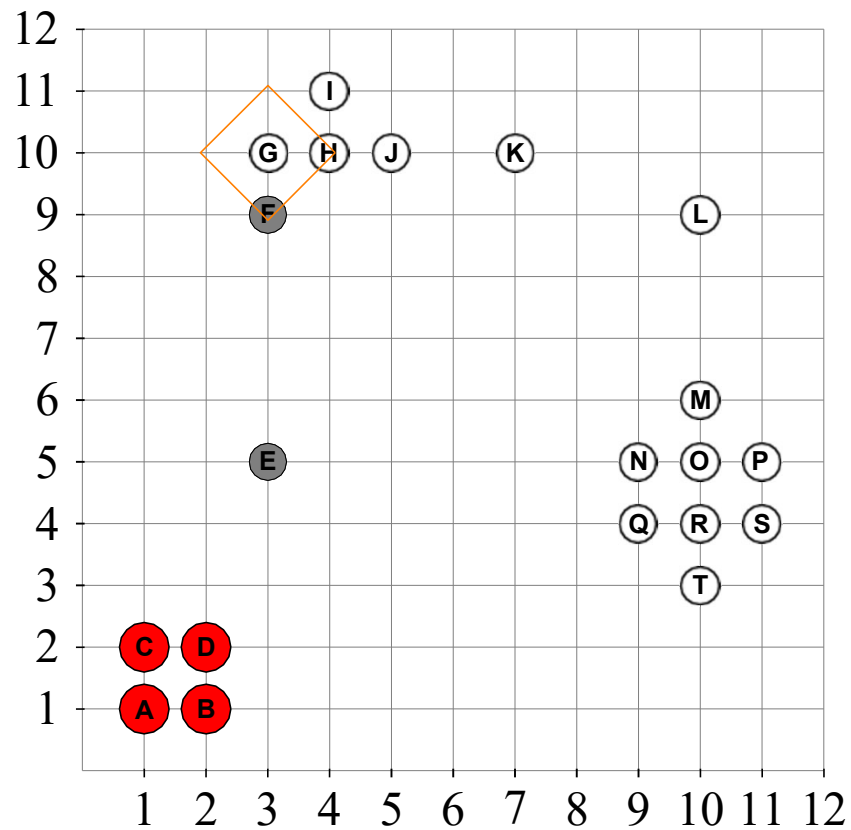
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

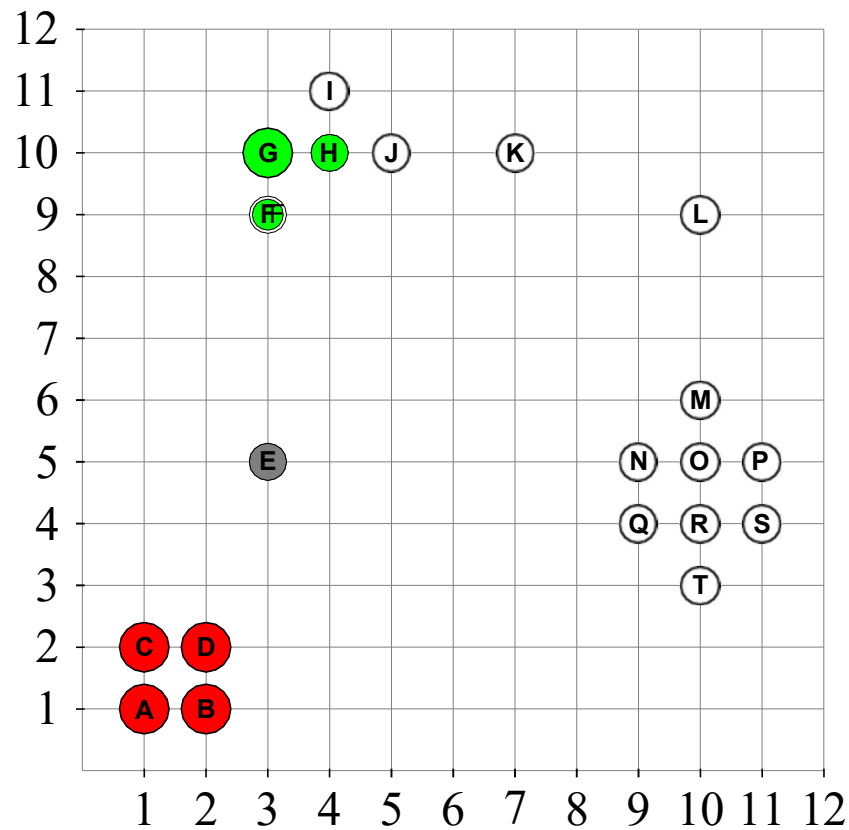
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 3$$

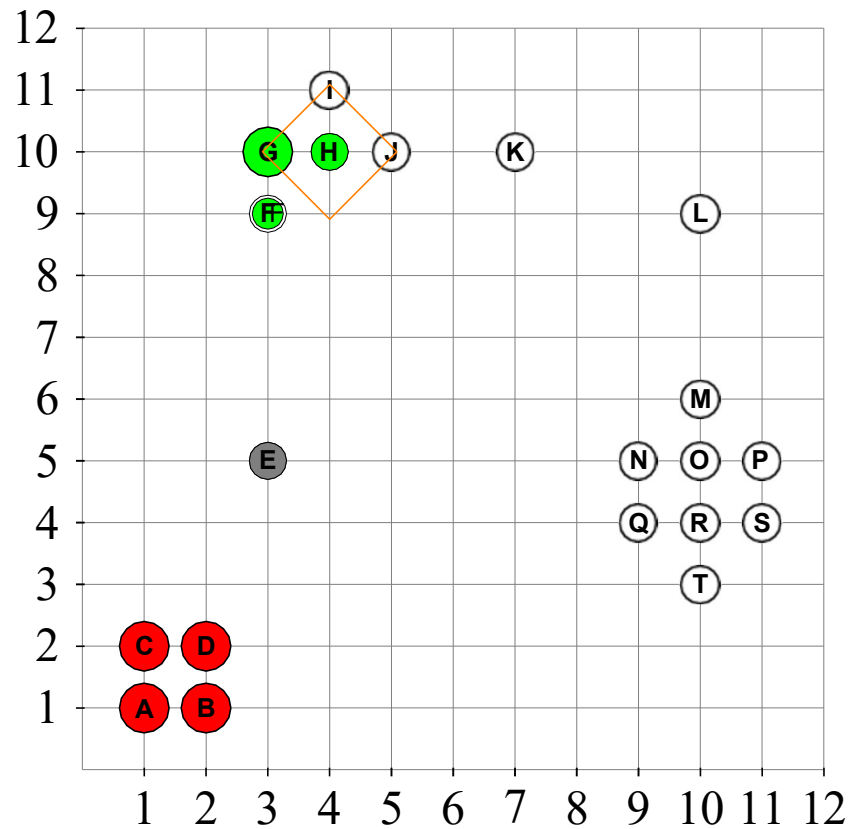
Seed list:

H

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

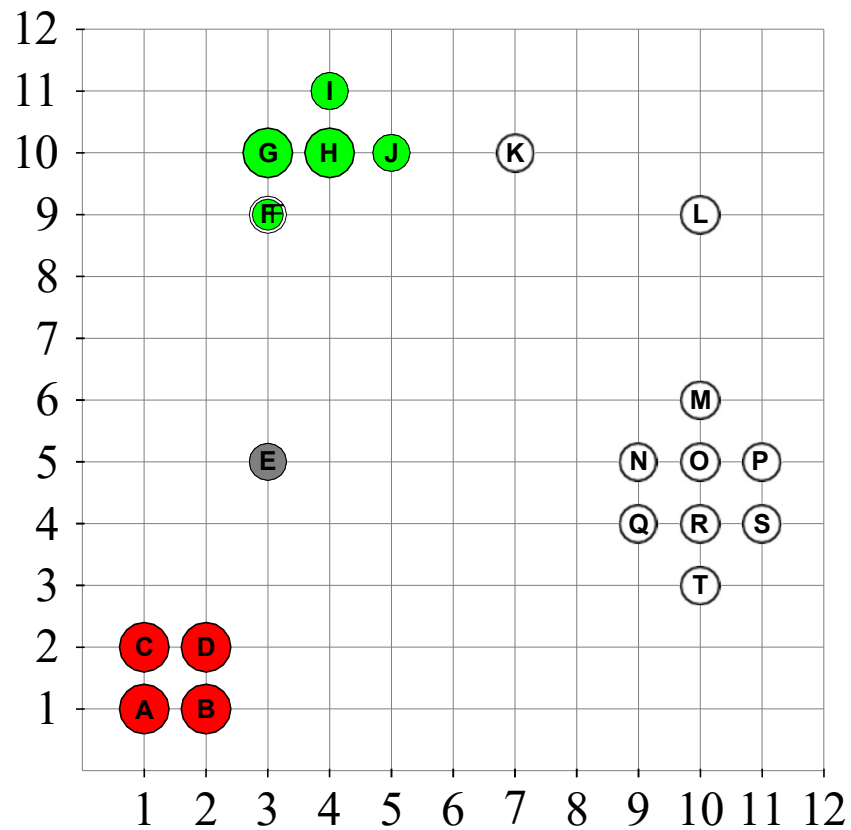
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 3$$

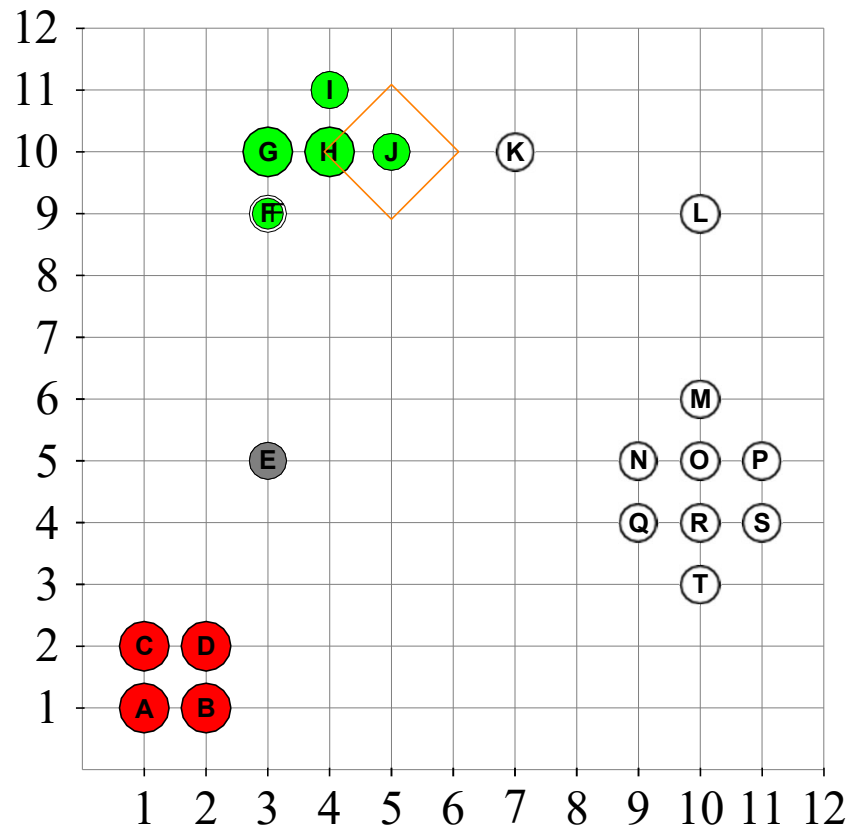
Seed list:

IJ

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 3$$

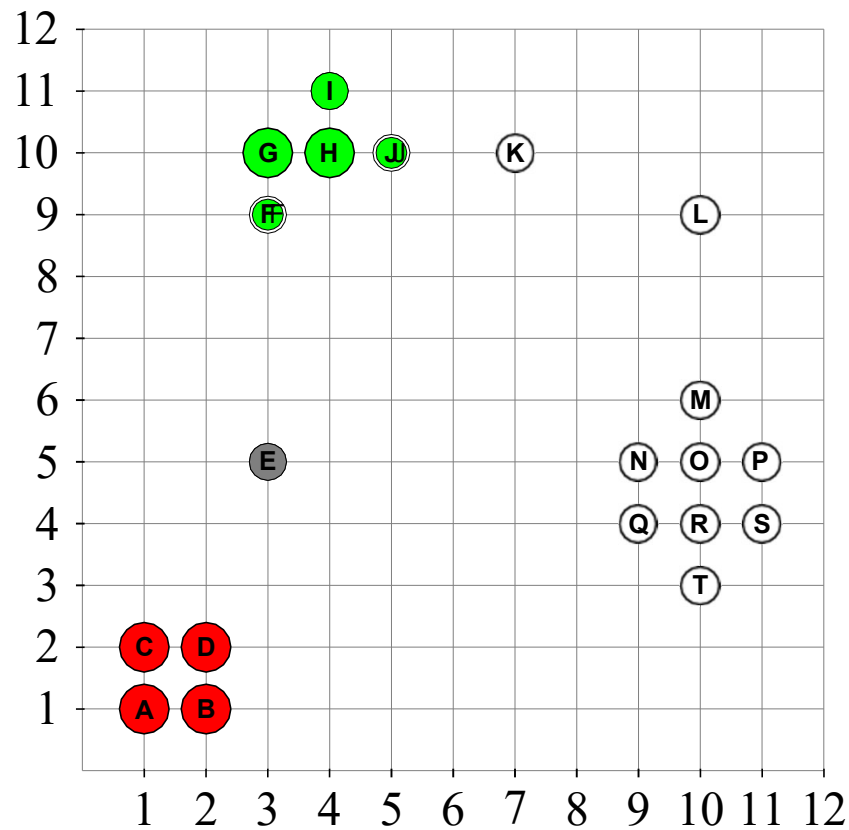
Seed list:

I

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 3$$

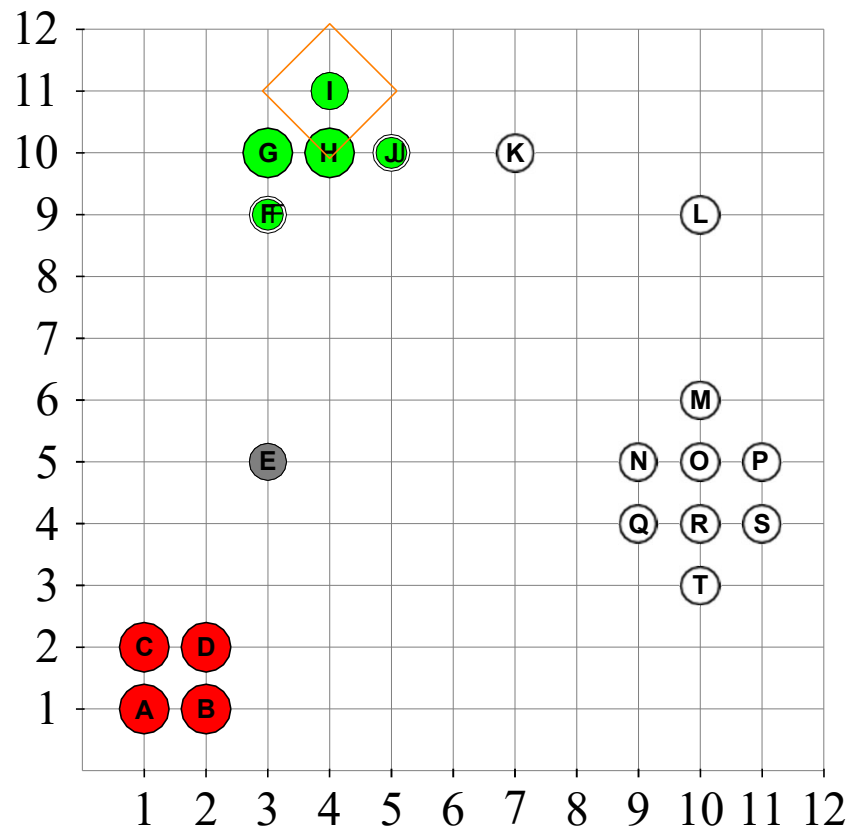
Seed list:

I

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

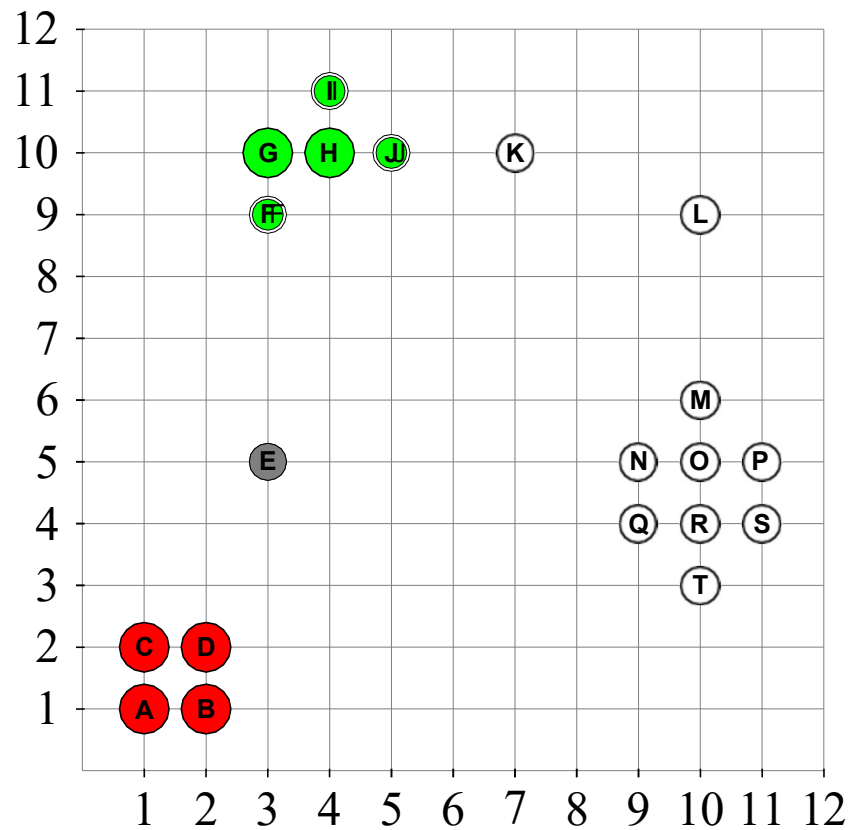
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

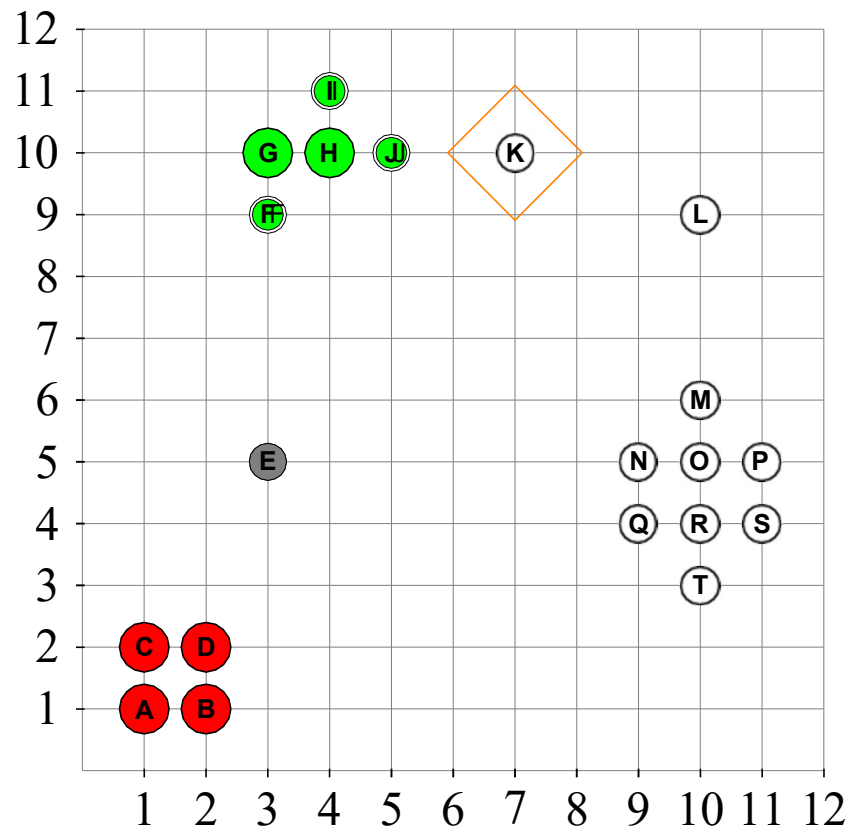
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

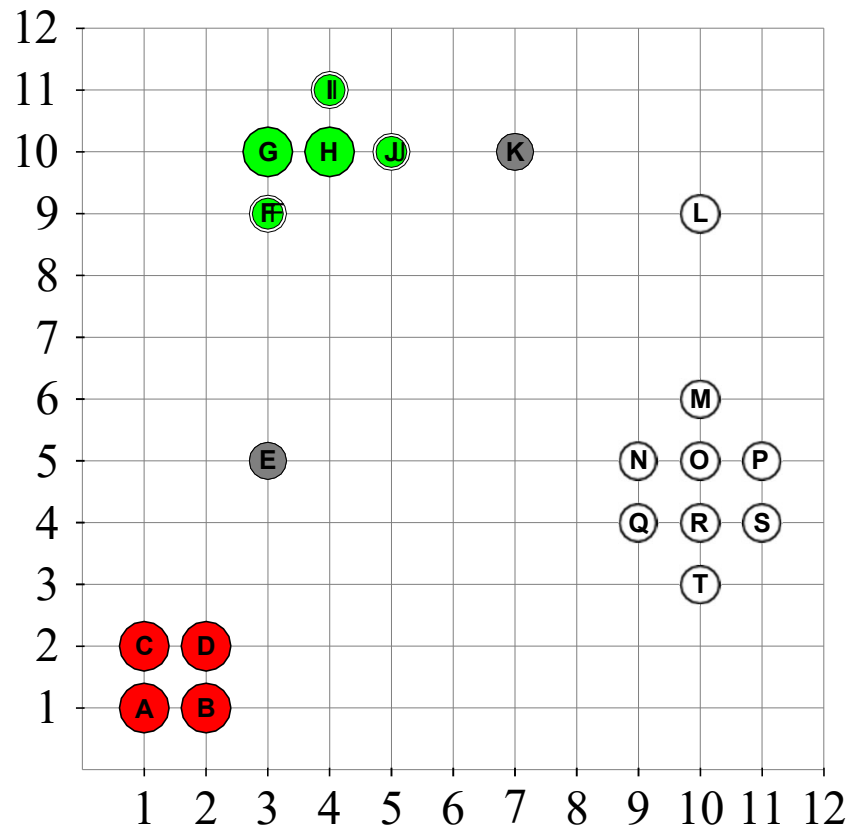
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

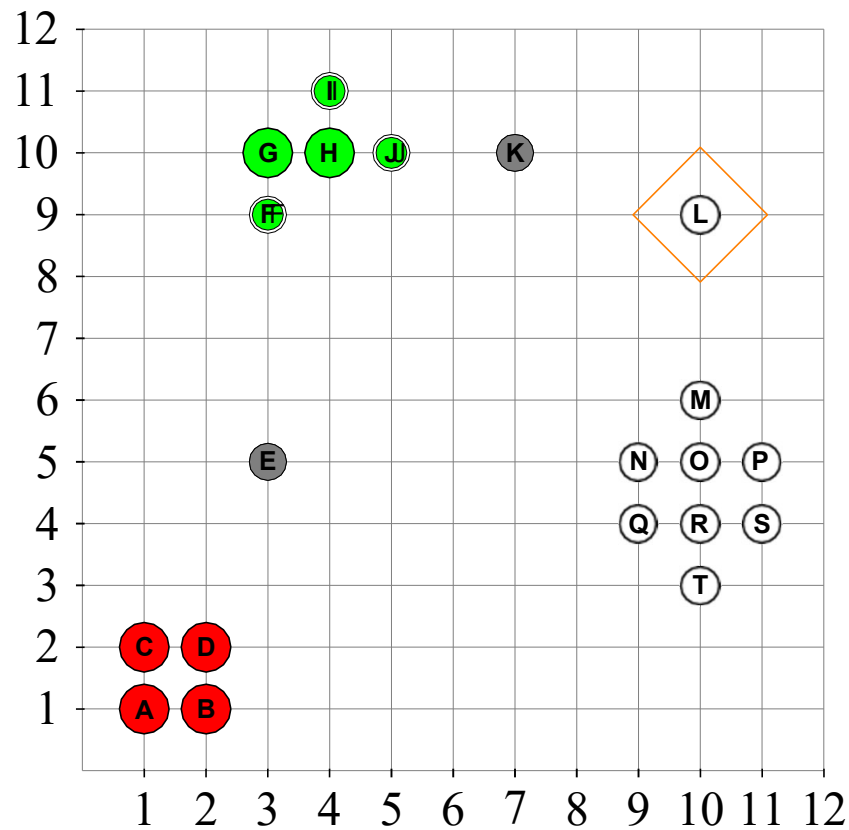
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

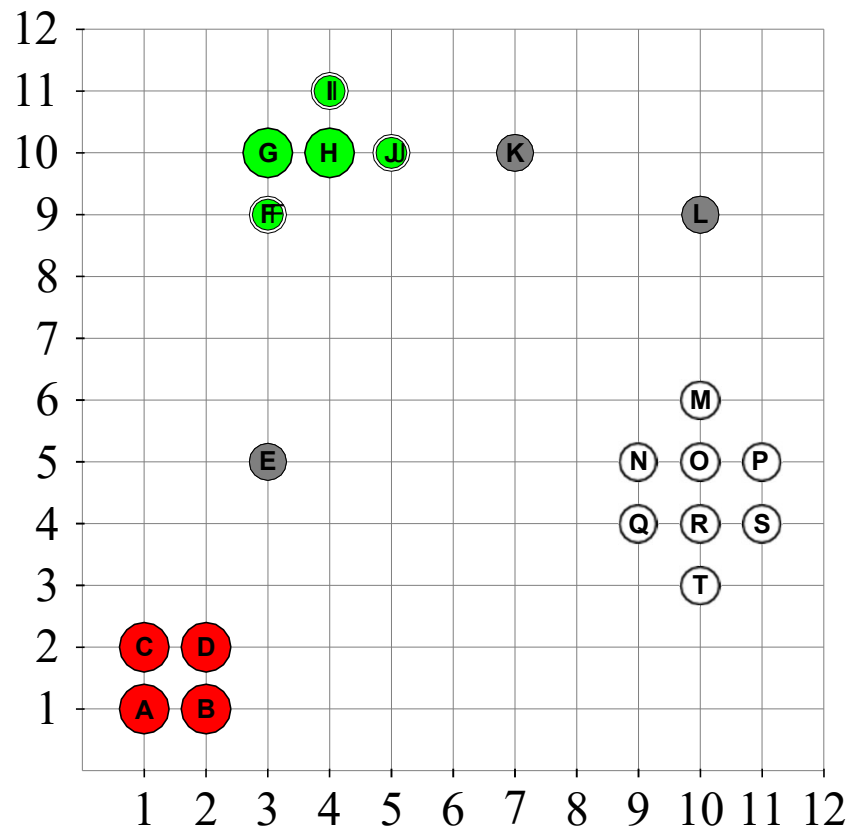
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

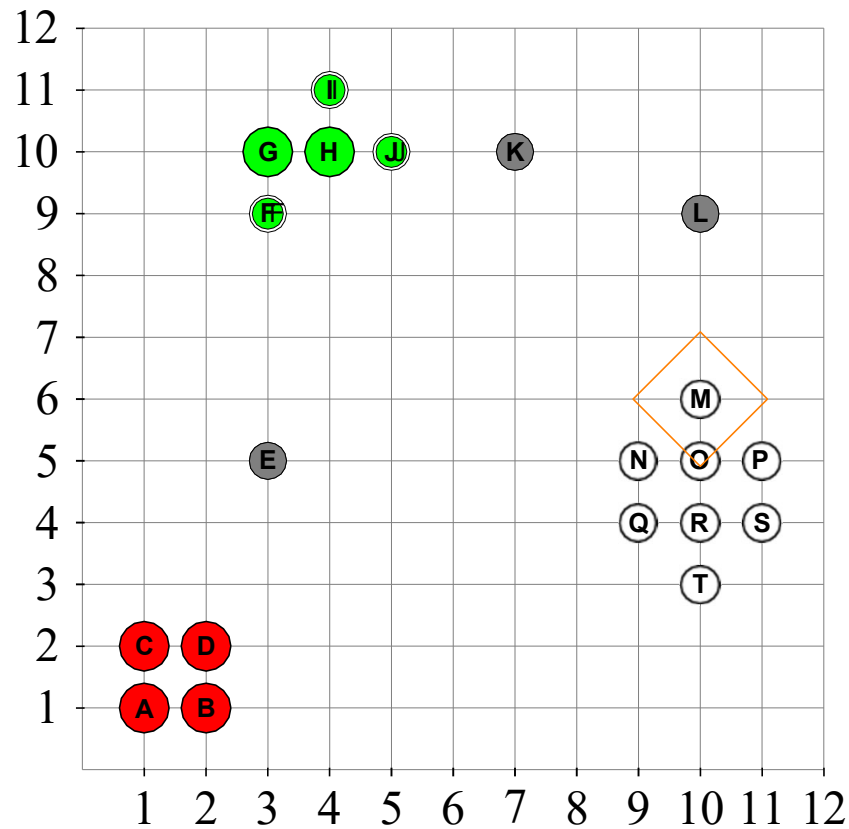
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

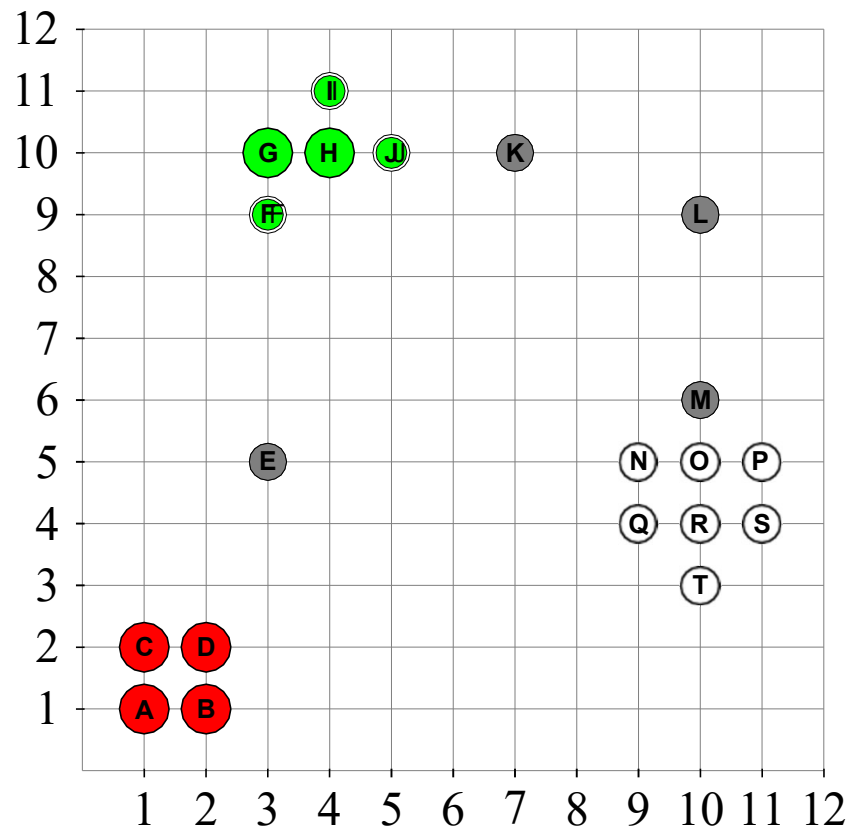
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

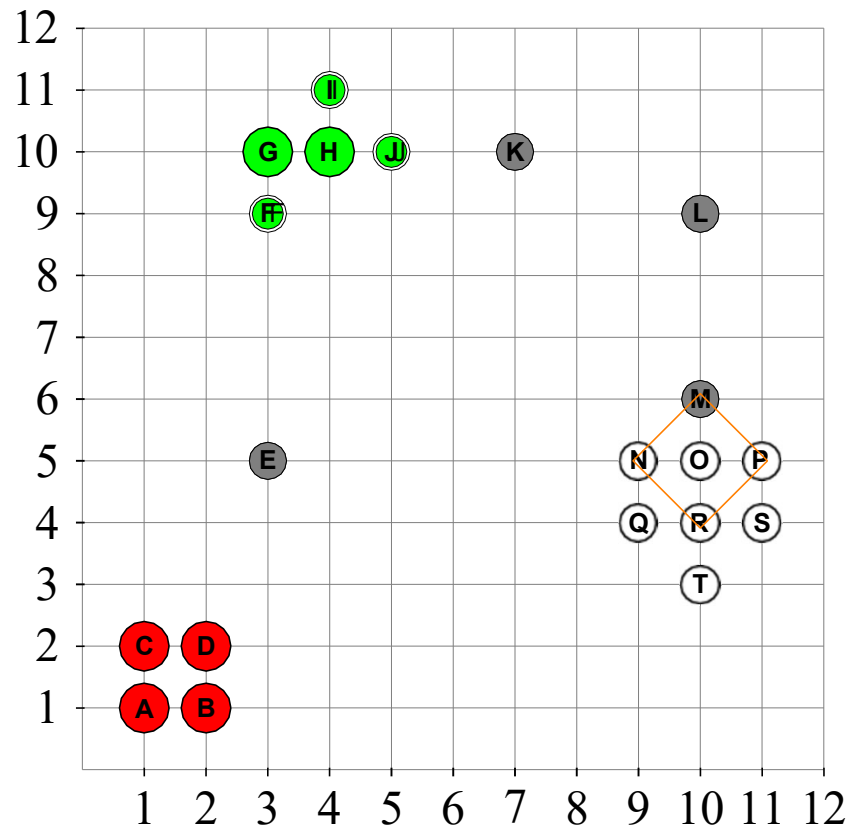
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

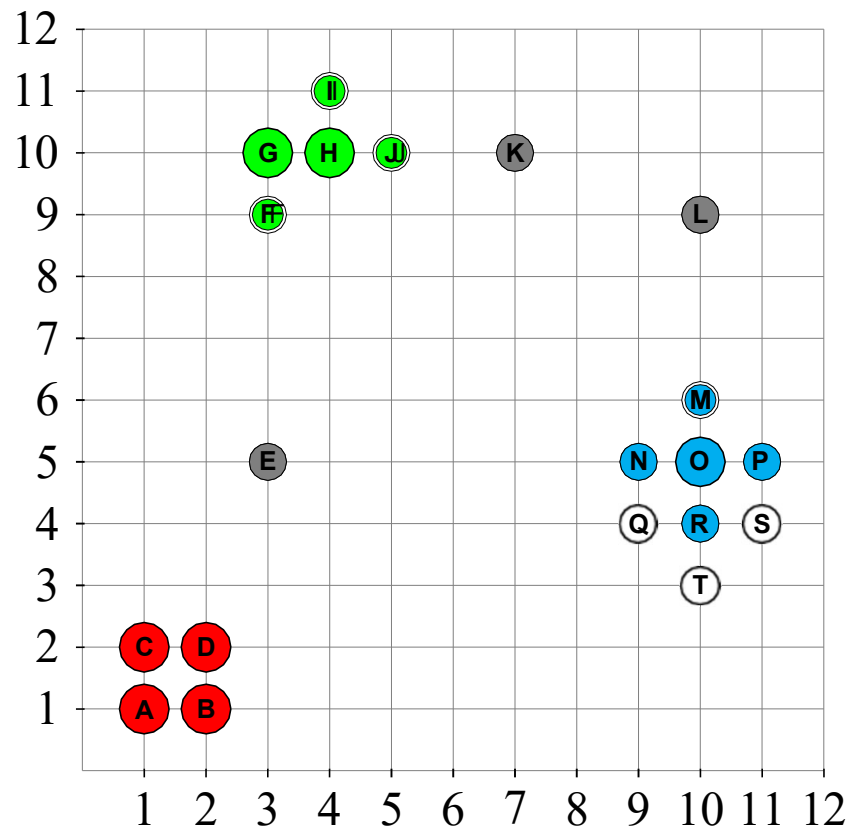
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

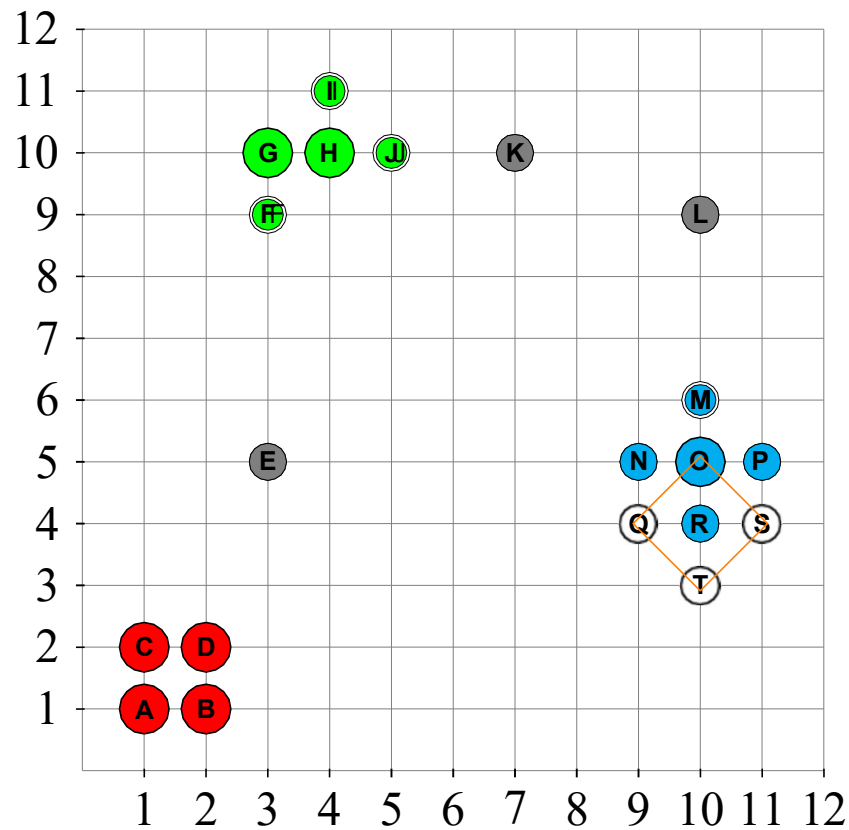
$$\text{minPts} = 3$$

Seed list:
NPR

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

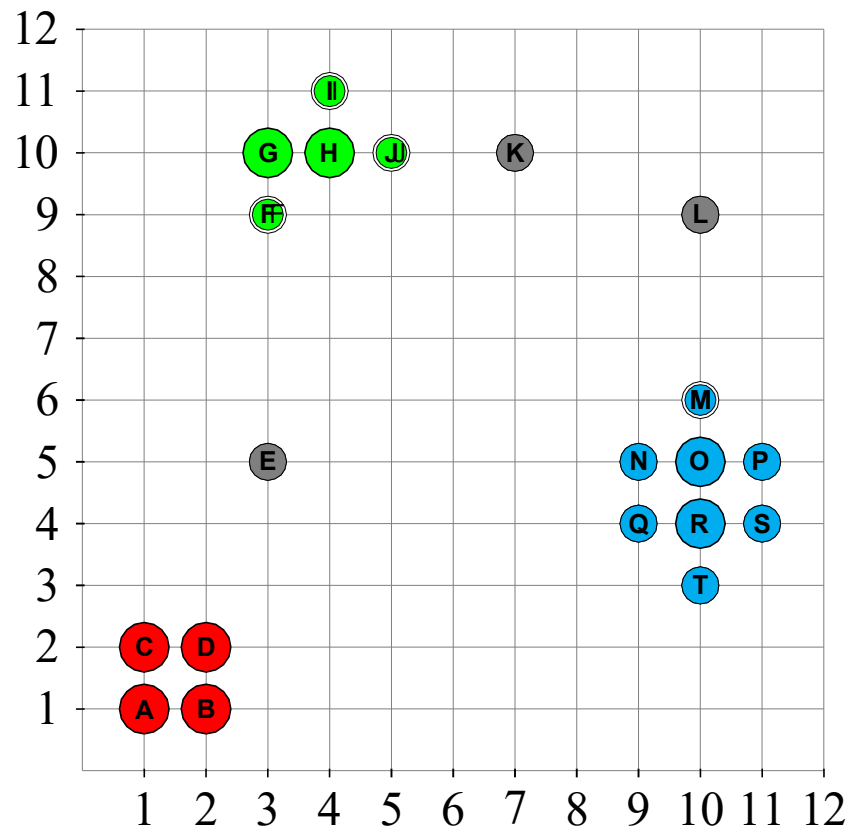
$$\text{minPts} = 3$$

Seed list:
NP

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

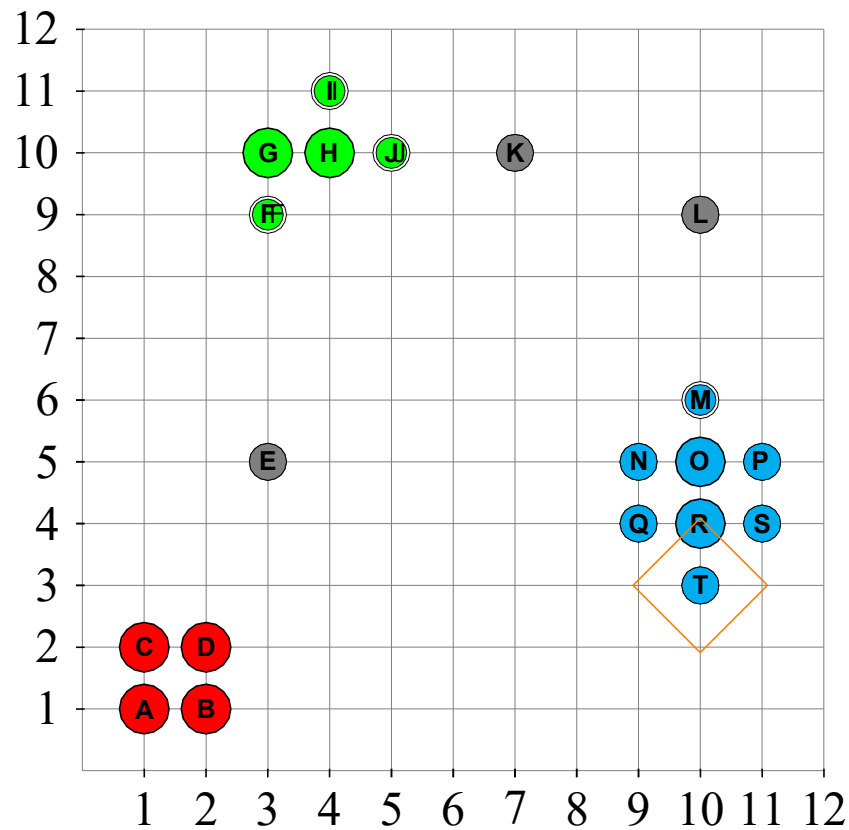
$$\text{minPts} = 3$$

Seed list:
NPQST

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

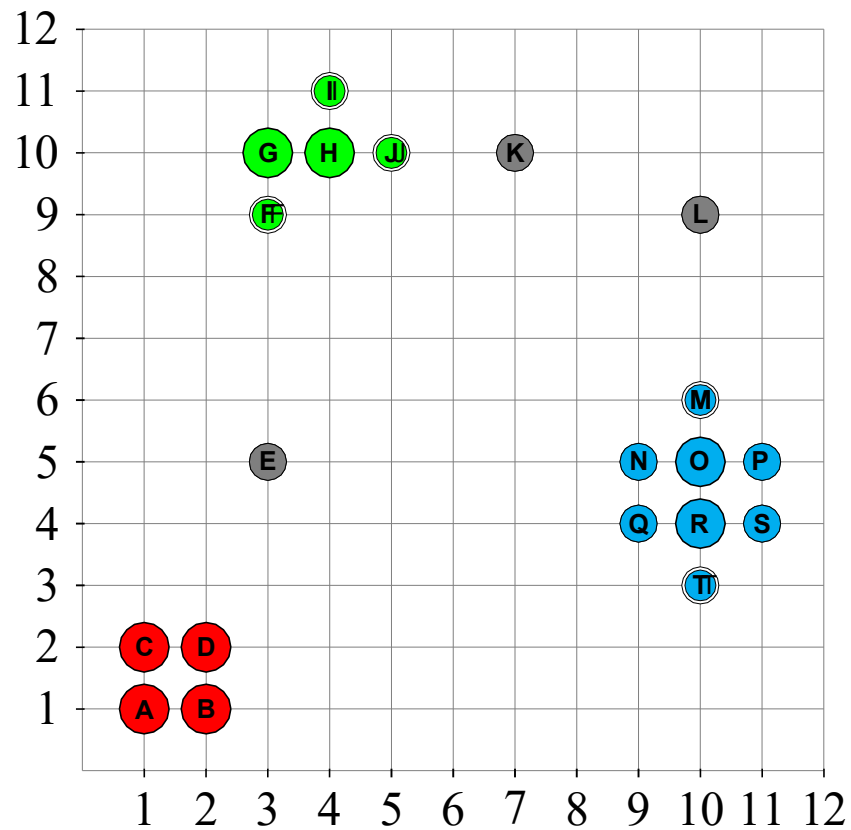
$$\text{minPts} = 3$$

Seed list:
NPQS

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

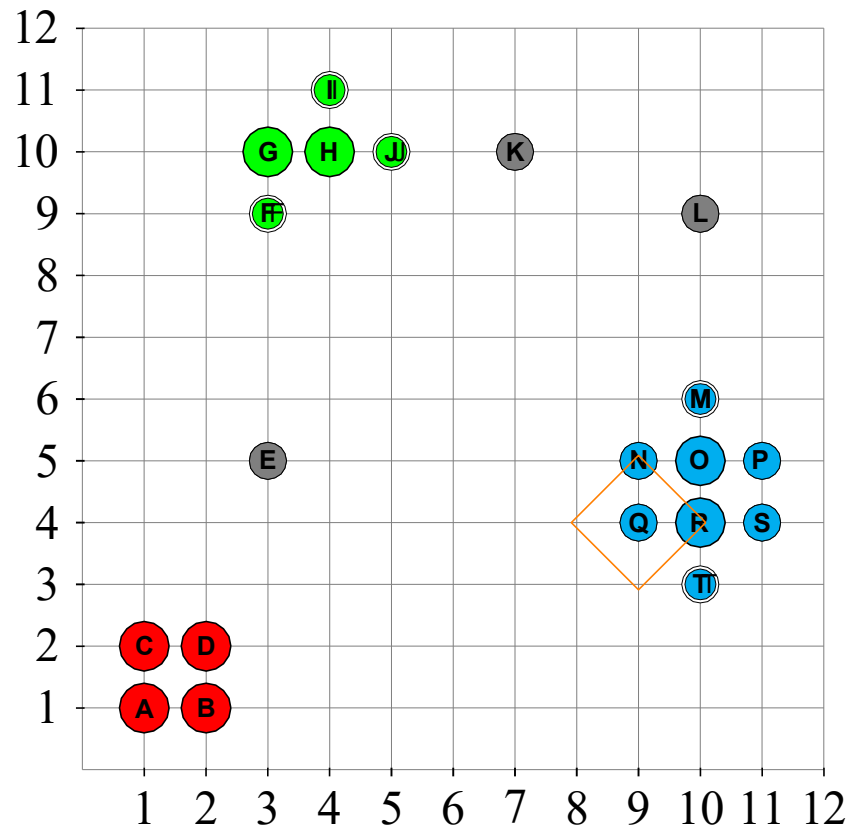
$$\text{minPts} = 3$$

Seed list:
NPQS

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

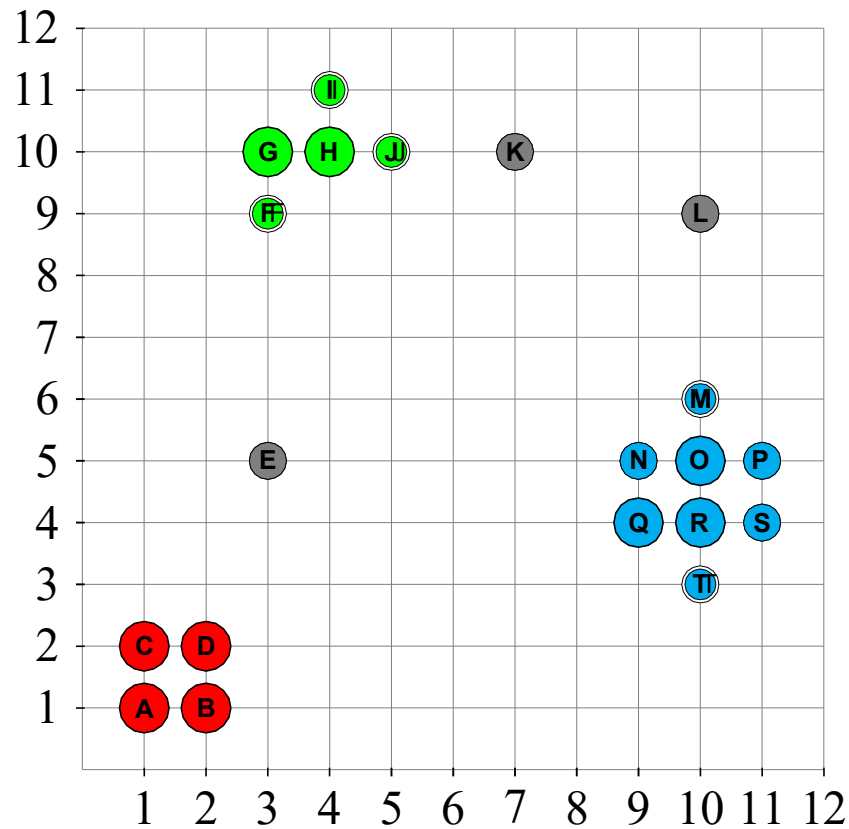
$$\text{minPts} = 3$$

Seed list:
NPS

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

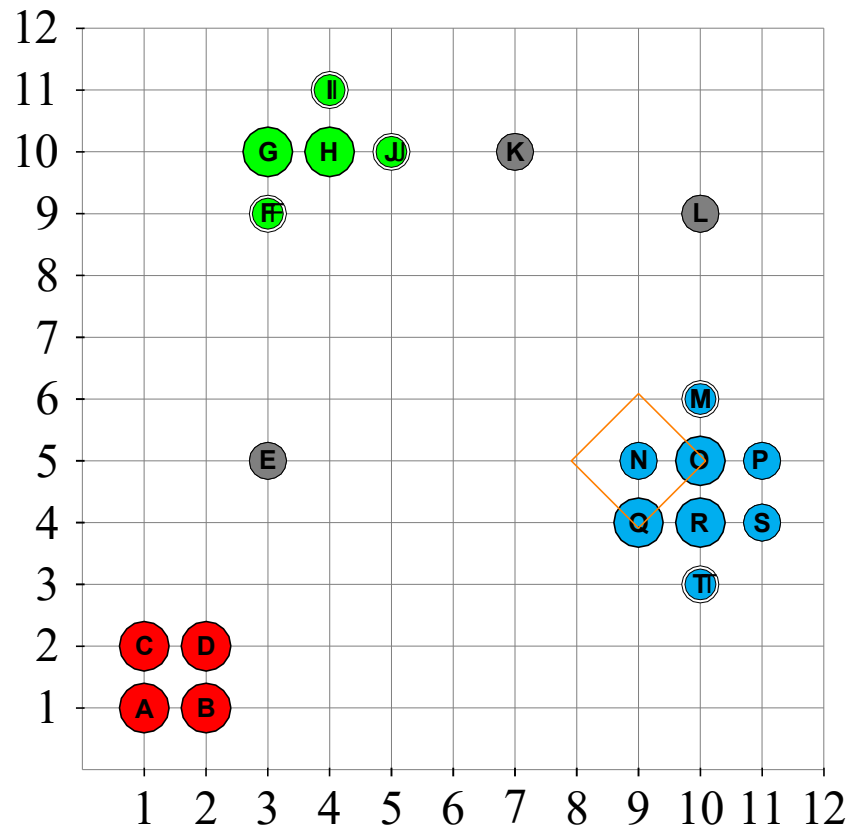
$$\text{minPts} = 3$$

Seed list:
NPS

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

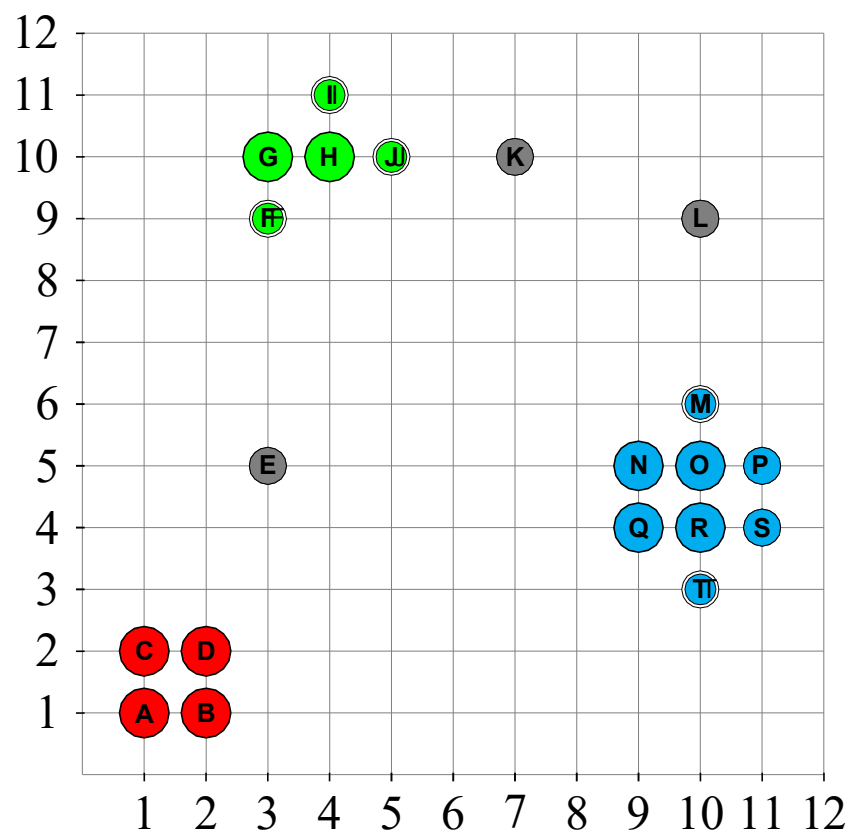
$$\text{minPts} = 3$$

Seed list:
PS

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

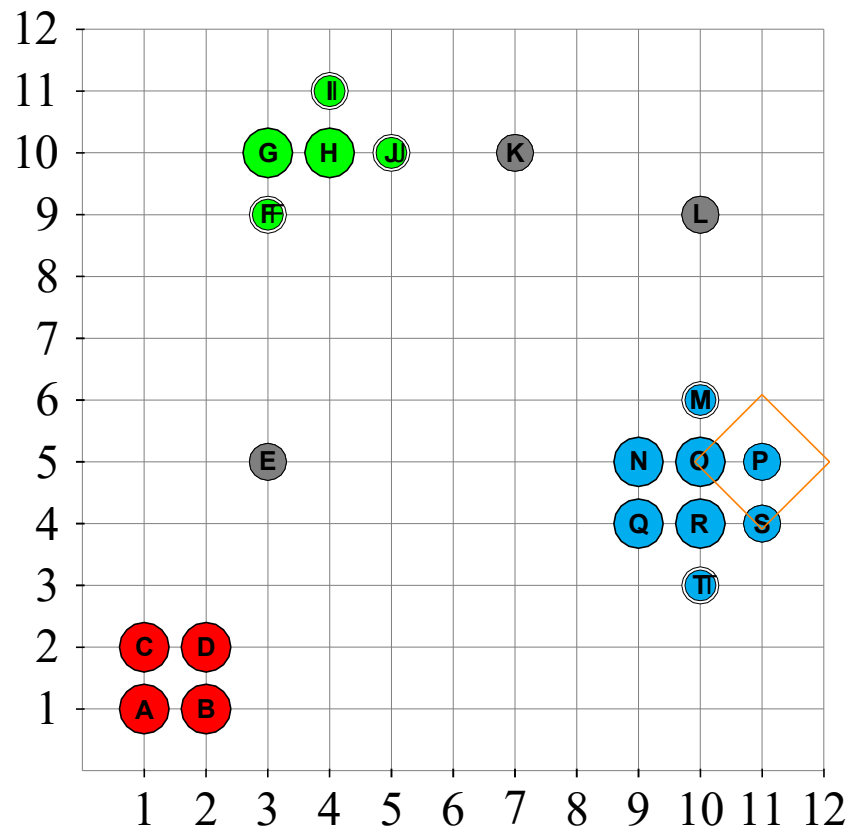
$$\text{minPts} = 3$$

Seed list:
PS

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 3$$

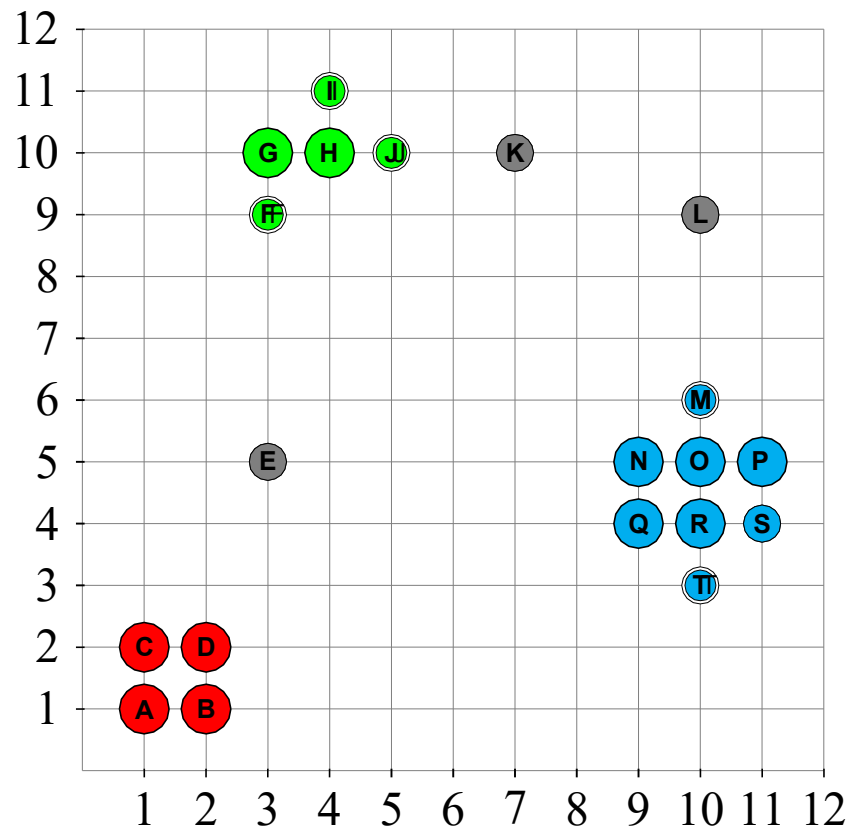
Seed list:

S

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 3$$

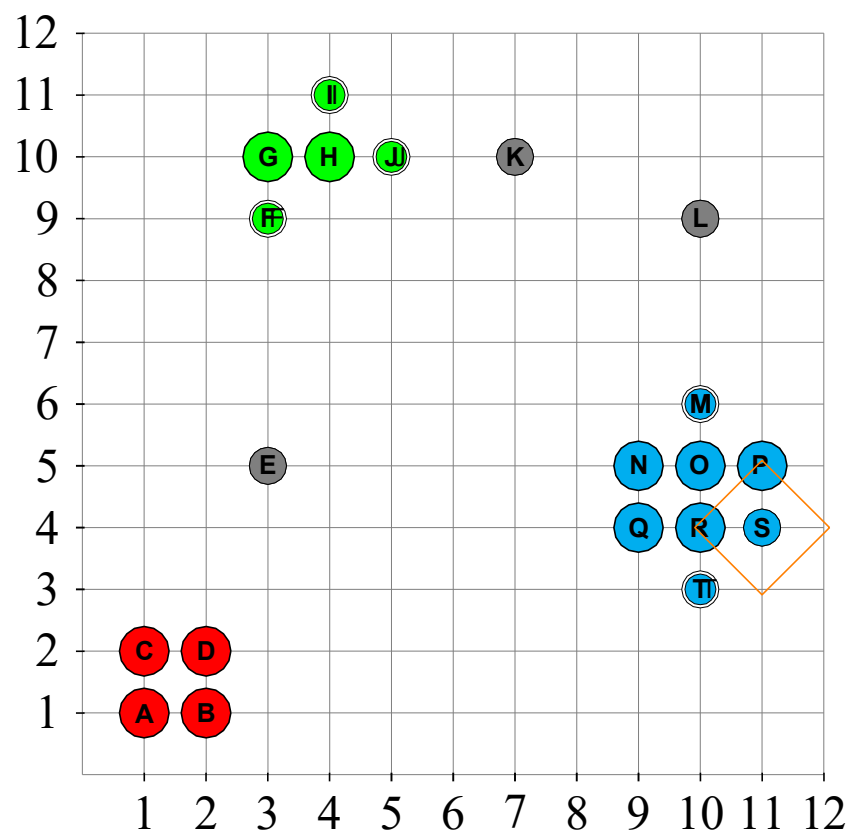
Seed list:

S

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

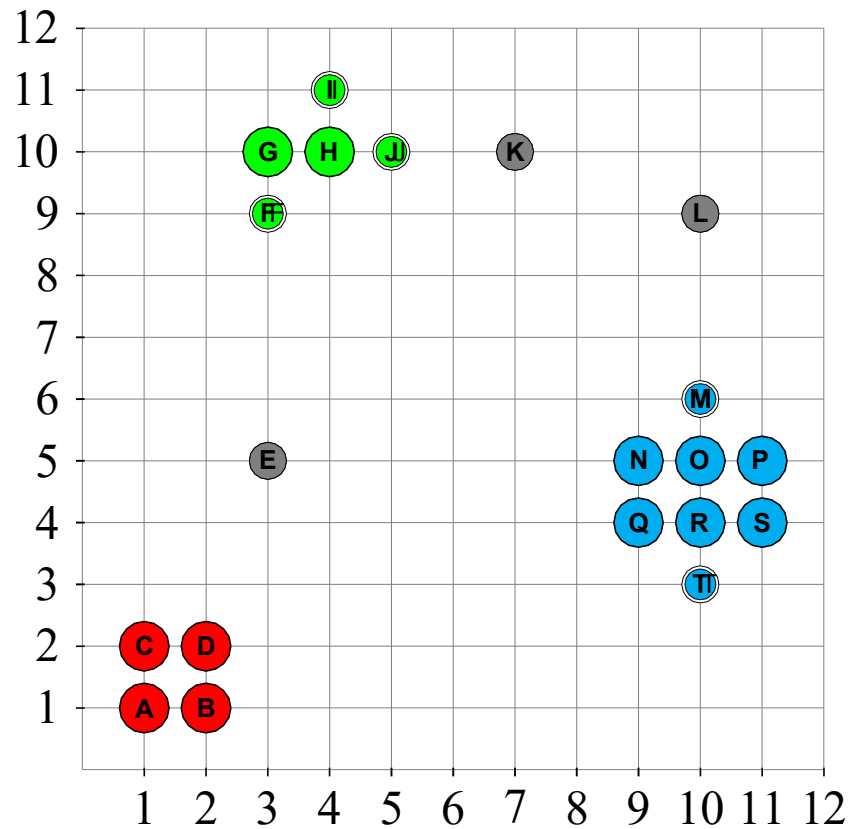
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

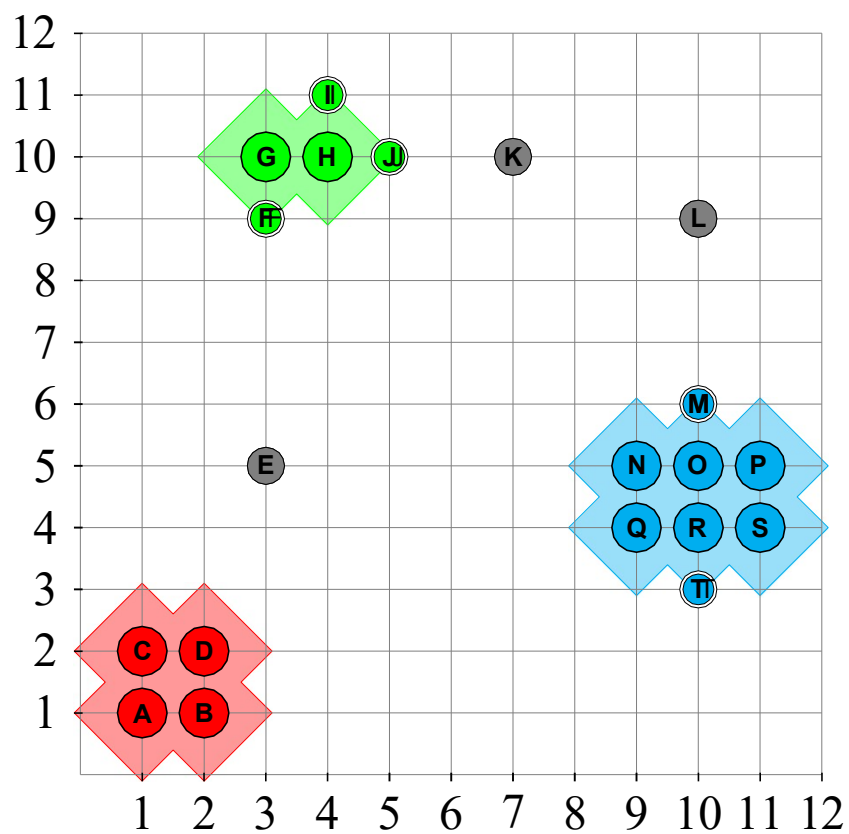
$$\text{minPts} = 3$$

Seed list:

Example 2

DM868 DM870
DS804

Arthur Zimek



$$\epsilon = 1.1$$

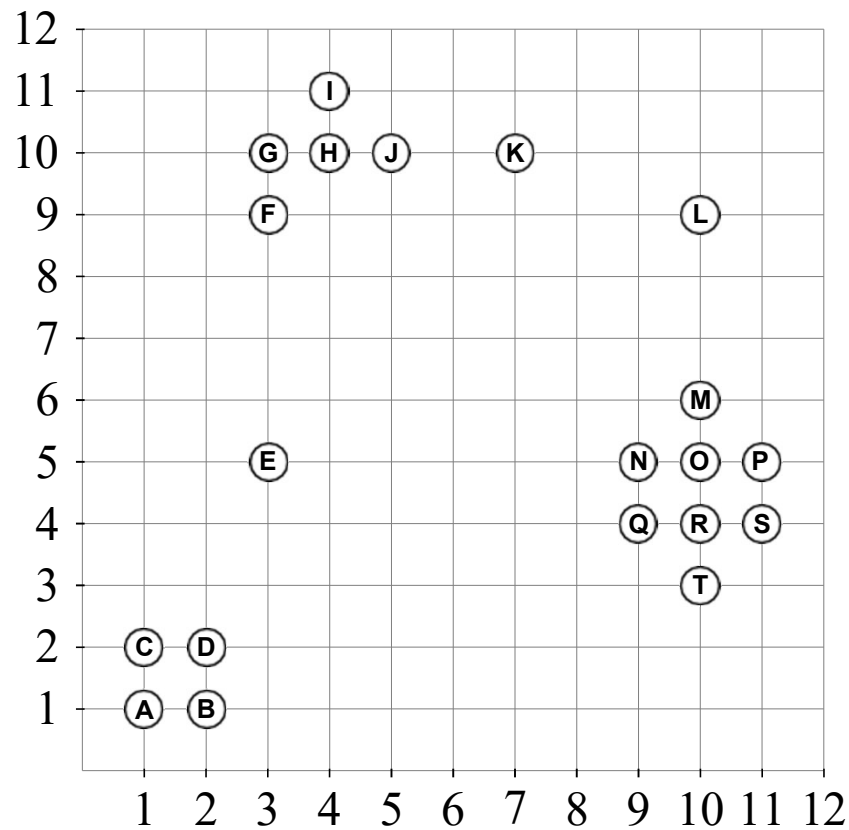
$$\minPts = 3$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

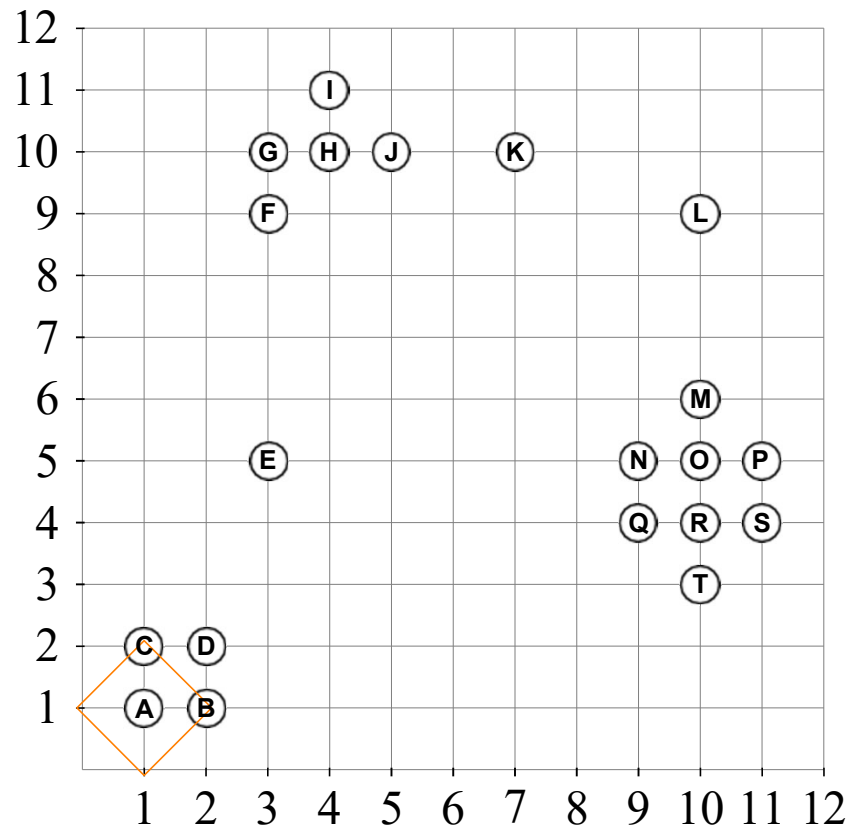
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

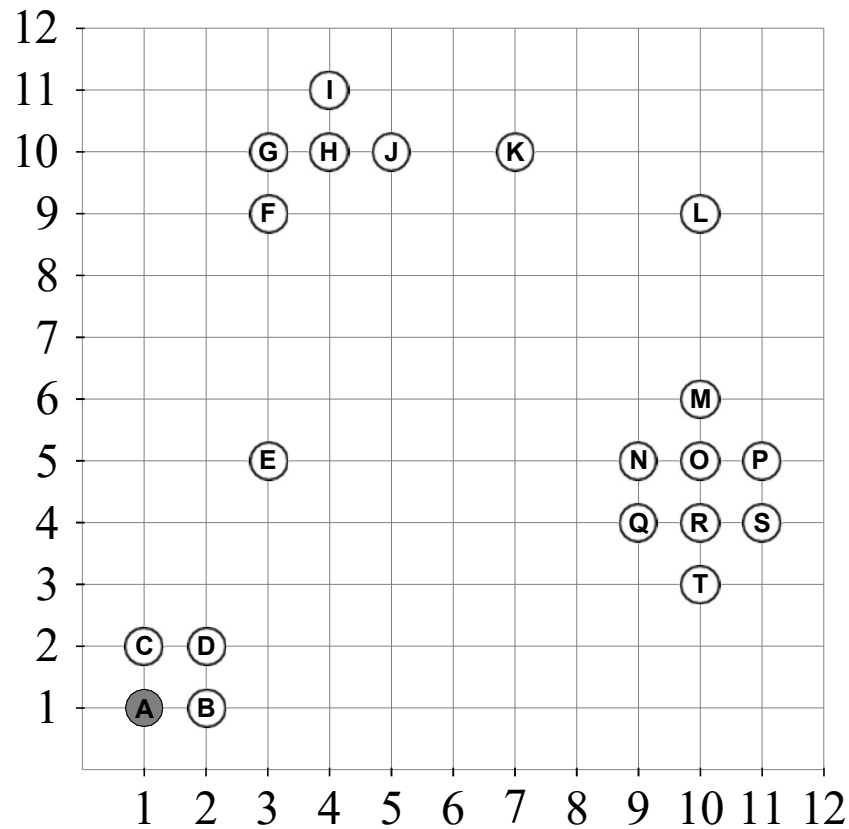
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

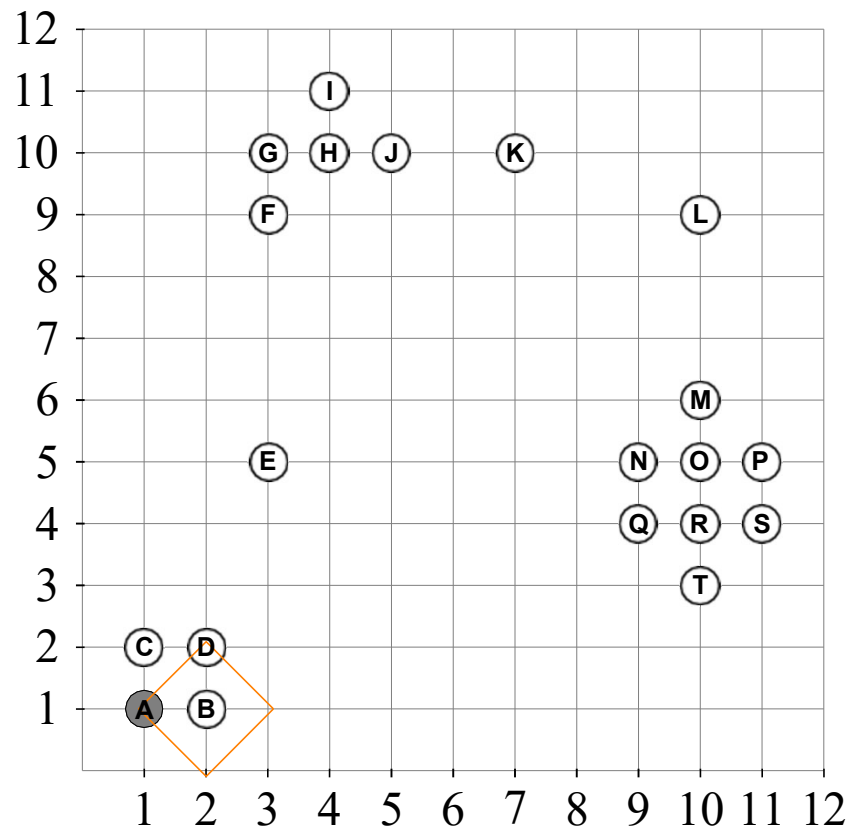
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

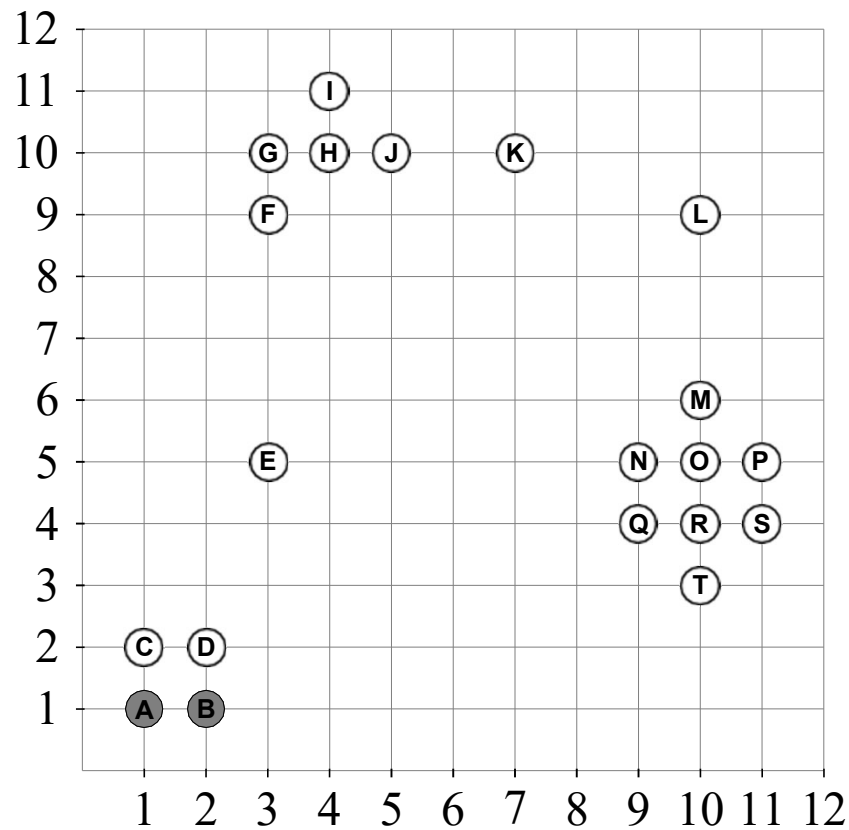
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

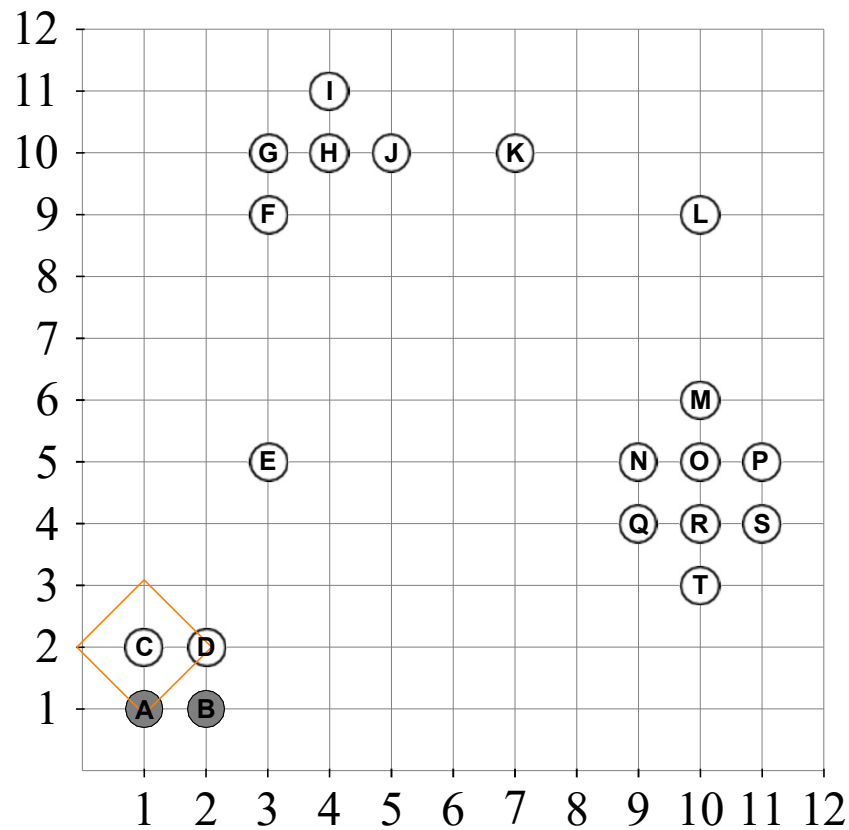
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

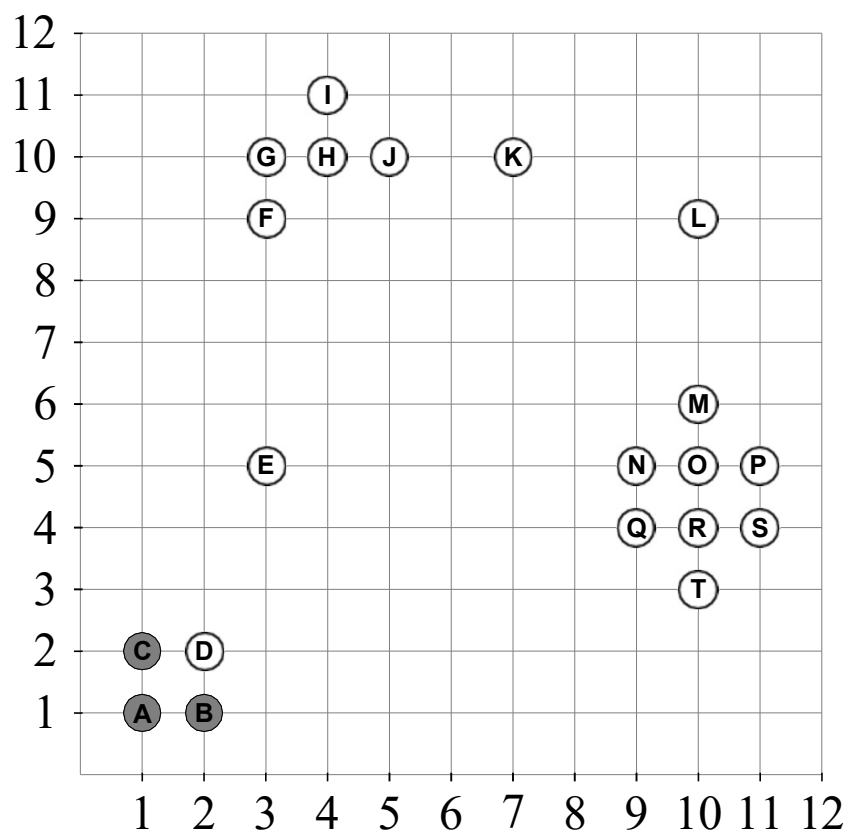
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

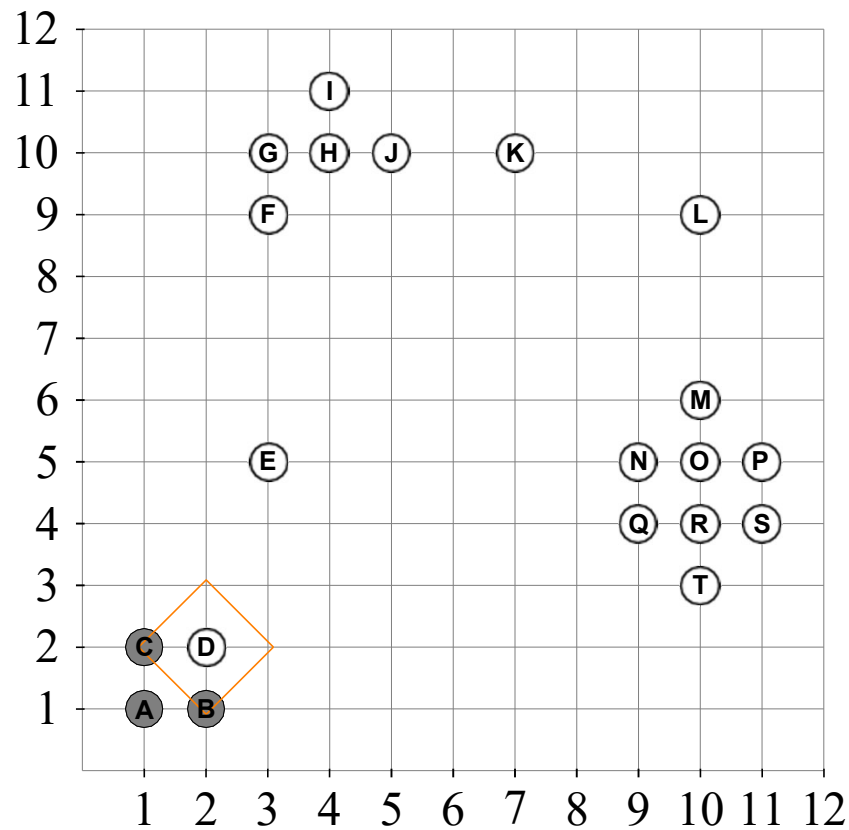
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

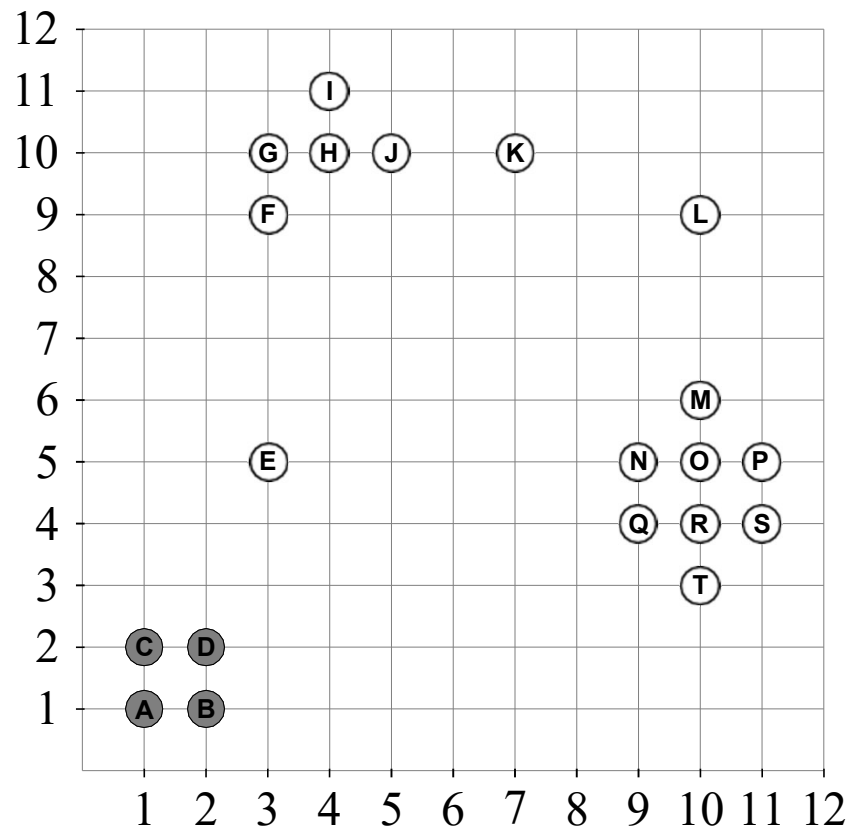
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

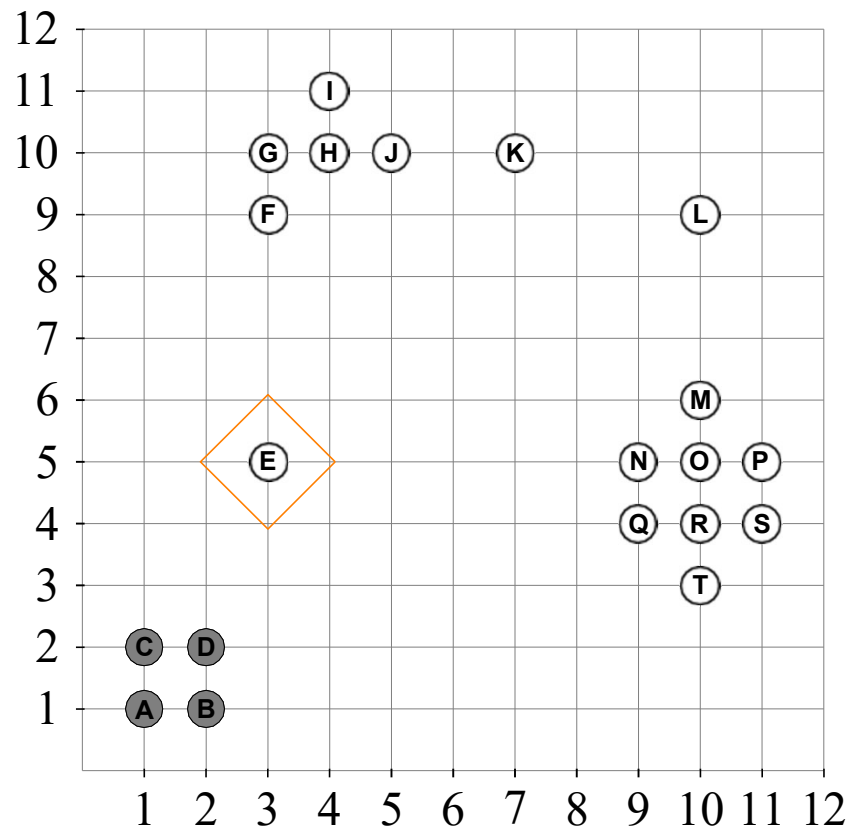
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

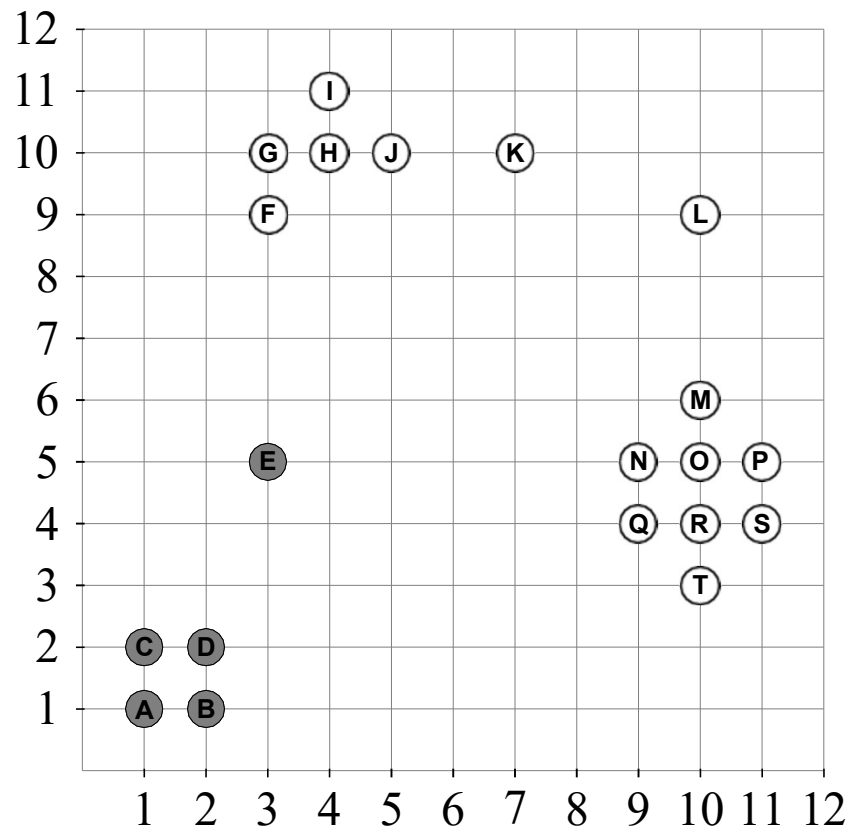
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

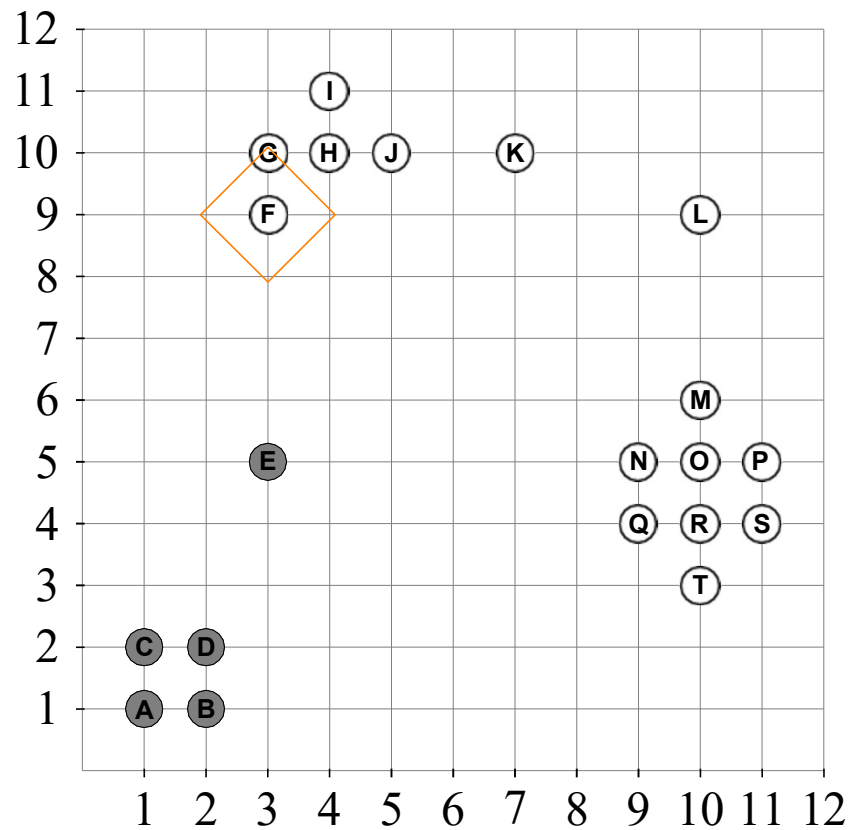
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

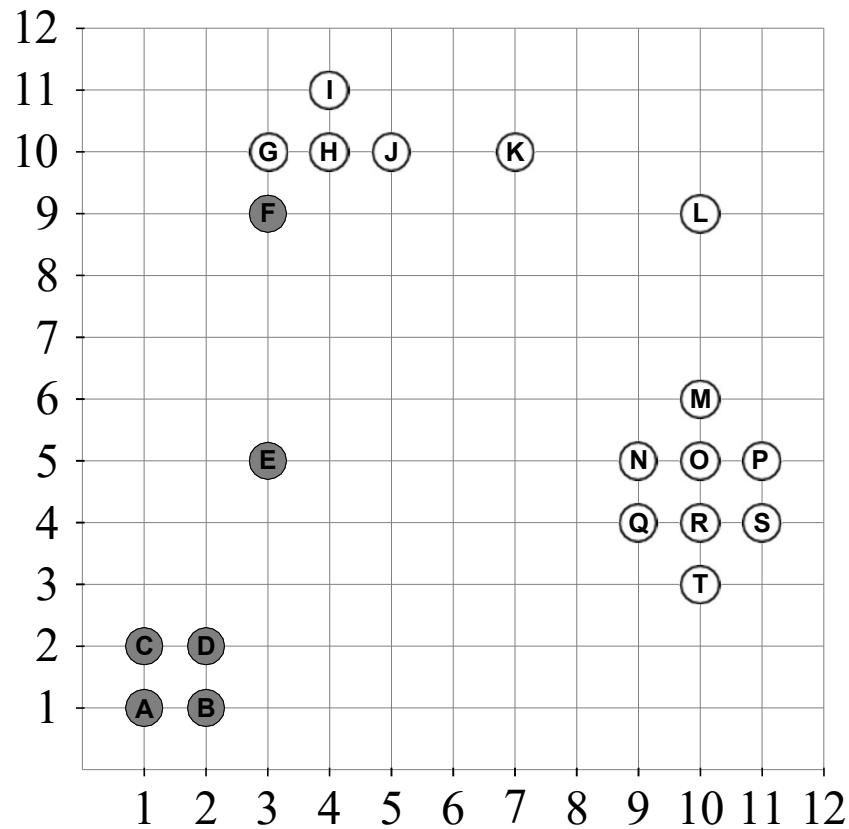
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

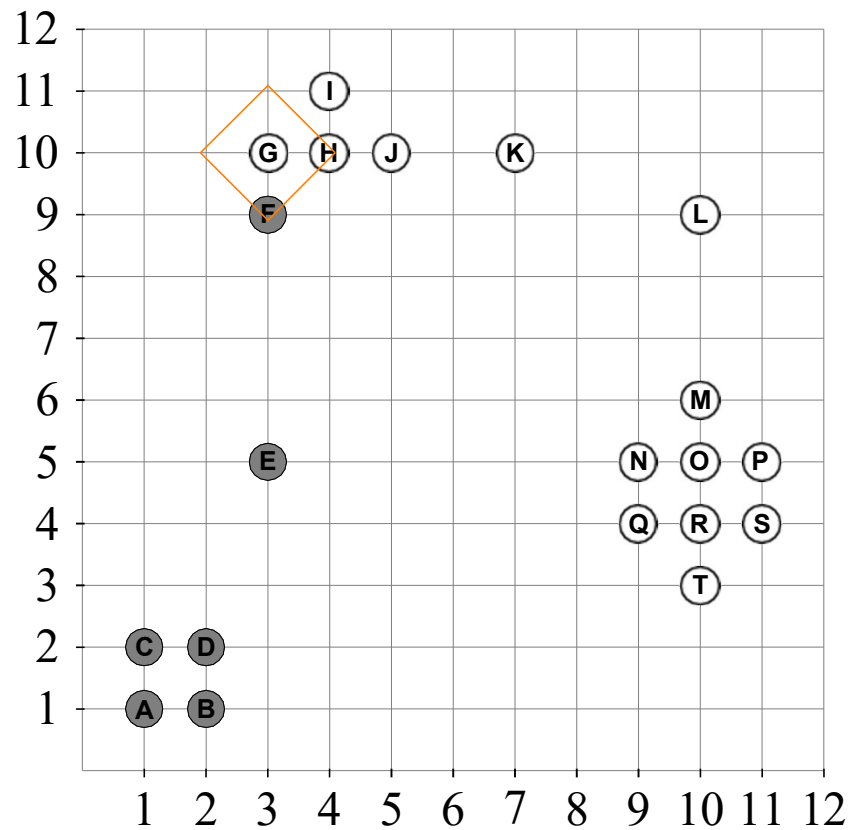
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

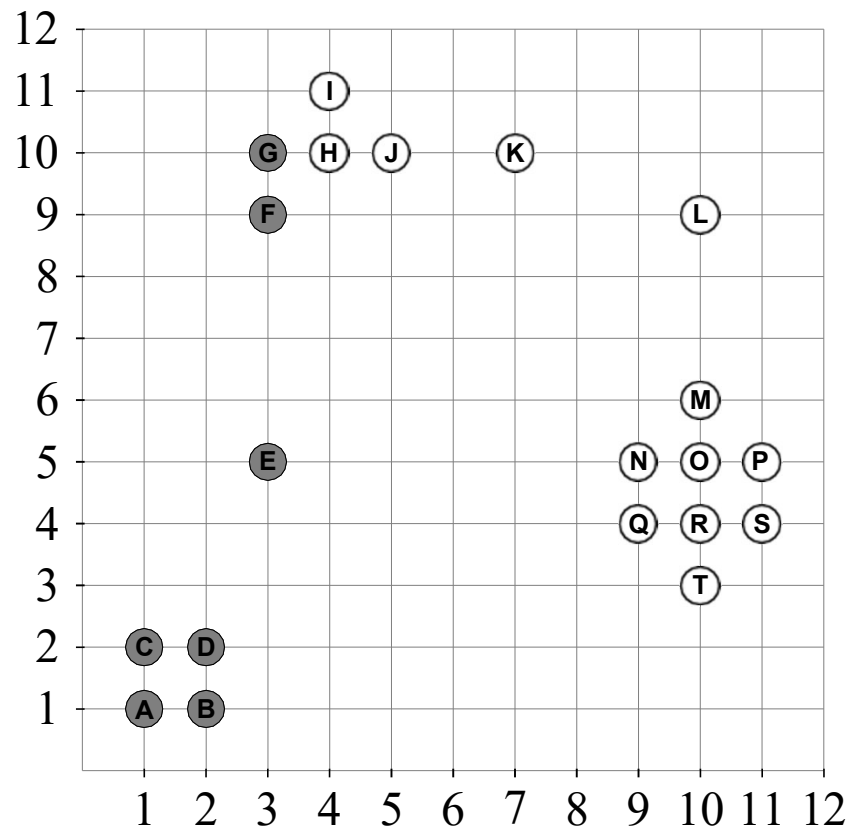
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

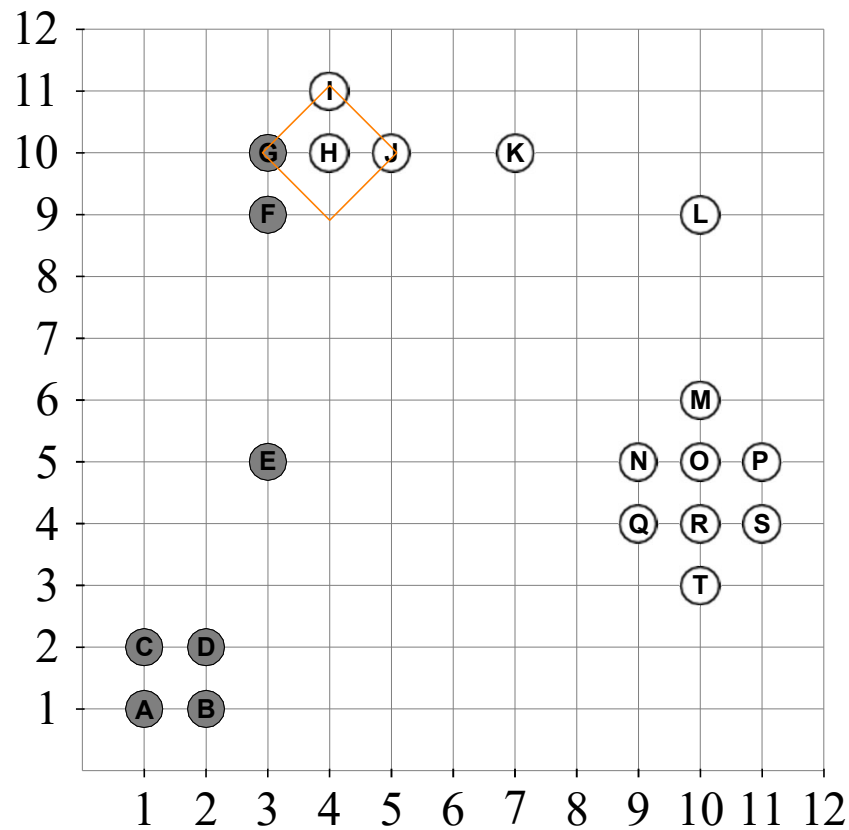
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

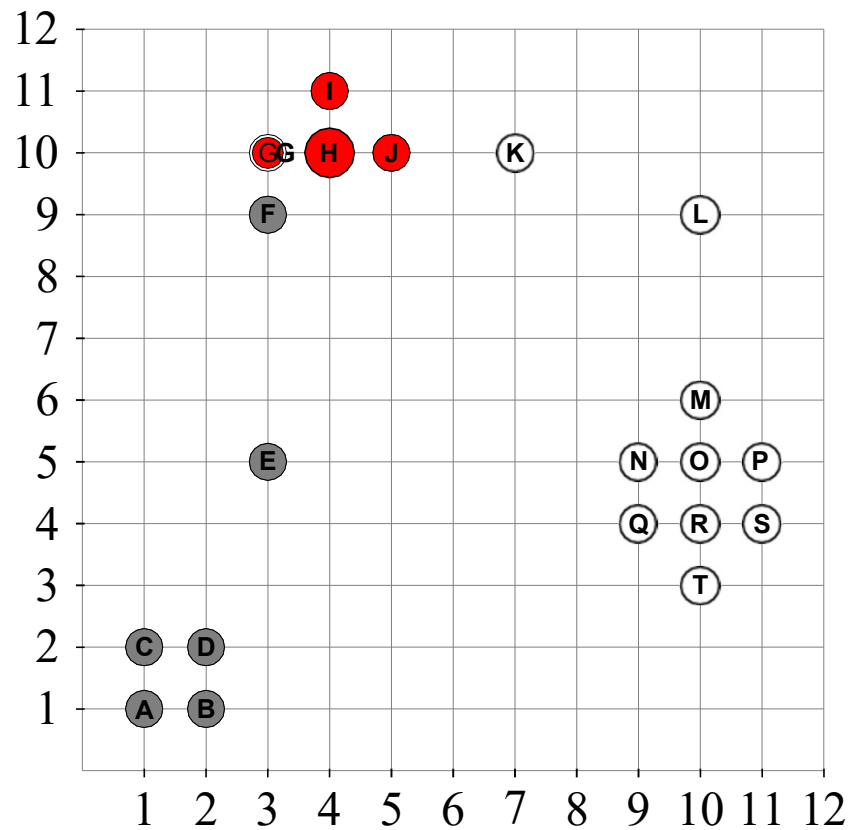
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 4$$

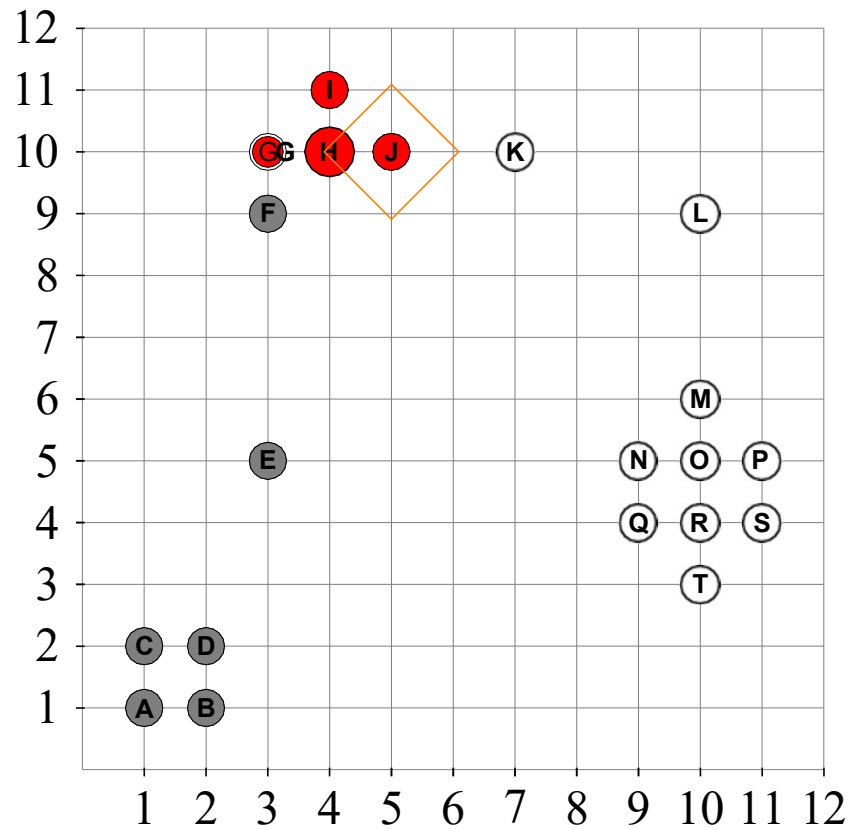
Seed list:

IJ

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

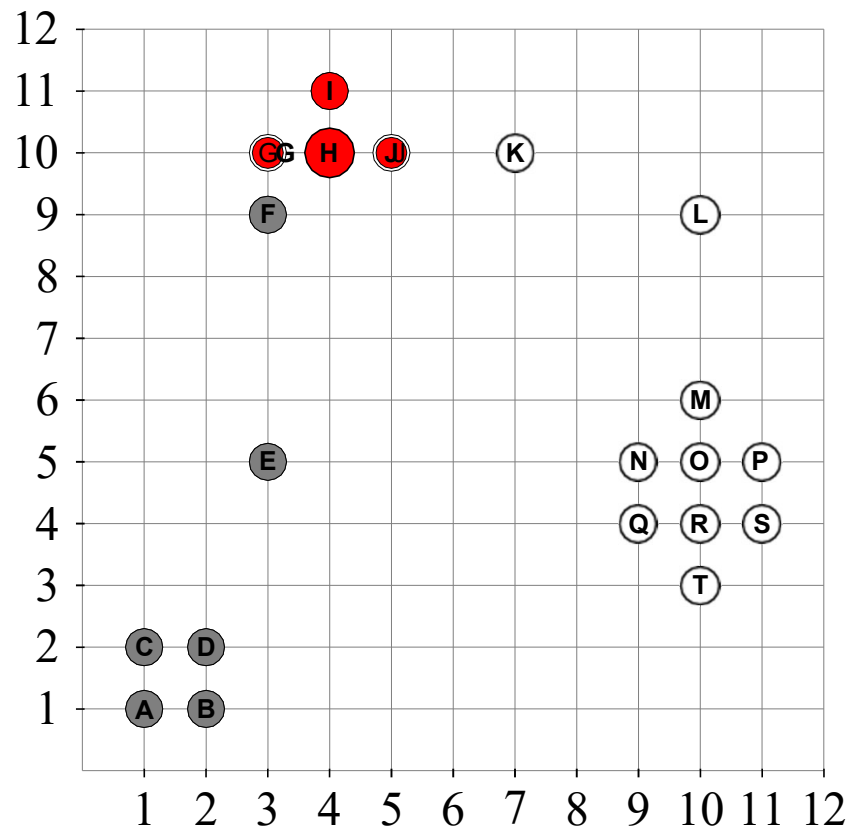
$$minPts = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 4$$

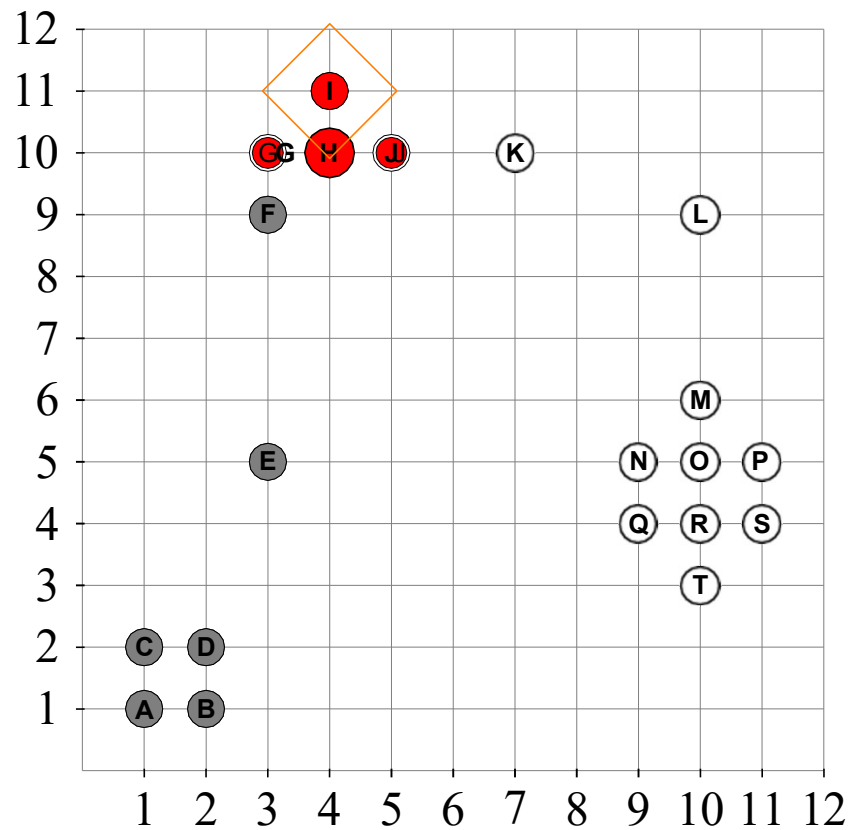
Seed list:

I

Example 3

DM868 DM870
DS804

Arthur Zimek

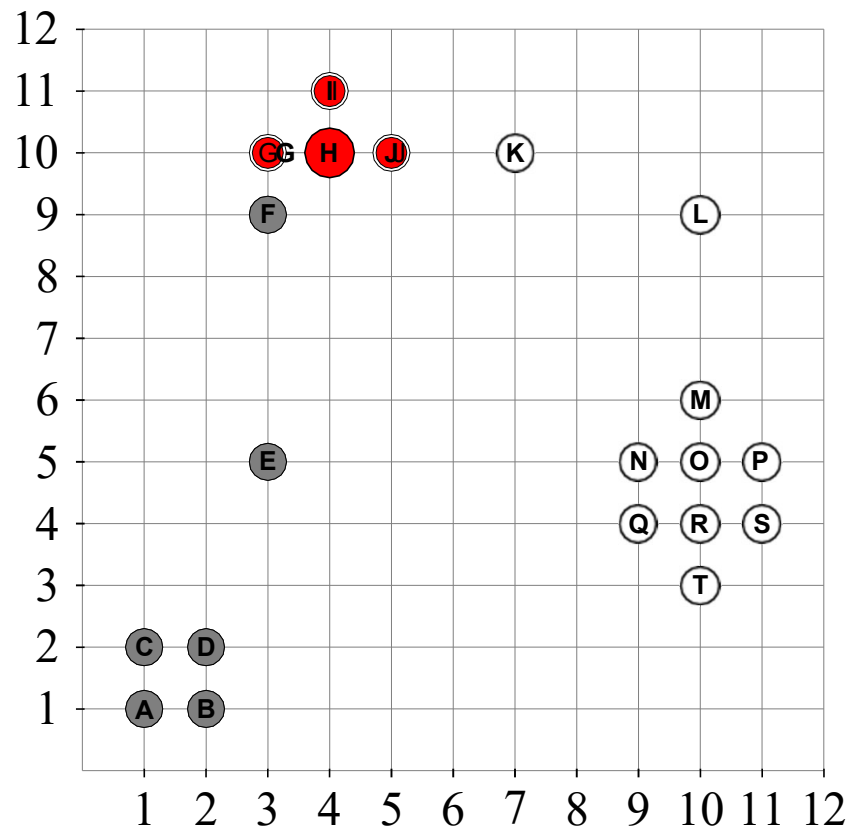

$$\varepsilon = 1.1$$
$$minPts = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

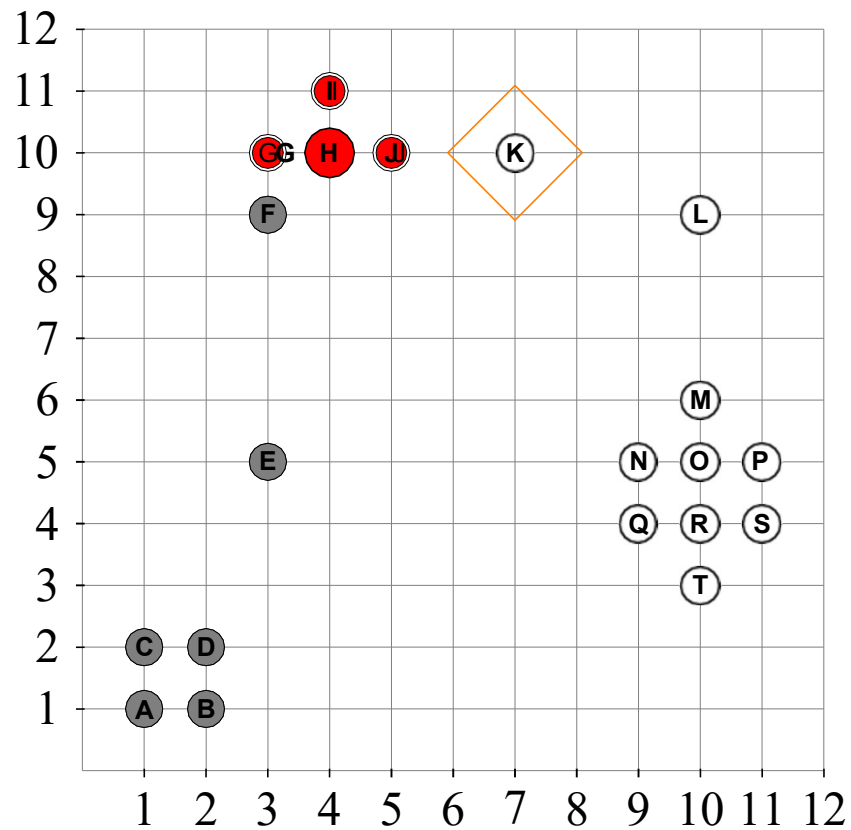
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek

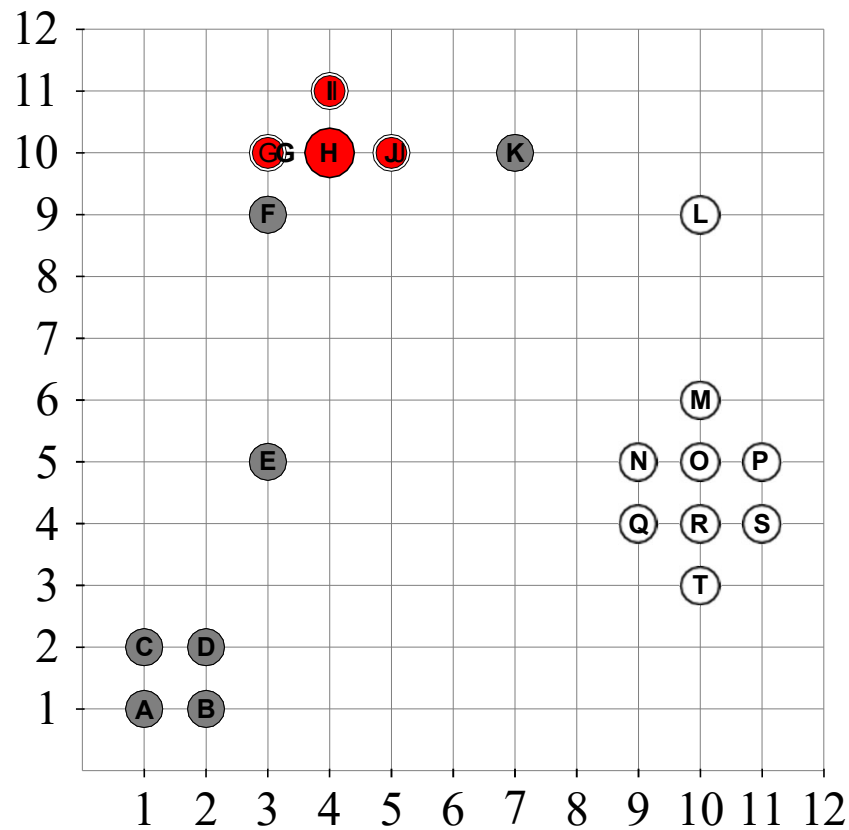

$$\varepsilon = 1.1$$
$$minPts = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

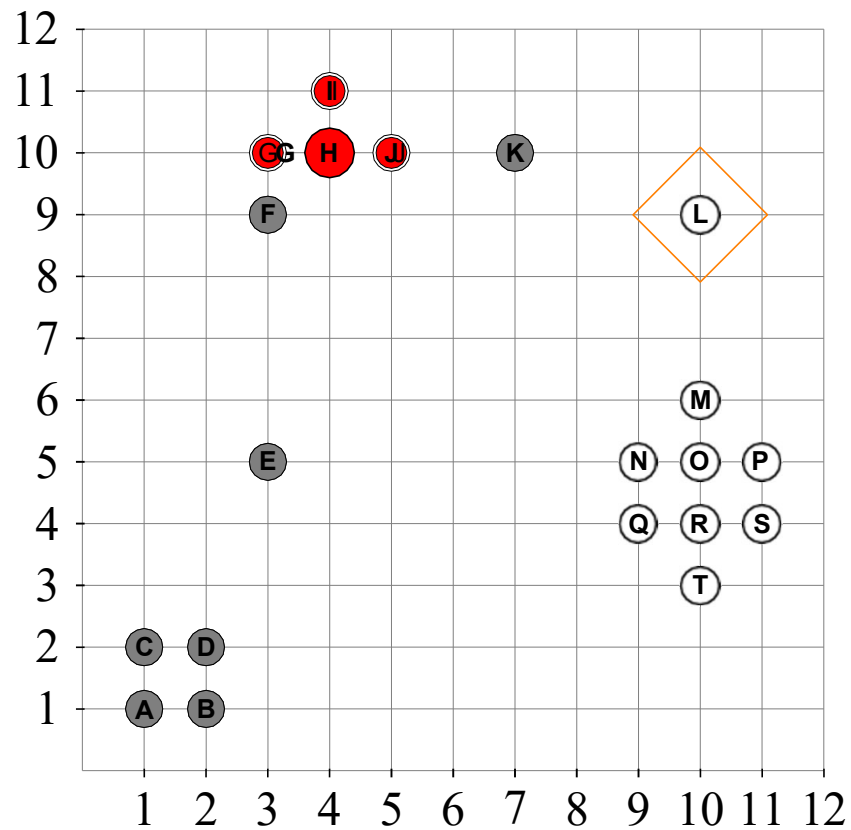
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek

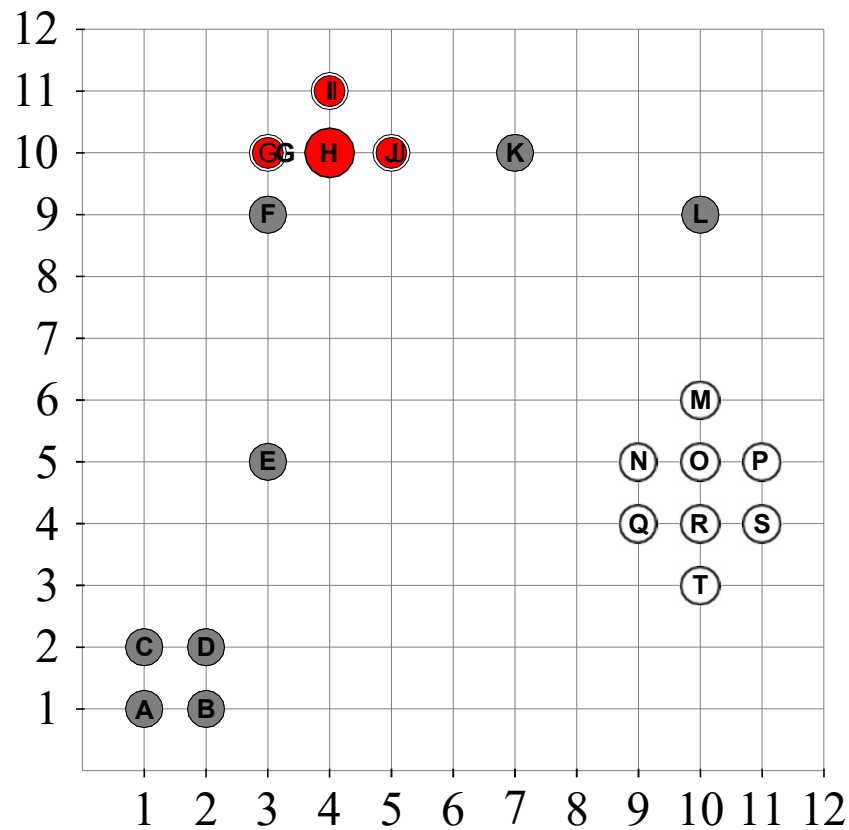

$$\varepsilon = 1.1$$
$$minPts = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

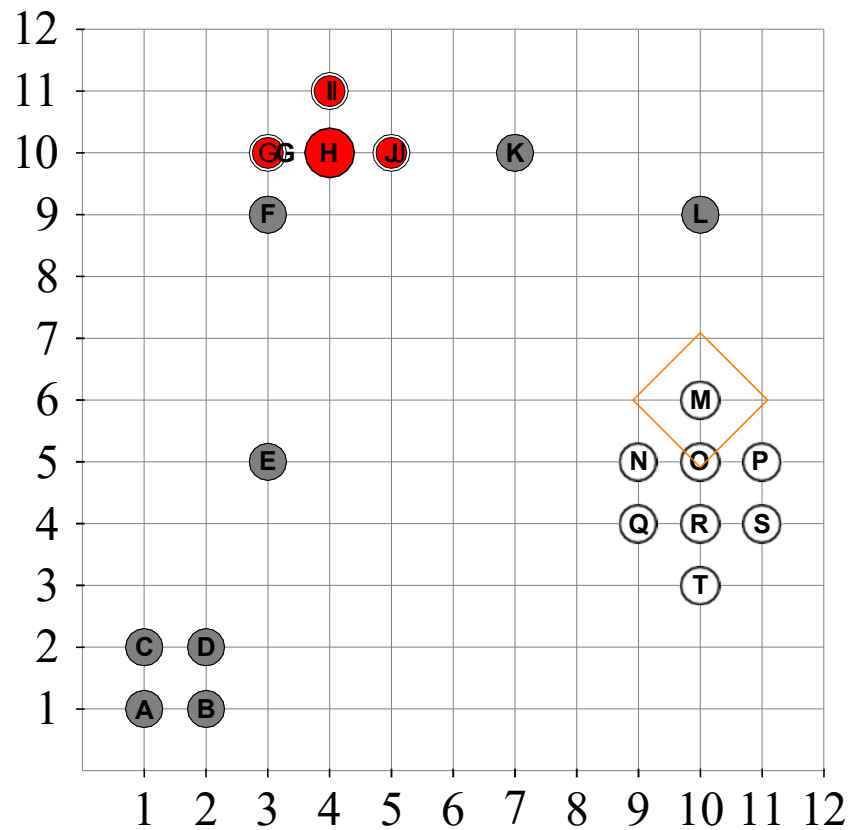
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek

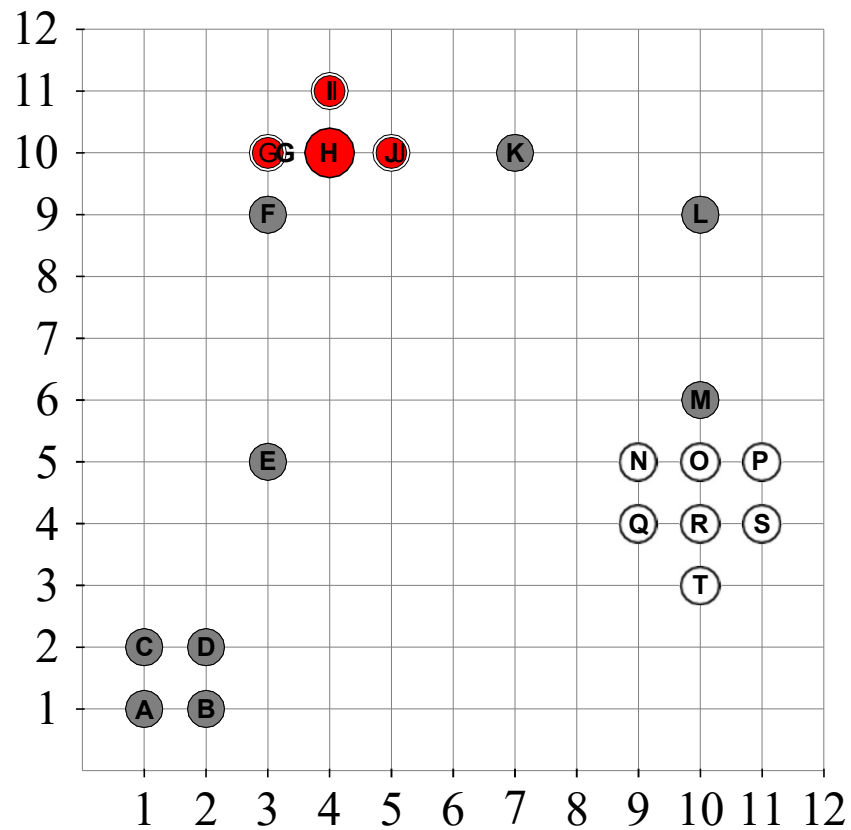

$$\varepsilon = 1.1$$
$$minPts = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

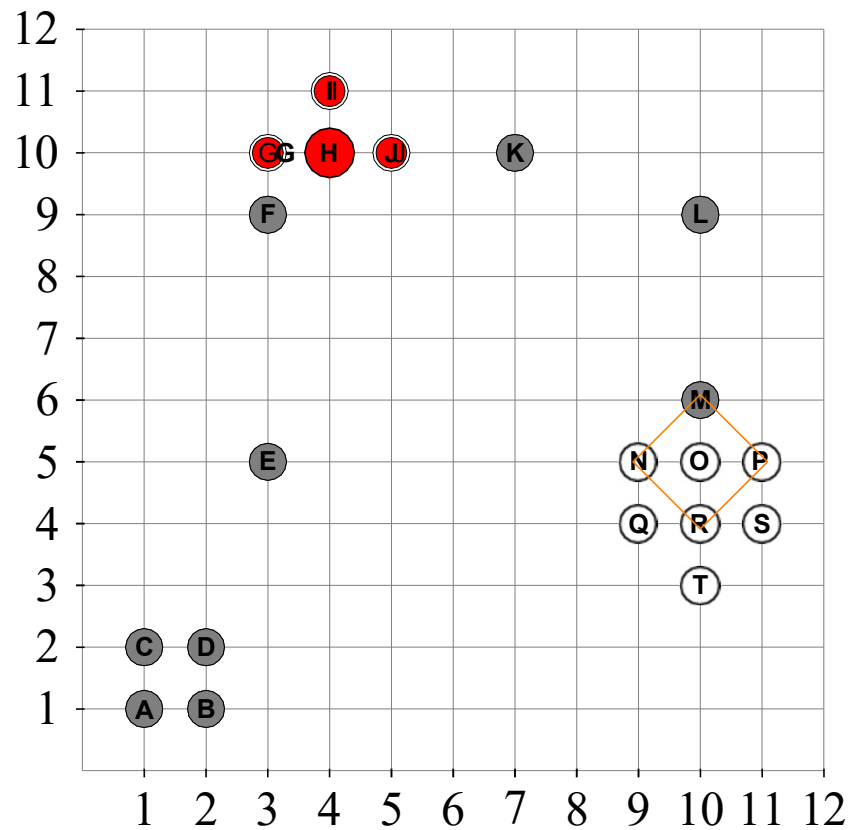
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek

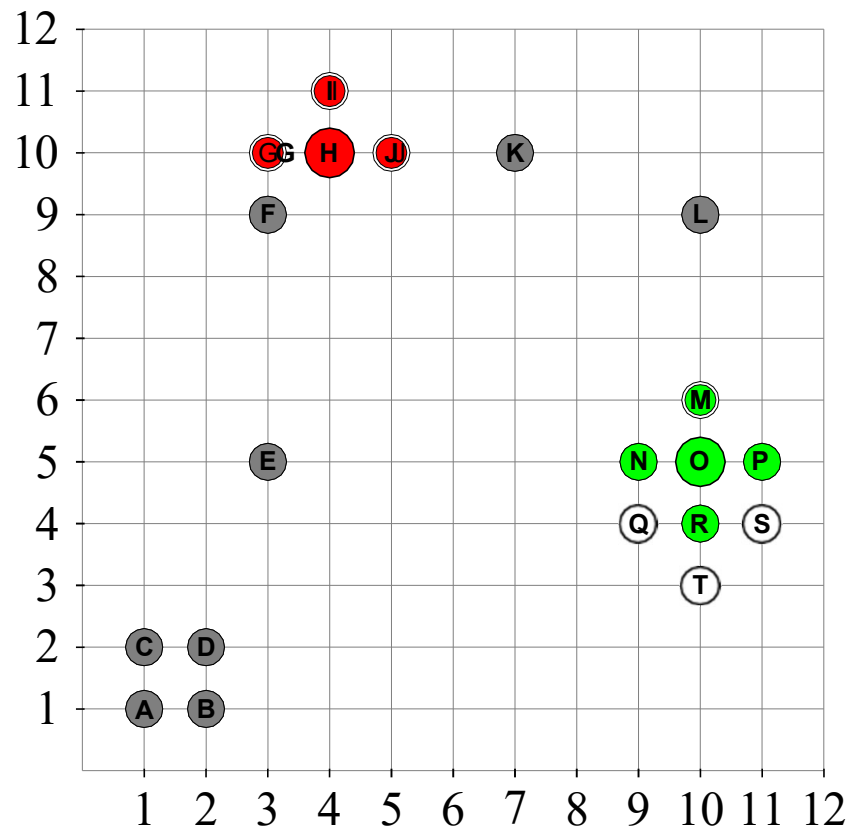

$$\varepsilon = 1.1$$
$$minPts = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

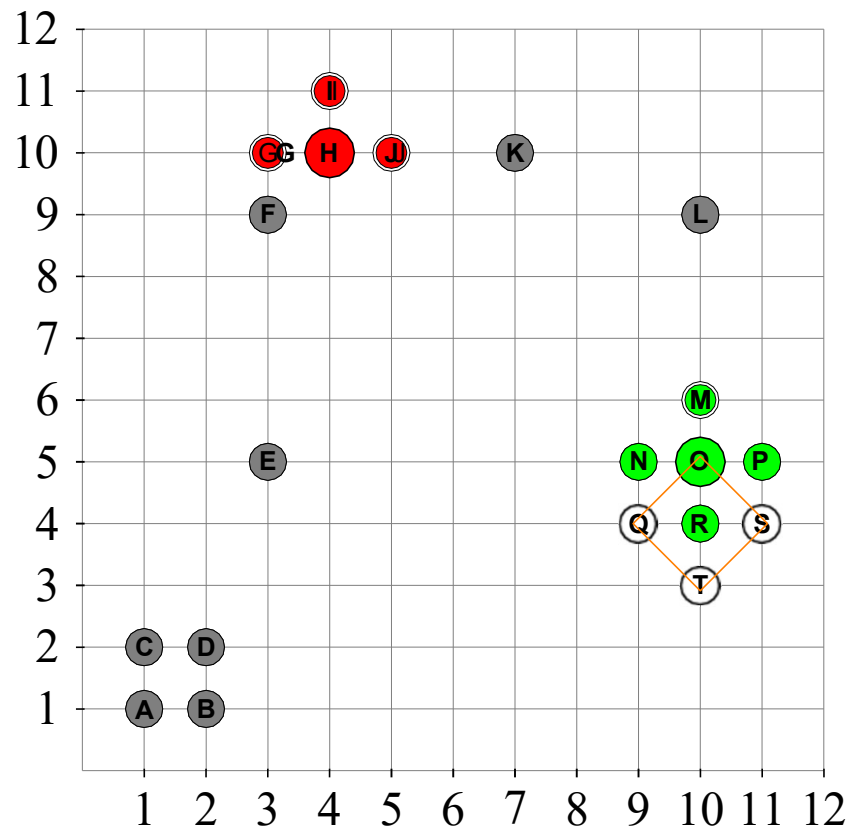
$$\text{minPts} = 4$$

Seed list:
NPR

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 4$$

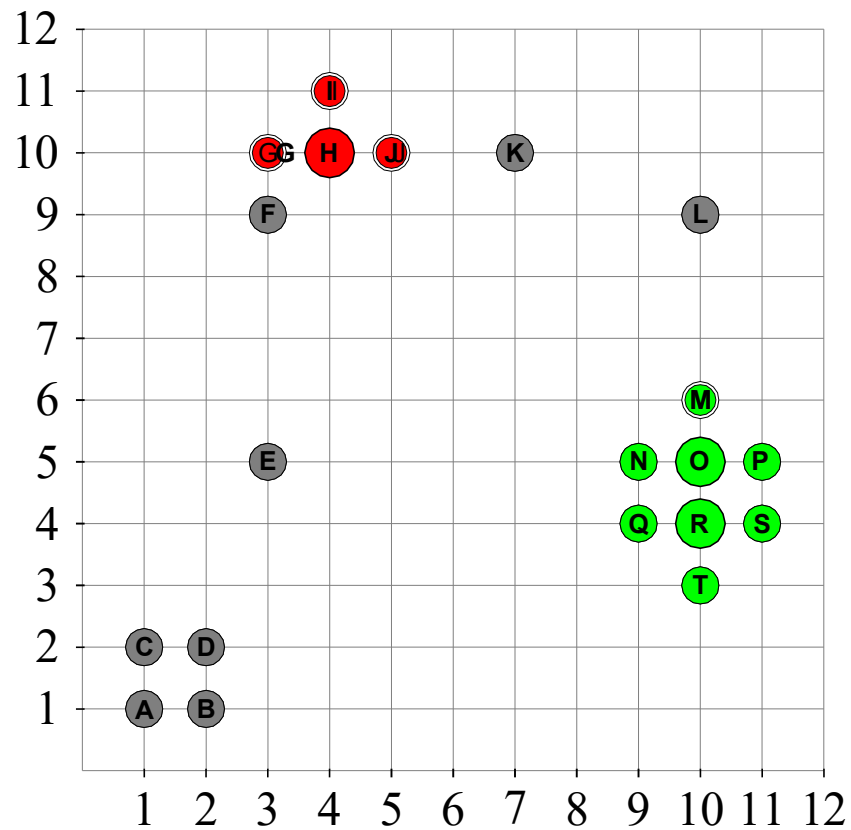
Seed list:

NP

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

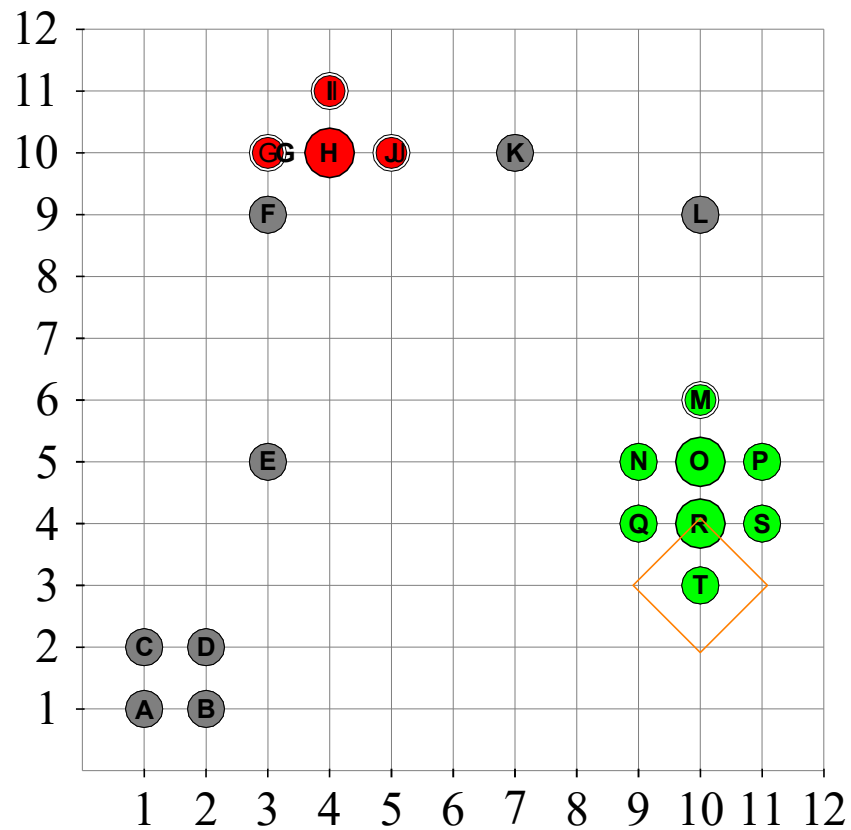
$$\text{minPts} = 4$$

Seed list:
NPQST

Example 3

DM868 DM870
DS804

Arthur Zimek

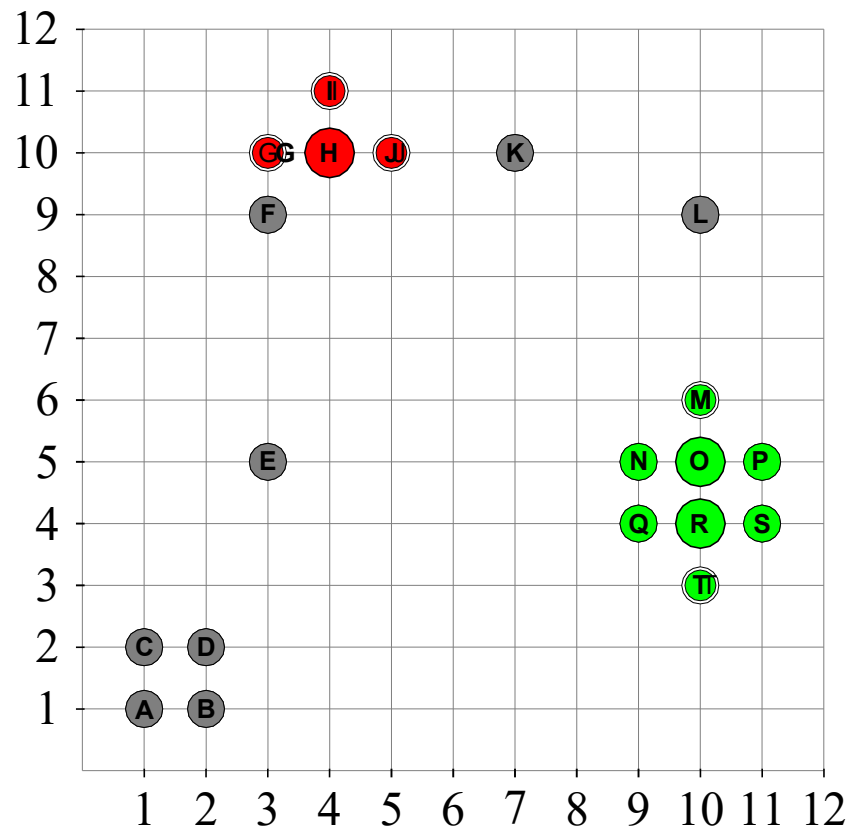

$$\varepsilon = 1.1$$
$$minPts = 4$$

Seed list:
NPQS

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

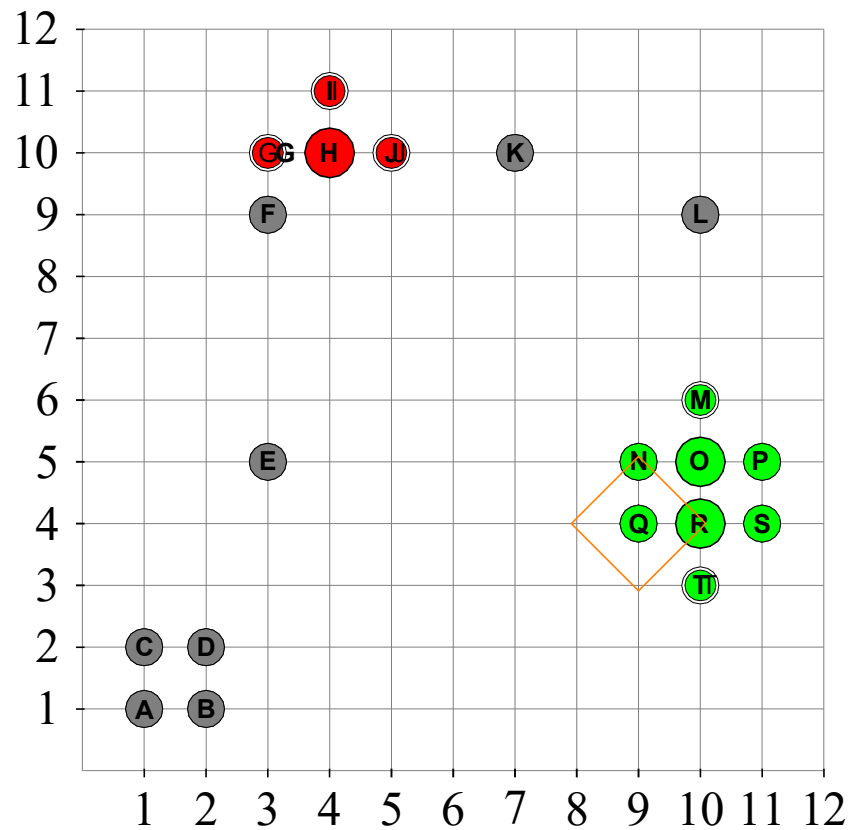
$$\text{minPts} = 4$$

Seed list:
NPQS

Example 3

DM868 DM870
DS804

Arthur Zimek

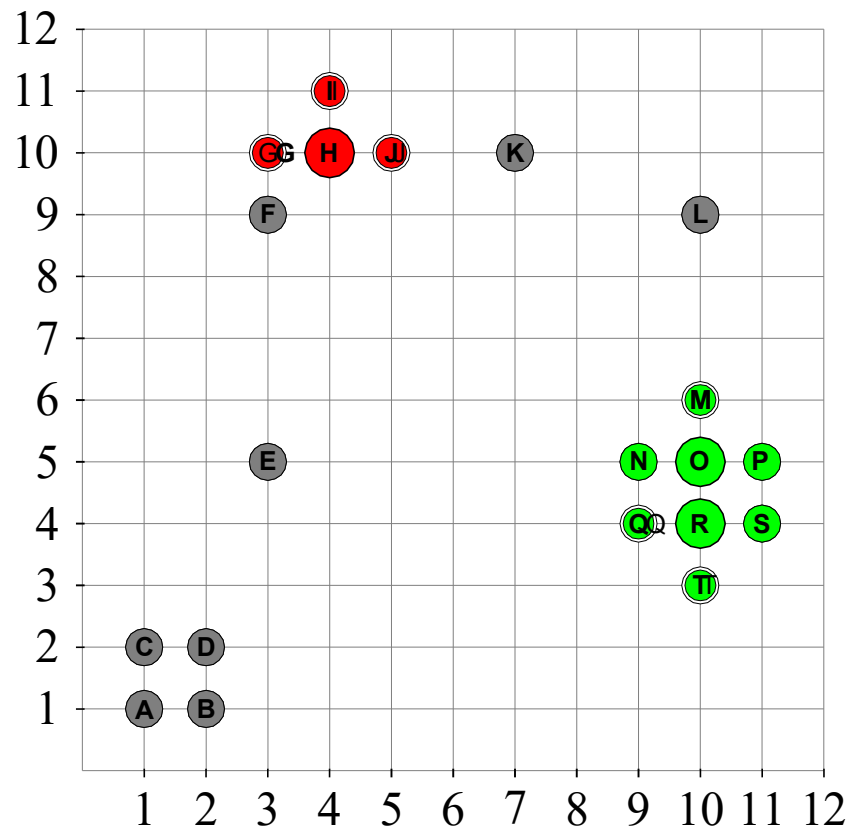

$$\varepsilon = 1.1$$
$$minPts = 4$$

Seed list:
NPS

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

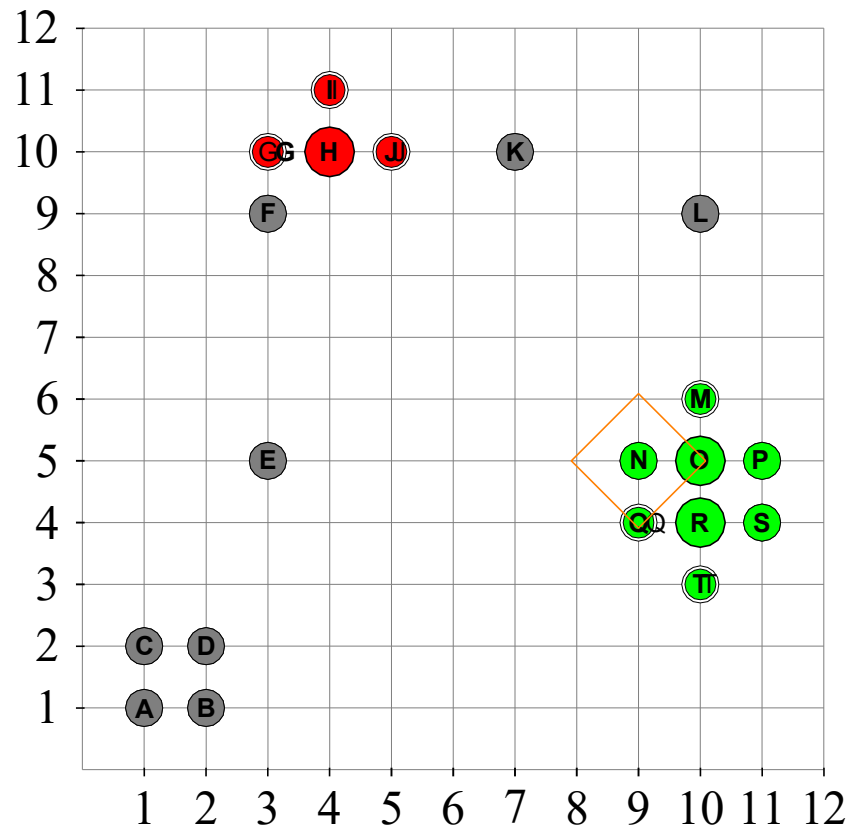
$$\text{minPts} = 4$$

Seed list:
NPS

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

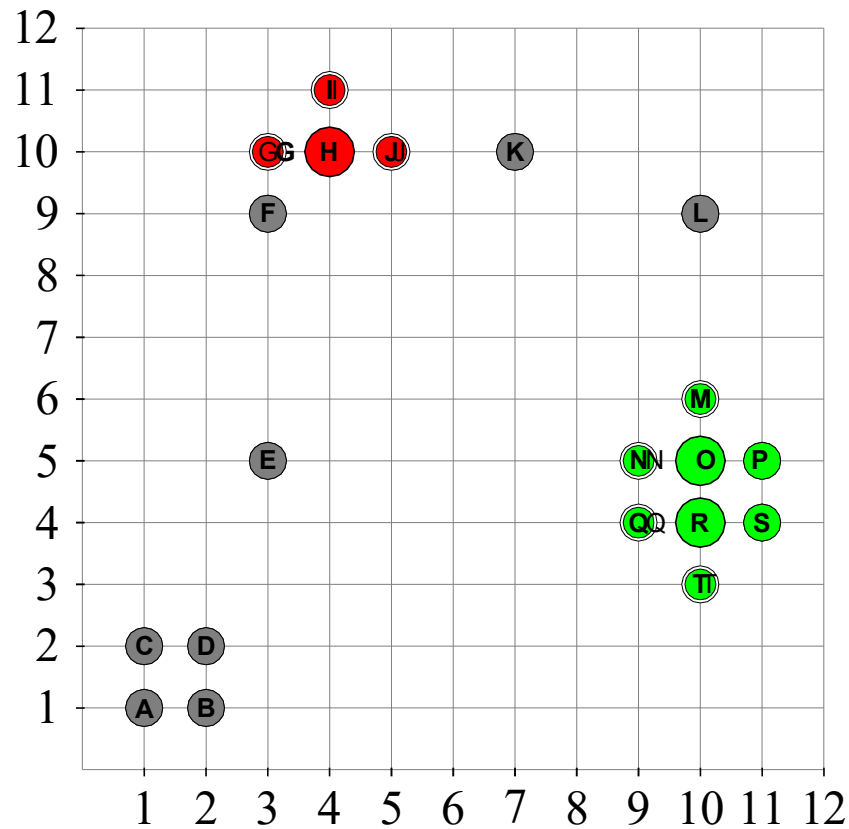
$$\text{minPts} = 4$$

Seed list:
PS

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 4$$

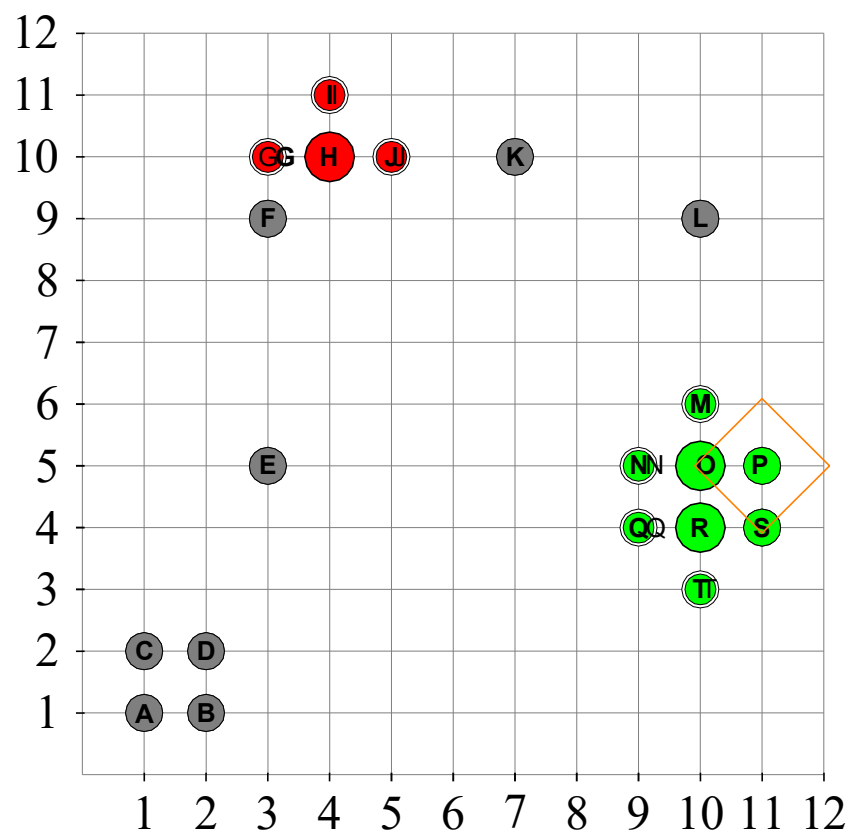
Seed list:

PS

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 4$$

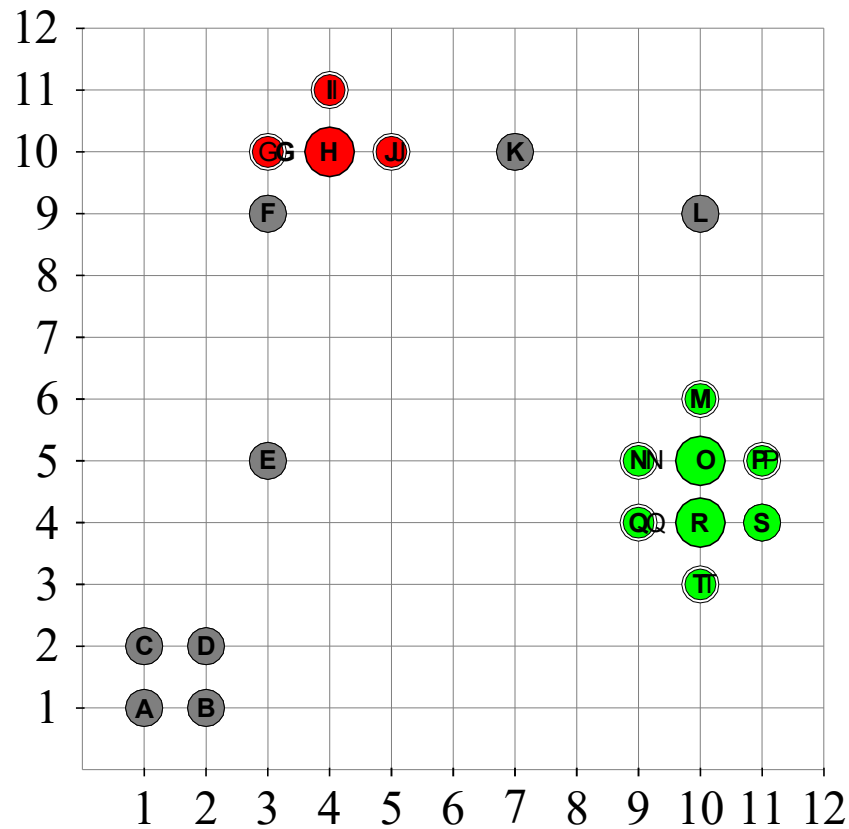
Seed list:

S

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

$$\text{minPts} = 4$$

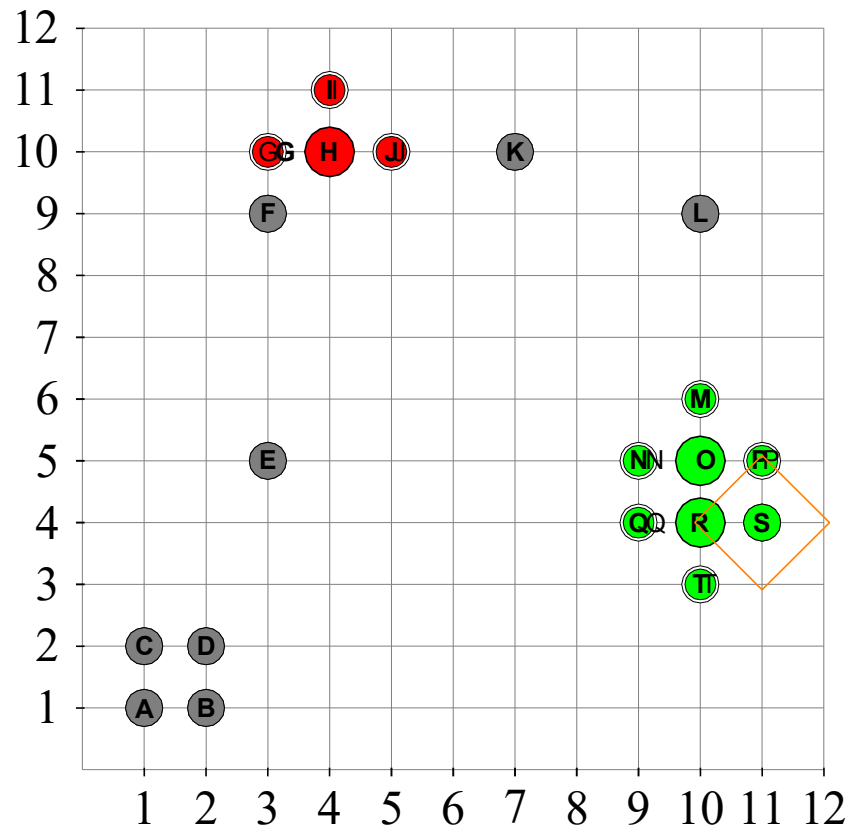
Seed list:

S

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

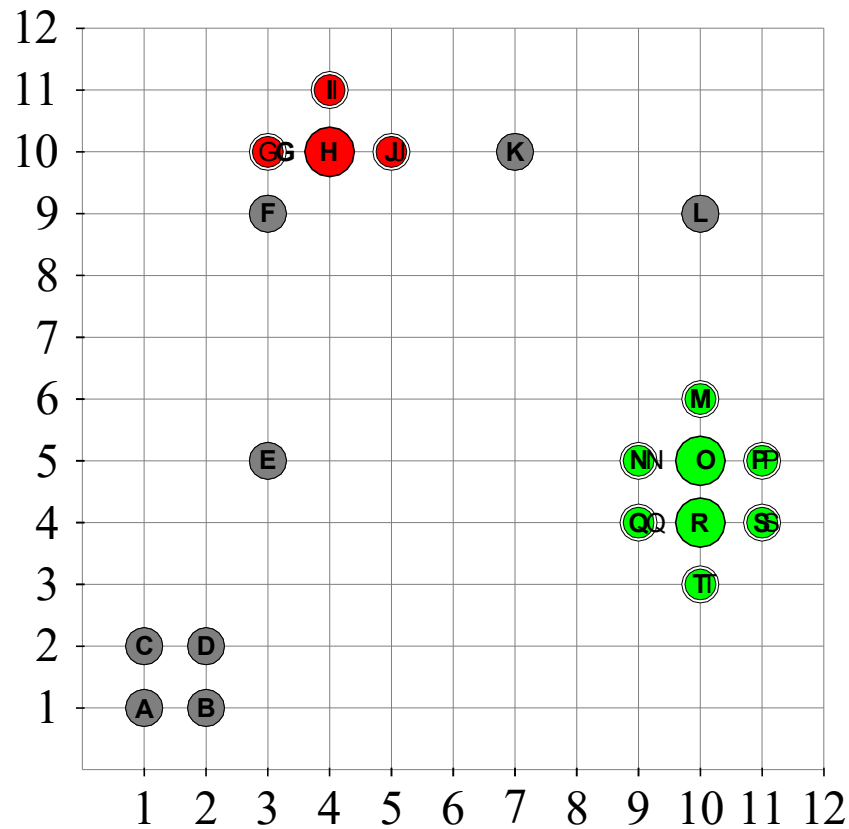
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

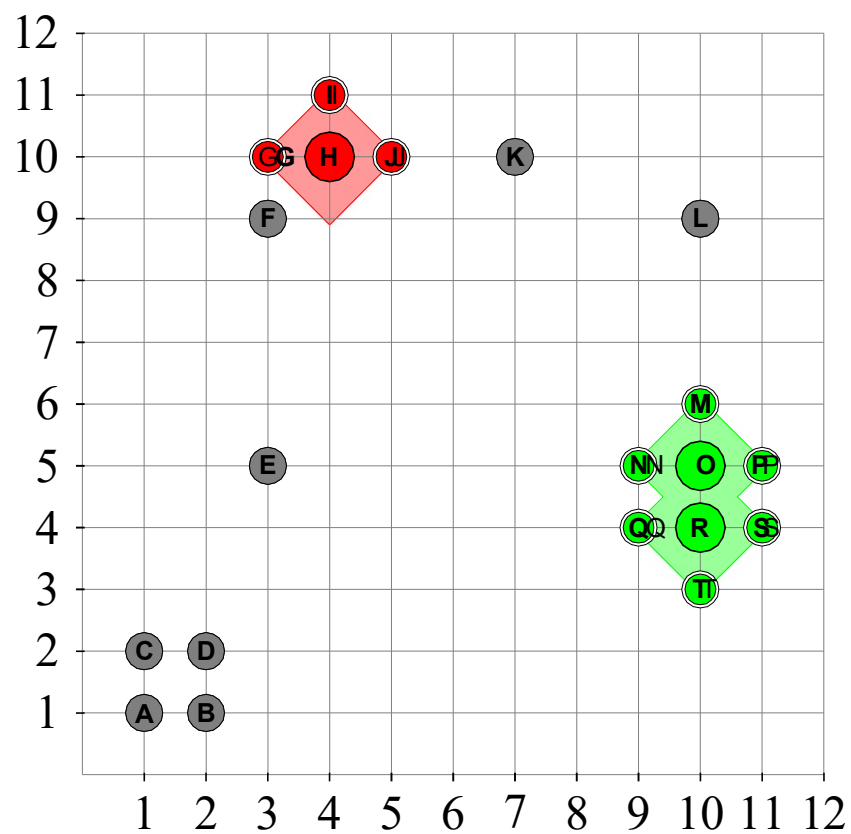
$$\text{minPts} = 4$$

Seed list:

Example 3

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 1.1$$

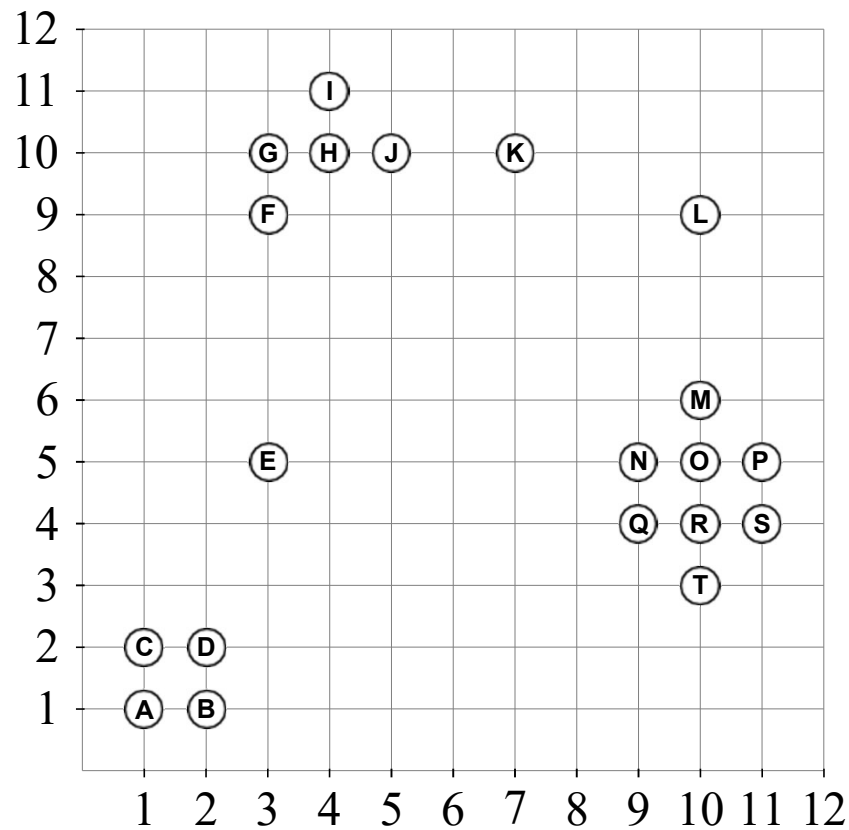
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

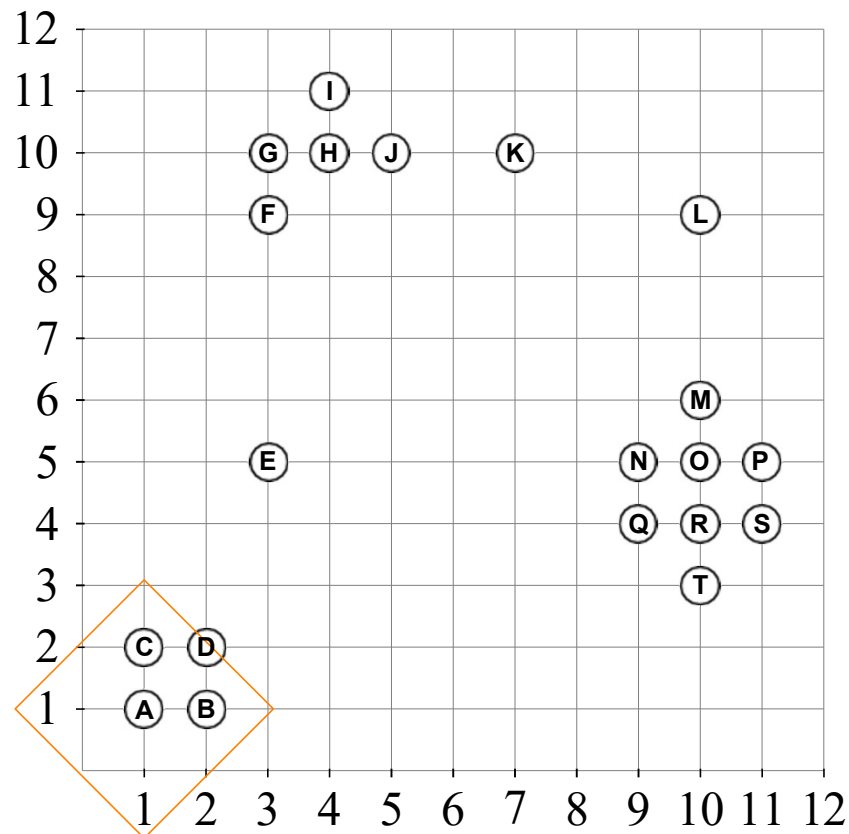
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

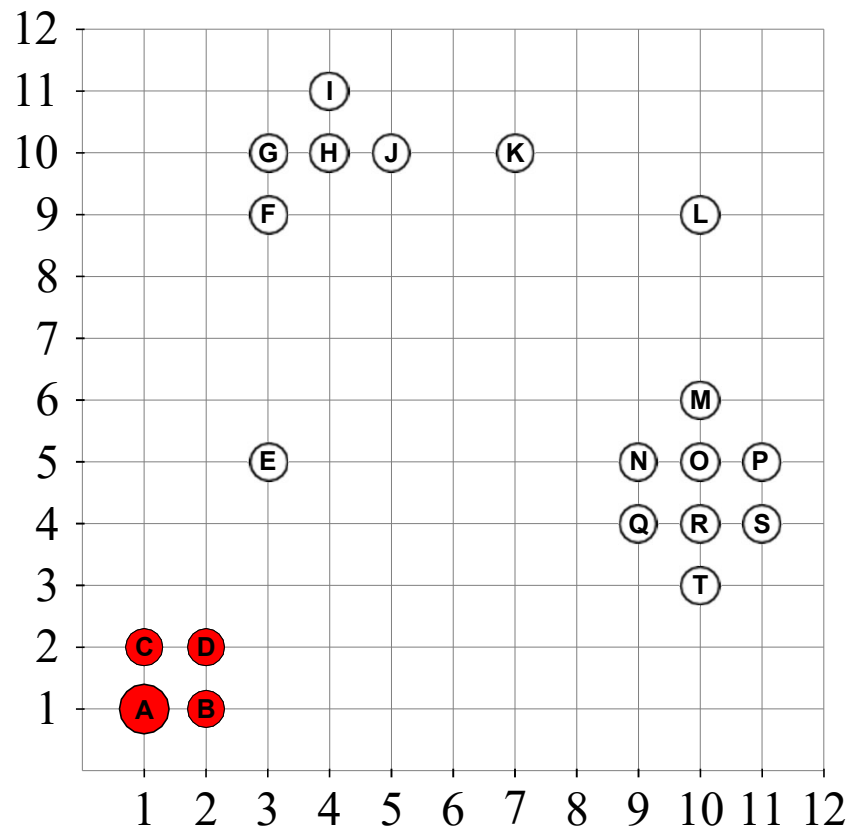
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

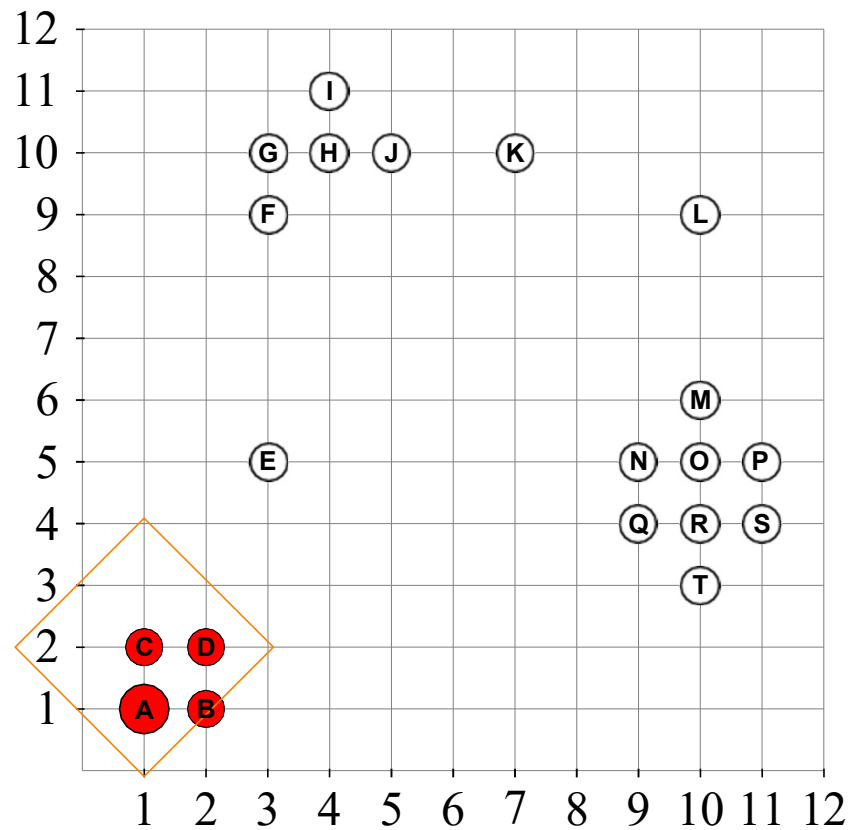
$$\text{minPts} = 4$$

Seed list:
BCD

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

$$\text{minPts} = 4$$

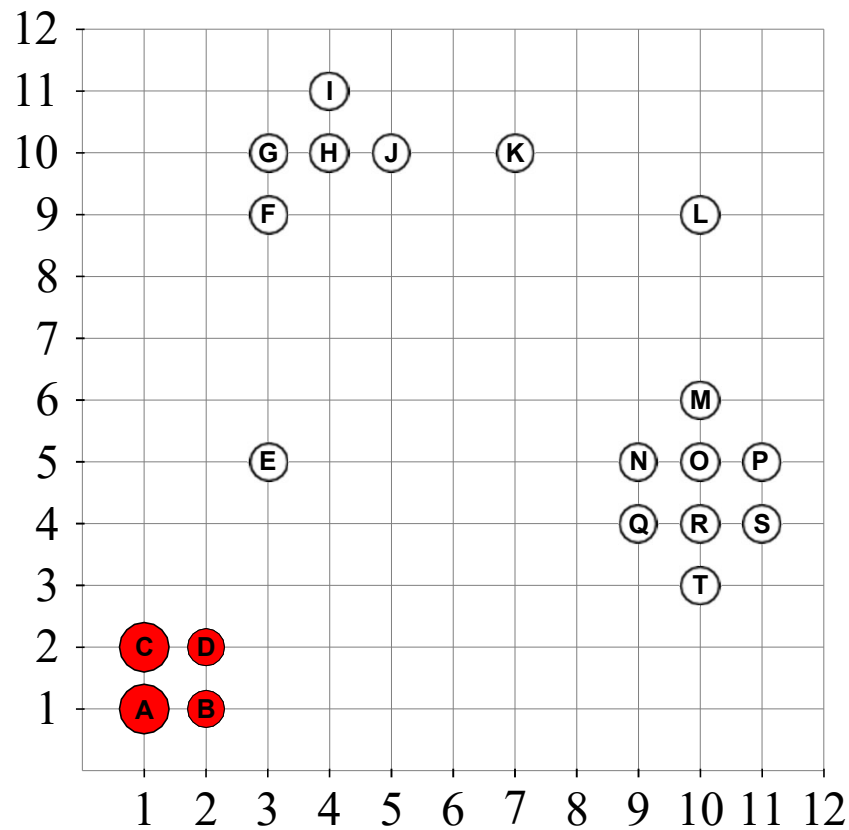
Seed list:

BD

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

$$\text{minPts} = 4$$

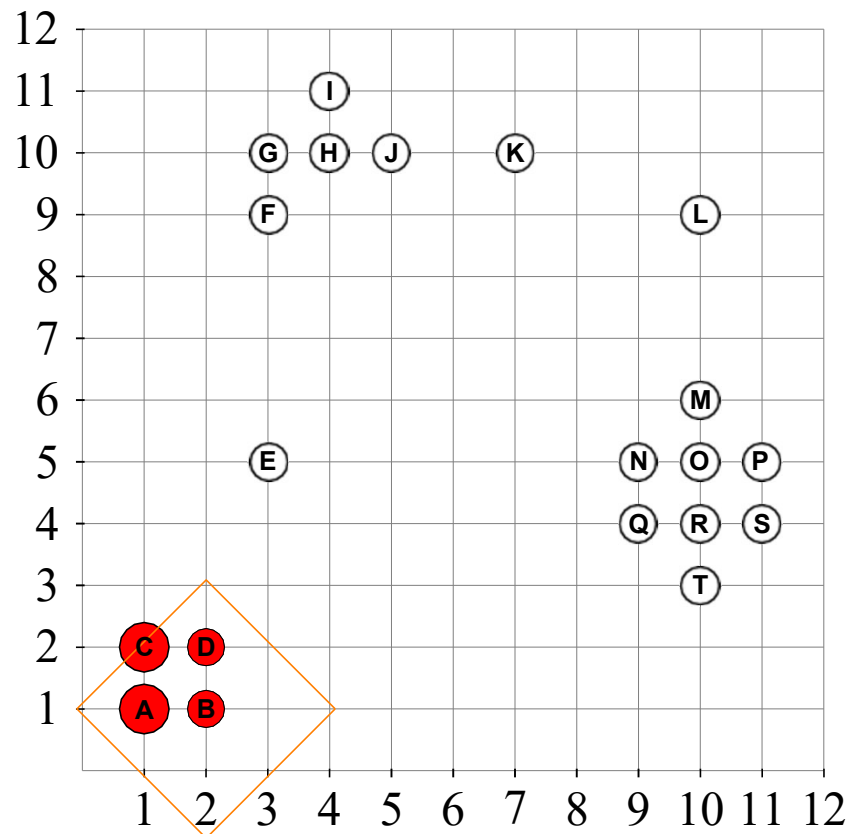
Seed list:

BD

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

$$\text{minPts} = 4$$

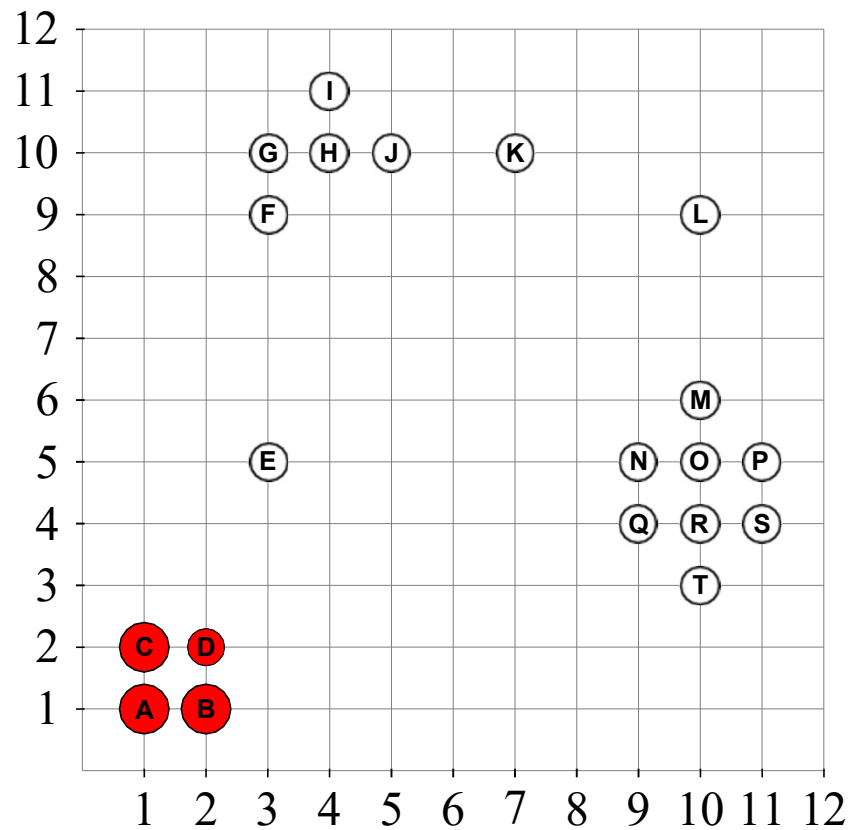
Seed list:

D

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

$$\text{minPts} = 4$$

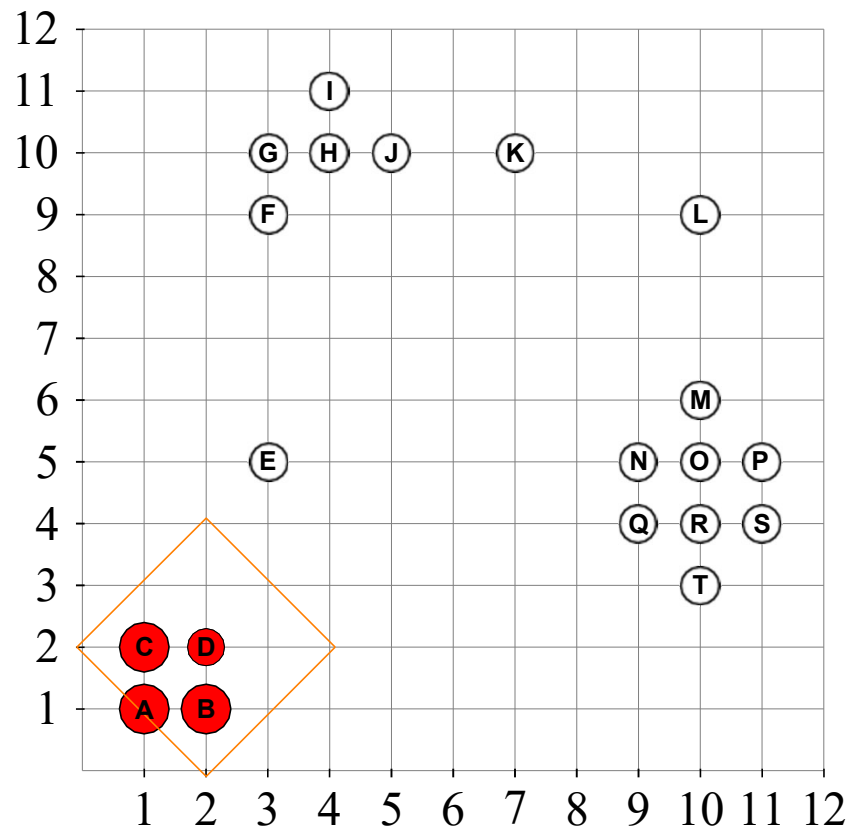
Seed list:

D

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

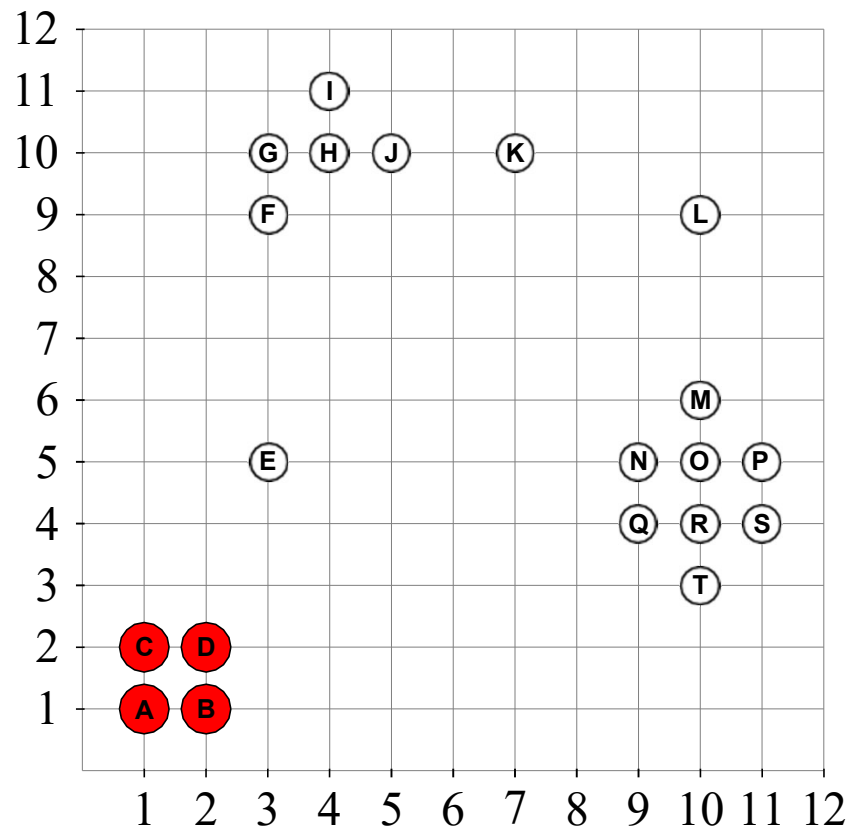
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

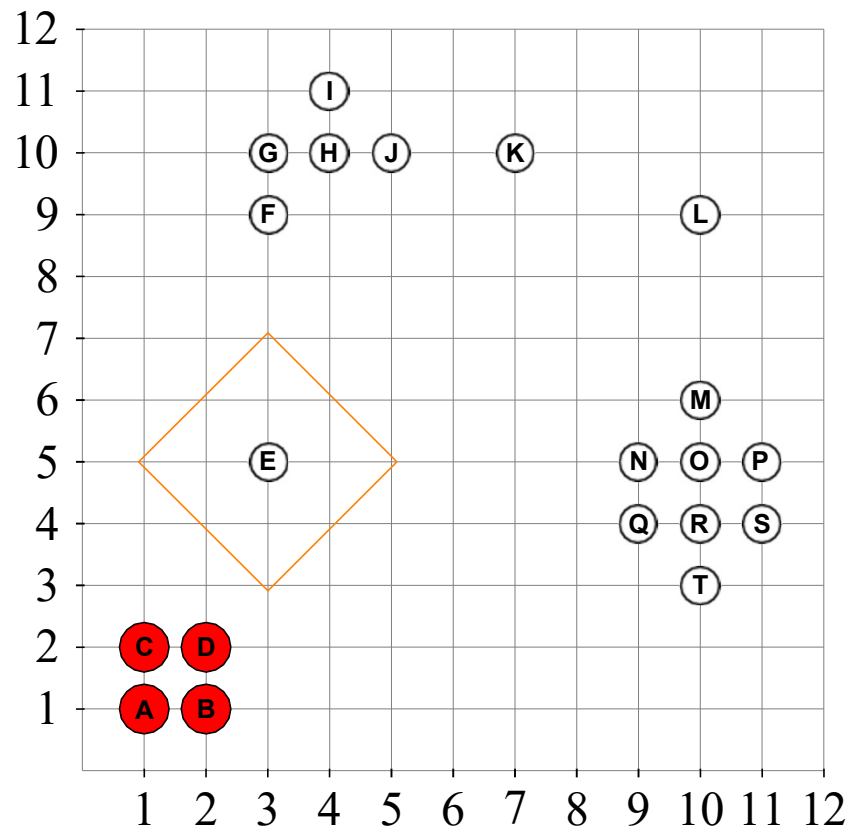
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

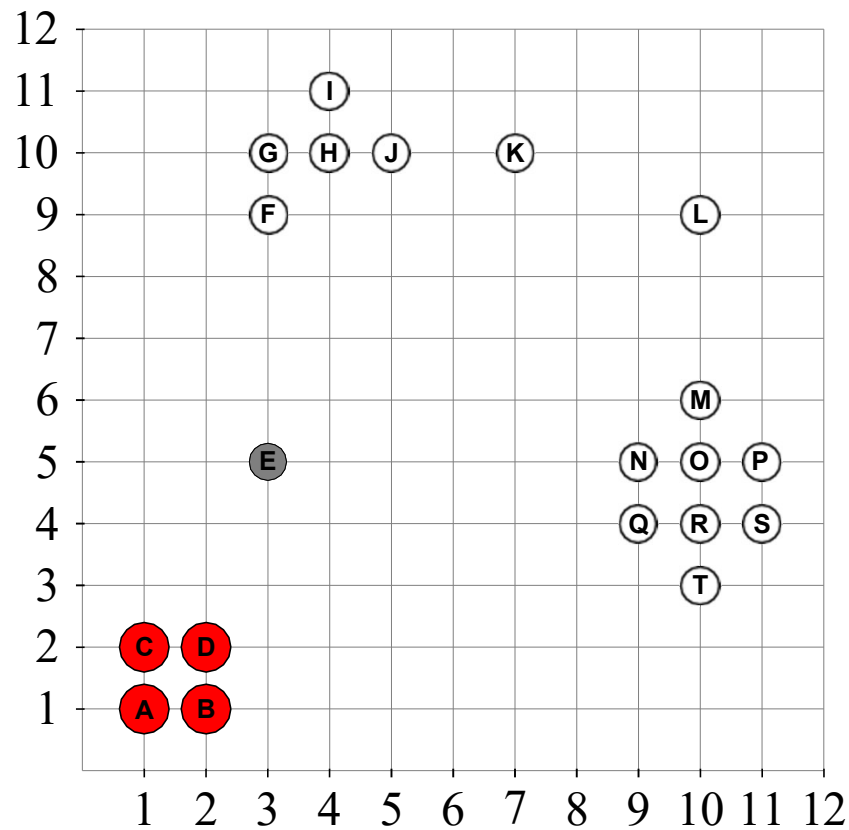
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

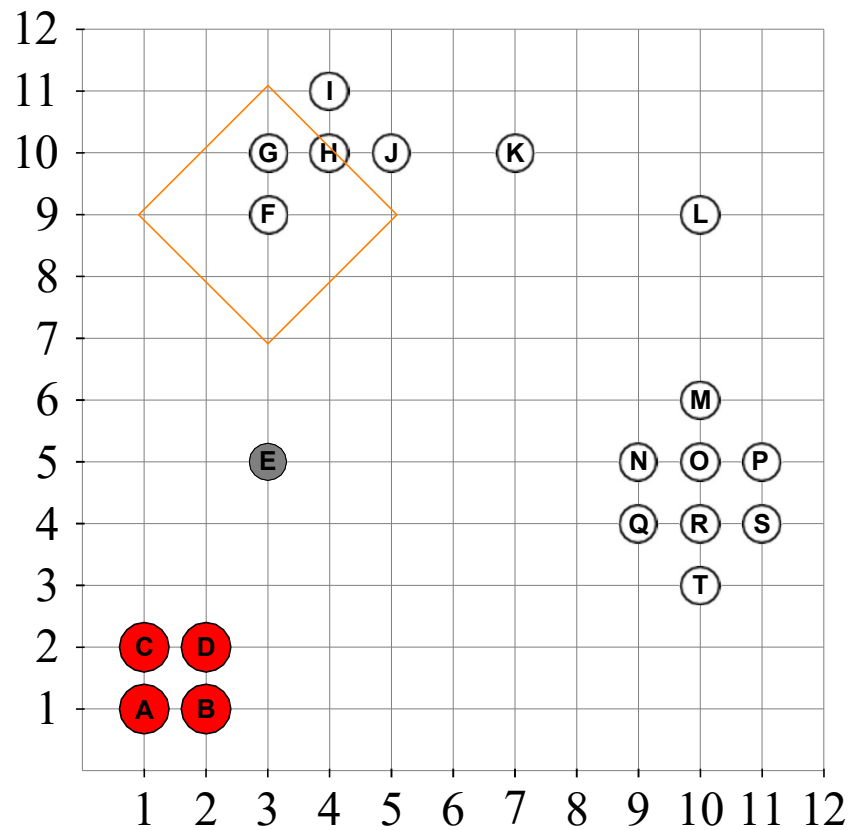
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

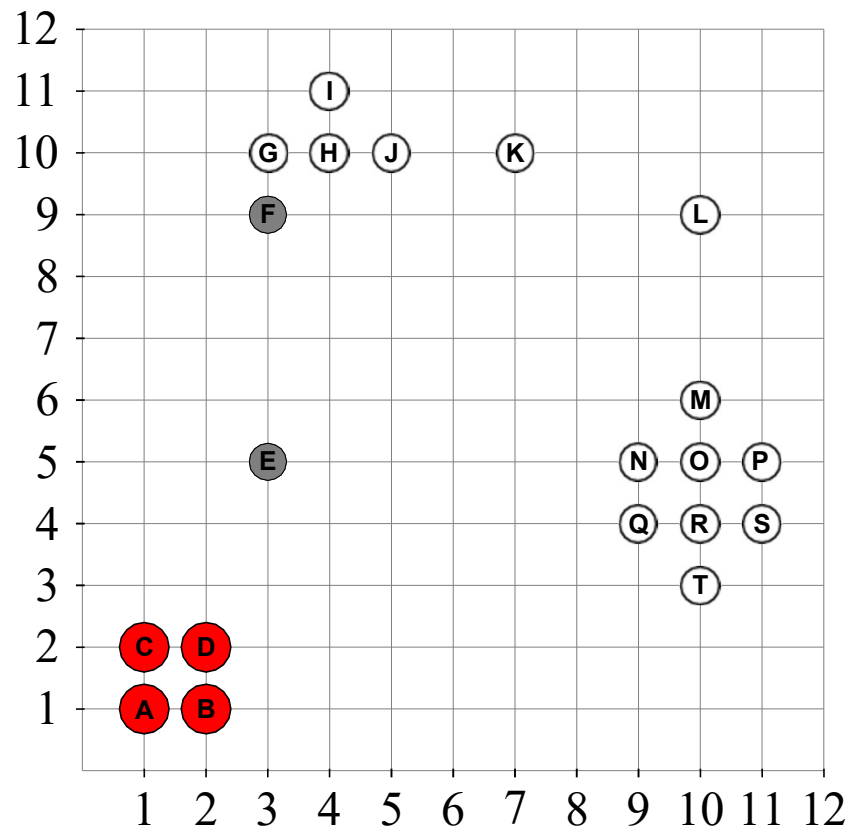
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

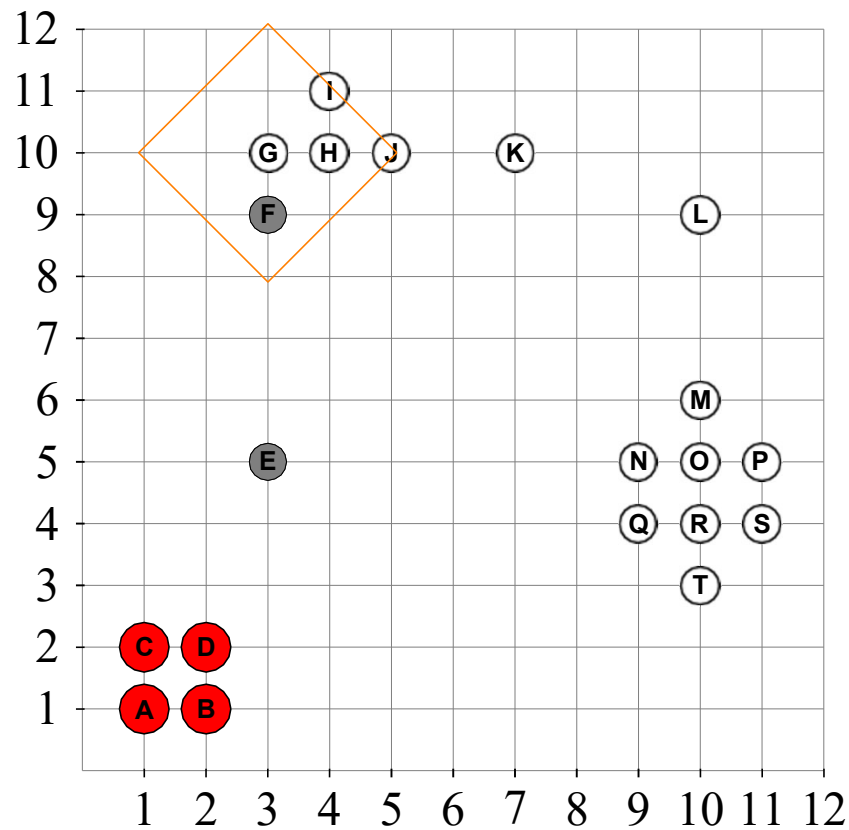
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

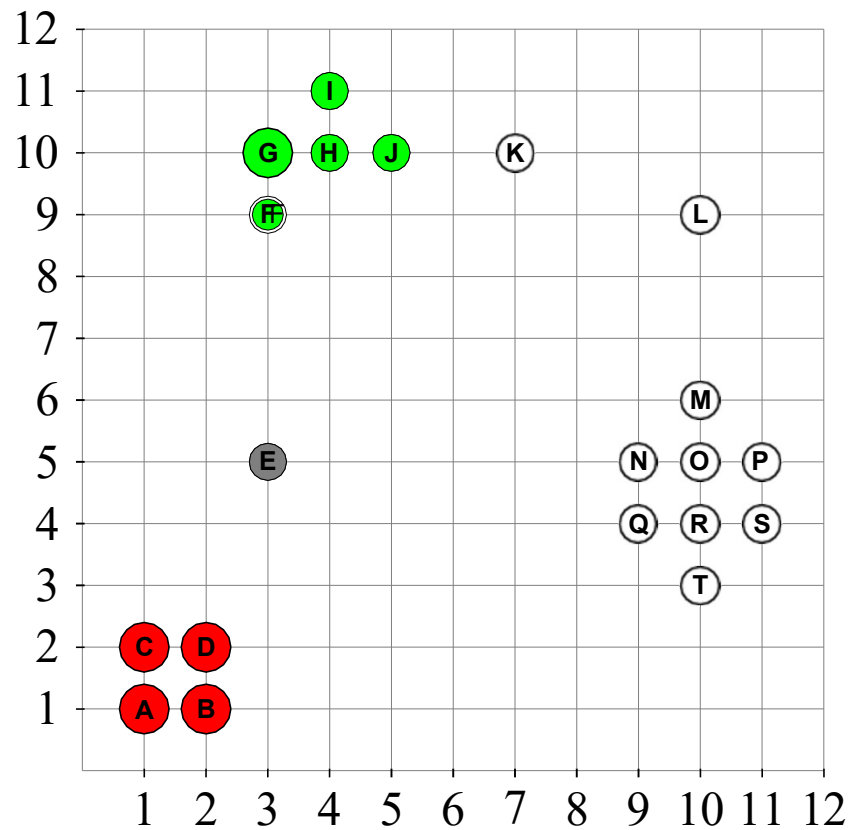
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

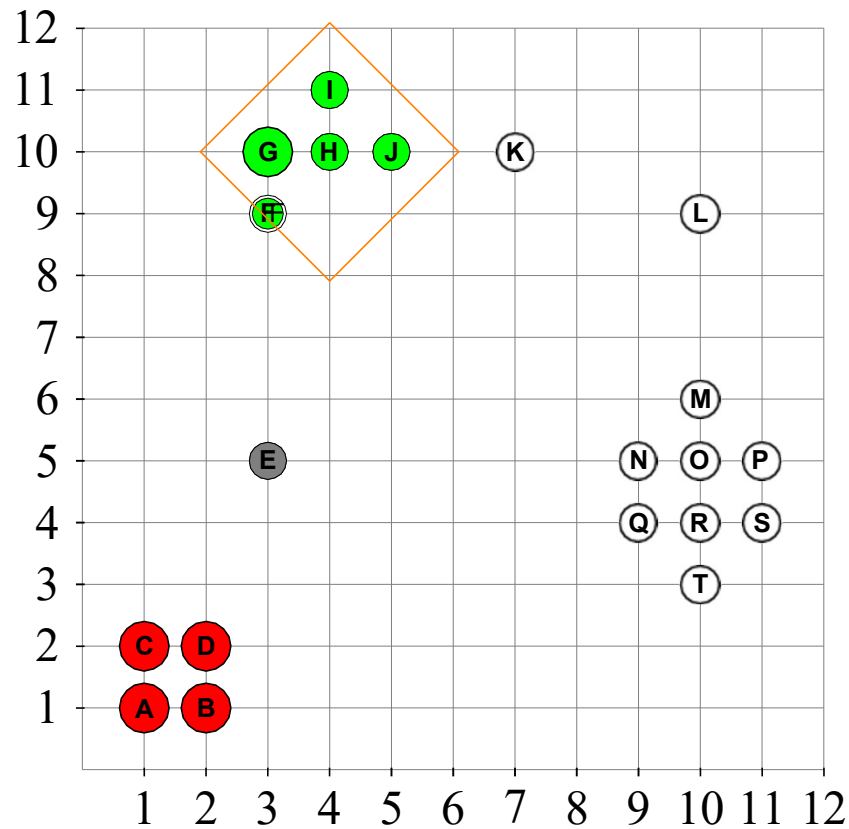
$$\text{minPts} = 4$$

Seed list:
HIJ

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

$$\text{minPts} = 4$$

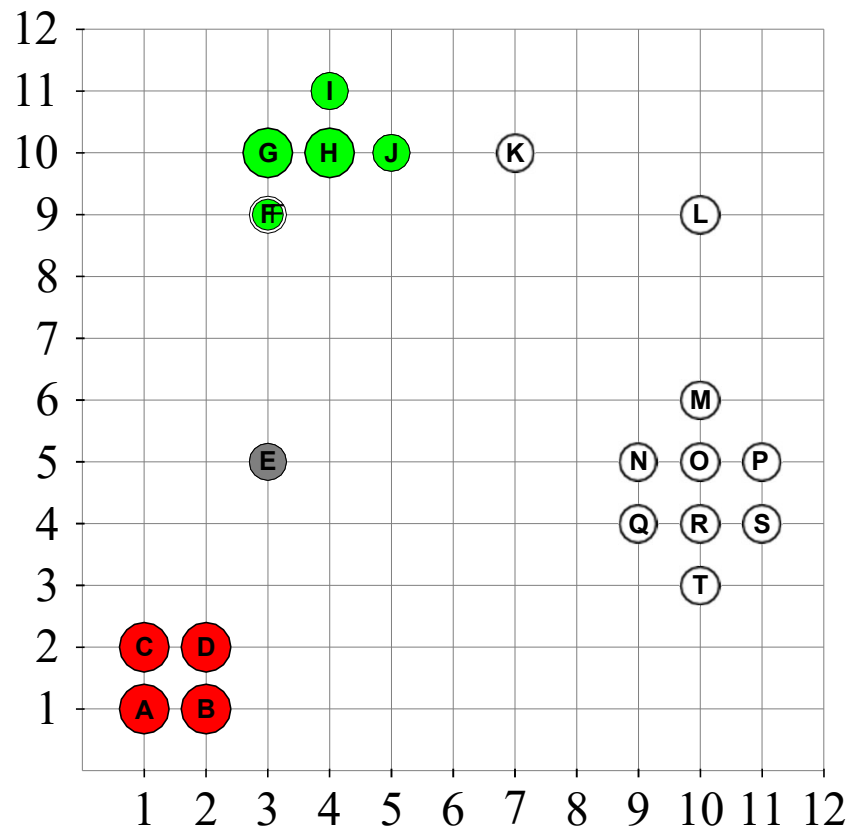
Seed list:

IJ

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

$$\text{minPts} = 4$$

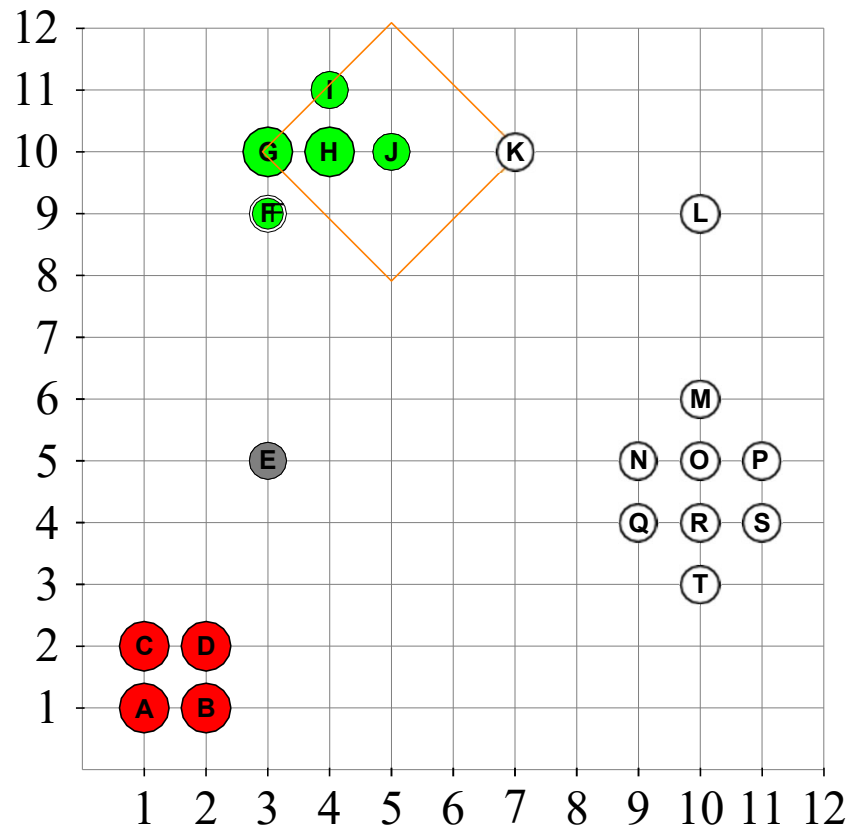
Seed list:

IJ

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

$$\text{minPts} = 4$$

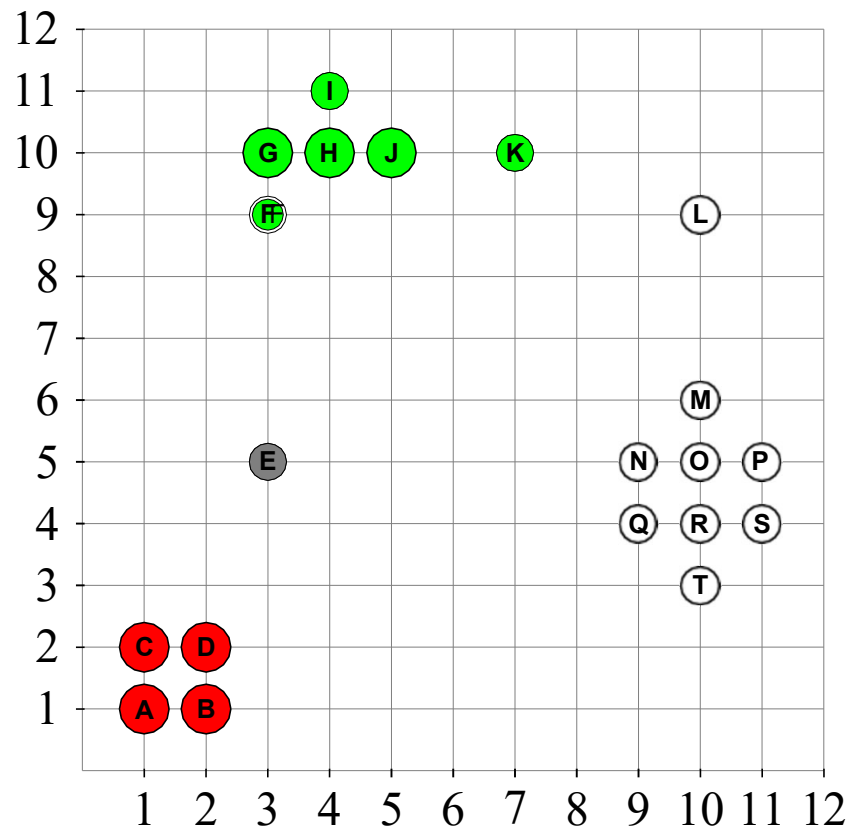
Seed list:

I

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

$$\text{minPts} = 4$$

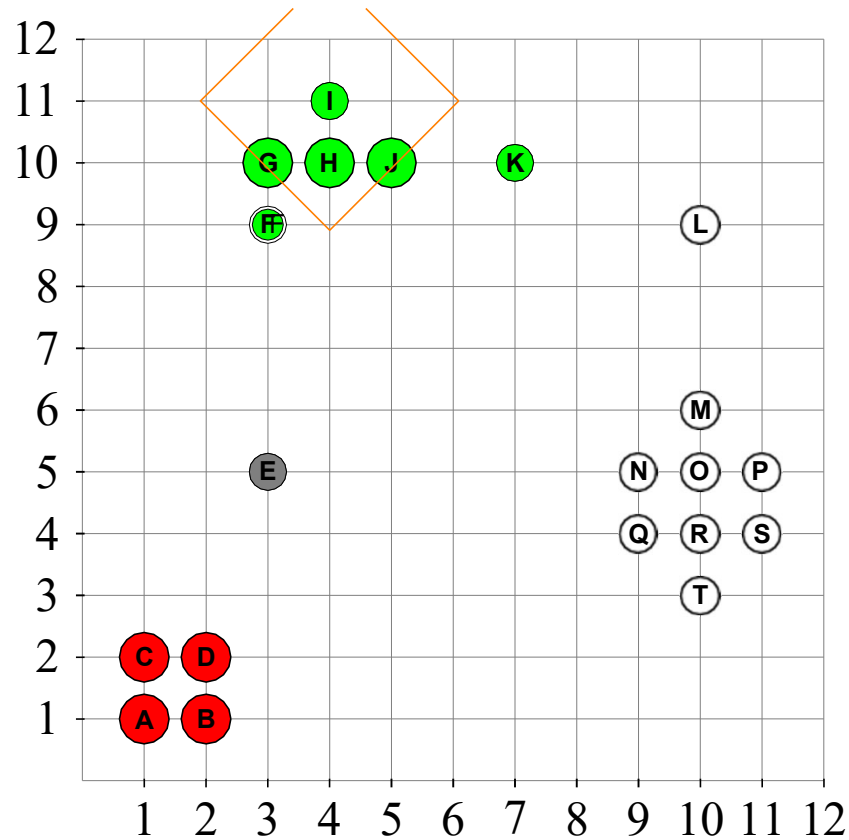
Seed list:

IK

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

$$\text{minPts} = 4$$

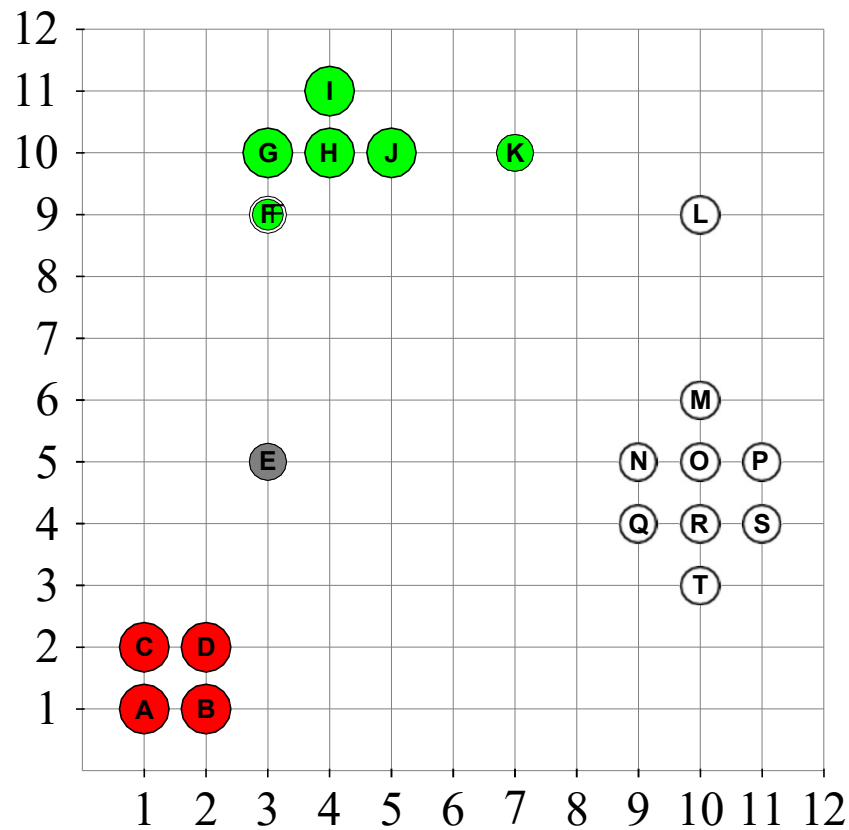
Seed list:

K

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

$$\text{minPts} = 4$$

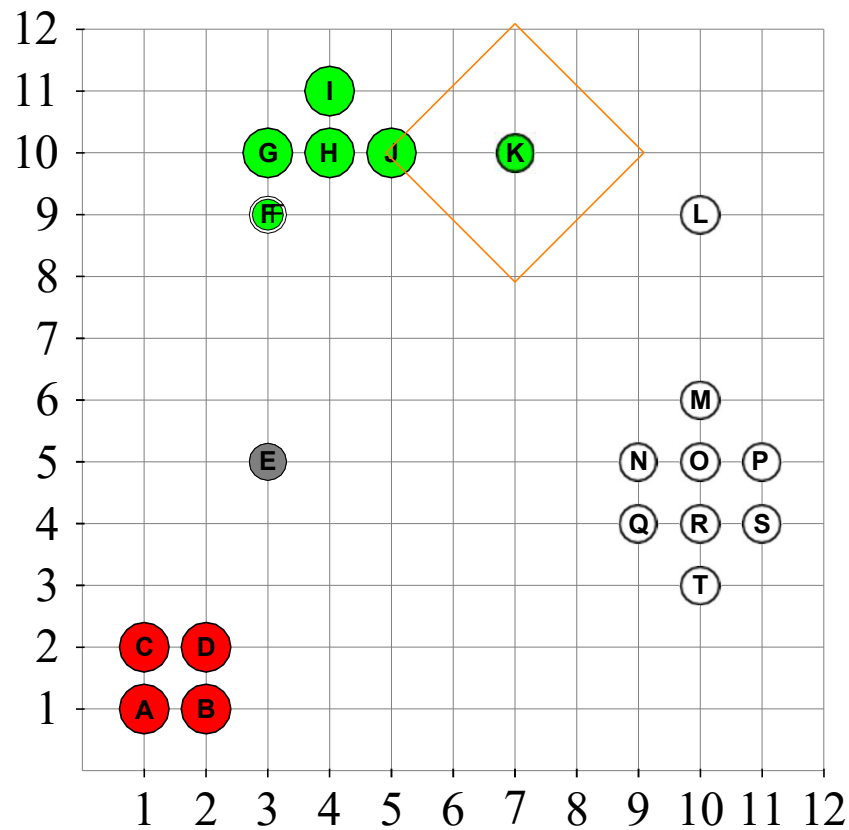
Seed list:

K

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

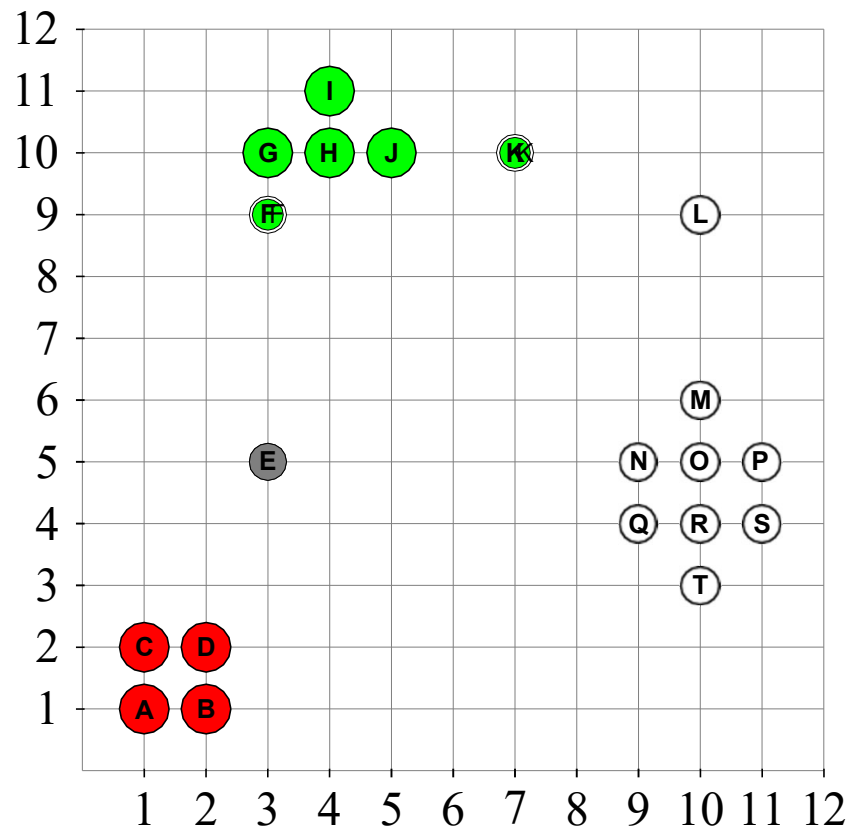
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

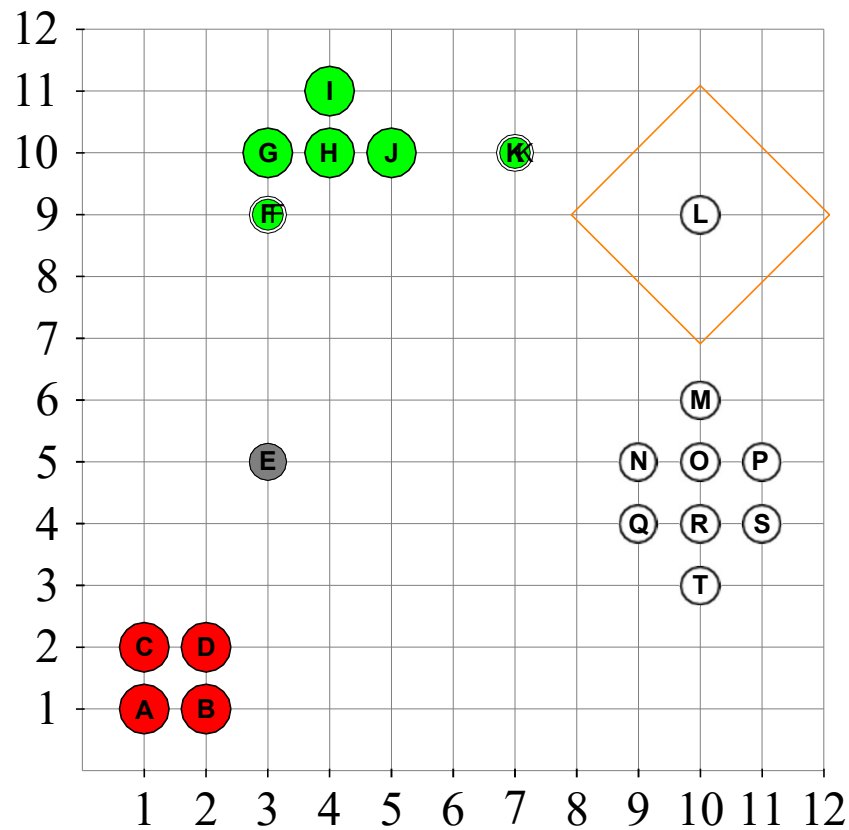
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

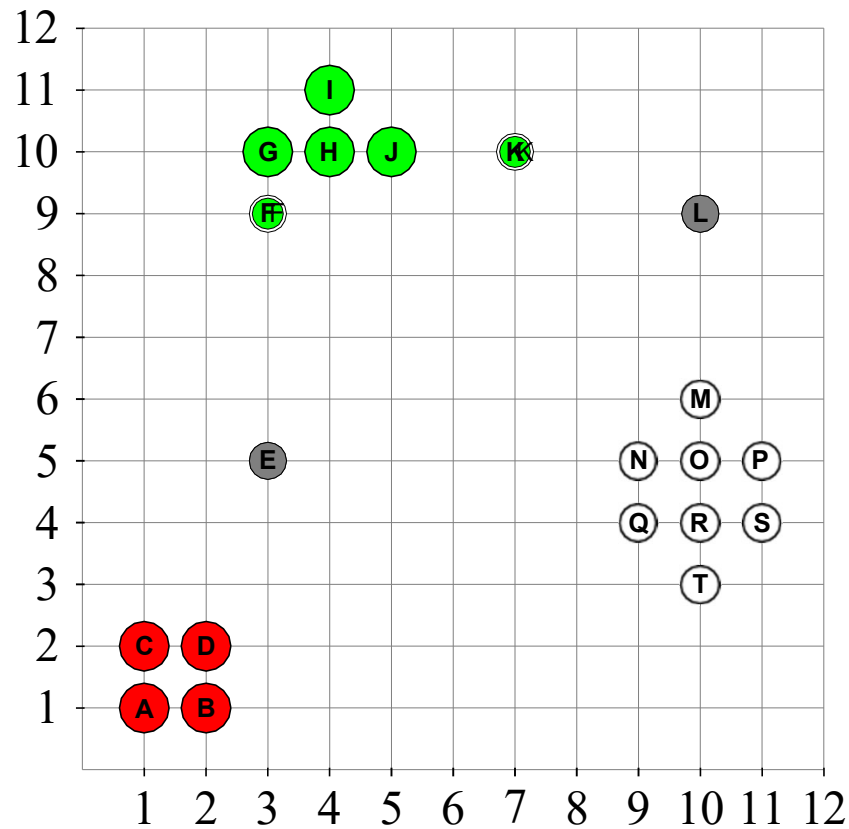
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

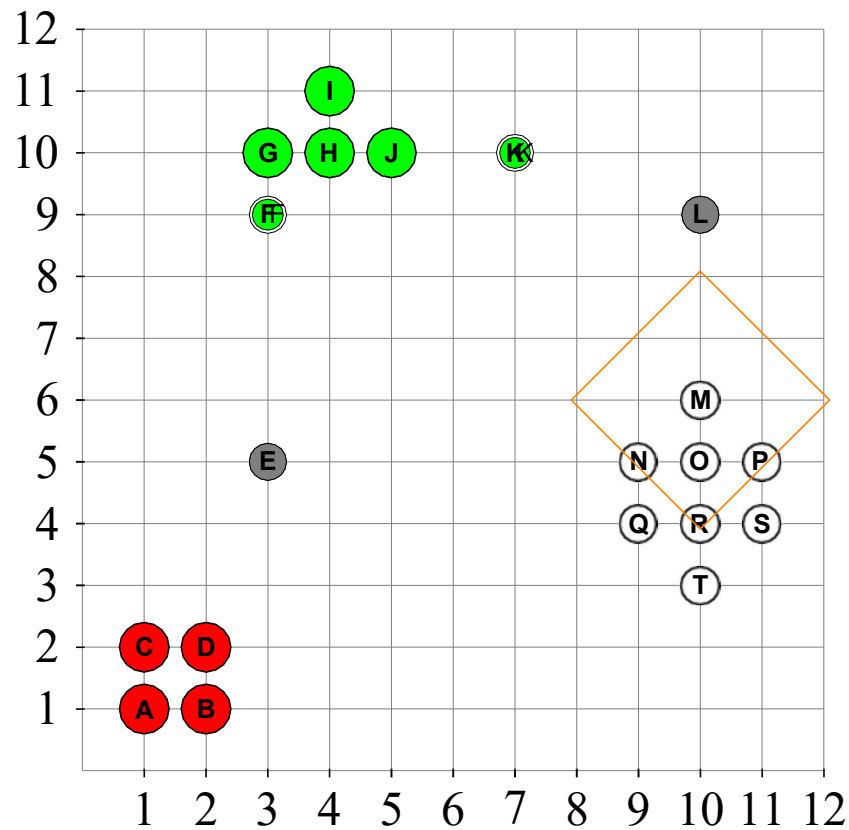
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

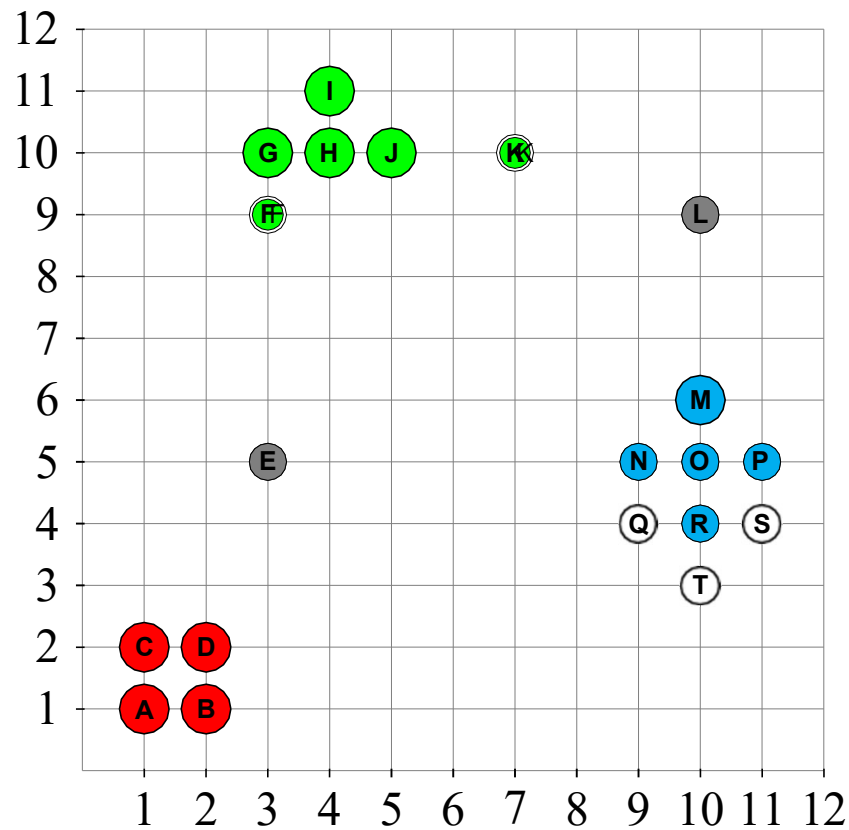
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

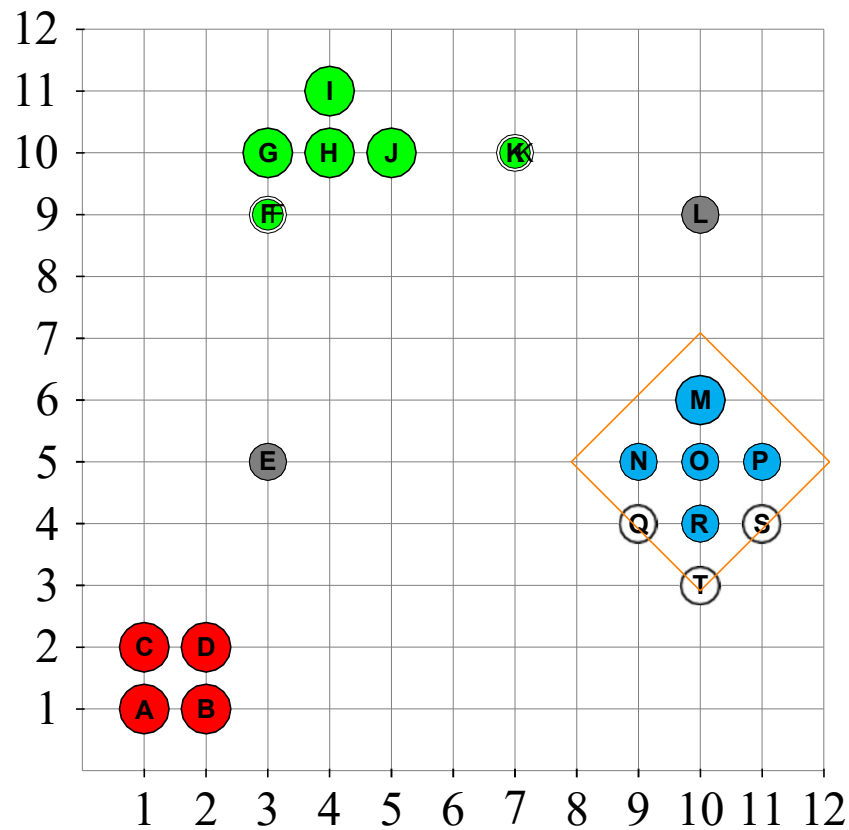
$$\text{minPts} = 4$$

Seed list:
ONPR

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

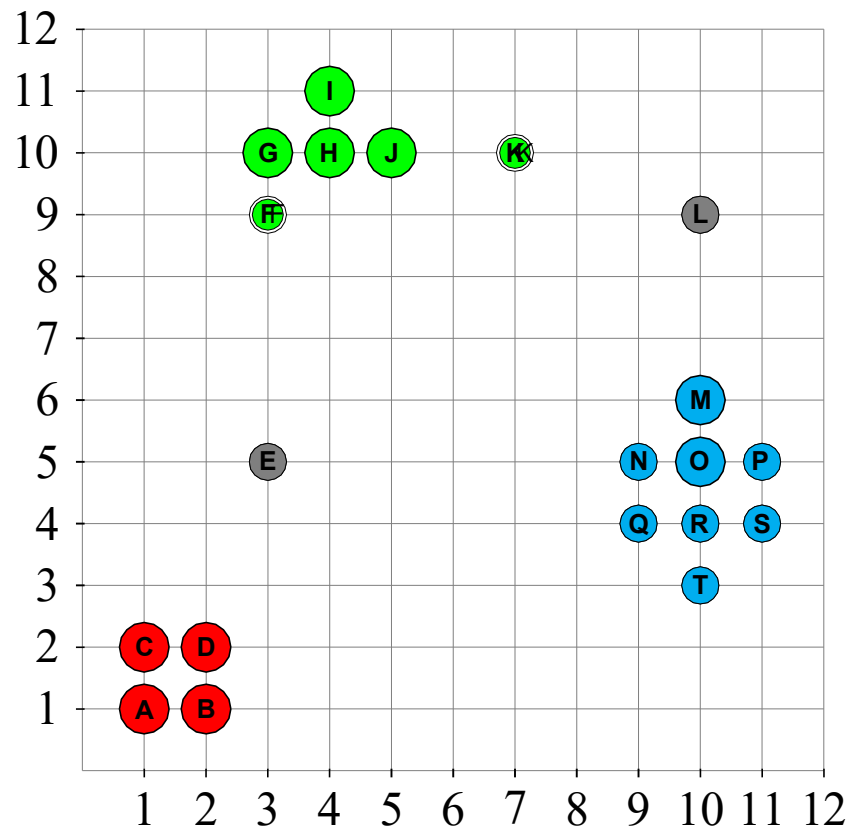
$$\text{minPts} = 4$$

Seed list:
NPR

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

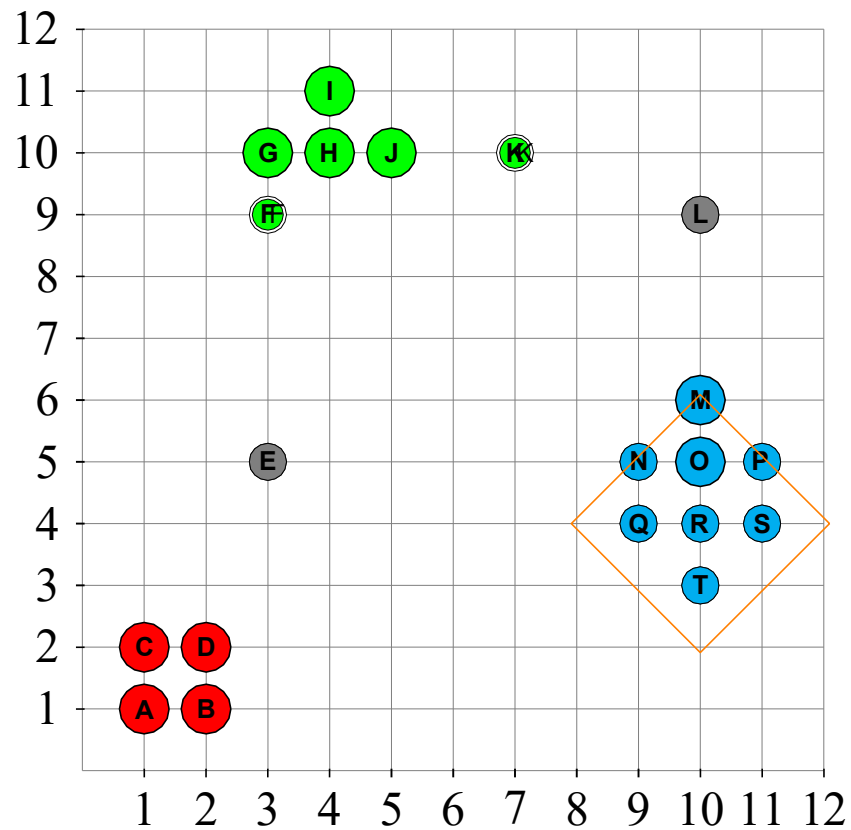
$$\text{minPts} = 4$$

Seed list:
NPRQST

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

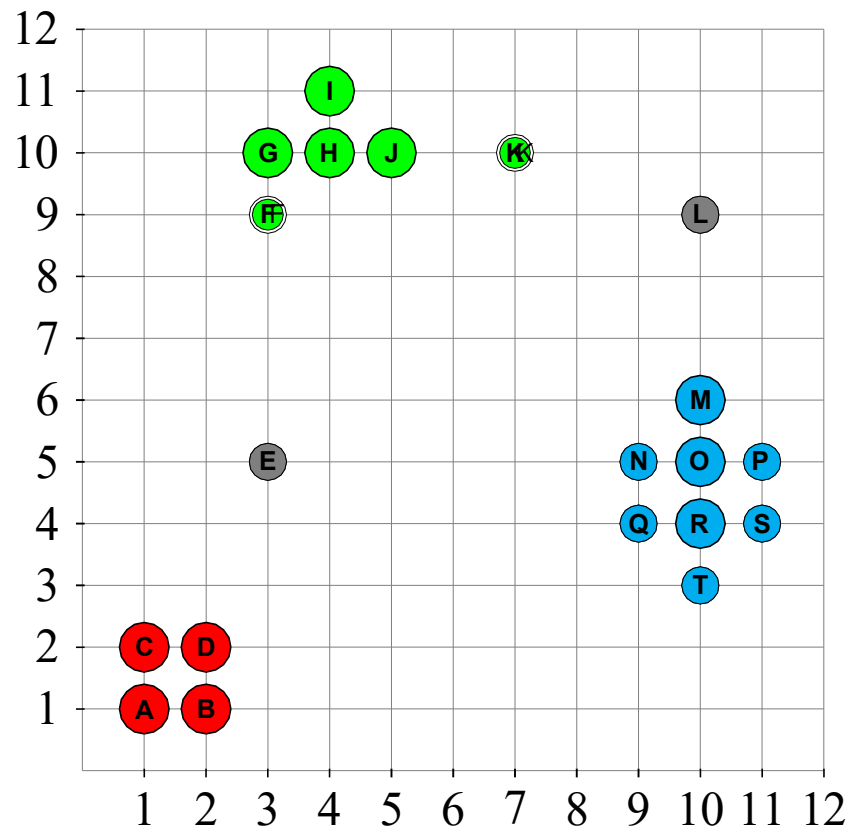
$$\text{minPts} = 4$$

Seed list:
NPQST

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

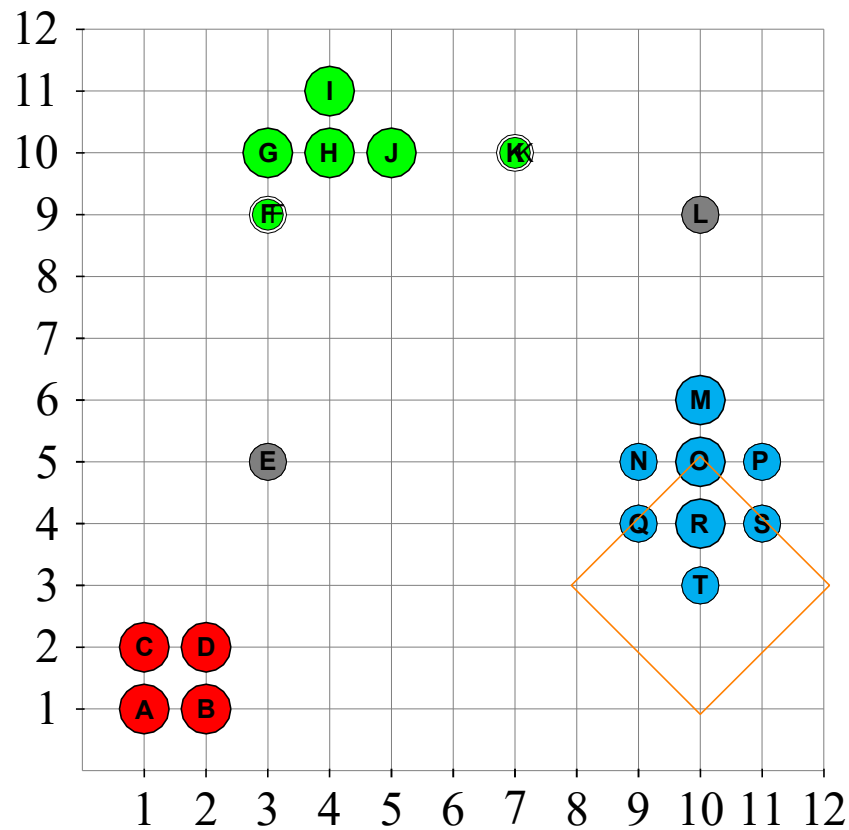
$$\text{minPts} = 4$$

Seed list:
NPQST

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

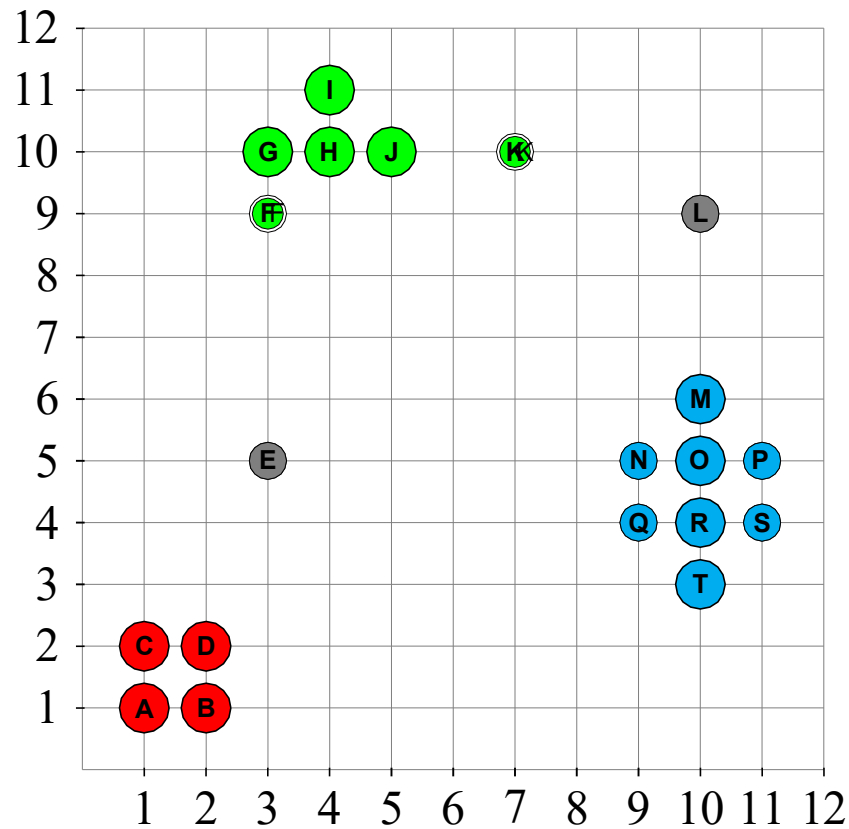
$$\text{minPts} = 4$$

Seed list:
NPQS

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

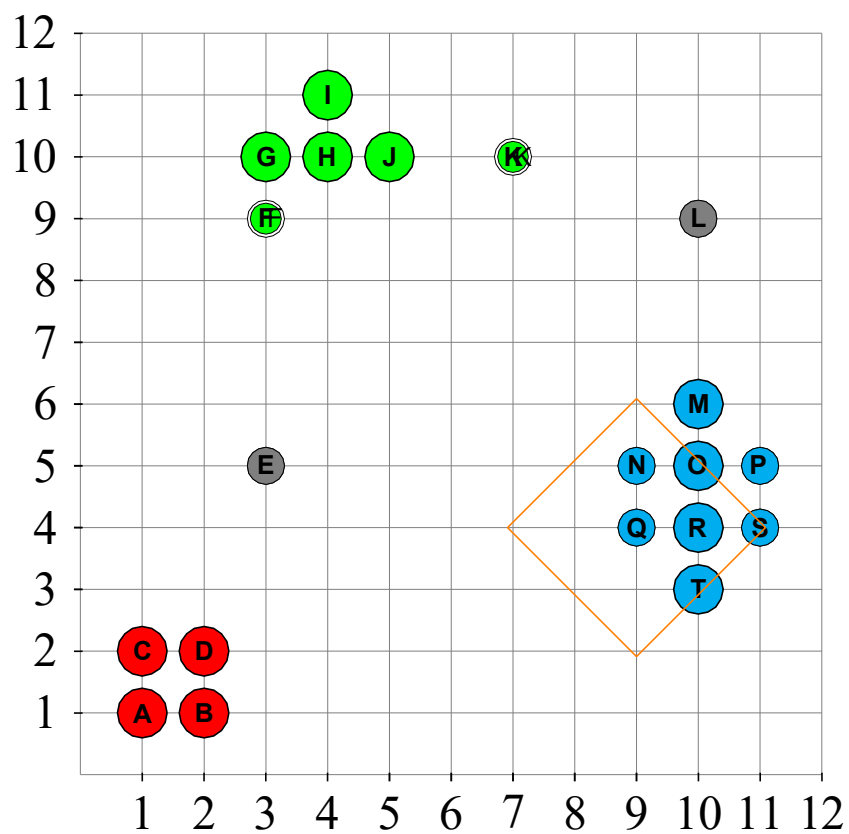
$$\text{minPts} = 4$$

Seed list:
NPQS

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

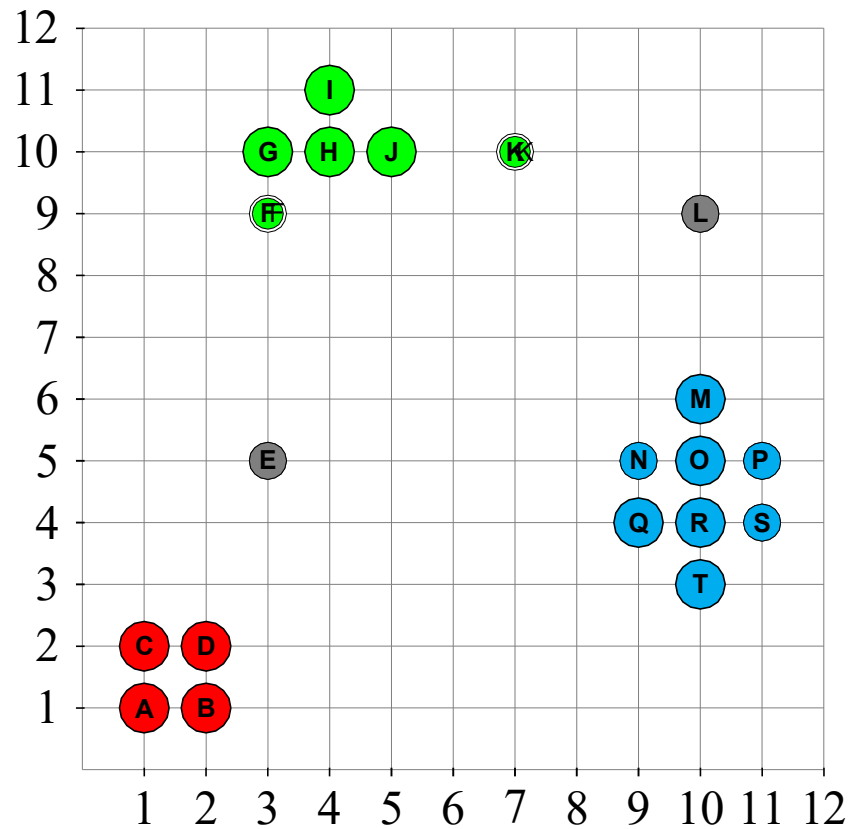
$$\text{minPts} = 4$$

Seed list:
NPS

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

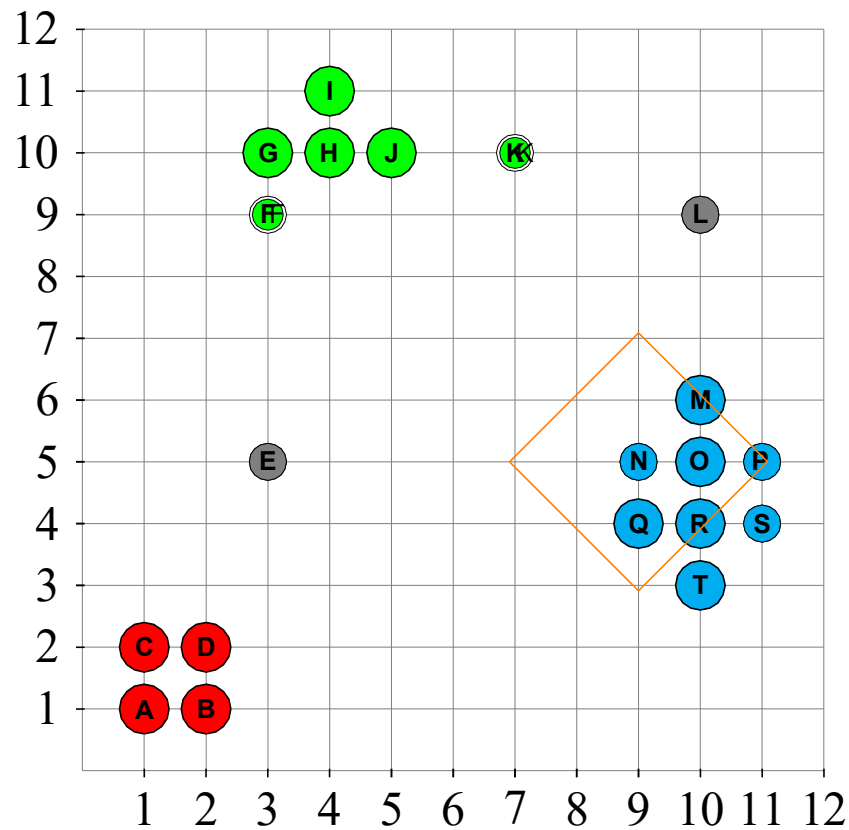
$$\text{minPts} = 4$$

Seed list:
NPS

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

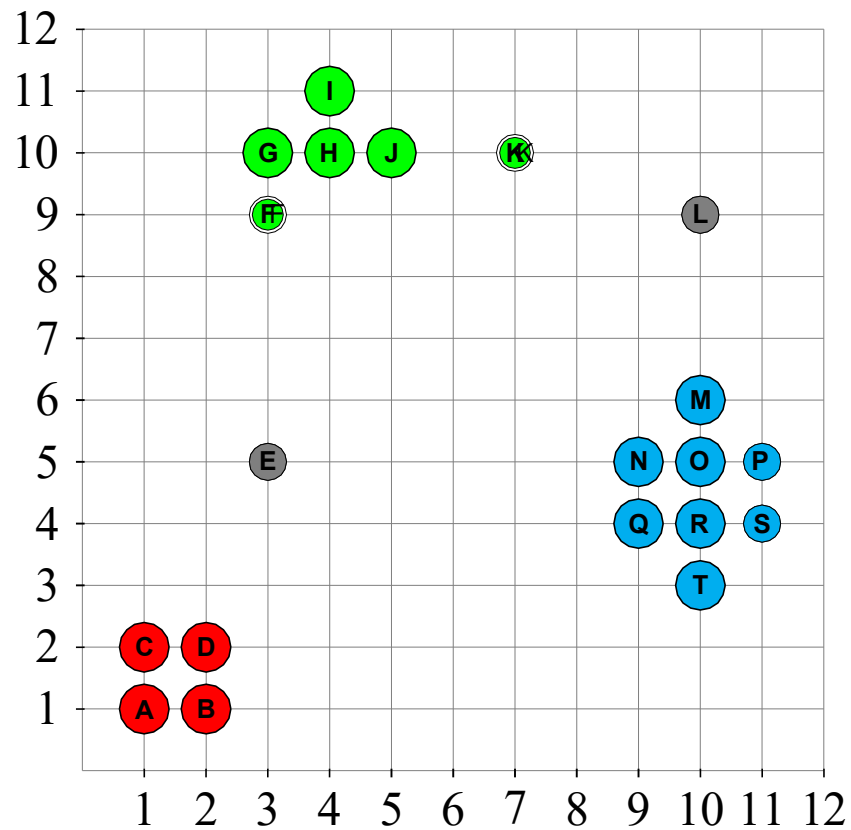
$$\text{minPts} = 4$$

Seed list:
PS

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

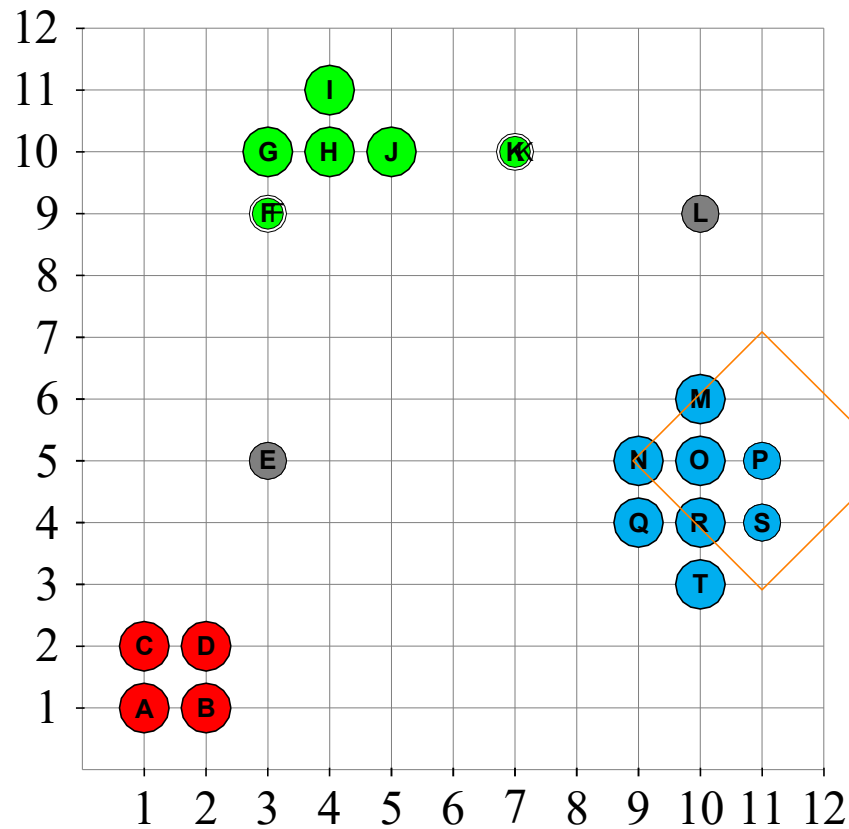
$$\text{minPts} = 4$$

Seed list:
PS

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

$$\text{minPts} = 4$$

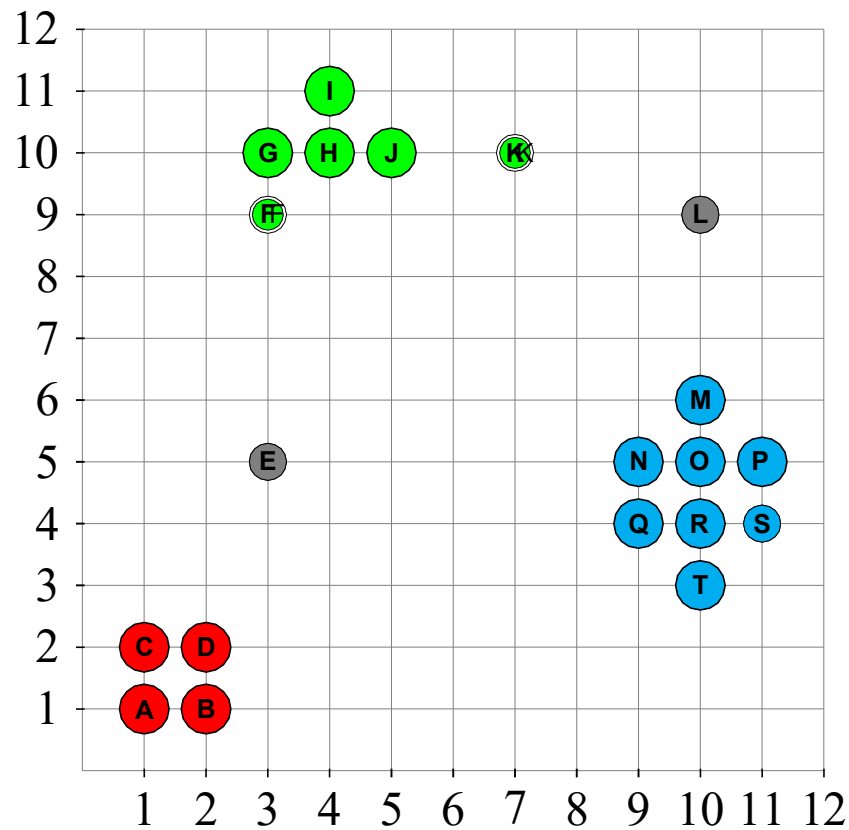
Seed list:

S

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

$$\text{minPts} = 4$$

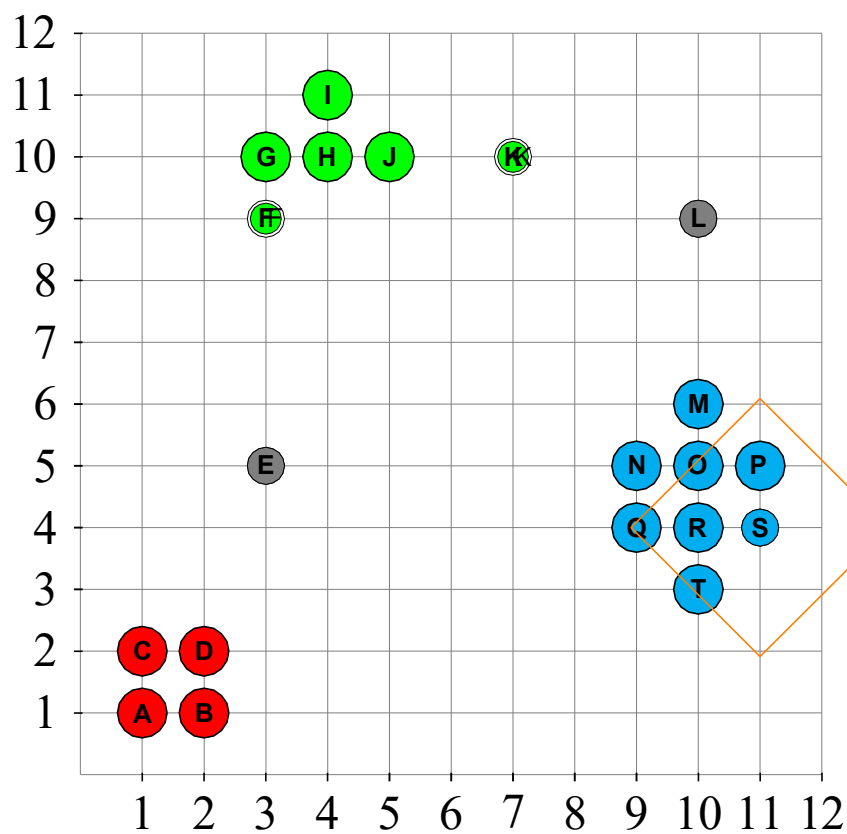
Seed list:

S

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

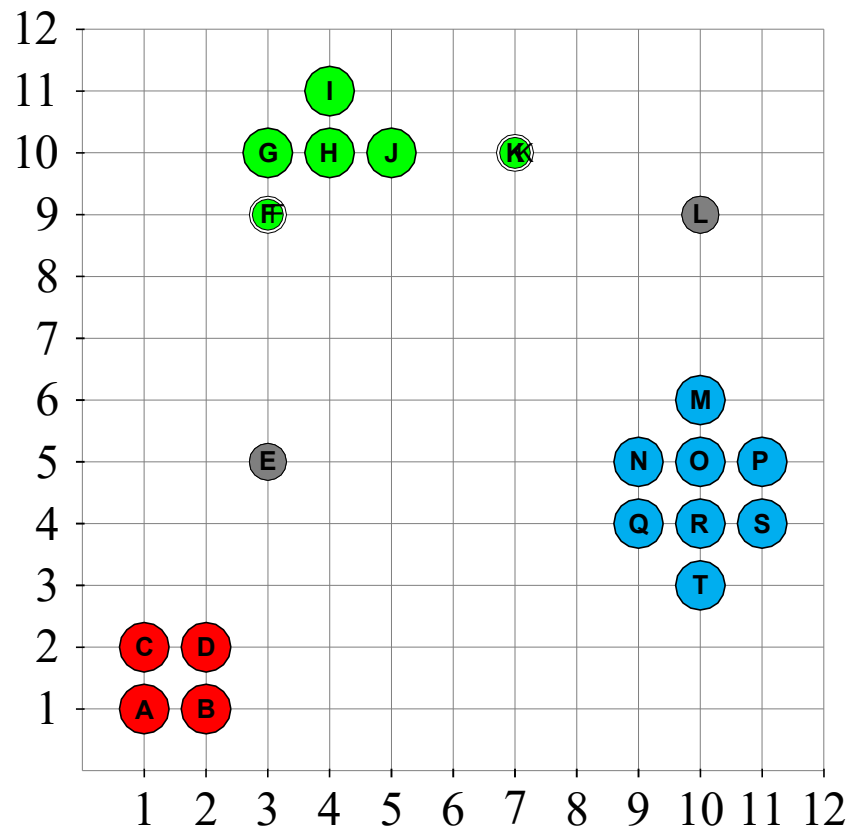
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 2.1$$

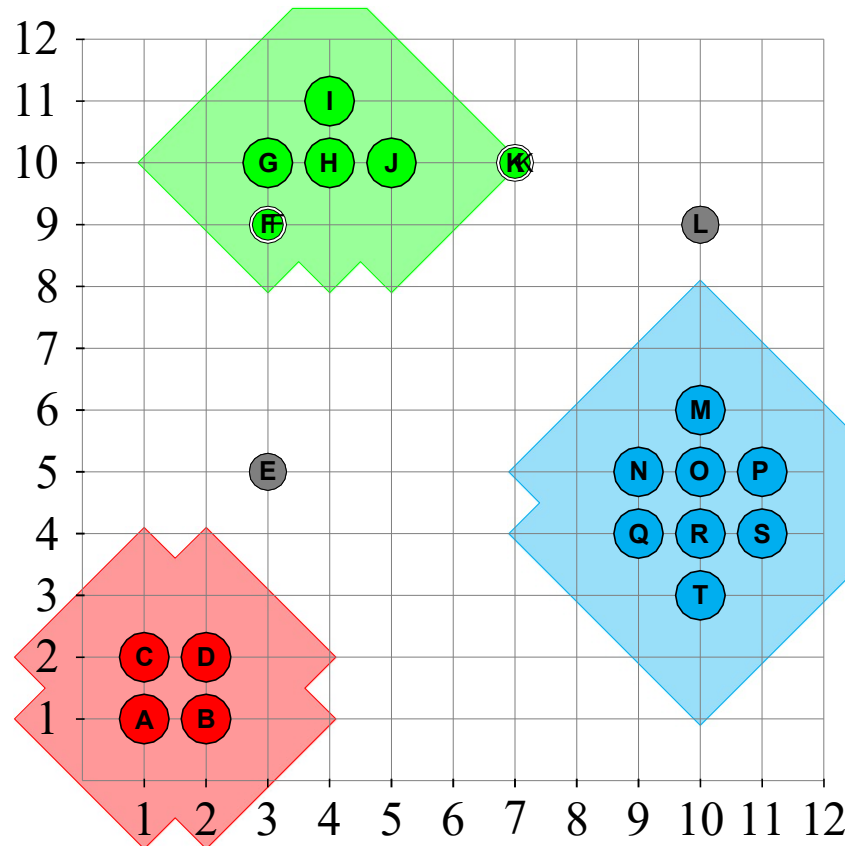
$$\text{minPts} = 4$$

Seed list:

Example 4

DM868 DM870
DS804

Arthur Zimek



$\epsilon = 2.1$

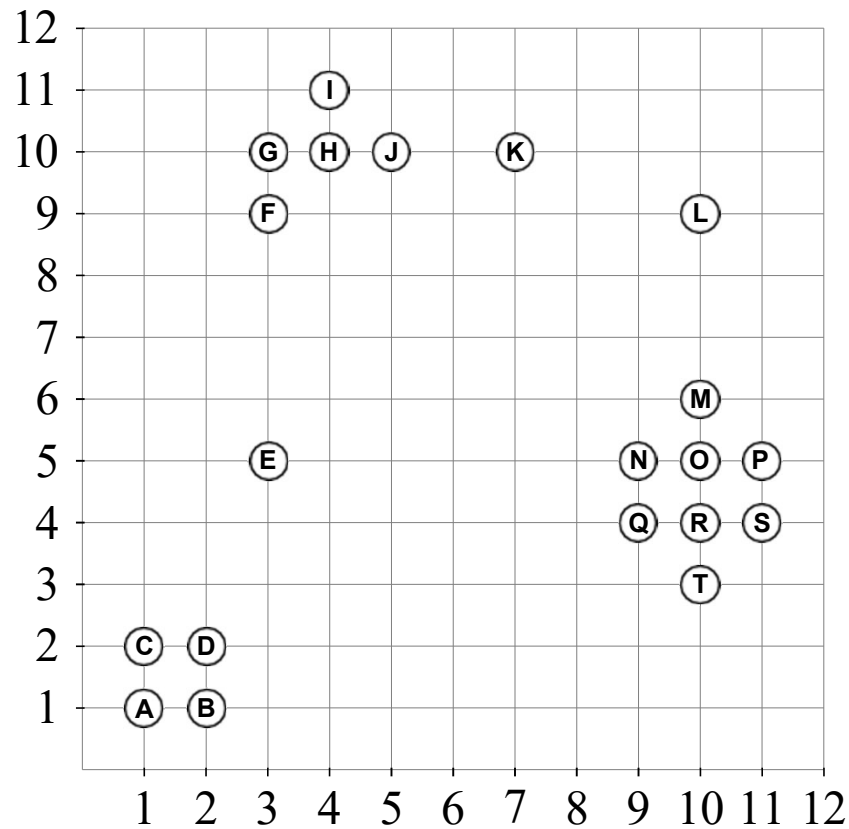
$minPts = 4$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

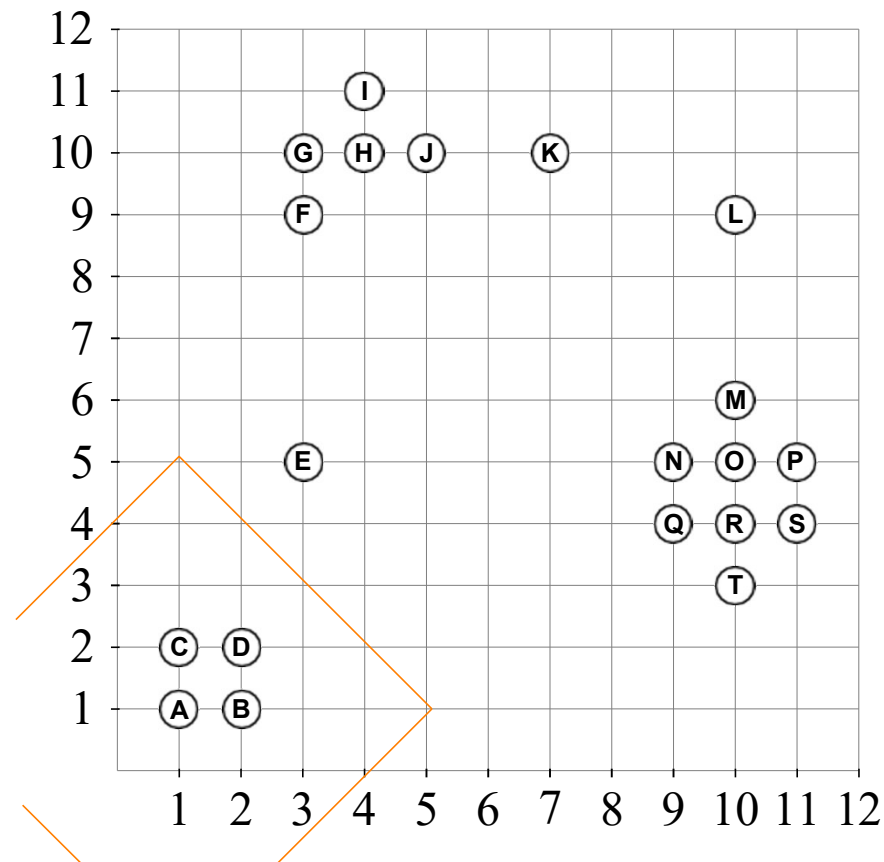
$$\text{minPts} = 5$$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

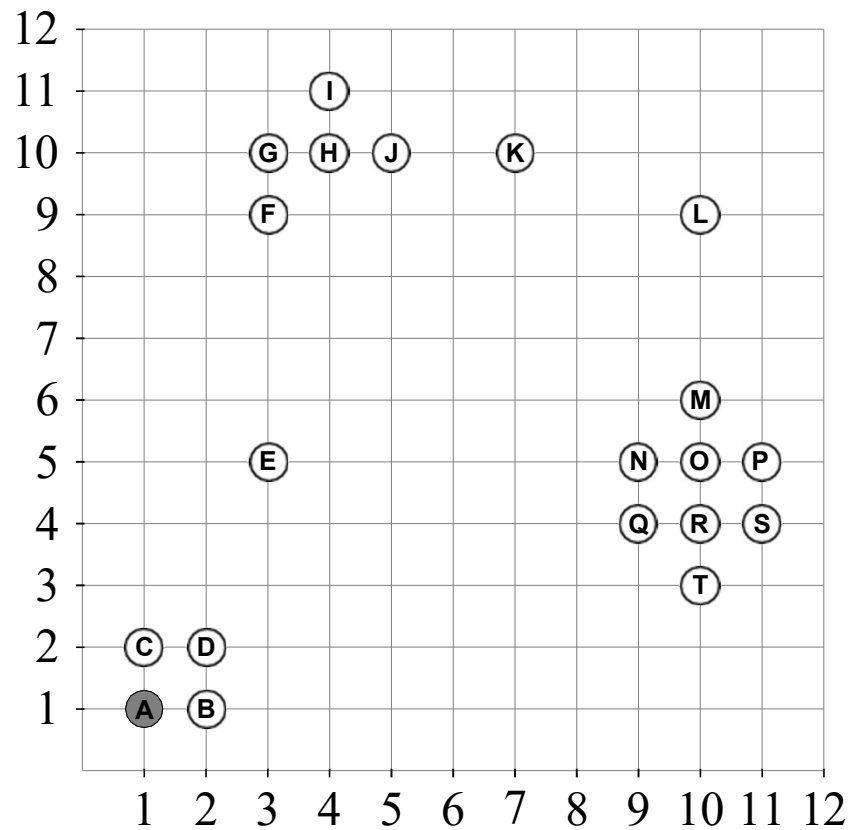
$$\text{minPts} = 5$$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

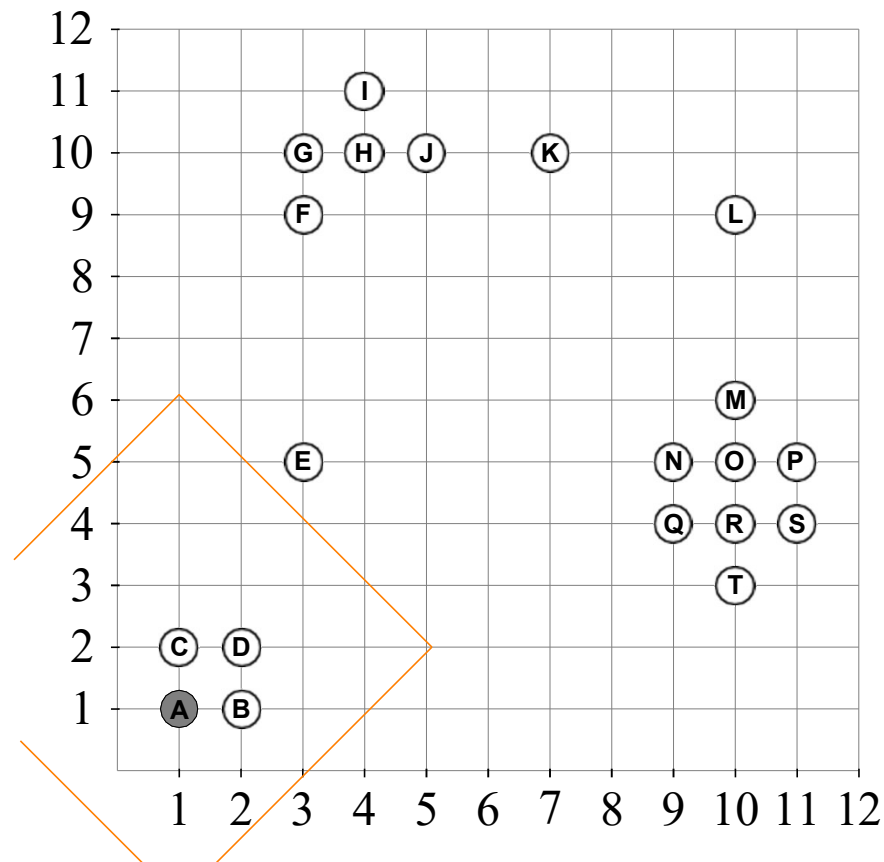
$$\text{minPts} = 5$$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

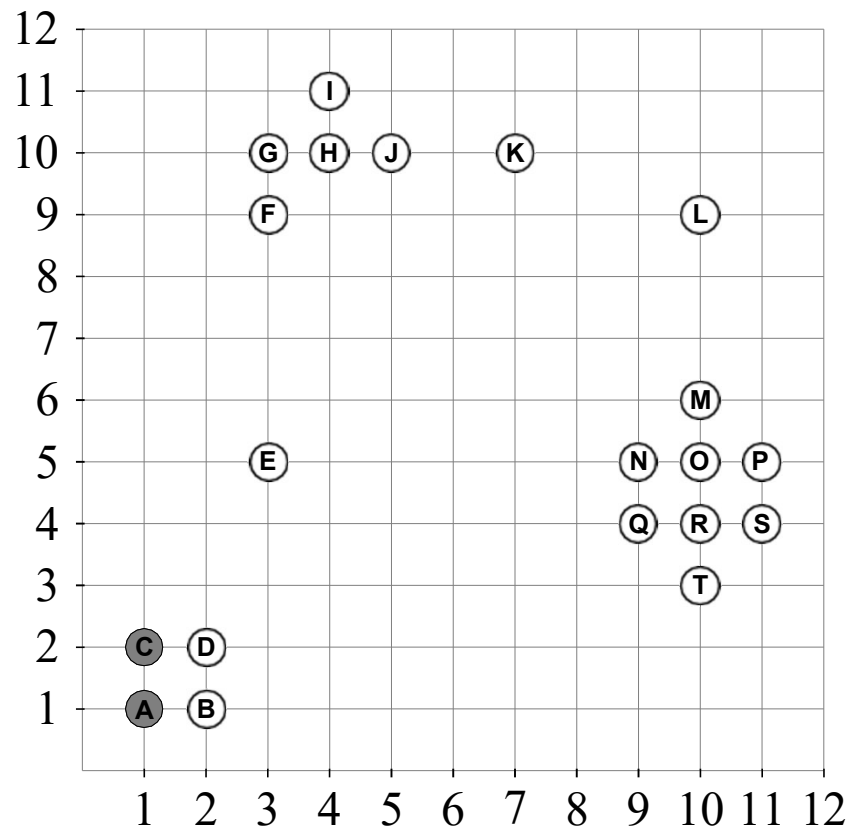
$$\text{minPts} = 5$$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

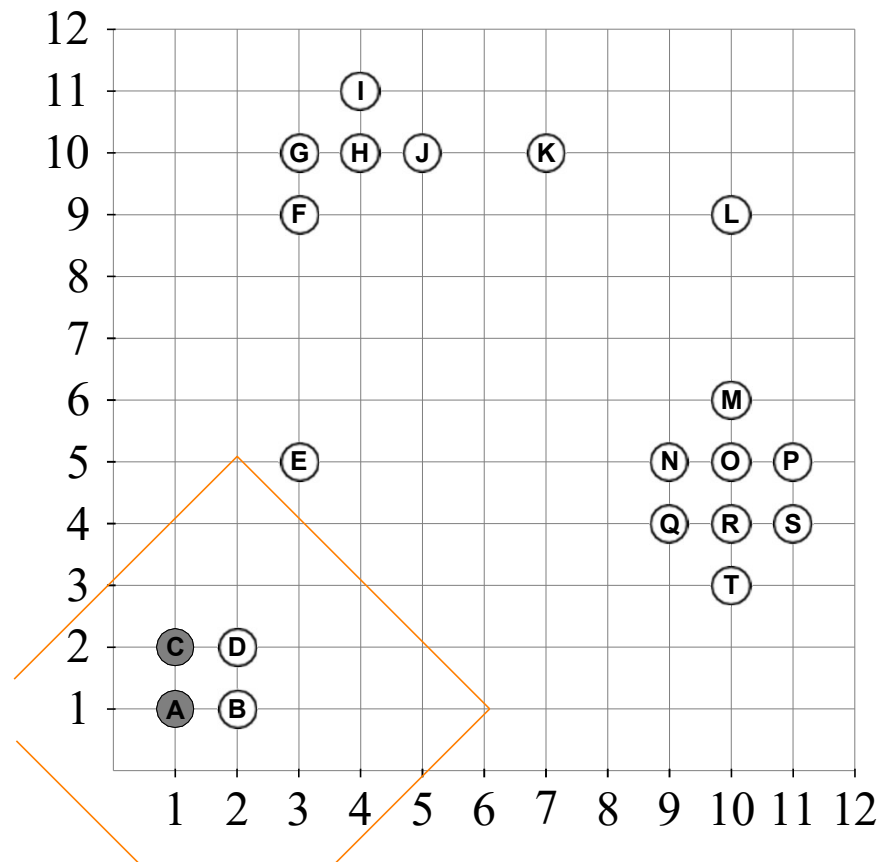
$$\text{minPts} = 5$$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

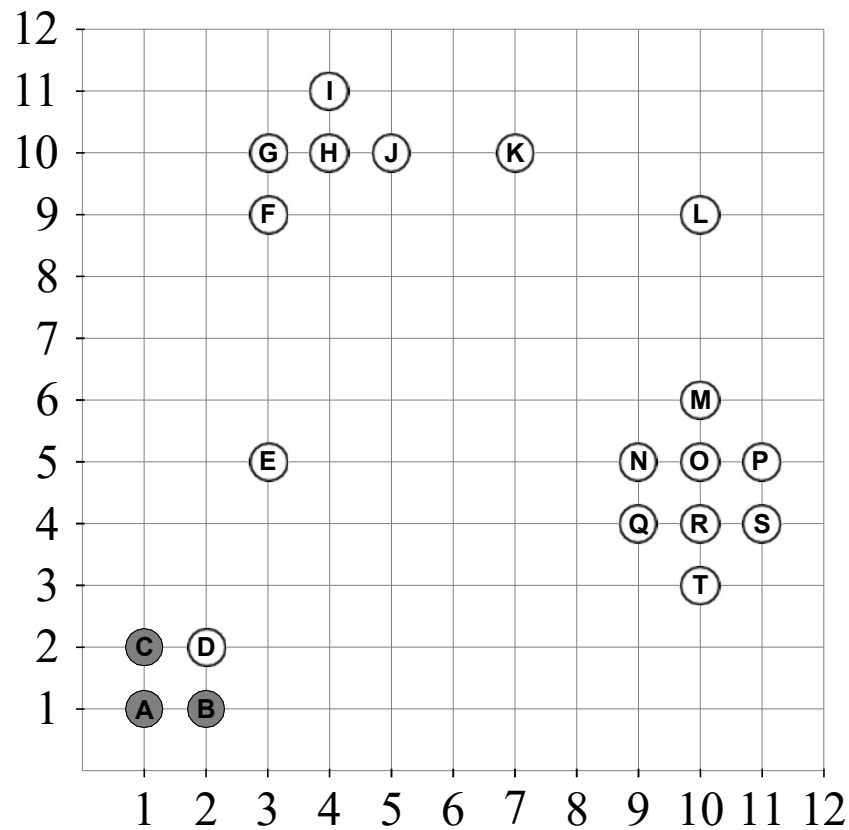
minPts = 5

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

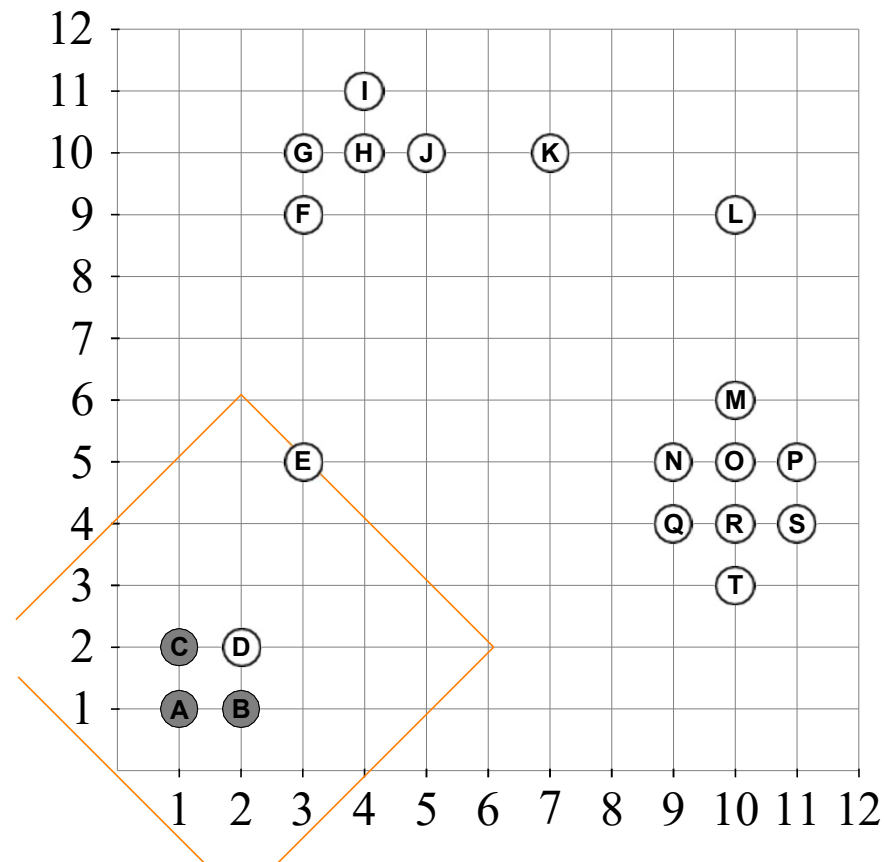
$$\text{minPts} = 5$$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

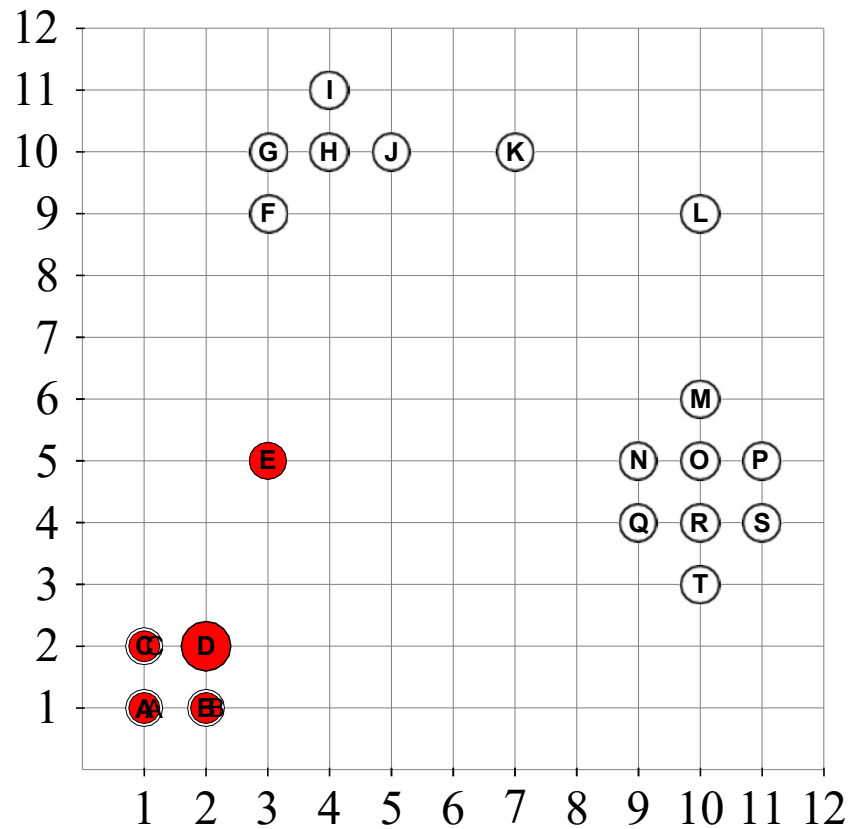
$$\text{minPts} = 5$$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

$$\text{minPts} = 5$$

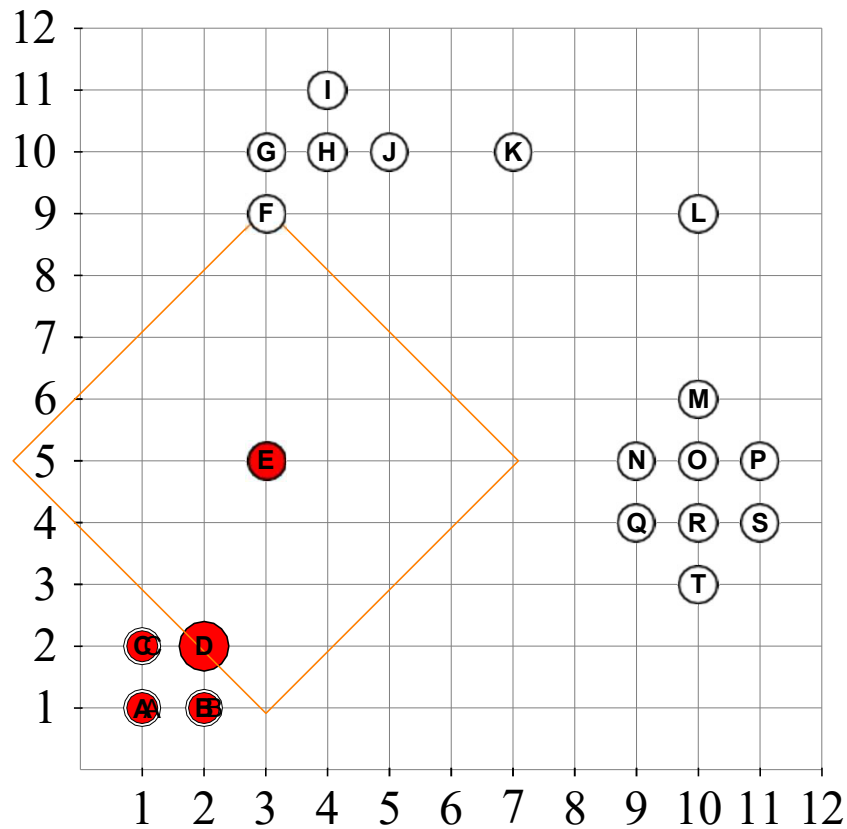
Seed list:

E

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

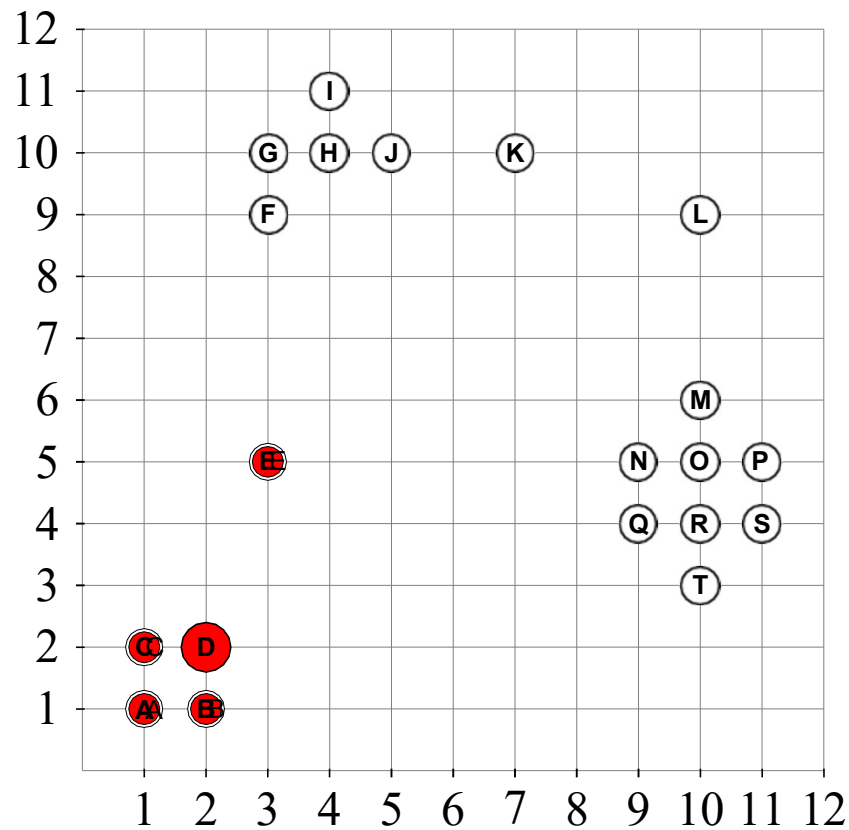
$\minPts = 5$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

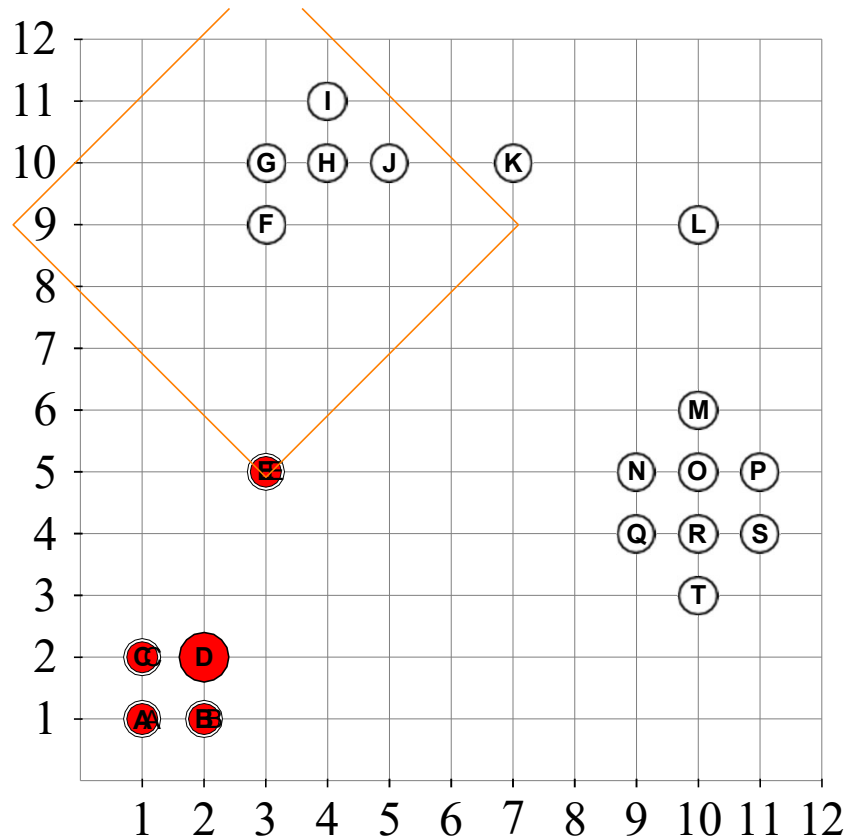
$$\text{minPts} = 5$$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

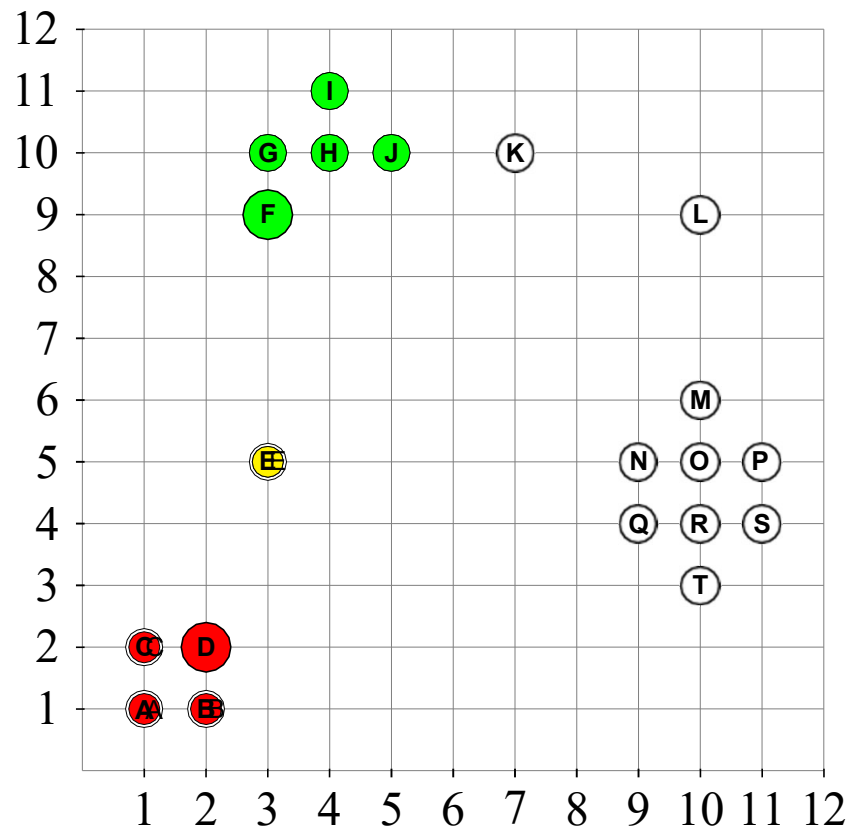
$$\text{minPts} = 5$$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

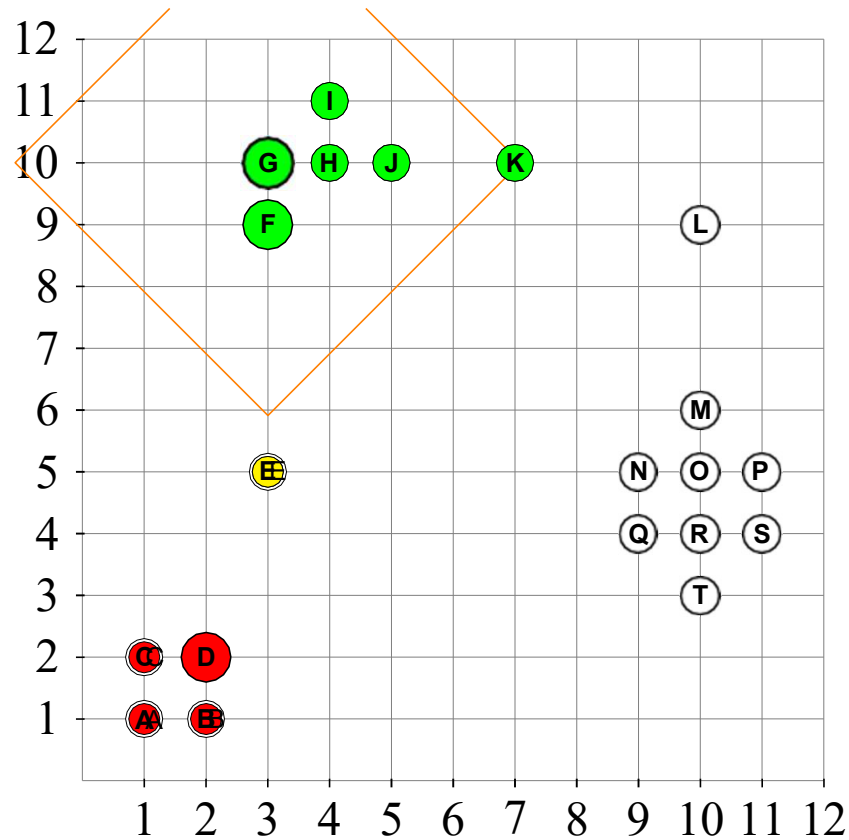
$$\text{minPts} = 5$$

Seed list:
GHIJ

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

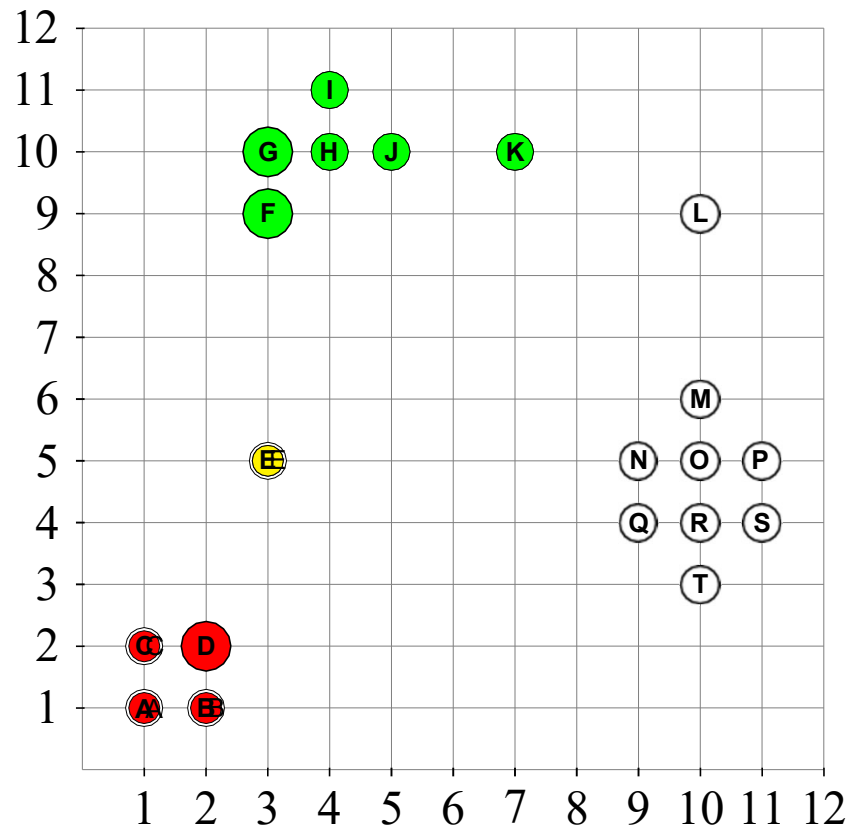
$$\text{minPts} = 5$$

Seed list:
HIJ

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

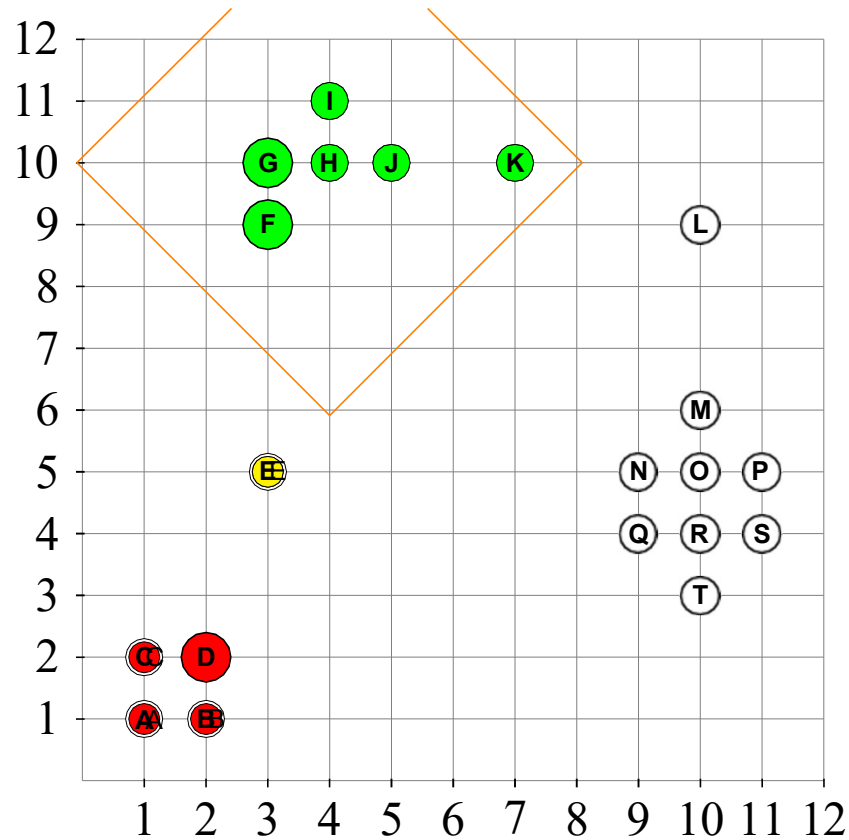
$$\text{minPts} = 5$$

Seed list:
HIJK

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

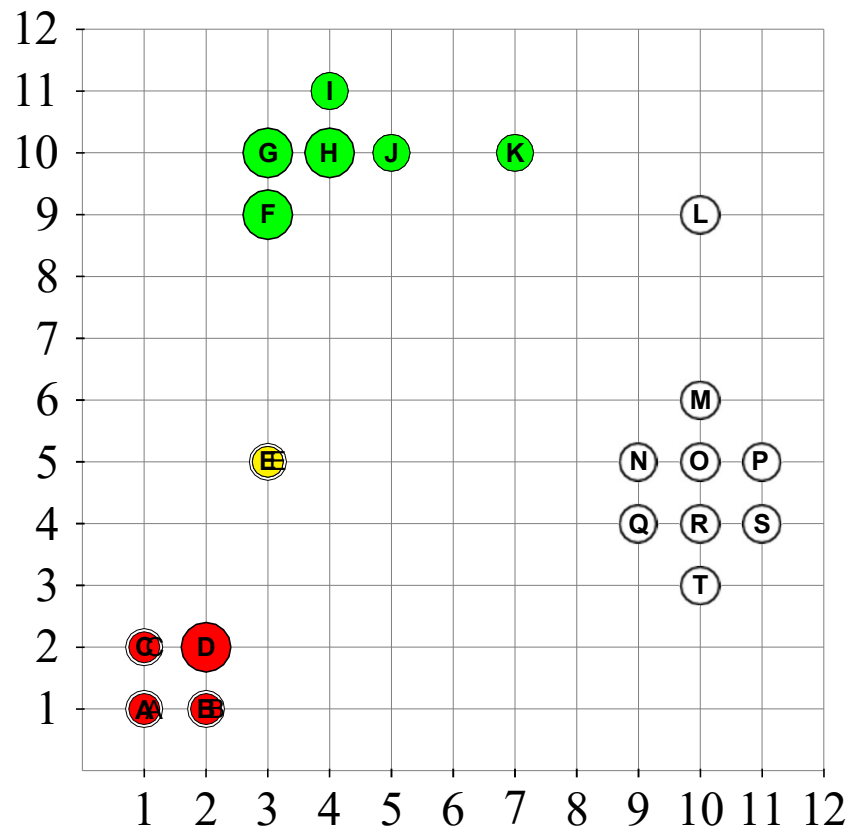
$$\text{minPts} = 5$$

Seed list:
IJK

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

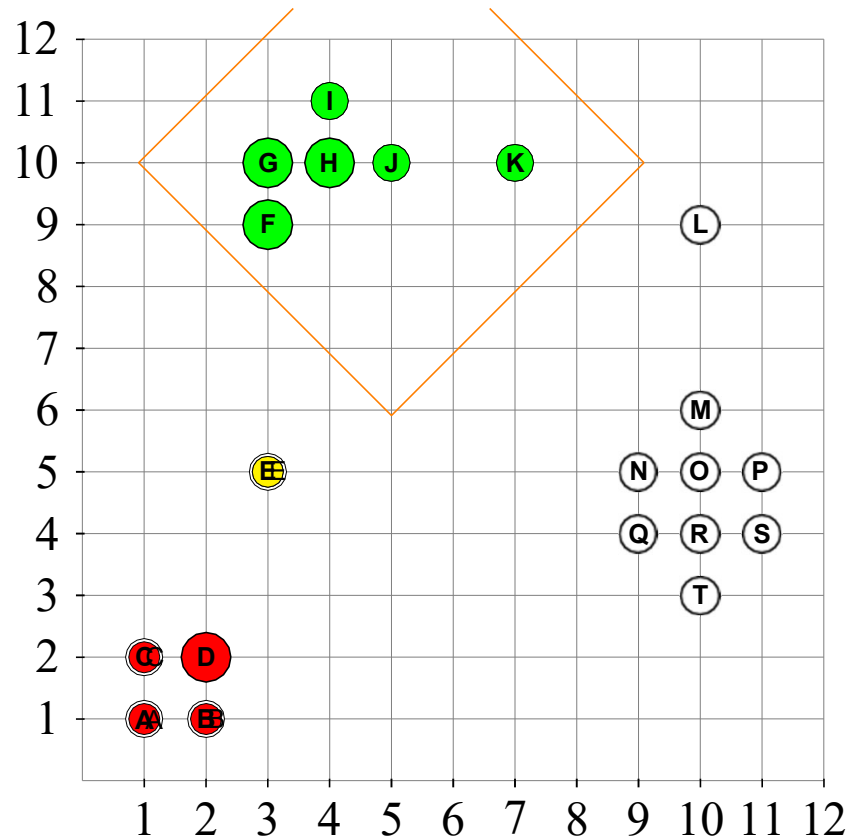
$$\text{minPts} = 5$$

Seed list:
IJK

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

$$\text{minPts} = 5$$

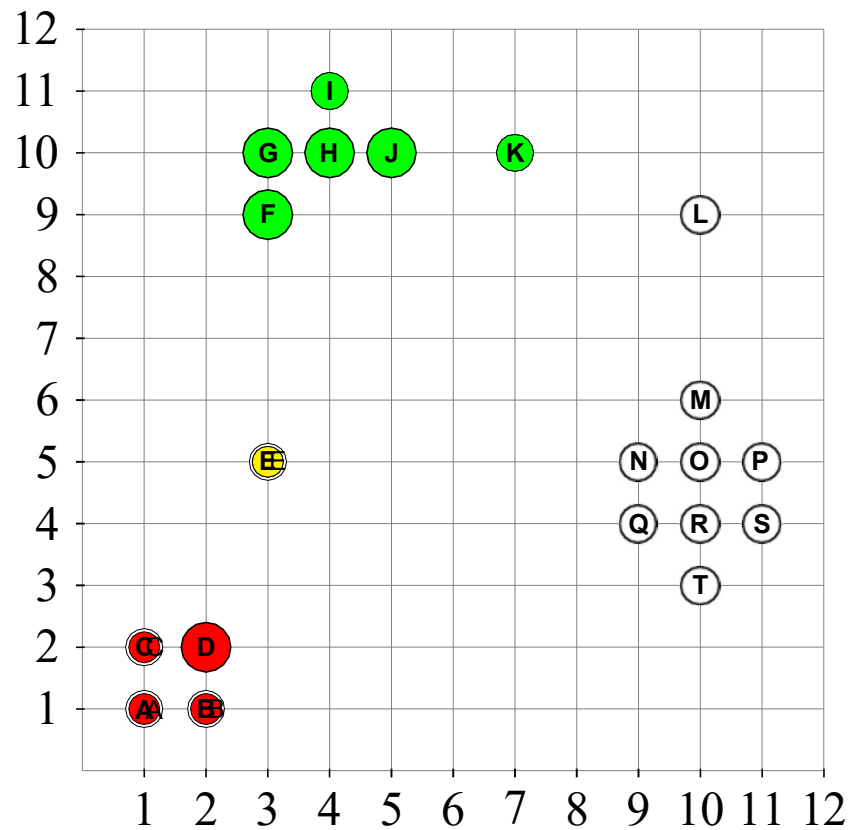
Seed list:

IK

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

$$\text{minPts} = 5$$

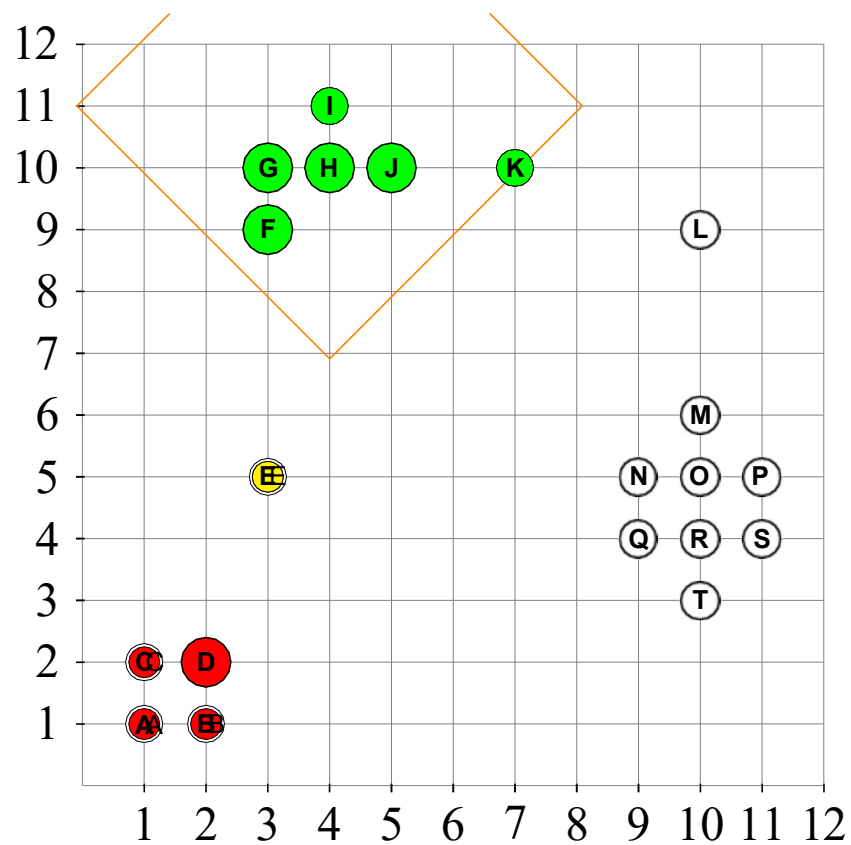
Seed list:

IK

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

$$\text{minPts} = 5$$

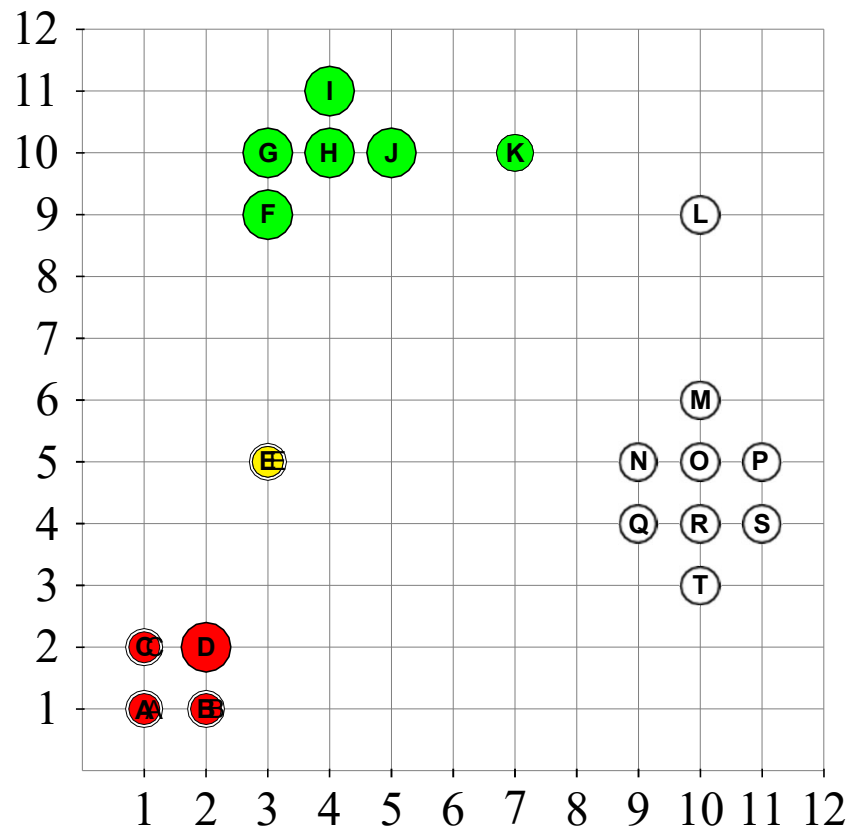
Seed list:

K

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

$$\text{minPts} = 5$$

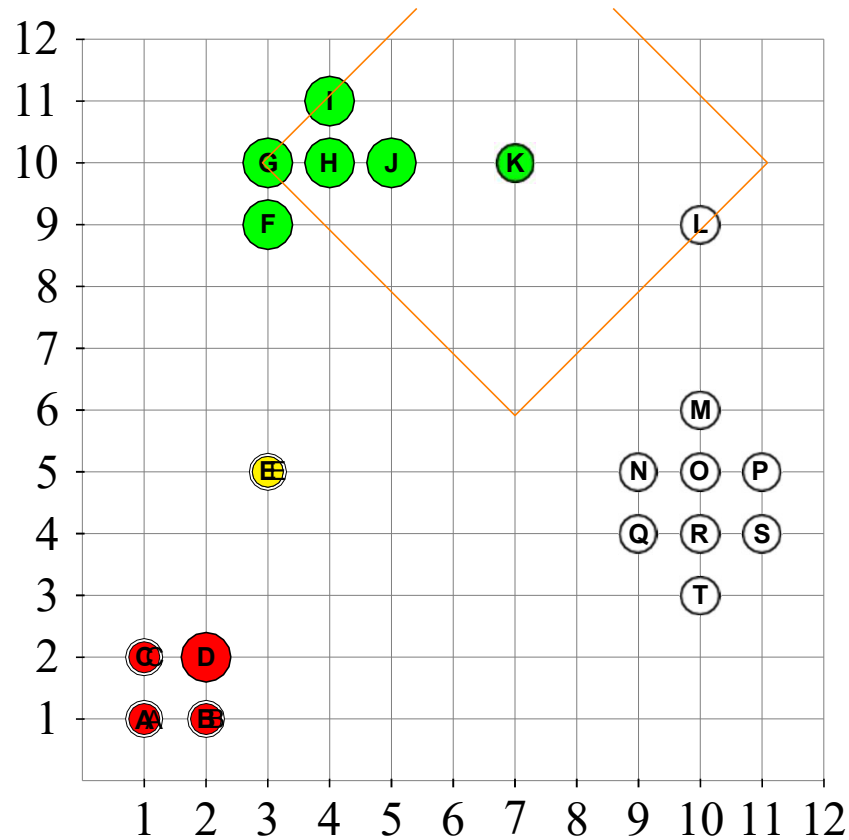
Seed list:

K

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

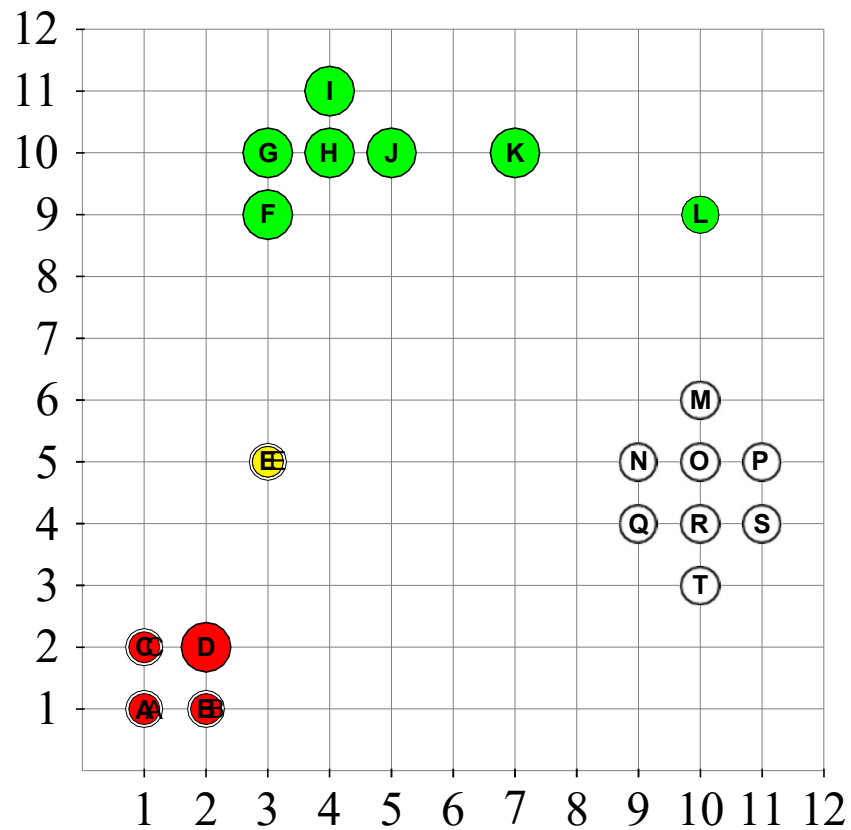
$$\text{minPts} = 5$$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

$$\text{minPts} = 5$$

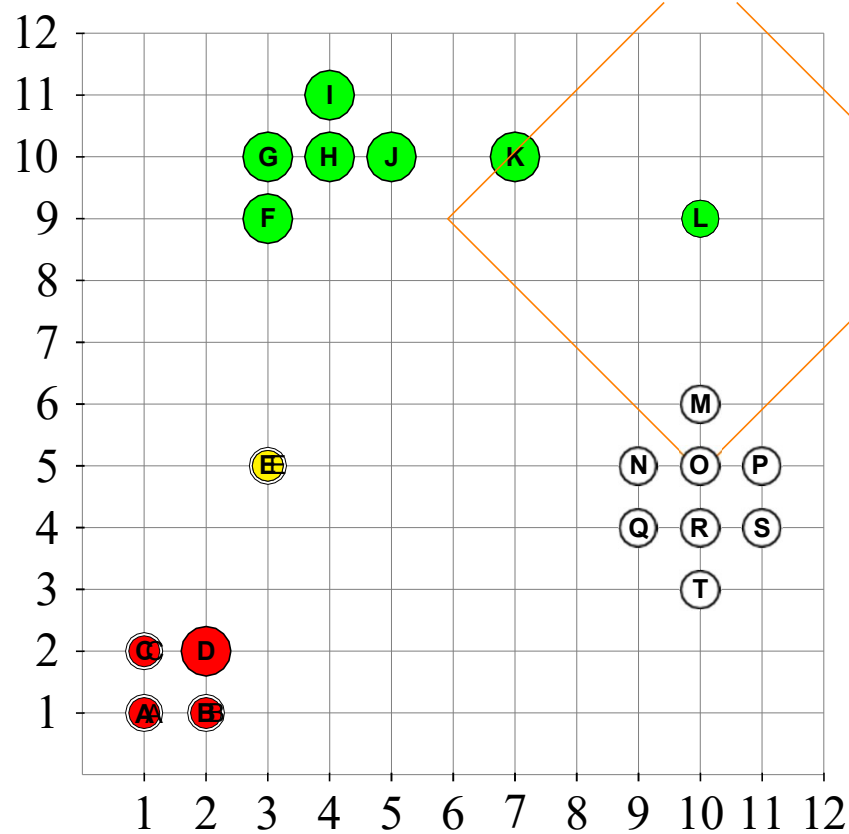
Seed list:

L

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

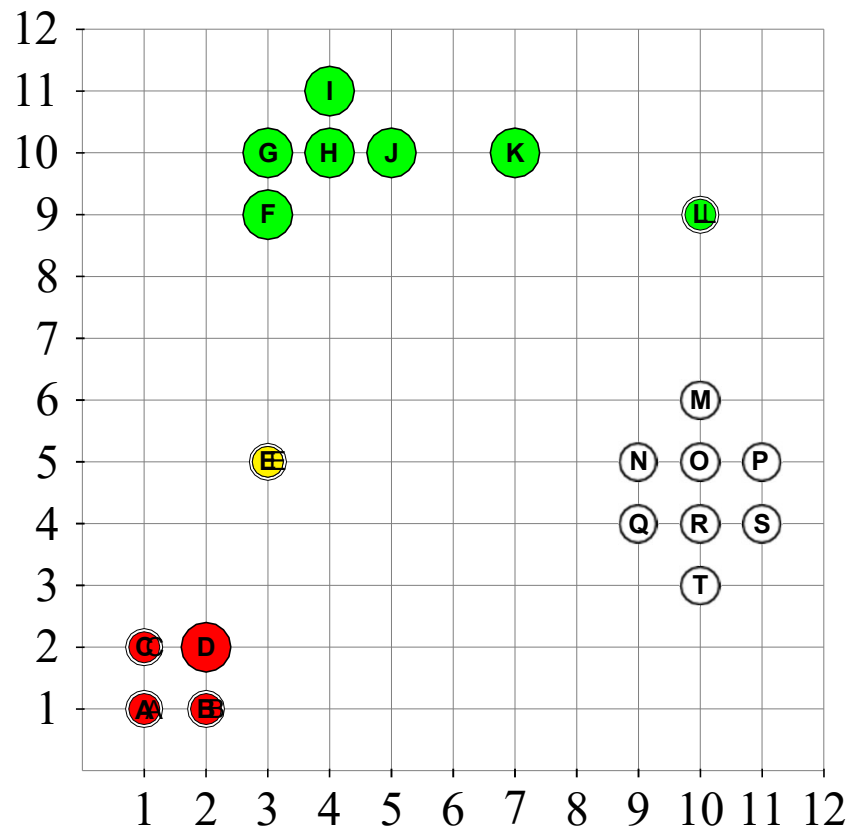
$minPts = 5$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

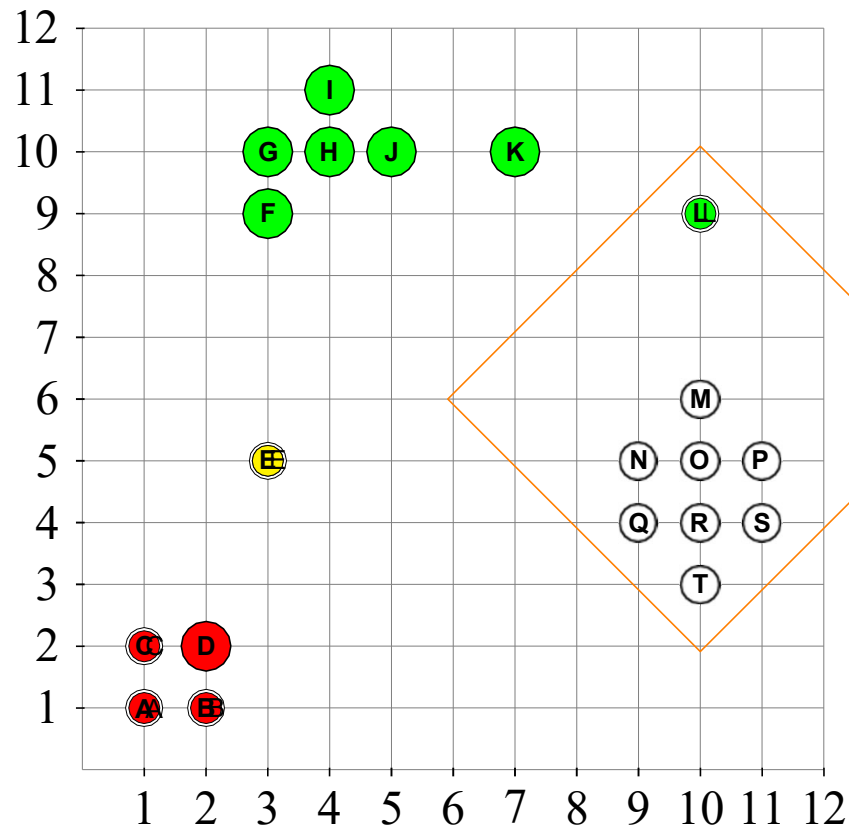
$$\text{minPts} = 5$$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

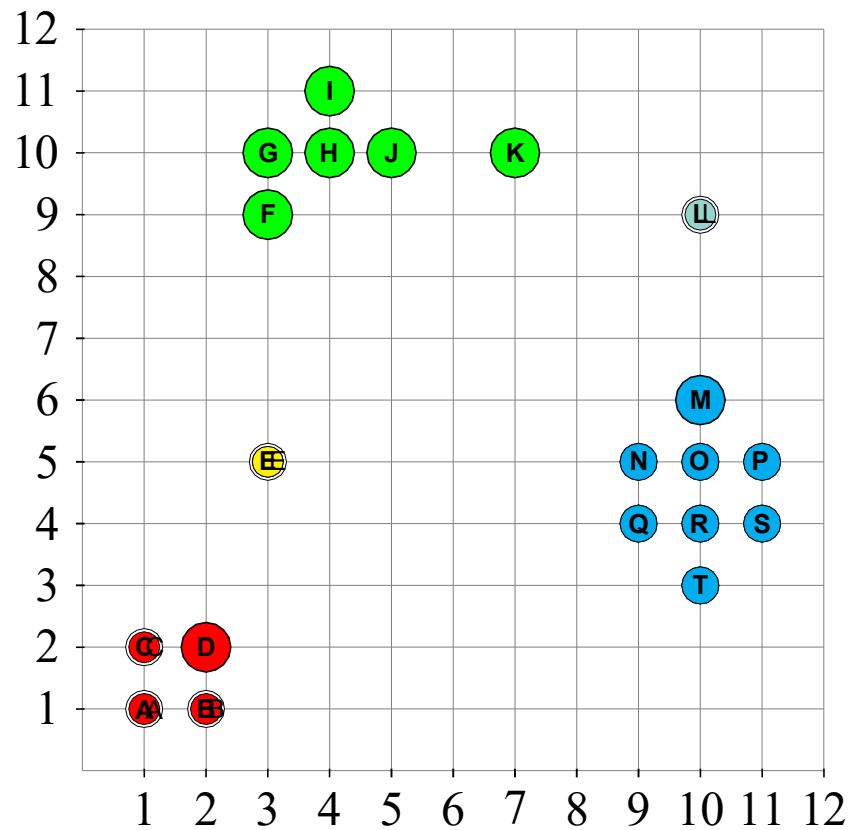
$\minPts = 5$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

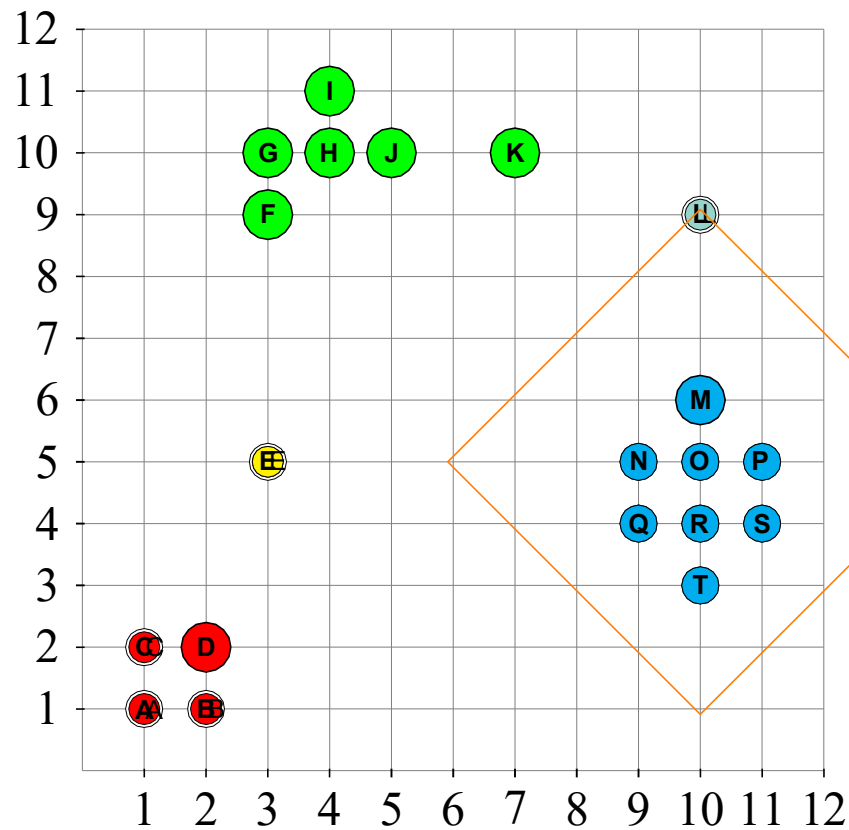
$$\text{minPts} = 5$$

Seed list:
ONPRQST

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

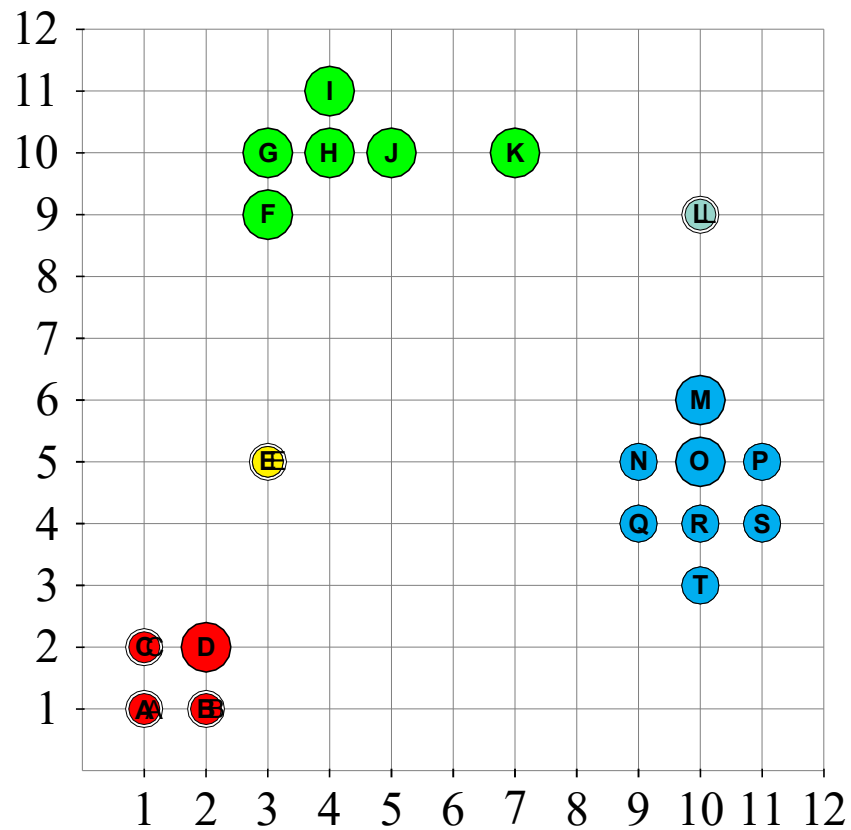
$$\text{minPts} = 5$$

Seed list:
NPQST

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

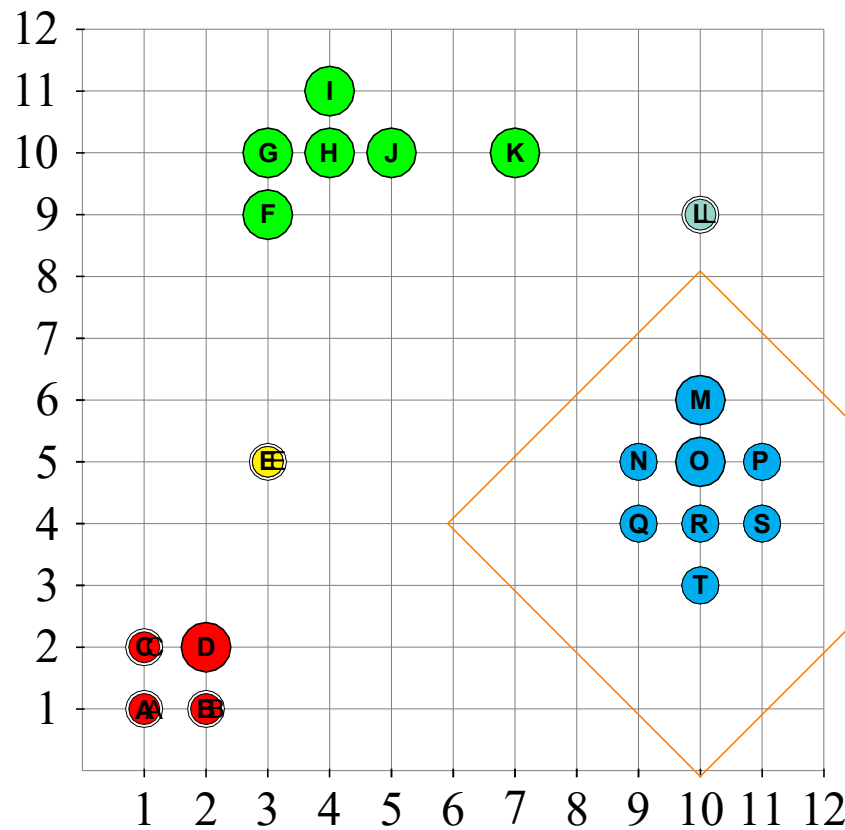
$$\text{minPts} = 5$$

Seed list:
NPQST

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

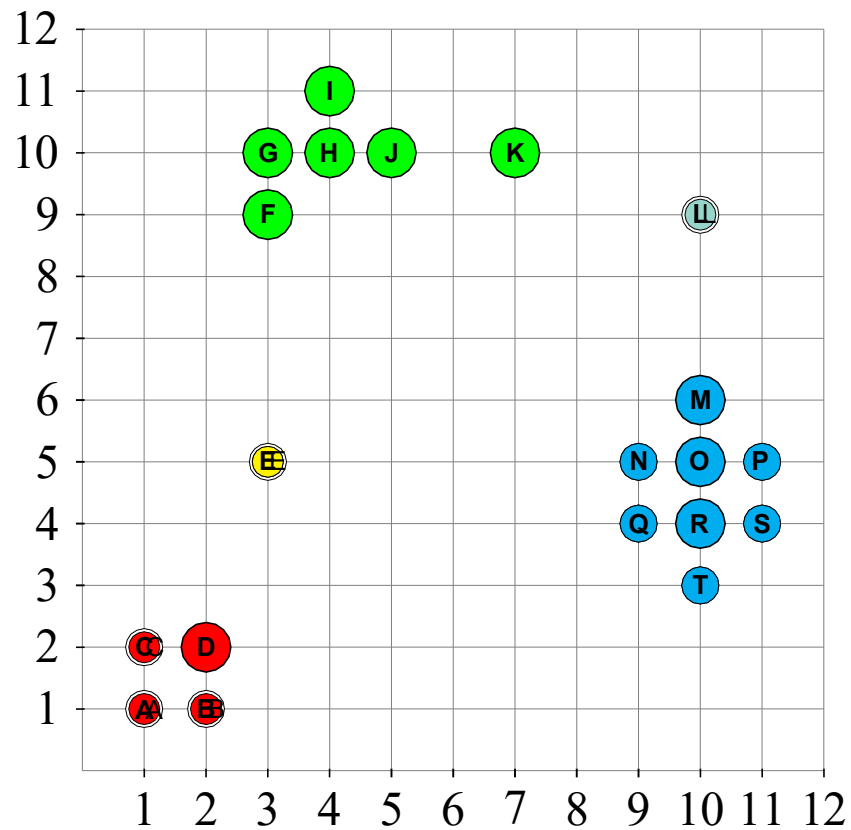
$\minPts = 5$

Seed list:
NPQST

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

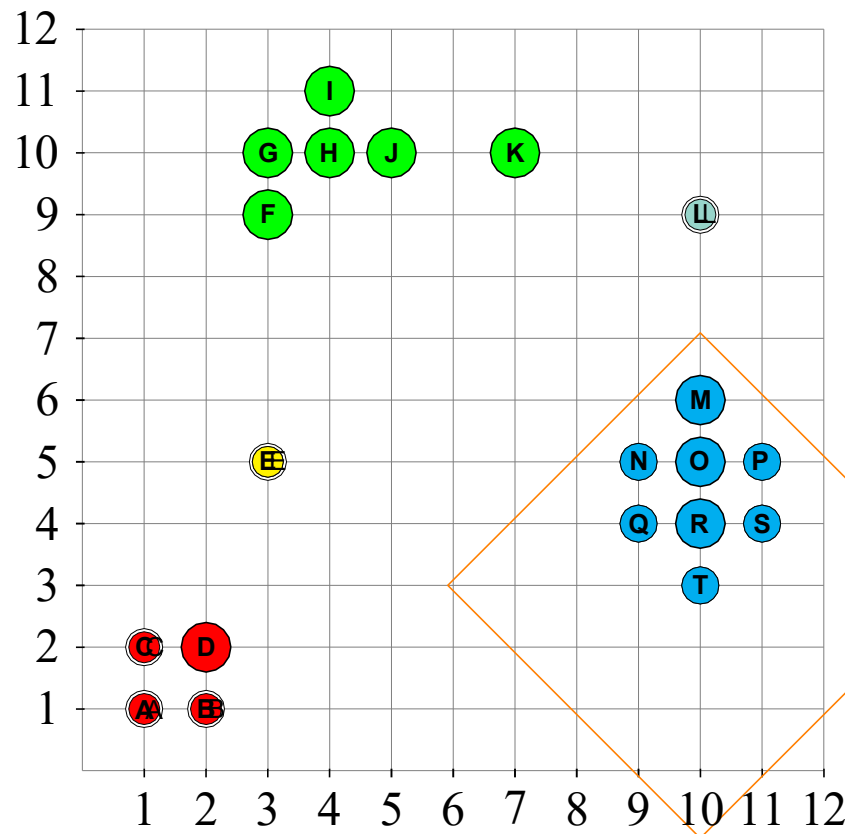
$$\text{minPts} = 5$$

Seed list:
NPQST

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

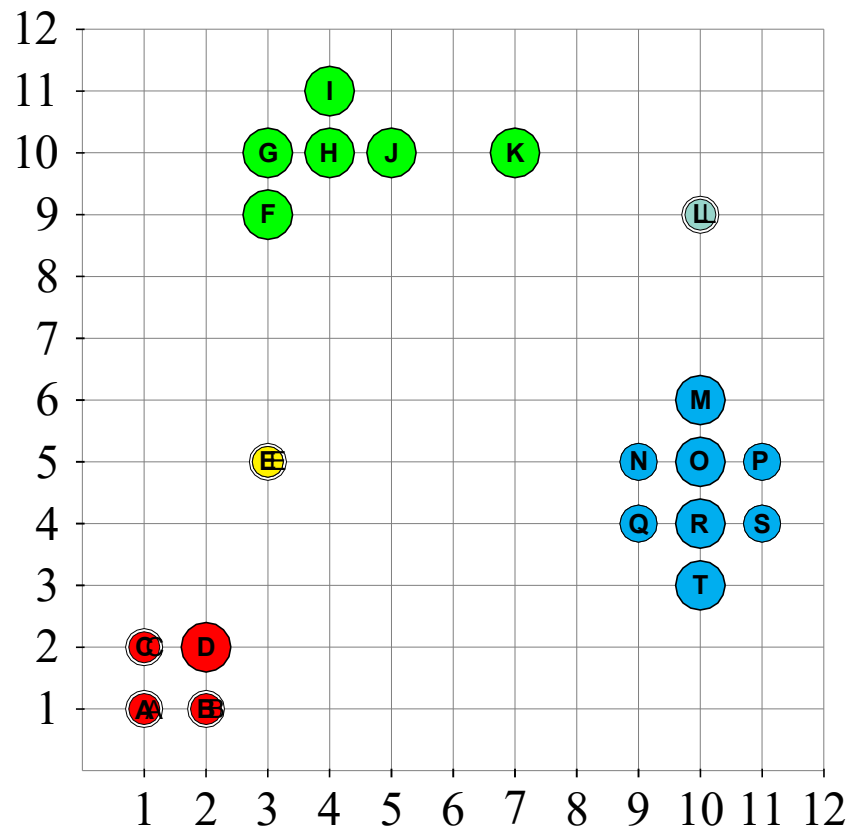
$\minPts = 5$

Seed list:
NPQS

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

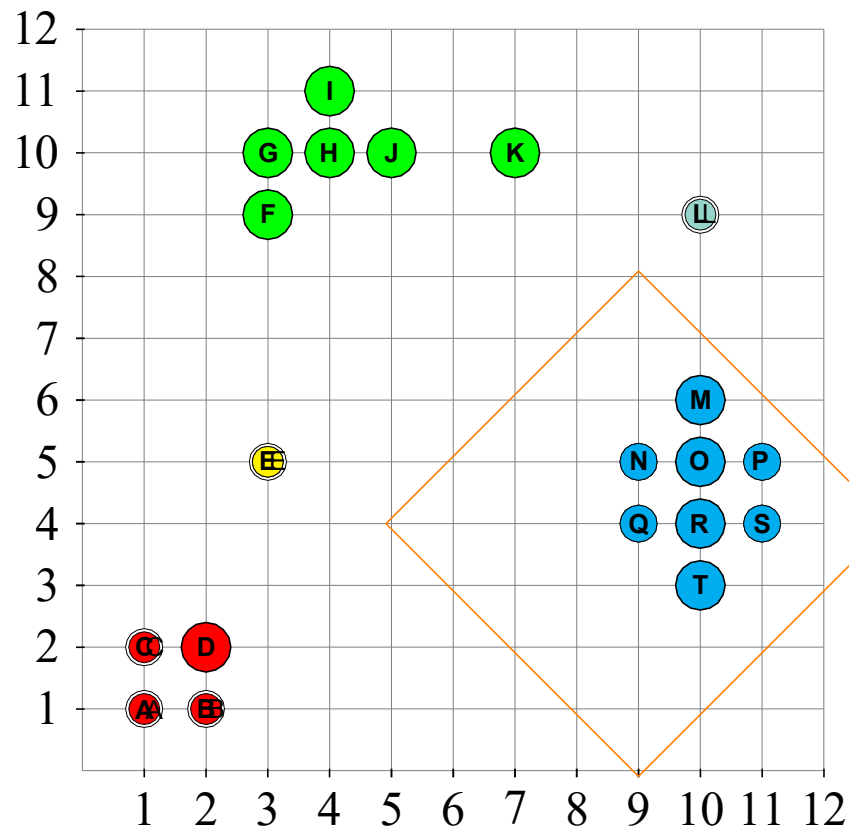
$$\text{minPts} = 5$$

Seed list:
NPQS

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

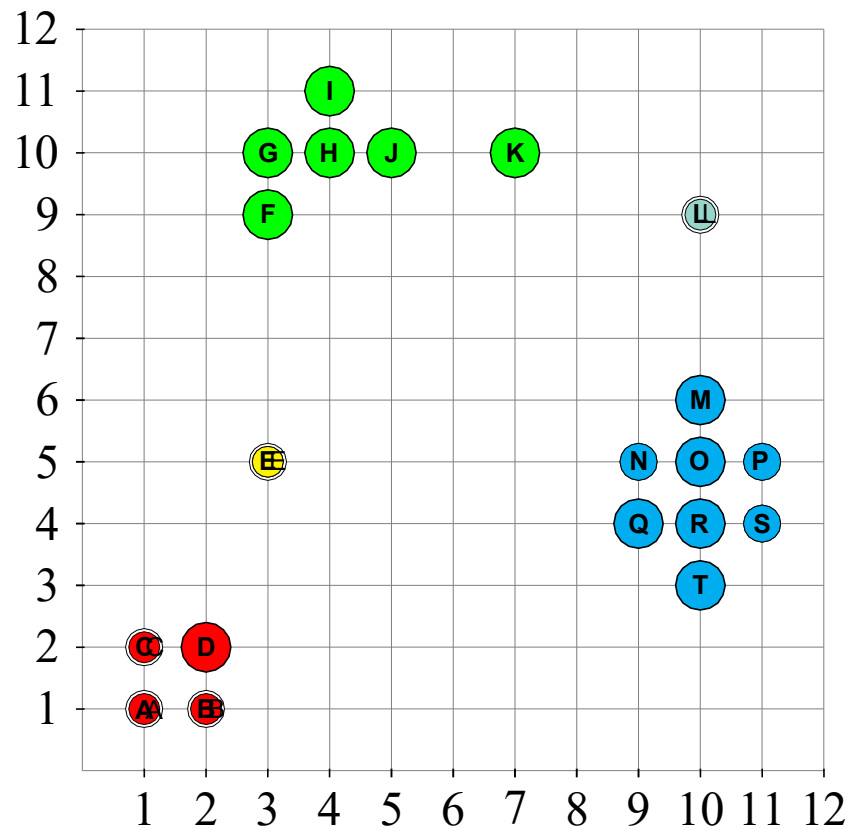
$minPts = 5$

Seed list:
NPS

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

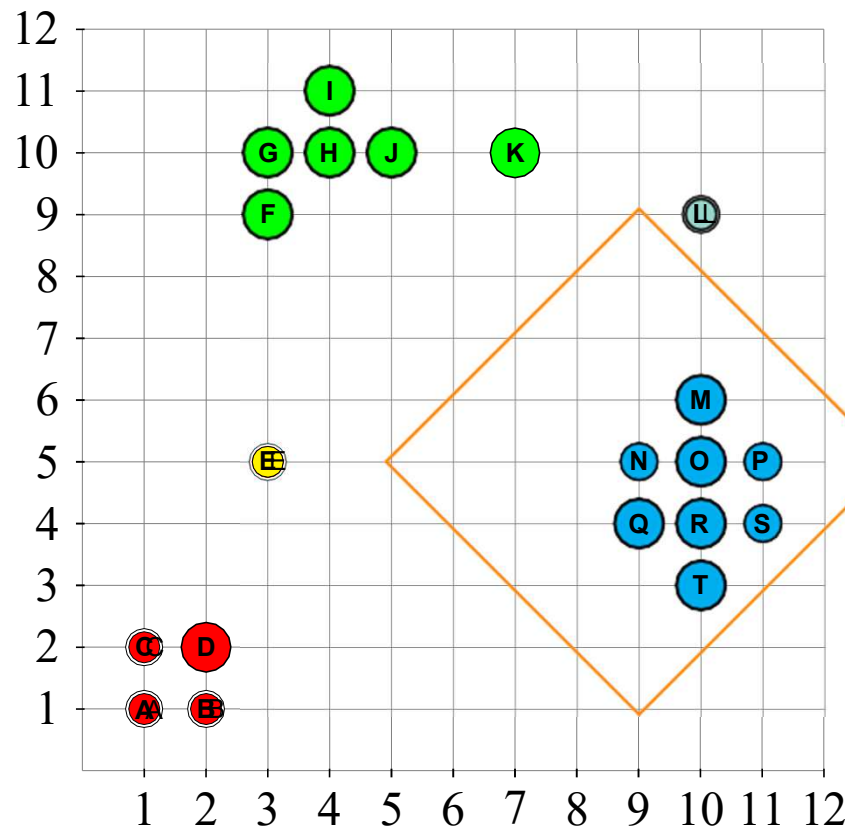
$$\text{minPts} = 5$$

Seed list:
NPS

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

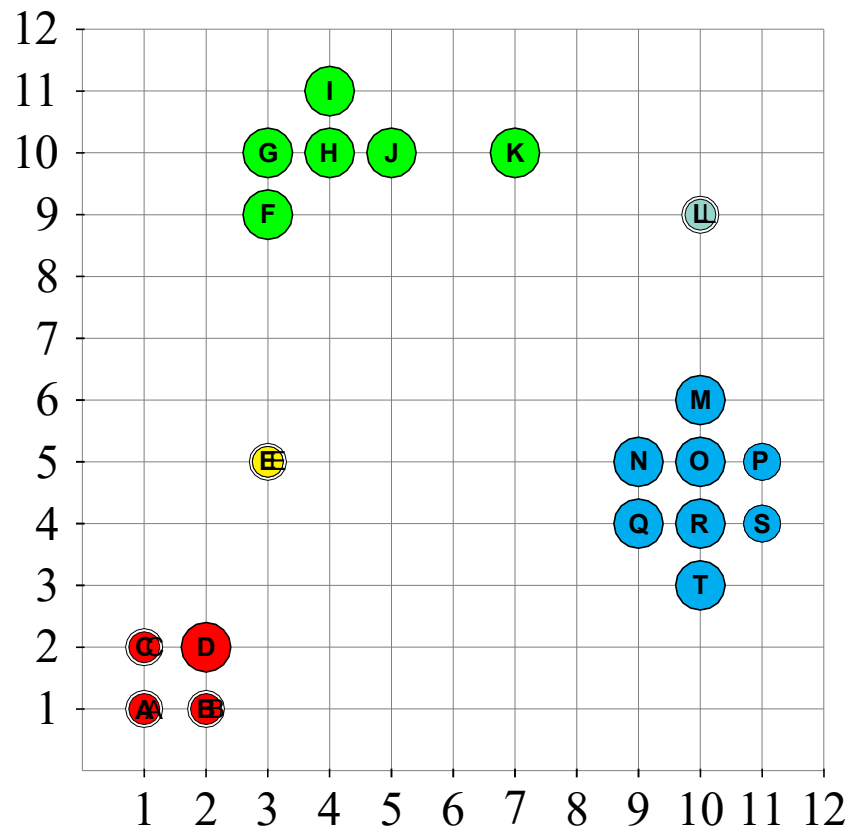
$minPts = 5$

Seed list:
PS

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

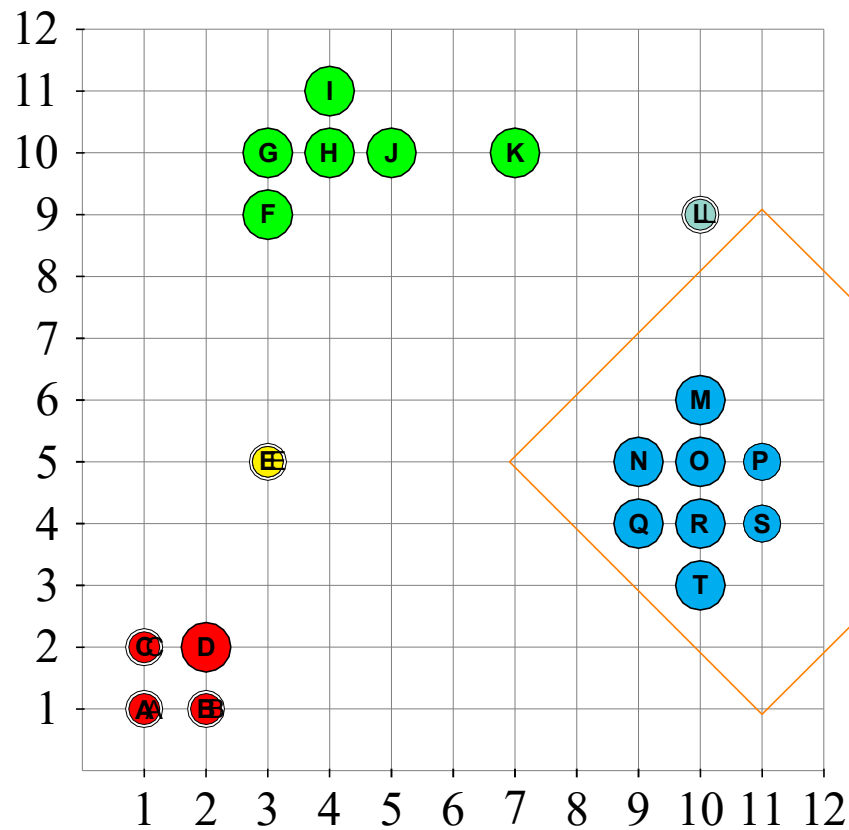
$$\text{minPts} = 5$$

Seed list:
PS

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

$minPts = 5$

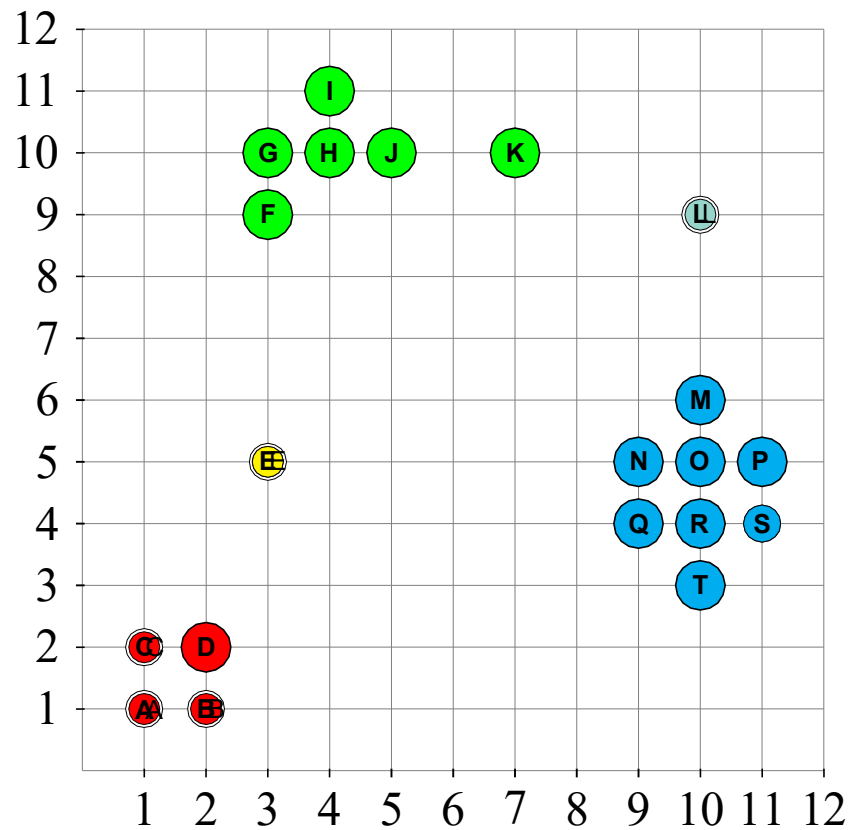
Seed list:

S

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

$$\text{minPts} = 5$$

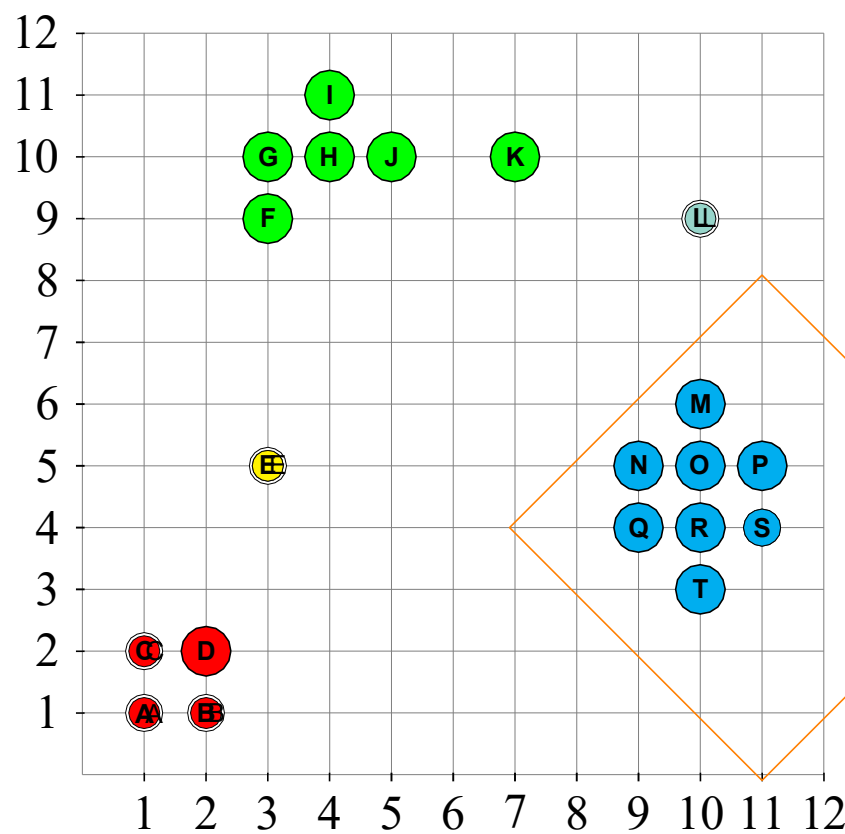
Seed list:

S

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

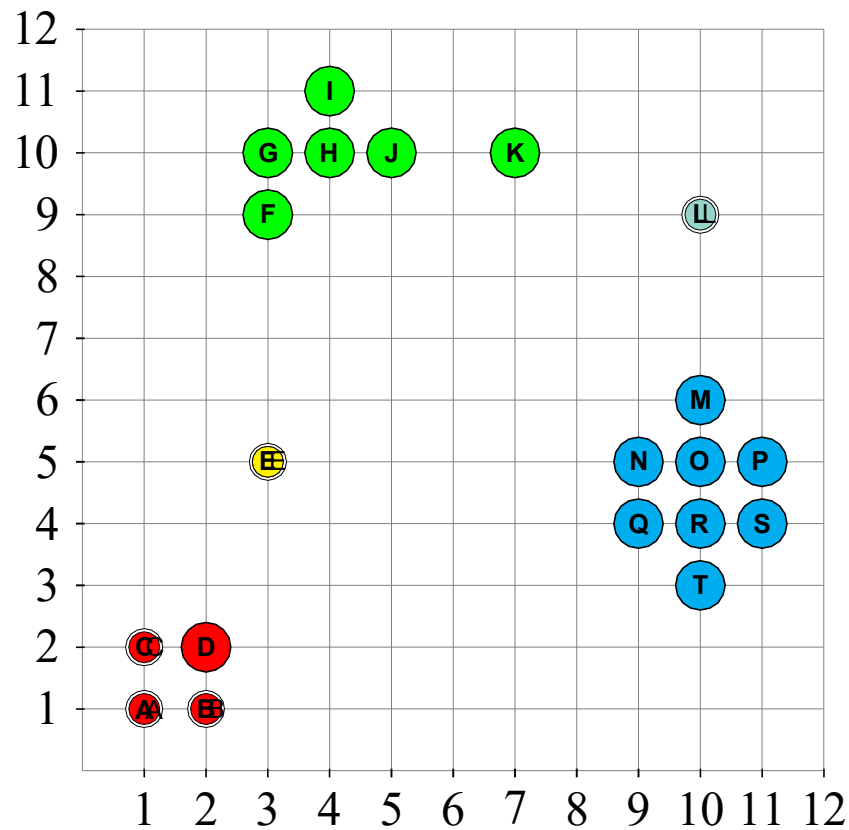
$minPts = 5$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

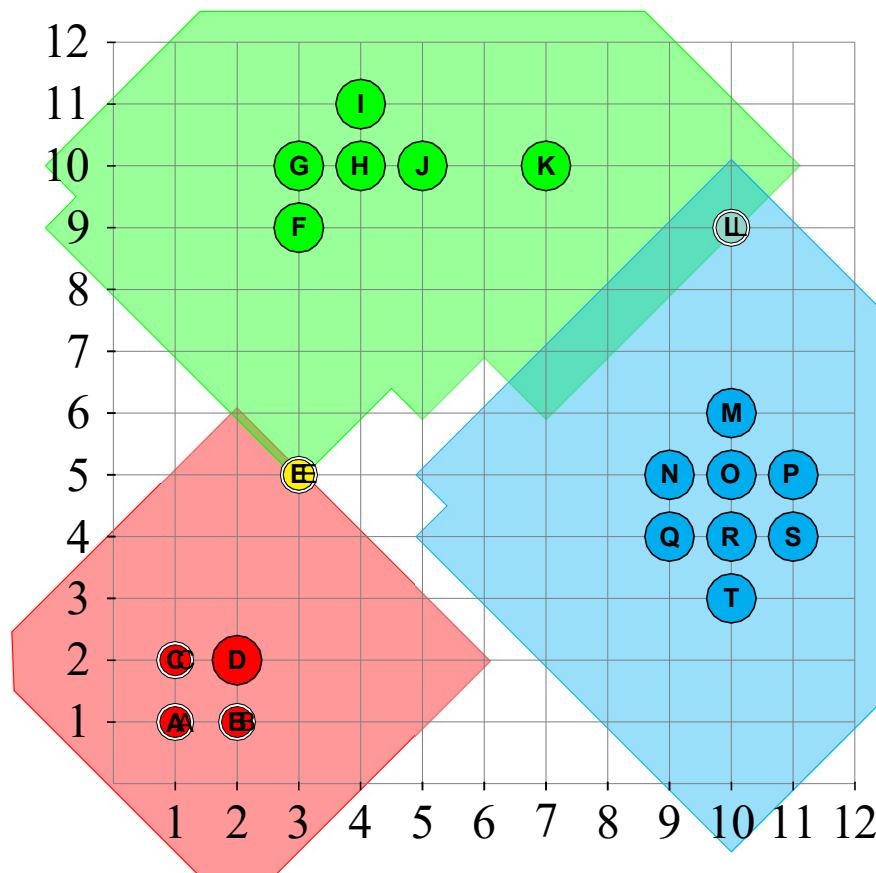
$$\text{minPts} = 5$$

Seed list:

Example 5

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

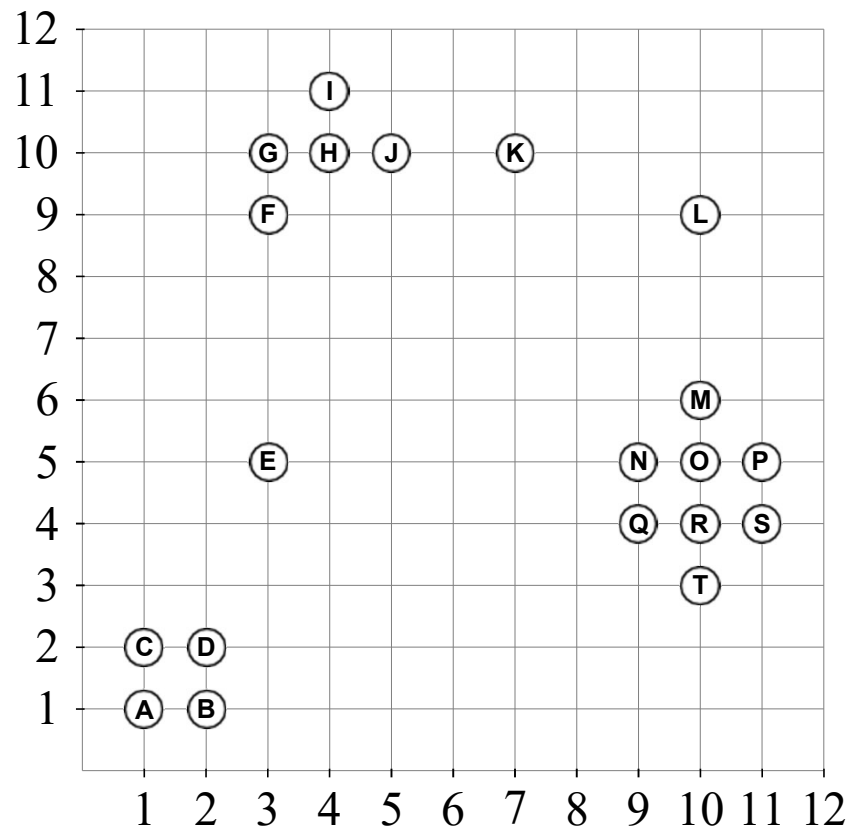
$\minPts = 5$

Seed list:

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

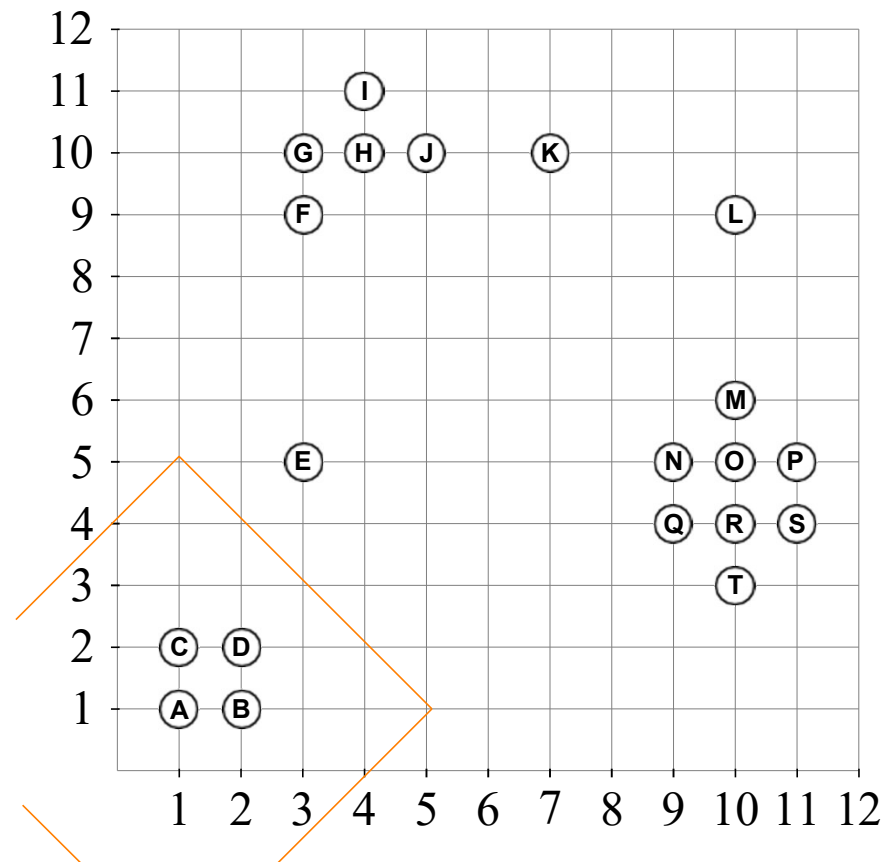
$$\text{minPts} = 4$$

Seed list:

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

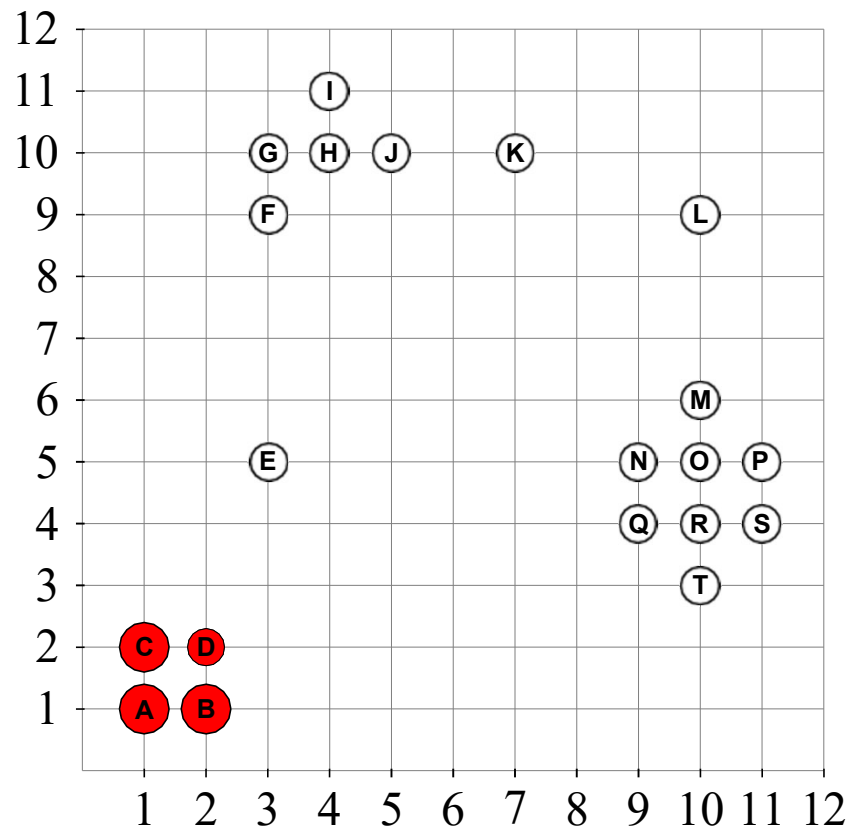
$$\text{minPts} = 4$$

Seed list:

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

$$\text{minPts} = 4$$

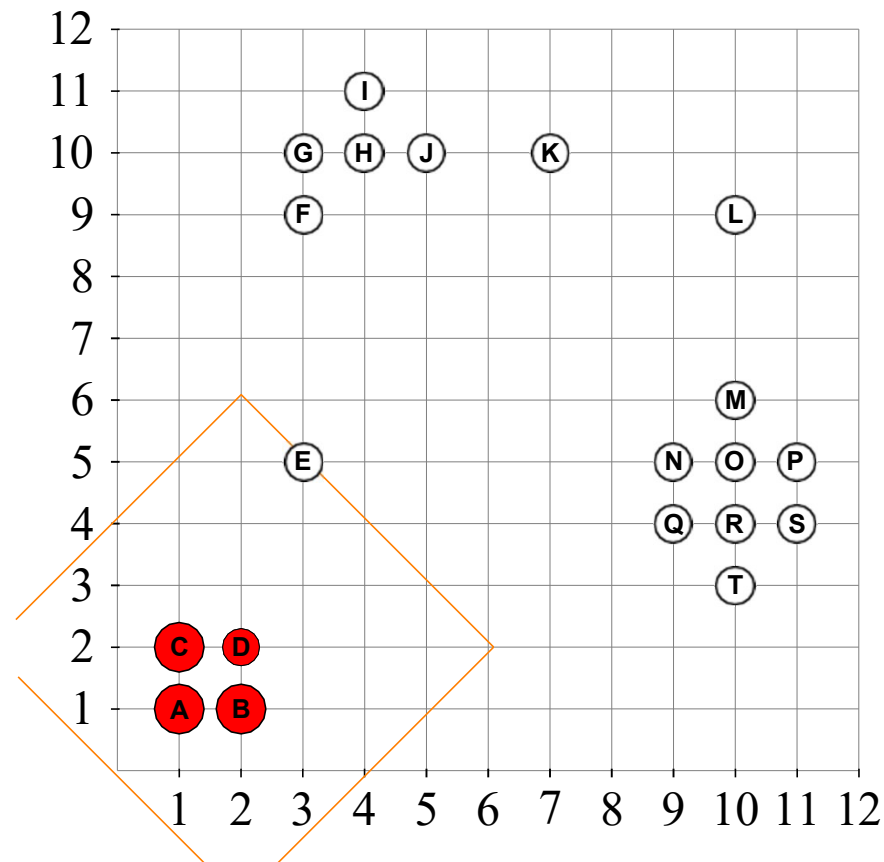
Seed list:

D

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

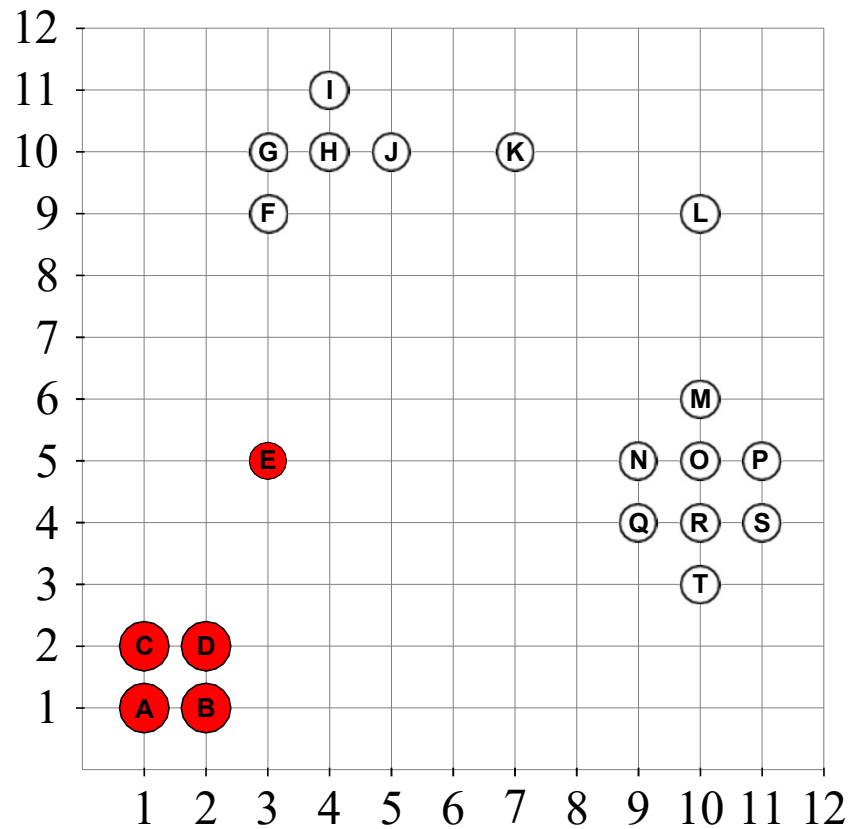
$$\text{minPts} = 4$$

Seed list:

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

$$\text{minPts} = 4$$

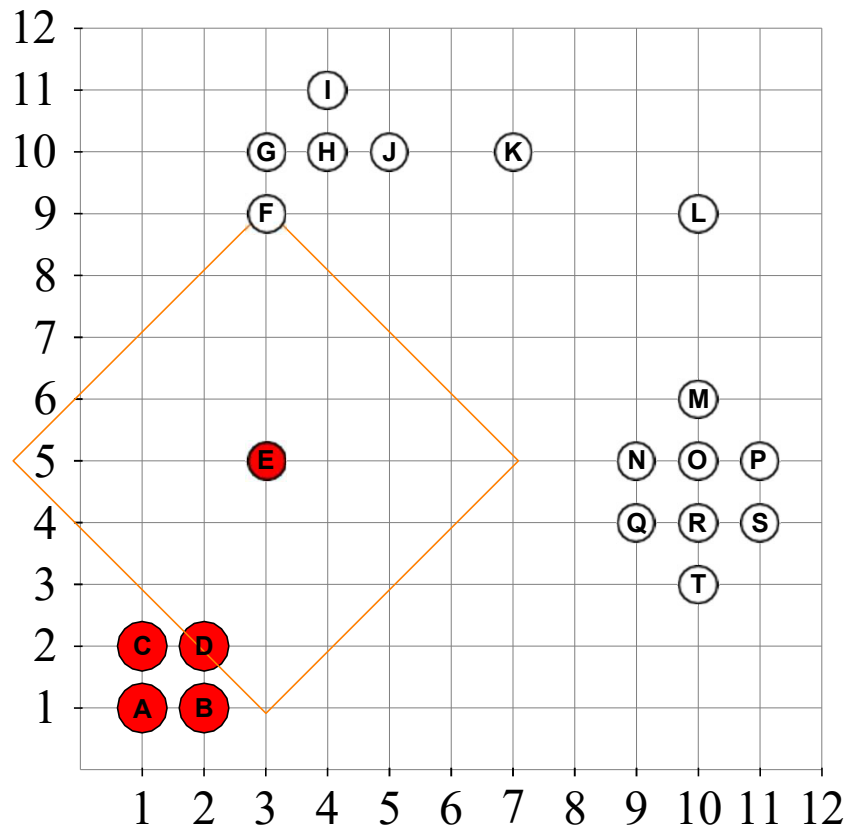
Seed list:

E

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

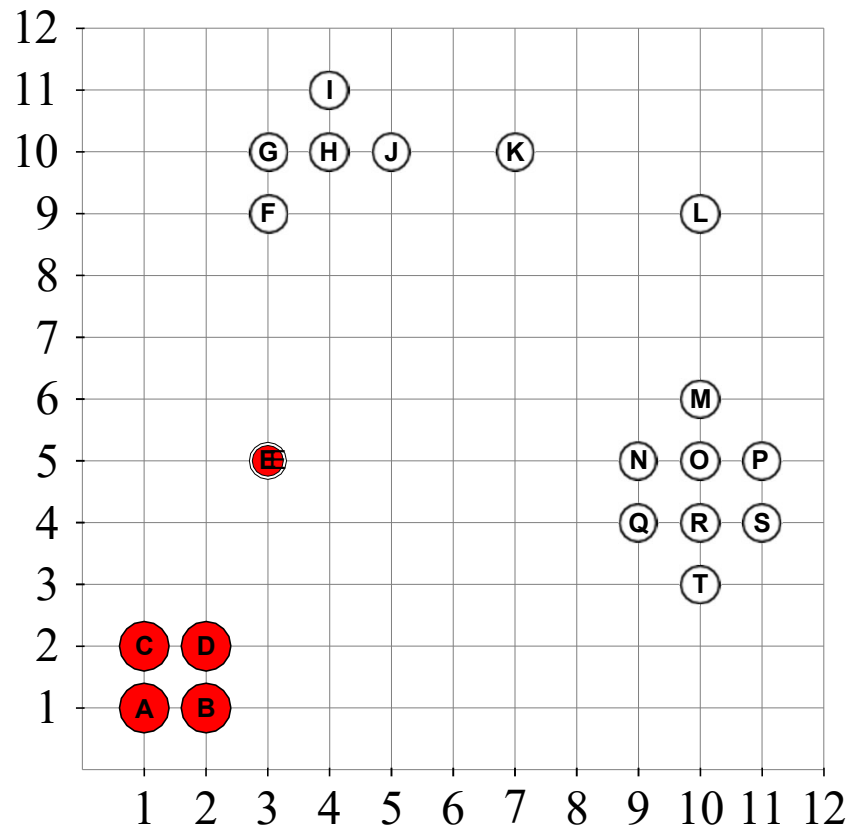
$$\text{minPts} = 4$$

Seed list:

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

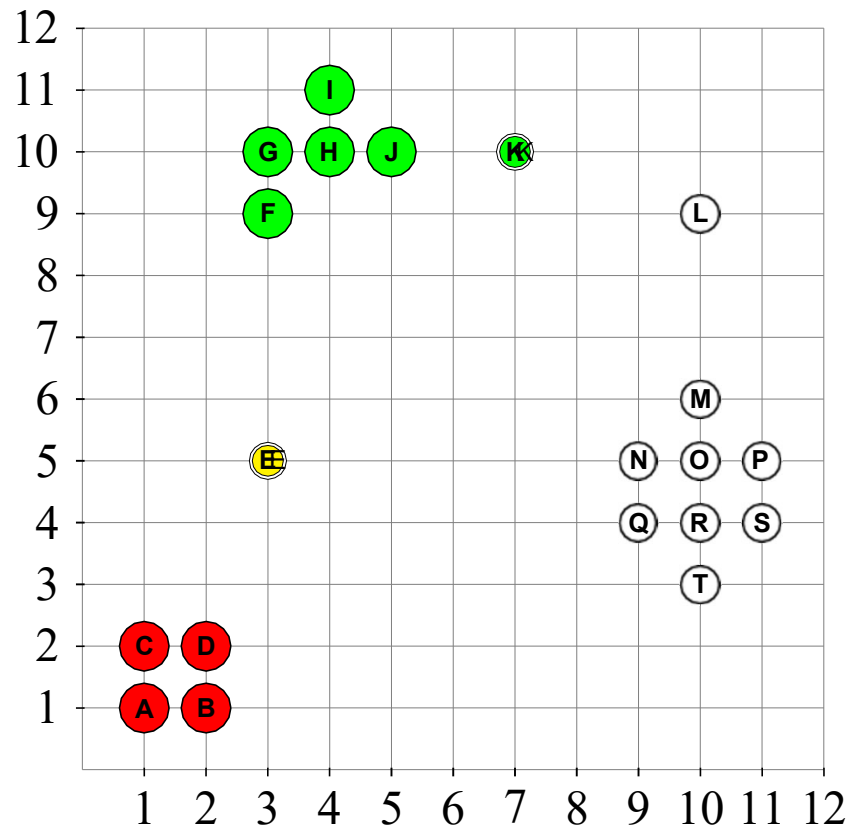
$$\text{minPts} = 4$$

Seed list:

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

$$\text{minPts} = 4$$

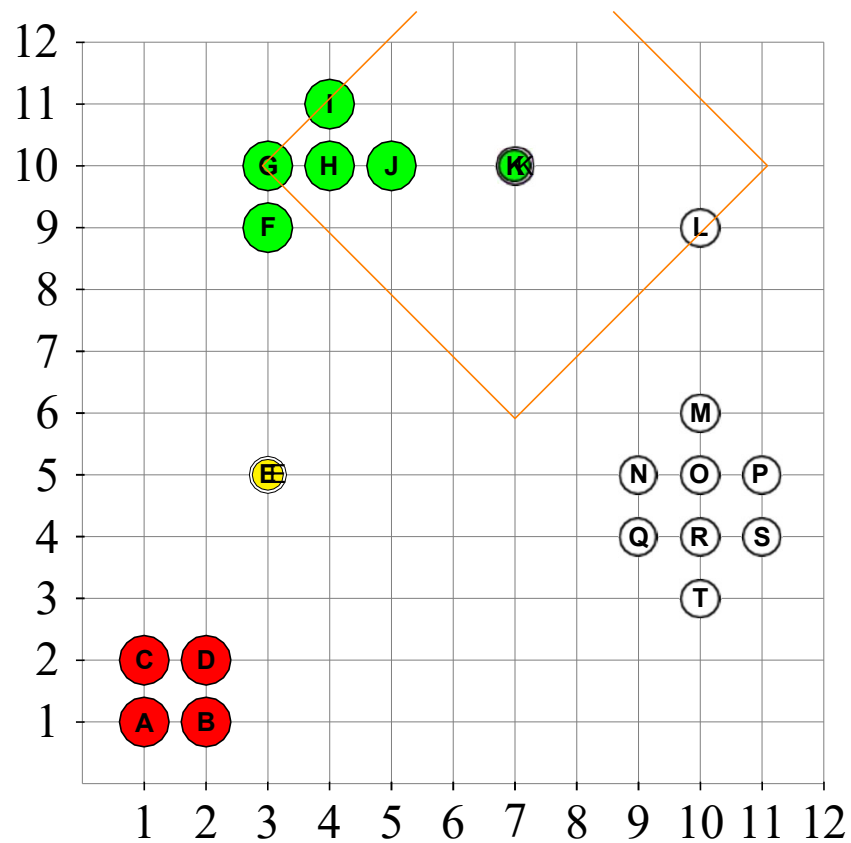
Seed list:

K

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

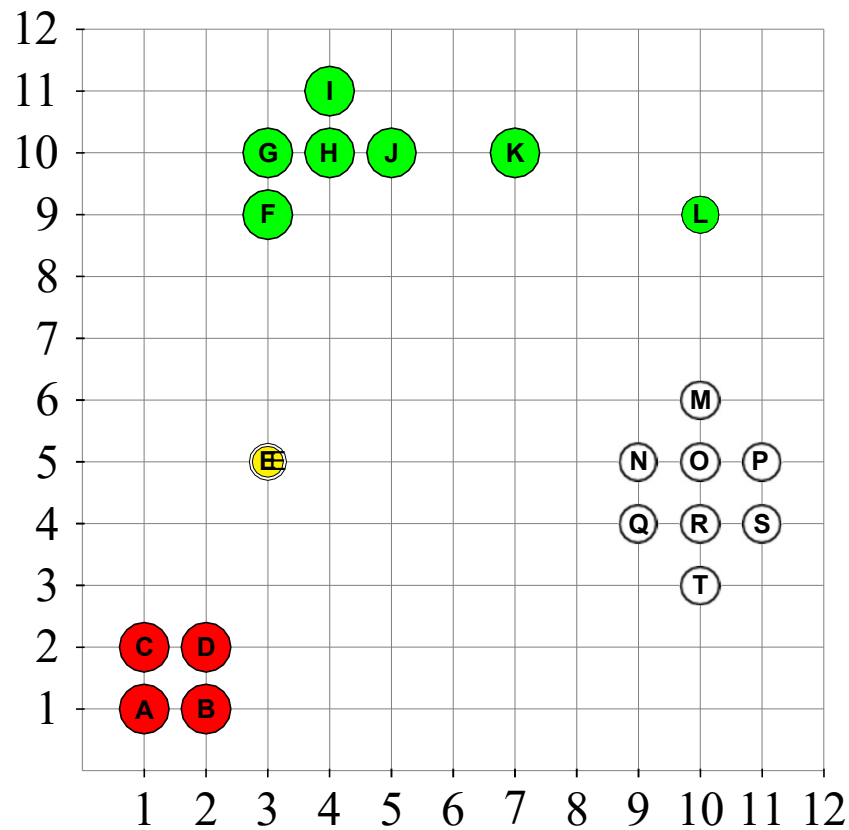
$$\text{minPts} = 4$$

Seed list:

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

$$\text{minPts} = 4$$

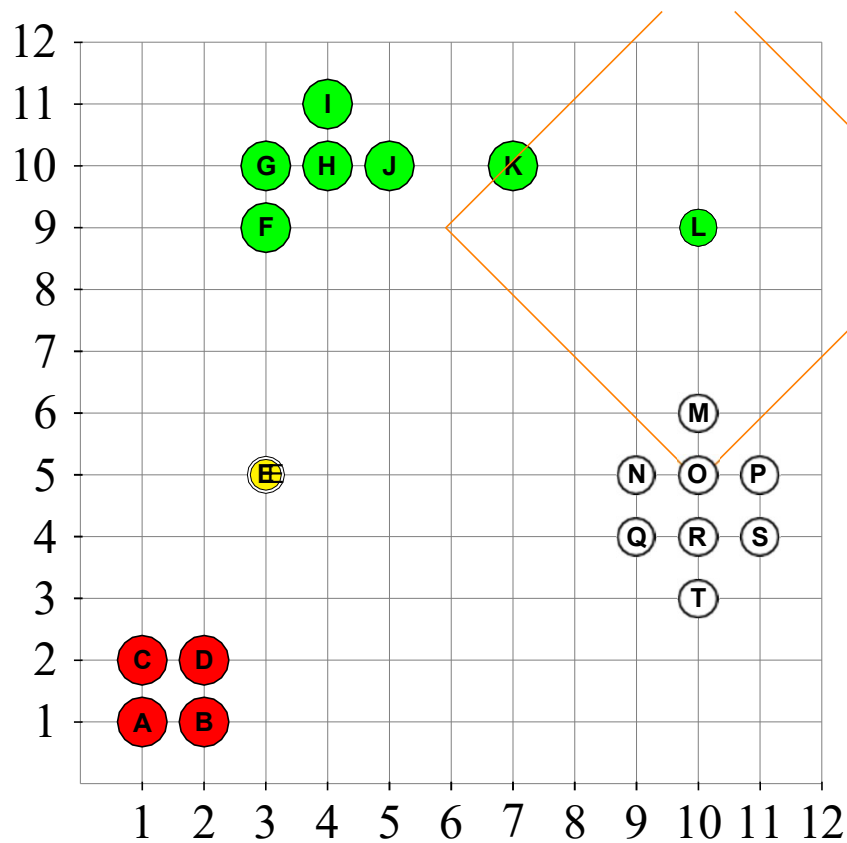
Seed list:

L

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

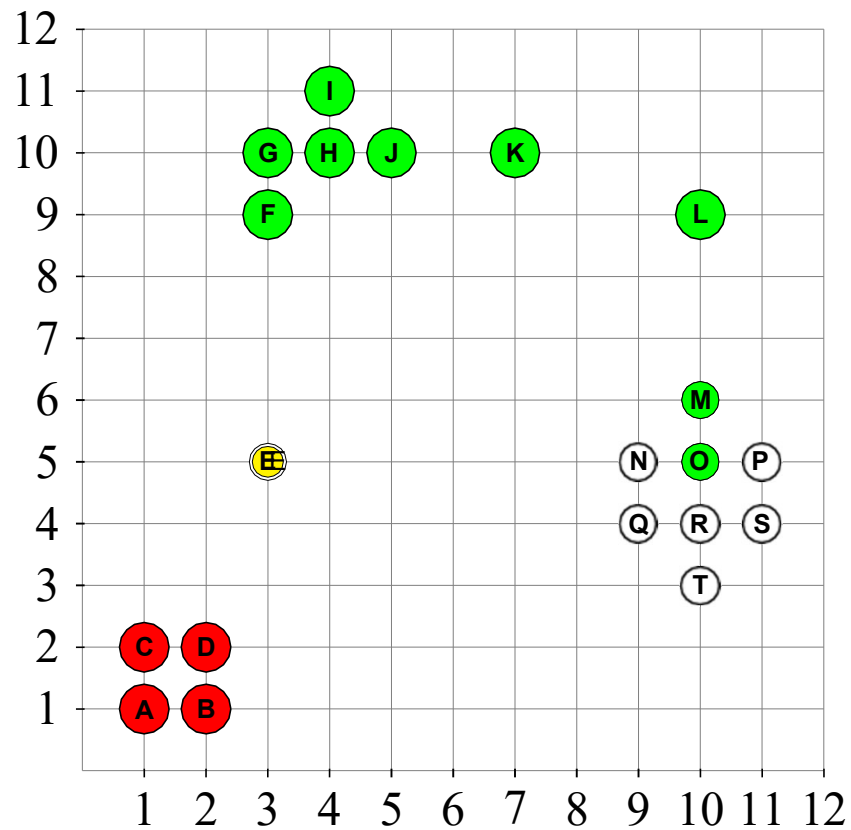
$$\text{minPts} = 4$$

Seed list:

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

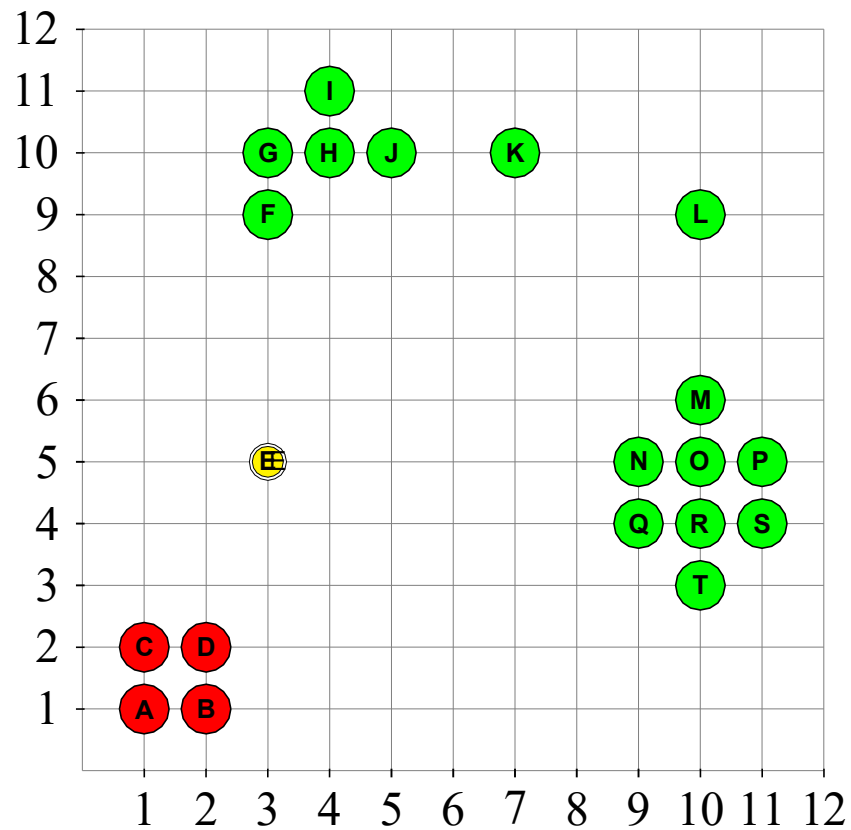
$$\text{minPts} = 4$$

Seed list:
MO

Example 6

DM868 DM870
DS804

Arthur Zimek



$$\varepsilon = 4.1$$

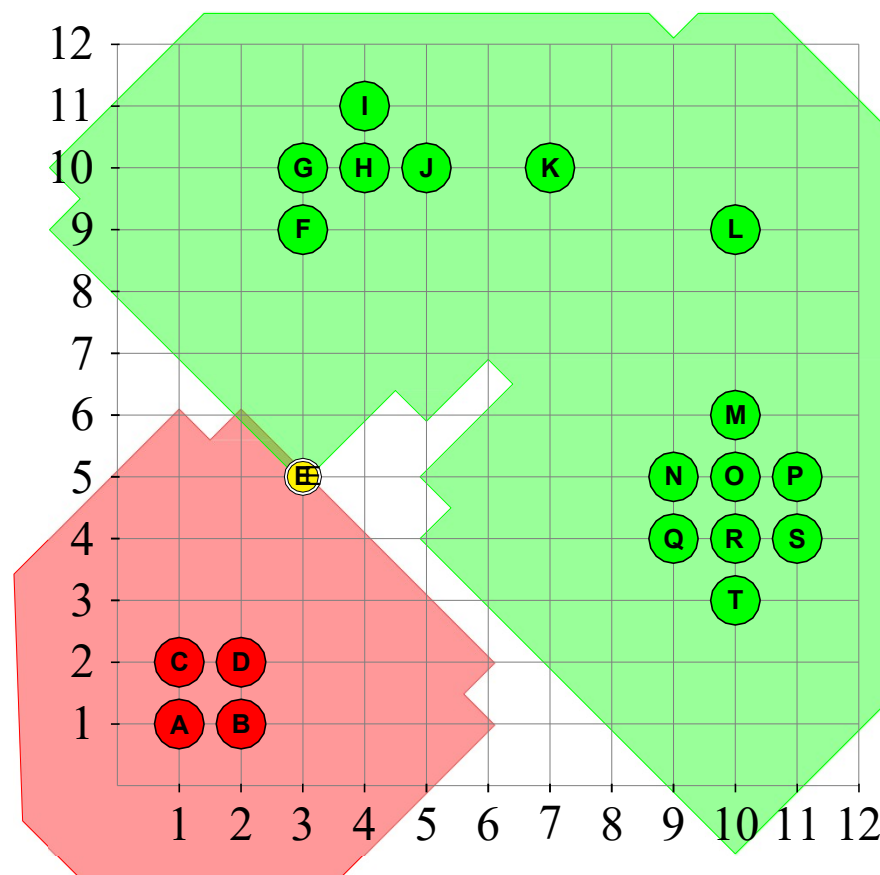
$$\text{minPts} = 4$$

Seed list:

Example 6

DM868 DM870
DS804

Arthur Zimek



$\epsilon = 4.1$

$minPts = 4$

Seed list:

DBSCAN from a Graph Perspective

- **Conceptual, Graph-Based Algorithm:**

1. Traverse the dataset and label objects as core, border or noise
2. Set aside objects labelled as **noise**
3. Insert an edge between every pair of neighbouring **core** objects
 - Two objects are neighbours if they are within each other's ϵ neighbourhood
4. Each *connected component* of the resulting graph is a cluster
5. Assign **border** objects to the cluster of the respective core objects
 - Ties must be resolved when there are core objects from different clusters associated with the same border object

Non-Euclidean DBSCAN

- DBSCAN is a "relational" algorithm in the sense that it only requires a measure of distance between data objects, not the objects themselves
 - It can be applied to data spaces other than $\mathbb{R}^D$, and distances other than Euclidean
- **Exercise:** Apply DBSCAN to the following distance matrix, with $\varepsilon = 5$ and $\text{MinPts} = 3$ (which includes the query object in question), indicating the final type of each object (core, border or noise) as well as the resulting clusters

$$\mathbf{D} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{matrix} & \begin{bmatrix} 0 & & & & & & \\ 1 & 0 & & & & & \\ 1 & 2 & 0 & & & & \\ 10 & 9 & 11 & 0 & & & \\ 11 & 10 & 12 & 2 & 0 & & \\ 21 & 20 & 22 & 18 & 19 & 0 & \\ 14 & 13 & 15 & 6 & 4 & 25 & 0 \end{bmatrix} \end{matrix}$$

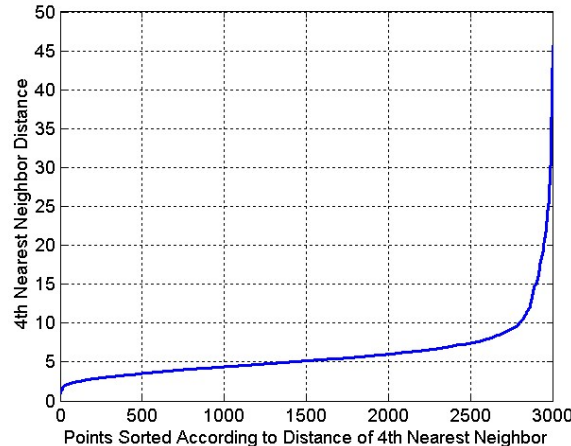
DBSCAN (Complexity)

- **Computational Complexity:**

- In practice, cluster labels can be assigned simultaneously to object type identification (single traversal of the dataset)
- Hence, runtime is $O(n \times \epsilon\text{-range search time})$
 - In general, this means $O(n^2)$, where n is the dataset size
 - For low- to moderate-dimensional datasets, $O(n \log n)$ runtime is achievable by using appropriate data structures
 - E.g. Kd-trees, R*-trees
- In terms of memory usage, if distances are dynamically computed, then only the labels need to be stored, so: $O(n)$

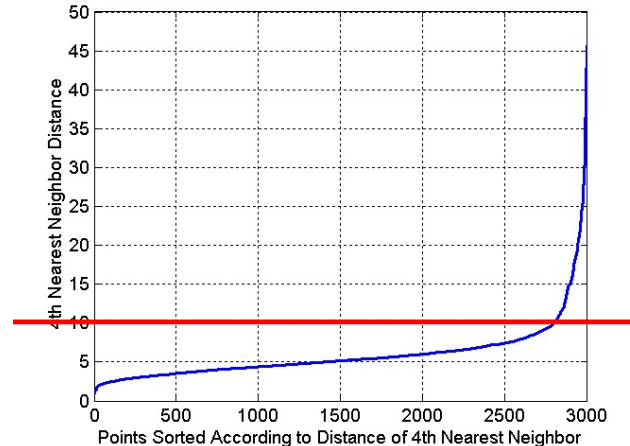
(Attempt at) Determining ε and MinPts

- ❑ Idea is that for points in a cluster, their k^{th} nearest neighbors (NN) are at roughly the same distance
- ❑ Noise points have the k^{th} NN at farther distance
- ❑ **Heuristic:**
 - ❑ plot sorted distance of every point to its k^{th} nearest neighbor



(Attempt at) Determining ε and MinPts

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- ❑ **Heuristic:**
 - ❑ plot sorted distance of every point to its k^{th} nearest neighbor

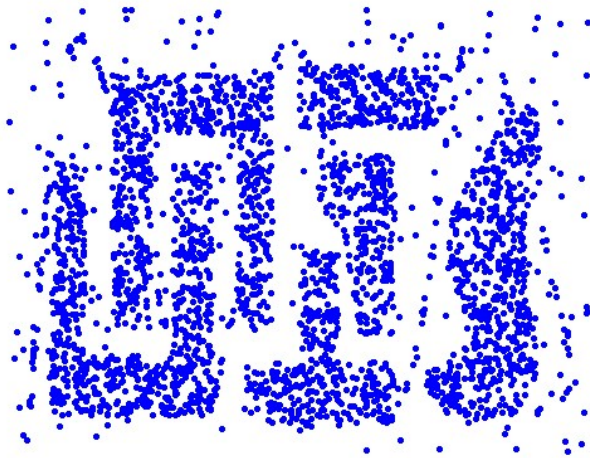


MinPts = 4, $\varepsilon = 10$

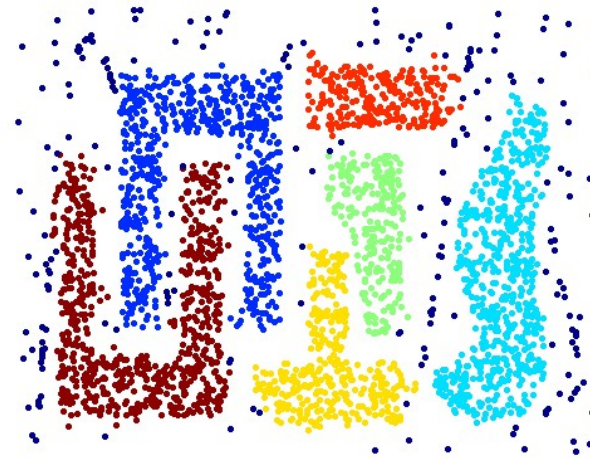
Problems:

- It requires KNN computations and sorting
- It only works if clusters have similar densities

When DBSCAN Works Well



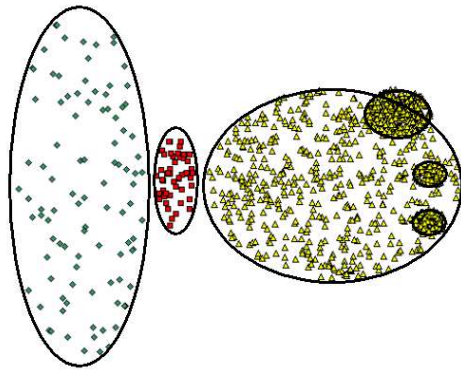
Original Points



Clusters

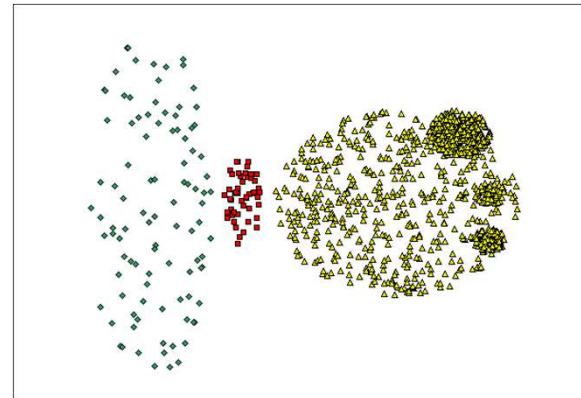
- Resistant to Noise
- Can handle clusters of different shapes and sizes

When DBSCAN Does NOT Work Well

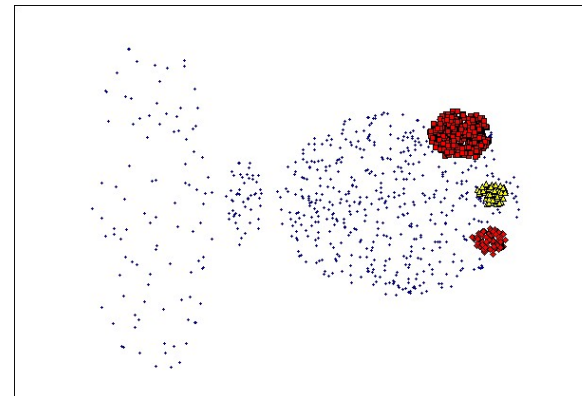


Original Points

- High-dimensional data
- Varying densities



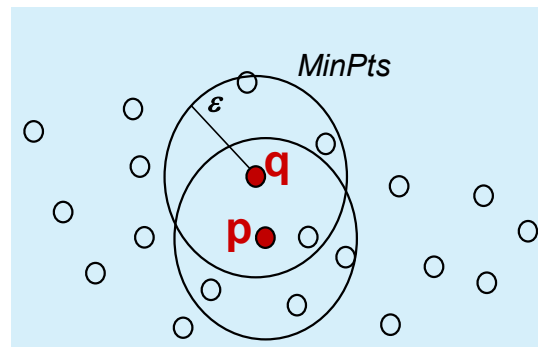
(MinPts=4, $\epsilon=9.92$).



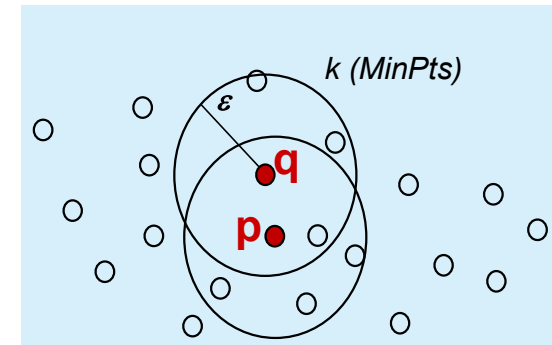
(MinPts=4, $\epsilon=9.75$)

Why Are Varied Densities a Problem?

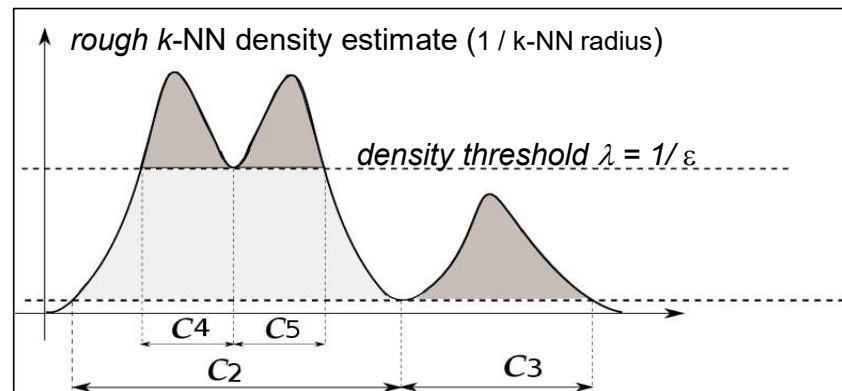
- For simplicity, consider DBSCAN's *simplified version without border points* (which are also labelled as noise), called **DBSCAN***:
 - Basically, *maximal subsets of density-connected core observations*
 - **Core (dense)**: the radius of the “ball” needed to reach at least *MinPts* observations is $\leq \epsilon$
 - **Density-connected**: directly or transitively ϵ -reachable (through core/dense observations)



Why Are Varied Densities a Problem?

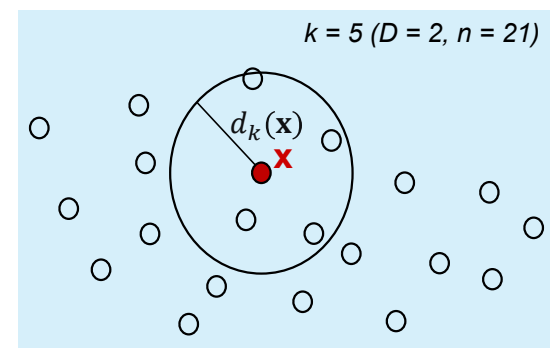


- Non-Parametric Density Estimate Interpretation of **DBSCAN***:
 - *MinPts* is a **smoothing factor** (parameter k of a rough/coarse KNN estimate)
 - The inverse of ε establishes the **density threshold**



K Nearest Neighbours (KNN) Density Estimate

- Definition: $f(\mathbf{x}) = \frac{k/n}{V_k(\mathbf{x})} = \frac{k/n}{v_u \cdot d_k(\mathbf{x})^D}$

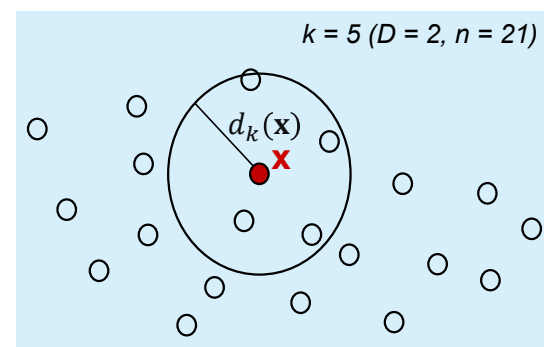


K Nearest Neighbours (KNN) Density Estimate

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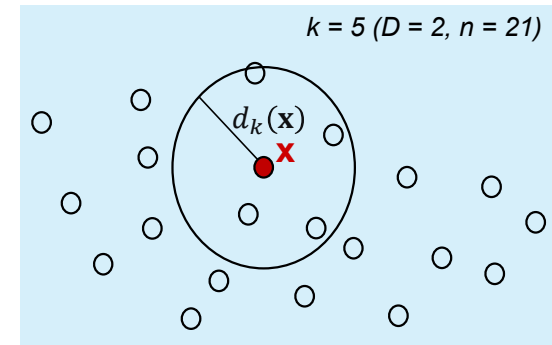
Annotations:

- user-defined neighborhood size (constant) → k
- dataset size (constant) → n
- dataset dimensionality (constant) → D
- normalized neighborhood volume – e.g., for Euclidean distance, this is the volume of the unit radius hypersphere (constant) → v_u



K Nearest Neighbours (KNN) Density Estimate

- Definition: $f(\mathbf{x}) = \frac{k/n}{V_k(\mathbf{x})} = \frac{k/n}{v_u \cdot d_k(\mathbf{x})^D}$
- Widely used rough approximation: $f(\mathbf{x}) \propto \frac{1}{d_k(\mathbf{x})}$
- Simple and does not require a feature space, just a distance function $d(\cdot)$ between pairs of observations, from which the distance between any observation $\mathbf{x}$ and its k th nearest neighbour, $d_k(\mathbf{x})$, can be computed
- Not a proper p.d.f. though
 - Not meaningful in absolute terms, just in relative terms



Relation to Density-Contour Clusters in DBSCAN*

- Arbitrary *Density Threshold* λ from an Arbitrary Distance (Ball Radius ε):

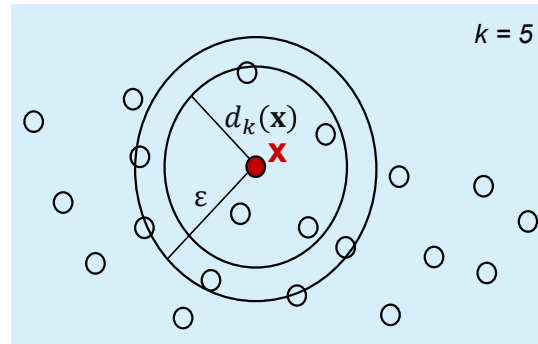
$$\lambda = \frac{1}{\varepsilon}$$

Relation to Density-Contour Clusters in DBSCAN*

- Arbitrary *Density Threshold* λ from an Arbitrary Distance (Ball Radius ε):

$$\lambda = \frac{1}{\varepsilon}$$

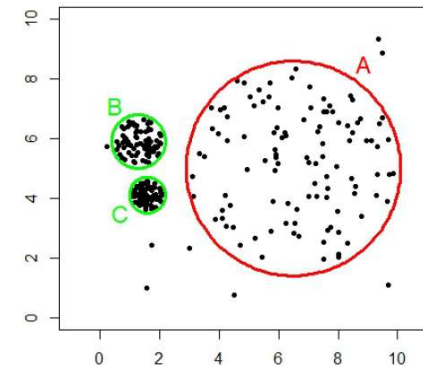
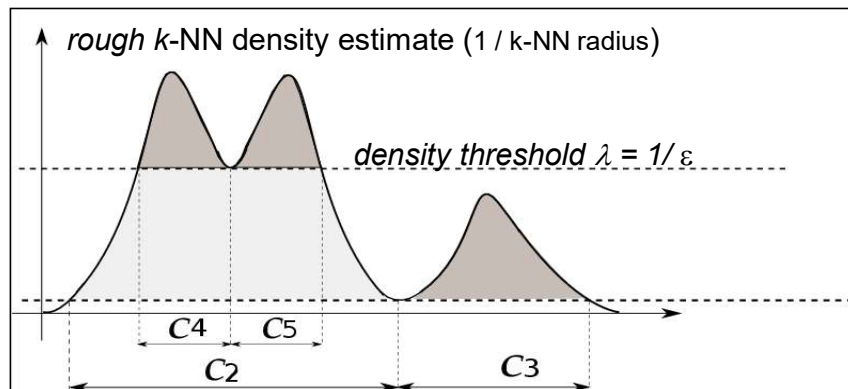
- **Level Set:** $\{\mathbf{x} \mid f(\mathbf{x}) \geq \lambda\} \rightarrow \{\mathbf{x} \mid 1/d_k(\mathbf{x}) \geq 1/\varepsilon\} \rightarrow \{\mathbf{x} \mid d_k(\mathbf{x}) \leq \varepsilon\}$



- This establishes the connection between the rough/coarse approximate KNN density estimate, the notion of density-contour clusters (which is defined in terms of level sets), and the concept of a core point in both DBSCAN and DBSCAN*

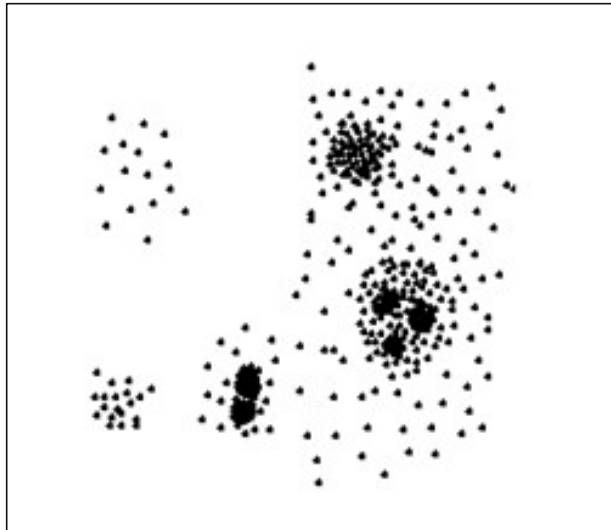
Limitations of a “Flat” Analysis (single resolution/density)

- Requires a user-defined density threshold (critical parameter ε)
- A single, global threshold may not capture patterns with varied densities



Limitations of a “Flat” Analysis (single resolution/density)

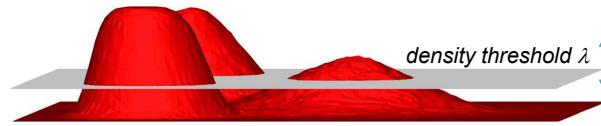
- It is a single “snapshot” at an arbitrary density-level
- It does not capture the density profile as a whole



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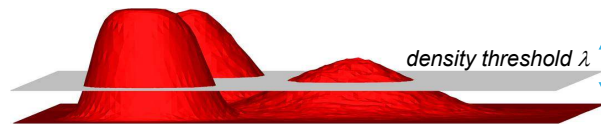
From “Flat” Models to Hierarchies

- “***Density-Contour Trees***” [J.A. Hartigan, “Clustering Algorithms”, JW&S, 1975]
 - **Hierarchical** relationships between level sets and density-contour clusters
 - across the entire range of possible density thresholds $\lambda \in [0, \infty)$

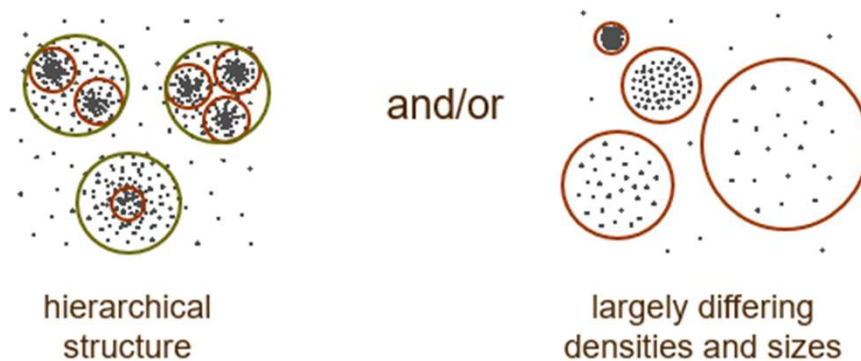


From “Flat” Models to Hierarchies

- “**Density-Contour Trees**” [J.A. Hartigan, “Clustering Algorithms”, JW&S, 1975]
 - **Hierarchical** relationships between level sets and density-contour clusters



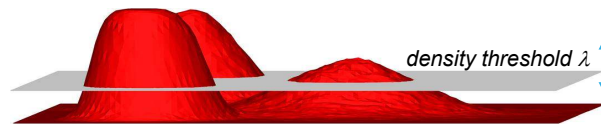
- *Conceptually* able to fully describe an underlying density profile, e.g.:



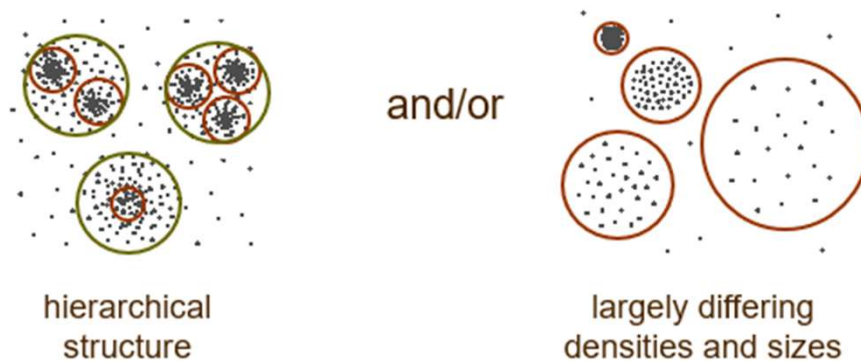
- In principle, difficult / computationally prohibitive to model

From “Flat” Models to Hierarchies

- “**Density-Contour Trees**” [J.A. Hartigan, “Clustering Algorithms”, JW&S, 1975]
 - **Hierarchical** relationships between level sets and density-contour clusters



- *Conceptually* able to fully describe an underlying density profile, e.g.:



- Computationally tractable materialization of the idea: **HDBSCAN*** algorithm

HDBSCAN*

ACM Transactions on Knowledge Discovery from Data, Vol. 10, No. 1, Article 5, Publication date: July 2015.

Hierarchical Density Estimates for Data Clustering, Visualization, and Outlier Detection

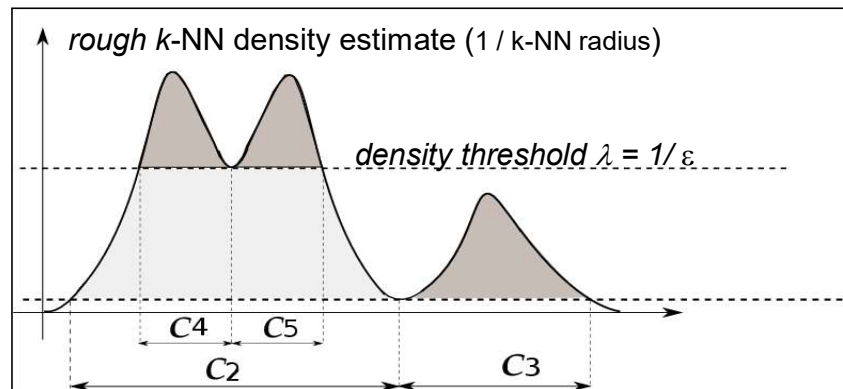
RICARDO J. G. B. CAMPELLO, Department of Computer Sciences, University of São Paulo, Brazil
DAVOUD MOULAVI, Department of Computing Science, University of Alberta, Canada
ARTHUR ZIMEK, Ludwig-Maximilians-Universität München, Germany
JÖRG SANDER, Department of Computing Science, University of Alberta, Canada

An integrated framework for density-based cluster analysis, outlier detection, and data visualization is introduced in this article. The main module consists of an algorithm to compute hierarchical estimates of the level sets of a density, following Hartigan's classic model of density-contour clusters and trees. Such an algorithm generalizes and improves existing density-based clustering techniques with respect to different aspects. It provides as a result a complete clustering hierarchy composed of all possible density-based clusters following the nonparametric model adopted, for an infinite range of density thresholds. The resulting hierarchy can be easily processed so as to provide multiple ways for data visualization and exploration. It can also be further postprocessed so that: (i) a normalized score of "outlierness" can be assigned to each data object, which unifies both the global and local perspectives of outliers into a single definition; and (ii) a "flat" (i.e., nonhierarchical) clustering solution composed of clusters extracted from local cuts through the cluster tree (possibly corresponding to different density thresholds) can be obtained, either in an unsupervised or in a semisupervised way. In the unsupervised scenario, the algorithm corresponding to this postprocessing module provides a global, optimal solution to the formal problem of maximizing the overall stability of the



HDBSCAN* Intuition (Summary)

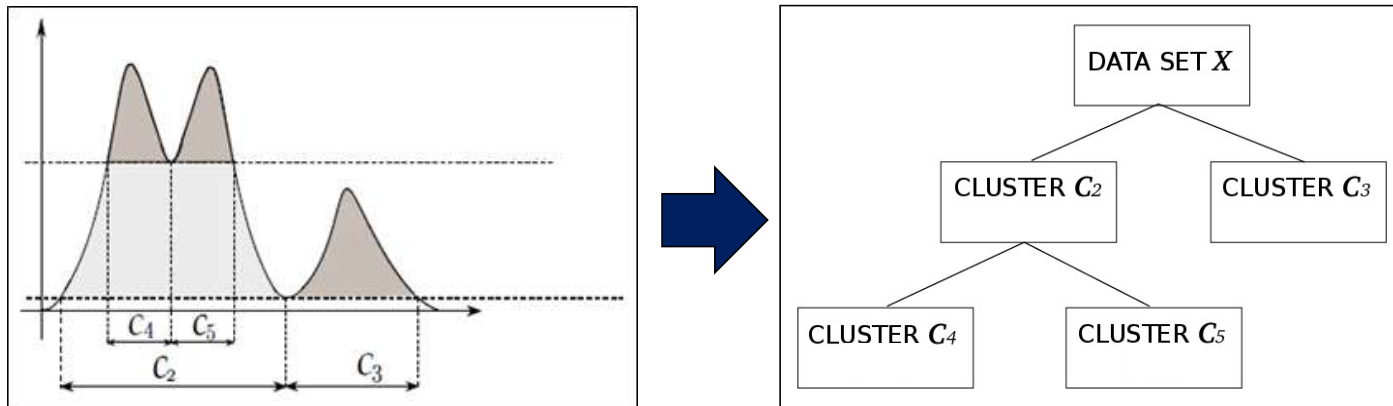
- **Compute the MST of a density-based weighted graph**
- Nested connected components (clusters) built hierarchically
 - for increasing values of density threshold ($\lambda = 1/\varepsilon$)



- **Automatically produces all DBSCAN* solutions for any $\varepsilon \in [0, \infty)$!!!**
- **Note:** DBSCAN* is DBSCAN without border objects (all objects are either labelled as core or noise only). Border objects can easily be identified as a post-processing step if desired.

Hierarchy Simplification

- **Noticing that...**
 - clustering hierarchy is built by increasing the density threshold
- **Main Idea:**
 - by merely loosing noise/spurious components, a cluster only *shrinks* (*not a split*)
 - simplifies hierarchy by orders of magnitude: **simplified cluster tree**



Visualization: Hierarchies

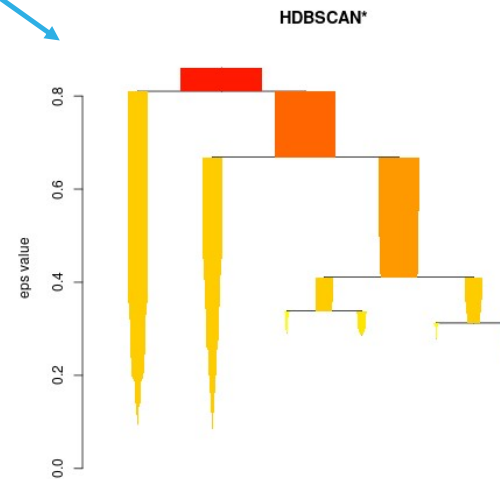
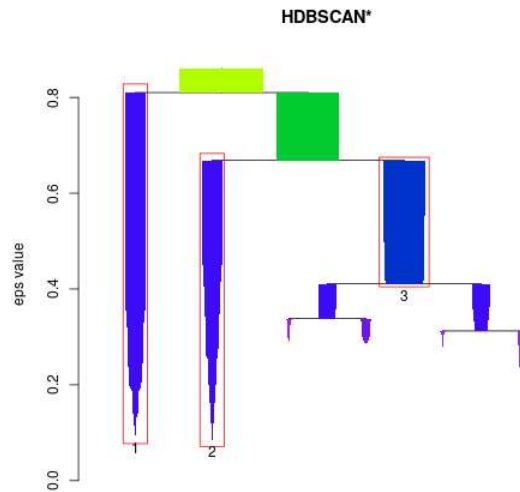
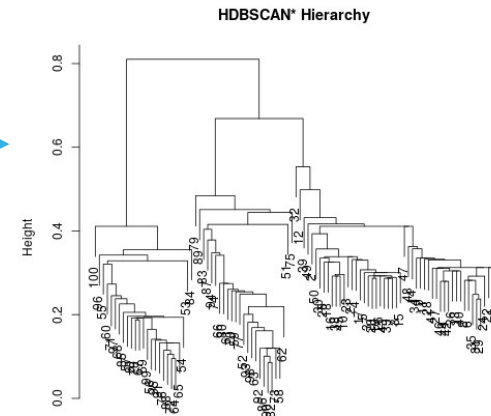
(Available e.g. in [R/CRAN – package ‘dbscan’](#))

```
myClust = hdbscan(x = data, minPts = 5)
plot(myClust$hc)
plot(myClust)
Plot(myClust, show_flat = T)
```

Visualization: Hierarchies

(Available e.g. in [R/CRAN – package ‘dbscan’](#))

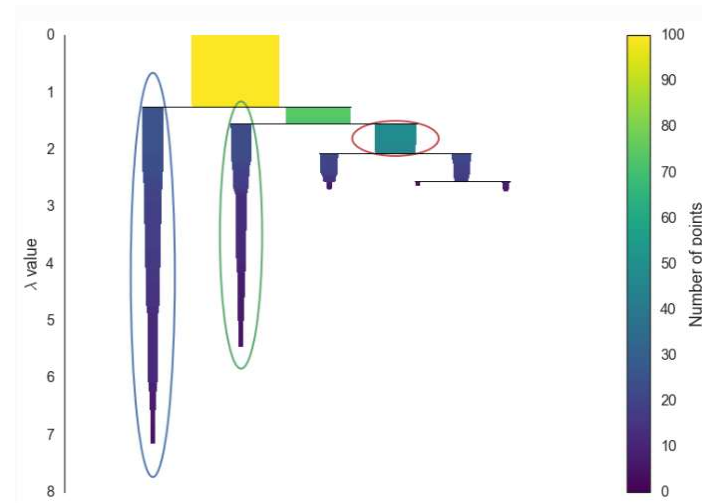
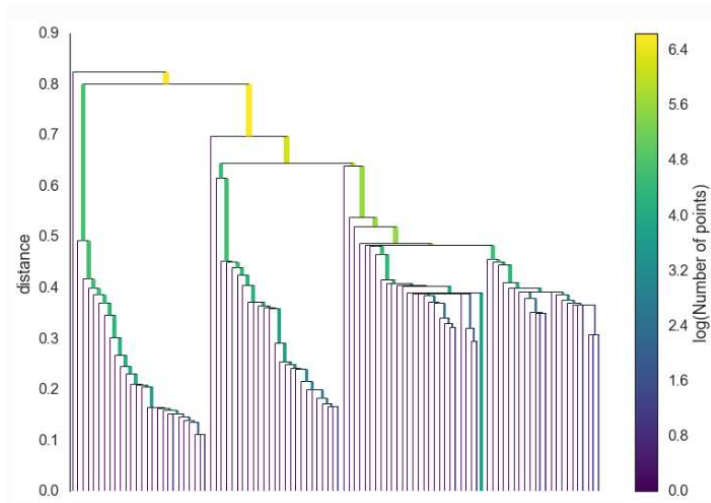
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myClust = hdbscan(x = data, minPts = 5)
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plot(myClust)
Plot(myClust, show_flat = T)
```



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Visualization: Hierarchies

(Available e.g. in [Python – Scikit-Learn](#))



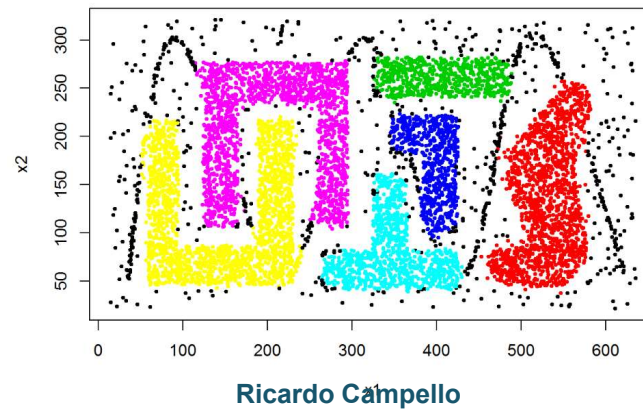
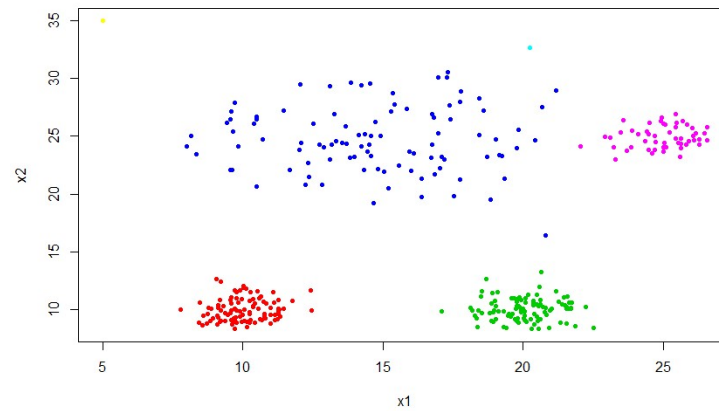
SciPy 2016 YouTube video by the creator of the Python package/notebook

<https://www.youtube.com/watch?v=AgPQ76Rli6A>

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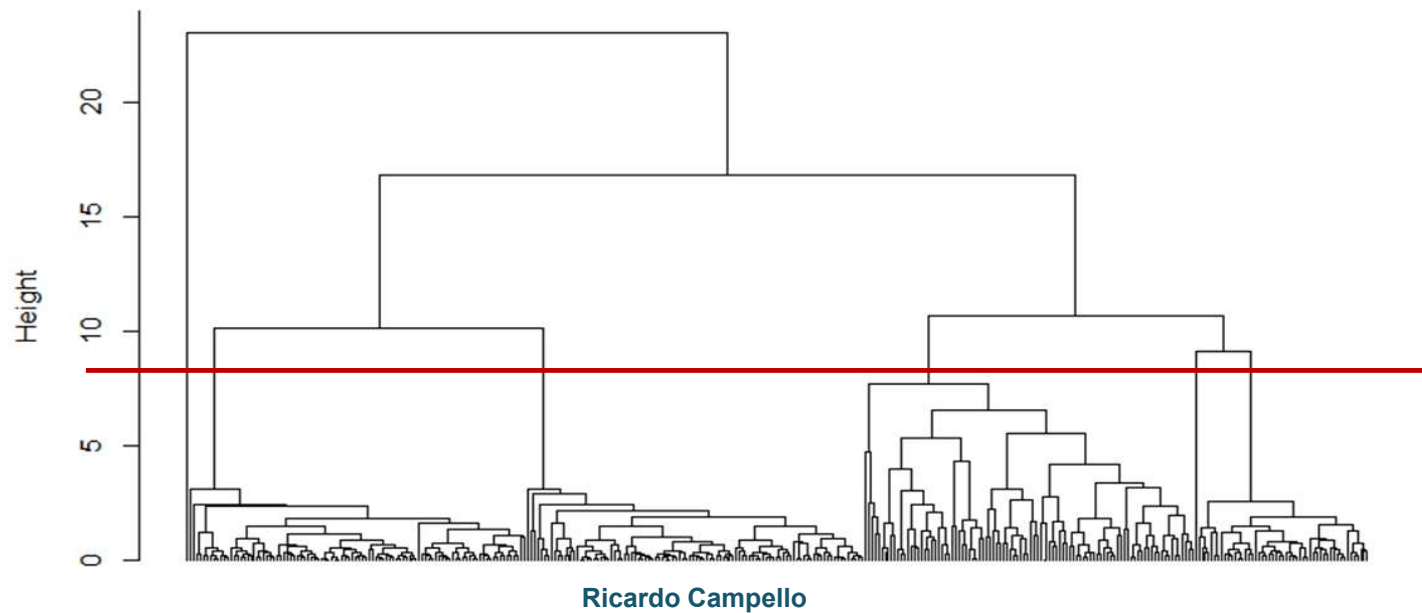
General Problem of Interest

How to automatically extract a “flat” clustering solution?



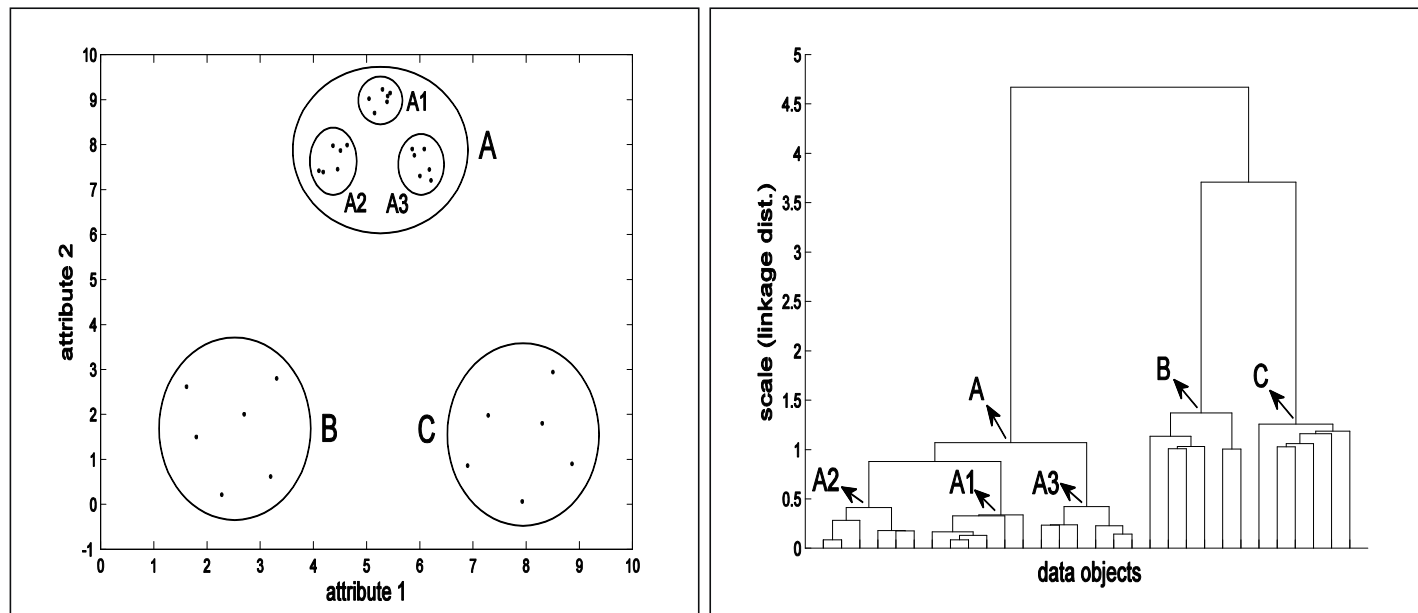
Traditional Approach

- ❑ A horizontal cut
- ❑ Choosing such a (**global**) cut level has been studied for decades



Limitation of the Traditional Approach

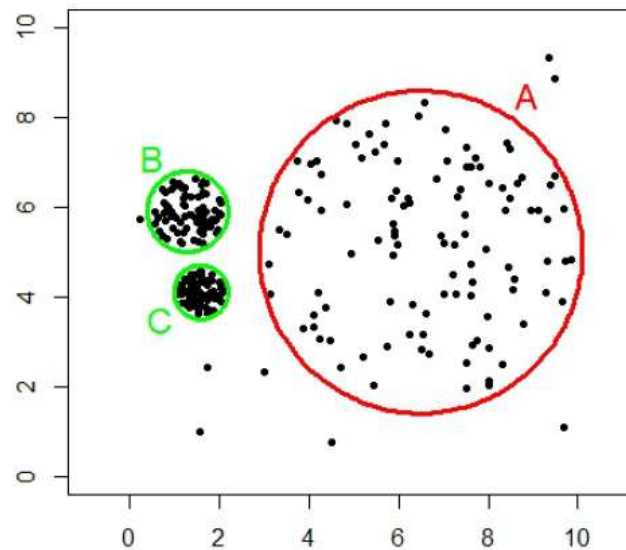
- ❑ It explores only a very limited subset of the valid candidate solutions!!!



- ❑ A1, A2, A3, B, and C would also be a viable candidate...

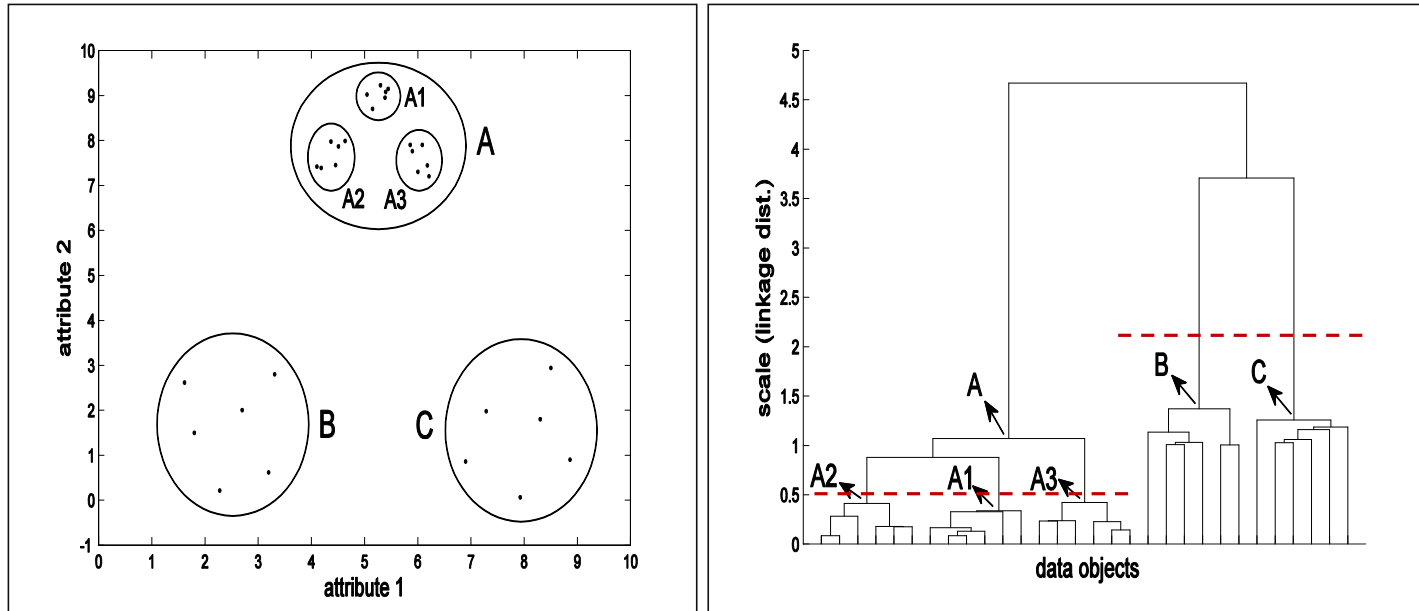
Limitation of the Traditional Approach

- ❑ Well-known issue in density-based clustering...
- ❑ But the connection with the hierarchy cut problem was long overlooked



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Better Approach: Local Cuts



- ❑ Surprisingly, long overlooked both in practice as well as in the literature

HDBSCAN* USES FOSC

Data Min Knowl Disc (2013) 27:344–371
DOI 10.1007/s10618-013-0311-4

A framework for semi-supervised and unsupervised optimal extraction of clusters from hierarchies

R. J. G. B. Campello · D. Moulavi · A. Zimek ·
J. Sander

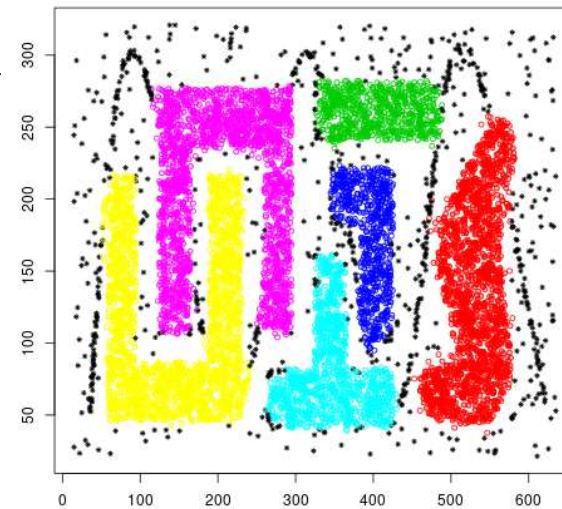
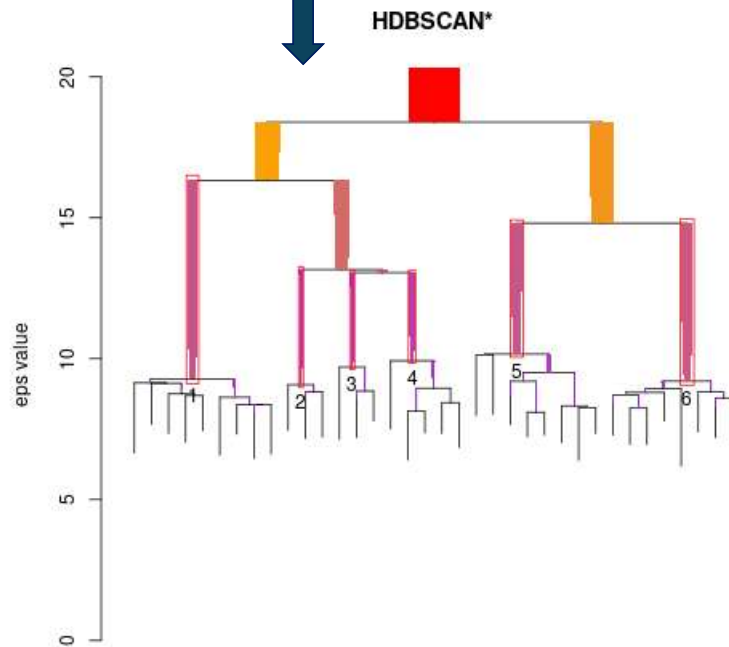
Received: 3 October 2012 / Accepted: 21 March 2013 / Published online: 4 April 2013
© The Author(s) 2013

Abstract We introduce a framework for the optimal extraction of flat clusterings from local cuts through cluster hierarchies. The extraction of a flat clustering from a cluster tree is formulated as an optimization problem and a linear complexity algorithm is pre-

Cluster Extraction Example

([R/CRAN – package ‘dbscan’](#))

```
myClust = hdbscan(x = DS3, minPts = 25)  
plot(DS3, col = color_fun(myClust$cluster))  
Plot(myClust, show_flat = T)
```

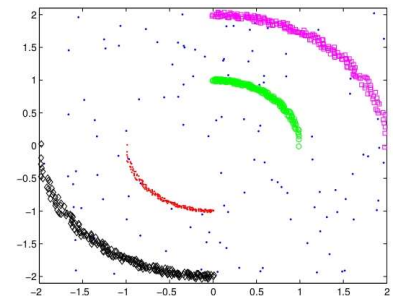


Suggested Reading

- Section 15.1 (DBSCAN):
 - M. J. Zaki and W. Meira Jr., “Data Mining and Machine Learning. Fundamental Concepts and Algorithms”. Cambridge University Press, 2nd edition, 2020
available at https://dataminingbook.info/book_html/

Summary of Density-Based Clustering

- Density-based clustering is a popular clustering paradigm
 - Statistically sound (Hartigan's model)
 - Clusters can have arbitrary shapes and sizes
 - Clusters are robust to and can be separated from noise
 - Allows for hierarchical and/or partitioning clustering, visualization, and outlier detection



Main References

- Tan, P.-N., Steinbach, M., and Kumar, V. (2006). *Introduction to Data Mining*, Addison-Wesley
- M. J. Zaki and W. Meira Jr., “Data Mining and Machine Learning. Fundamental Concepts and Algorithms”. Cambridge University Press, 2nd edition, 2020
- J. A. Hartigan (1975). *Clustering Algorithms*, J. Wiley & Sons
- M. Ester, H.-P. Kriegel, J. Sander, & X. Xu (1996). *A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise*, KDD, 226–231
- R. J. G. B. Campello, P. Kröger, J. Sander, & A. Zimek (2020). *Density-Based Clustering*, Wiley
Interdisciplinary Reviews: Data Mining and Knowledge Discovery, 10(2), e1343
- R. J. G. B. Campello, D. Moulavi, A. Zimek, & J. Sander (2015). *Hierarchical Density Estimates for Data Clustering, Visualization, and Outlier Detection*, ACM Trans. on Knowledge Discovery from Data, V. 10, 1-51
- R. J. G. B. Campello, D. Moulavi, A. Zimek, & J. Sander (2013). *A Framework for Semi-Supervised and Unsupervised Optimal Extraction of Clusters from Hierarchies*, Data Mining and Knowledge Discovery, V. 27, 344-371